Internal Use Only

MOBILE PHONE
SERVICE MANUAL

CAUTION
BEFORE SERVICING THE UNIT, READ THE "SAFETY PRECAUTIONS" IN THIS MANUAL

MODEL : LG-K520

Date: Apr. , 2016 / Issue 1.0

Table Of Contents

1. INTRODUCTION

1.1 Purpose

This manual provides the information necessary to repair, calibration, description and download the features of this model.

1.2 Regulatory Information

A. Security

This material is prohibited to share and release to unauthorized person, in accordance with the regulations, LG Electronics, Civil / criminal responsibility in accordance with the relevant provisions violate.

B. Precautions for repair

- In case of Disassembly or Assembly to repair product, be careful of a product failure caused by RF signals and Static electricity.
- When using Magnetic tool for the Phone's SVC repair, you should check affect the Electric parts according to effect of Magnet.
- When fastening the screw, be careful not to damage the head of screw and even product.

C. Attention

Boards, which contain Electrostatic Sensitive Device (ESD), are indicated by the  **sign.**

Following information is ESD handling:

- Service personal should ground themselves by using a wrist, strap when exchange system board.
- When repair are made to a system board, they should spread the floor with anti-static mat which is also grounded.
- Use a suitable, grounded soldering iron.
- Keep sensitive parts in these protective packages until these are used.
- When returning system board or parts like EEPROM to the Factory, use the protective package as described.

2.1 Band Specification

Support Band	TX Freq (MHz)	RX Freq (MHz)
WCDMA(FDD1)	1920 – 1980	2110 – 2170
WCDMA(FDD2)	1850-1910	1930-1990
WCDMA(FDD5)	824 – 849	869 – 894
WCDMA(FDD8)	880 – 915	925 – 960
EGSM	880 – 915	925 – 960
GSM850	824 – 849	869 – 894
DCS1800	1710 – 1785	1805 – 1880
PCS1900	1850 – 1910	1930 – 1990
LTE B1	1920 – 1980	2110 – 2170
LTE B3	1710 – 1785	1805 – 1880
LTE B7	2500 – 2570	2620 – 2690
LTE B8	880 – 915	925 – 960
LTE B20	832 – 862	791 – 821

2.2 HW Features

List	Type / Spec.	
1. Phone Type	DOP Type	
2. Size	155 x 79.6 x 7.4mm	
3. Weight	145 g (with Battery)	
4. Battery	**3000mAh(Typ) (Li-Ion)**, 2900mAh(Min.)	
5.Chipset	MSM8916 1.2GHz Quad core	
6. Memory	16GB(EMMC) + 1.5GB(LPDDR) External Memory(SD Card) : Up to 128GB	
7. LCD	Size	5.7 inch
	Display Type	IPS incell
	Color	16.7M colors
	Resolution	720(H) X 1280(V), 258ppi
8. Touch	Type	In-cell Capacitive type
9. Main Camera (13M)	Type	CMOS image sensor
	Resolution	4160x3120 pixels.
	Image Scaling Down	1:1 (3120x3120) 16:9 (4160x2340)
	Format	**Image :** JPG, **Video :** MP4

2.2 HW Features

10. VT Camera	Type	CMOS image sensor
	Resolution	3264x2448 pixels.
	Image Scaling Down	1:1 (2448x2448) 16:9 (3264x1836)
	Format	**Image :** JPG, **Video :** Normal - H.264/MP4, MMS - MPEG4/3GP
11. Audio	Receiver	11 X 06 X 2.65T Receiver
	Speaker	16 X 12 X 2.65T Speaker
	Format	WMA, MP3, AAC, Vorbis, AAC+, AMR, PCM, FLAC
12. Bluetooth	Standard	Bluetooth 4.1
	Effective Distance	10M
	Distance	0 m ~ 10 m (depend on environment)
123. WLAN	Standard	IEEE 802.11 b/g/n
	Throughput	up to 72 Mbps data rate
	Depend on environment	0 ~ 50m (depend on environment)
14. GPS	type	A-GPS
15. FM	type	FM Radio, 3.5pi Ear-jack

2.3 RSSI Display

Antenna BAR	Specification			Unit
	WCDMA(RSCP)	GSM(RSSI)	LTE (Unit : dBm/15khz)	
5→4	- 87dBm± 3dB	- 90dBm± 3dB	-81dBm ~ -91dBm	dBm
4→3	- 93dBm± 3dB	- 96dBm± 3dB	-92dBm ~ -101dBm	
3→2	- 99dBm± 3dB	- 98dBm± 3dB	-102dBm ~ -111dBm	
2→1	- 103dBm± 3dB	- 102dBm± 3dB	-112dBm ~ -121dBm	
1→0	- 109dBm± 3dB	- 104dBm± 3dB	-122dBm ~ -133dBm	

2.4 Current consumption

Mode	Specification		
	LTE	WCDMA	GSM
Sleep Mode	7mA	7mA	7mA
Sleep + Earjack	8mA	8mA	8mA
Standby (sleep+idle)	10mA @ DRX 2.56sec	10mA @ DRX 7	10mA @ PP5
Standby with BT connected	11mA	11mA	11mA
Talk (LTE=Connection)	300 @ TX 10dBm	700mA @ Max pwr 300mA @ TX 10dBm	400mA @ GSM LVL 5
No SVC	300mA	300mA	300mA
Power Off Mode	0.3mA		

2.5 Battery bar

Battery Bar	Specification	Battery Bar	Specification
Bar 20(Full)	98% over	Bar 9 -> Bar 8	43% -> 42%
Bar 20 -> Bar 19	98% -> 97%	Bar 8 -> Bar 7	38% -> 37%
Bar 19 -> Bar 18	93% -> 92%	Bar 7 -> Bar 6	33% -> 32%
Bar 18 -> Bar 17	88% -> 87%	Bar 6 -> Bar 5	28% -> 27%
Bar 17 -> Bar 16	83% -> 82%	Bar 5 -> Bar 4	23% -> 22%
Bar 16 -> Bar 15	78% -> 77%	Bar 4 -> Bar 3	16% -> 15%
Bar 15 -> Bar 14	73% -> 72%	Bar 3 -> Bar 2	13% -> 12%
Bar 14 -> Bar 13	68% -> 67%	Bar 2 -> Bar 1	8% -> 7%
Bar 13 -> Bar 12	63% -> 62%	Bar 1 -> Bar 0	3% -> 2%
Bar 12 -> Bar 11	58% -> 57%	Power off	1% under
Bar 11 -> Bar 10	53% -> 52%	Power off voltage	3.35~3.55V
Bar 10 -> Bar 9	48% -> 47%	Low battery pop-up	15% , 5%

2.6 SW Specification

Item	Feature	Comment
RSSI	0 ~ 5 Levels	
Battery Charging	0 ~ 20 Levels	
Key Volume	0 ~ 7 Level	
Audio Volume	0 ~ 15 Level	
Time / Date Display	Yes	
Multi-Language	Yes	depending on build language
Quick Access Mode	Phone / Messaging / Browser/ Applications	Phone / Contact / Messaging / Applications
PC Sync	Yes	
Speed Dial	Yes	Voice mail center -> 1 key
Profile	Yes	not same with feature phone setting
CLIP / CLIR	Yes	
Phone Book	Name / Number / Email / Groups / Postal addresses / Organizations / IM / Note / Nickname / Website / Event /	There is no limitation on the number of items. It depends on available memory amount.
Last Dial Number	Yes	
Last Received Number	Yes	
Last Missed Number	Yes	
Search by Number/Name	Yes	
Group	Yes	There is no limitation on the number of items. It depends on available memory amount.
Fixed Dial Number	Yes	
Service Dial Number	No	
Own Number	Yes	My Profile (add/edit/delete are supported)

2.6 SW Specification

Voice Memo	Yes	Support voice recorder
Call Reminder	No	
Network Selection	Automatic	
Mute	Yes	
Call Divert	Yes	
Call Barring	Yes	
Call Charge (AoC)	No	
Call Duration	Yes	
SMS (EMS)	There is no limitation on the number of items . It depends on available memory amount.	EMS does not support.
SMS Over GPRS	Yes	
EMS Melody / Picture Send / Receive / Save	No	
MMS MPEG4 Send / Receive / Save	Yes	➢ **Send / Receive** : Yes ➢ **Save** : depends on content type Support video content type li st 1. video/mp4 2. video/h263 3. video/3gpp2 video/3gpp
Long Message	MAX 2000 characters	The standard of Open vender
Cell Broadcast	Yes	Supported CB app.
Download	Over the Web	
Game	No	
Calendar	Yes	
Memo	Yes	There is no limitation on the number o f items. It depends on available memory amo unt.
World Clock	Yes	

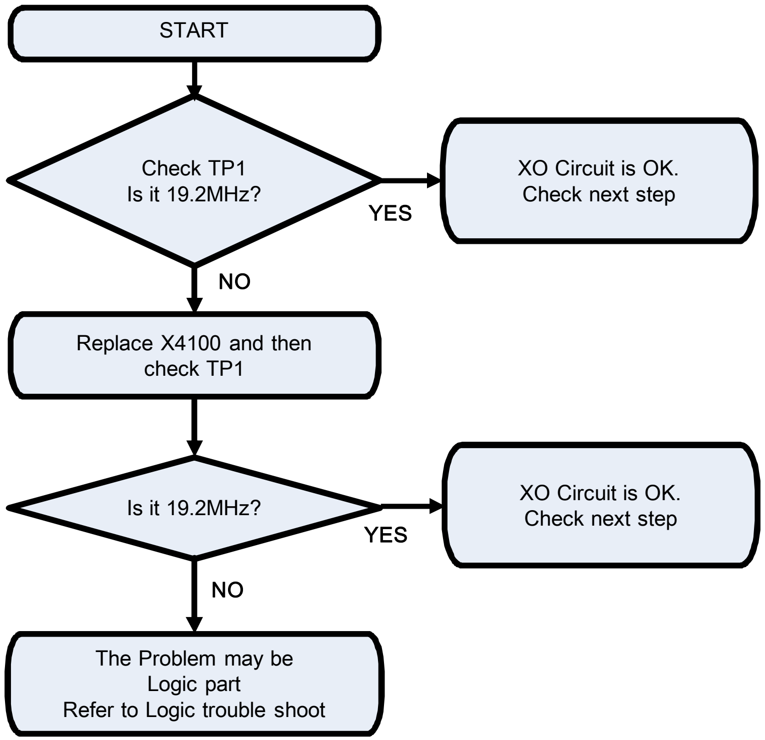
2.6 SW Specification

Unit Convert	No	
Stop Watch	Yes	
Wall Paper	Yes	
WAP Browser	No	Support only web browser based on webkit. WAP stack and wml are not supported.
Download Melody / Wallpaper	Yes	Over web browser
SIM Lock	No	
SIM Toolkit	Yes	
MMS	Yes	Google MMS Client
EONS	Yes	
CPHS	Yes	V4.2
ENS	No	
Camera	Yes	13M AF / Digital Zoom : x4
JAVA	No	
Voice Dial	No	US English only
IrDa	No	IrRC
Bluetooth	Yes	Ver. 4.0
FM radio	Yes	
GPRS	Yes	Class 33
EDGE	Yes	Class 33
Hold / Retrieve	Yes	
Conference Call	Yes	Max. 6
DTMF	Yes	
Memo pad	No	
TTY	No	
AMR	Yes	
SyncML	No	
IM	Yes	Google Hangout
Email	Yes	

3.1 Checking XO Block

The output frequency(19.2MHz) of XO(X4100) is used as the reference one of WTR4905 and PM8916 internal VCO

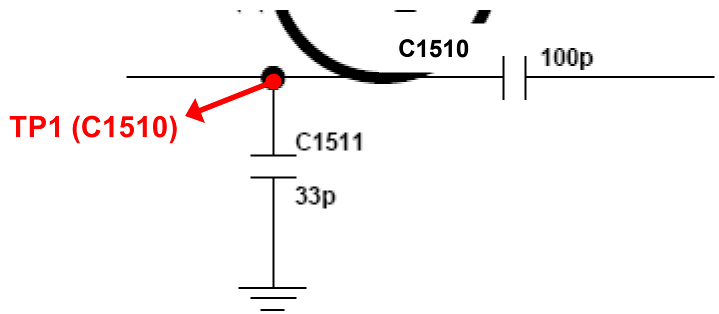
Checking Flow



Main Top

Image

Circuit Diagram



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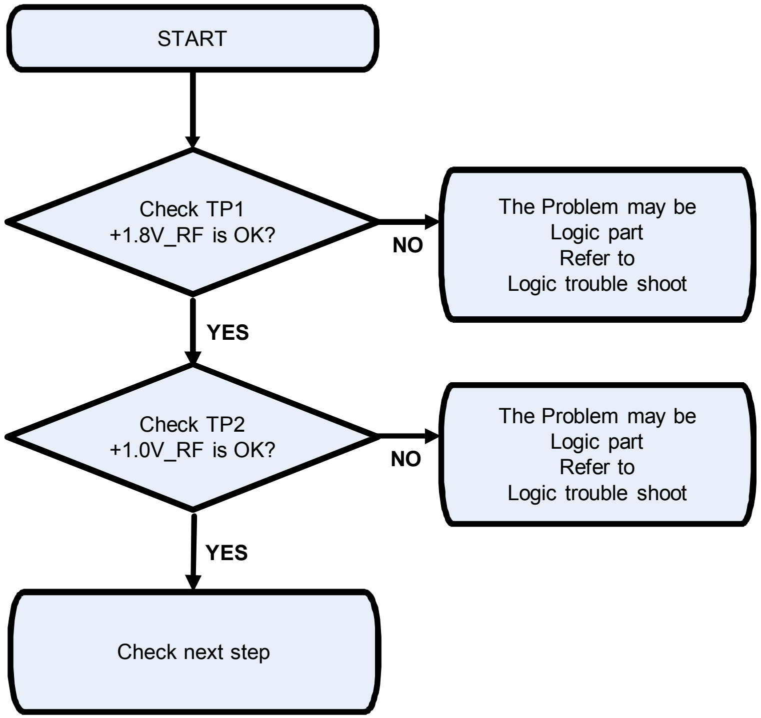
3.2 Transceiver DC Power Supply Circuit Block

The WTR4905 operating voltages used two voltage sources 1.0V and 1.8V

Checking Flow



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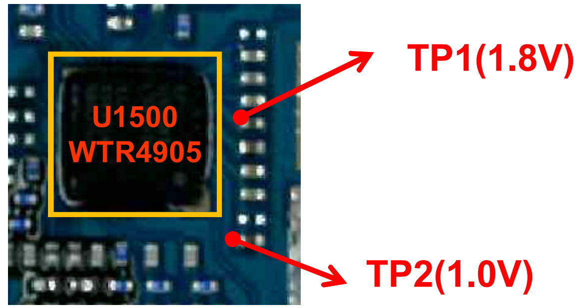
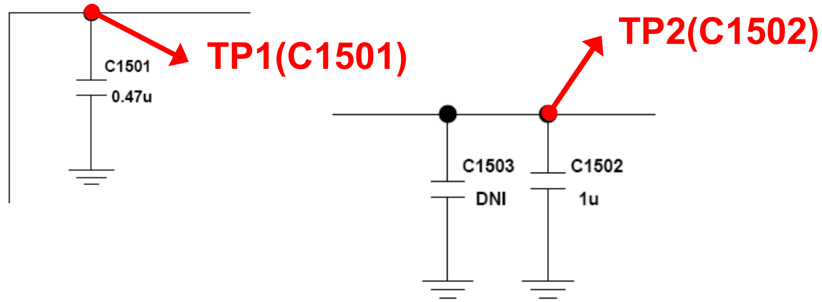


Circuit Diagram



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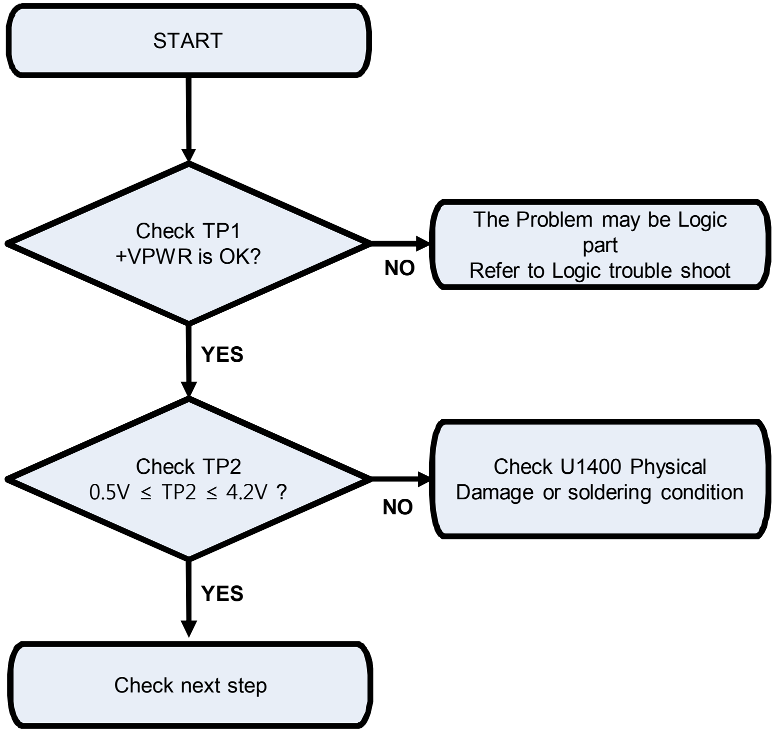
3.3 DC-DC Block

The DC-DC(SKY87006, U1400) output voltages is used as the reference one of WTR4905

Checking Flow



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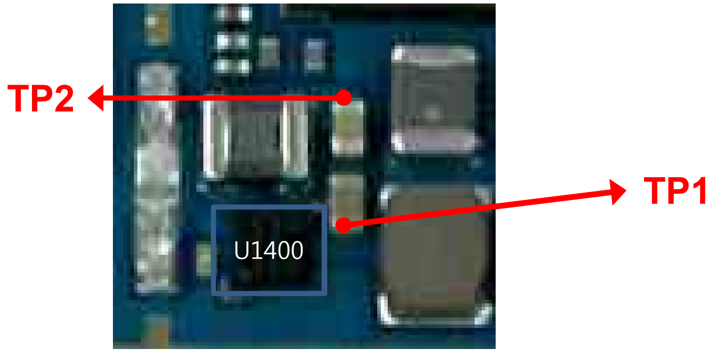
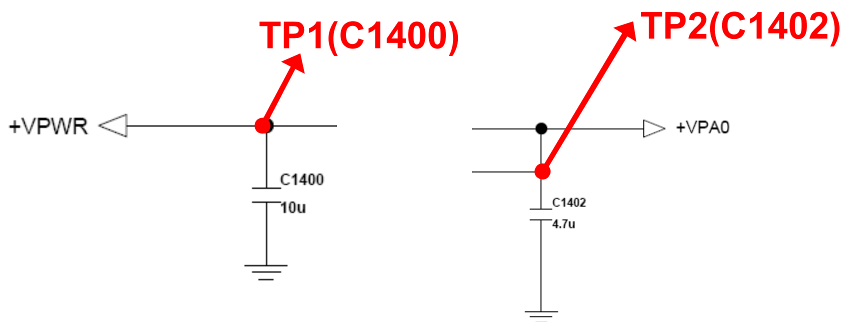


Circuit Diagram



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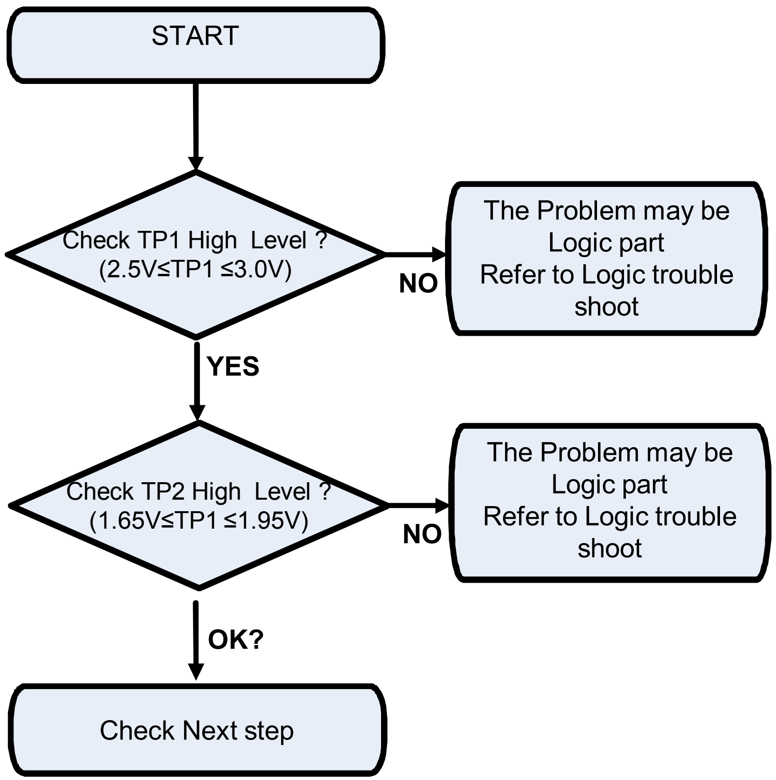
3.4 ASM Block

3.4.1 Checking ASM (EGSM/G850,W 850/W900, LTE B7/8/20) Block

Checking Flow



Image

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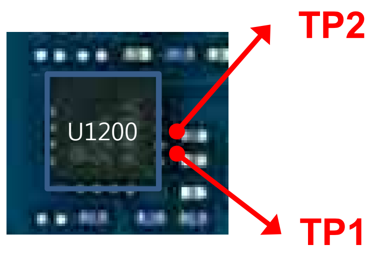
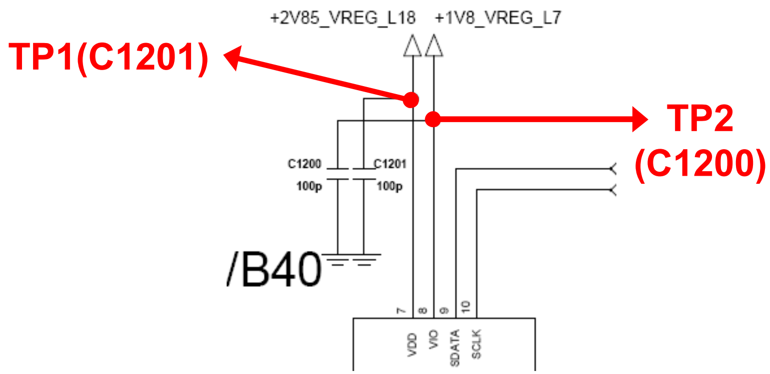


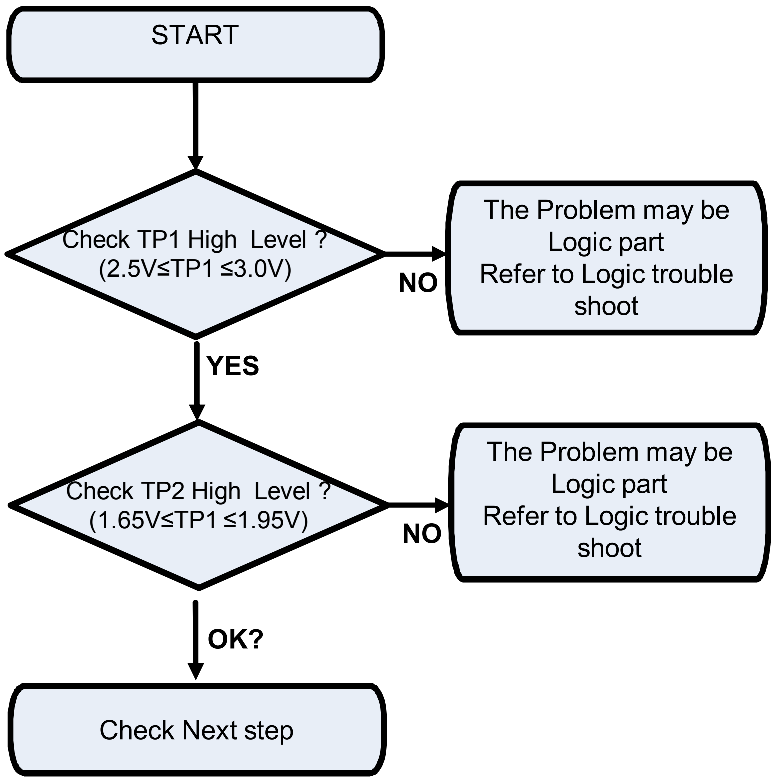
Circuit Diagram



3.4 ASM Block

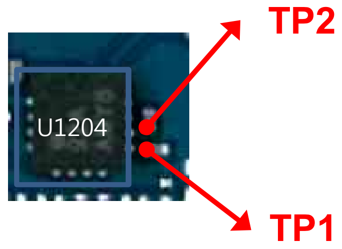
3.4.2 Checking ASM (G1800/G1900,W 2100/W1900, LTE B1/3) Block

Checking Flow

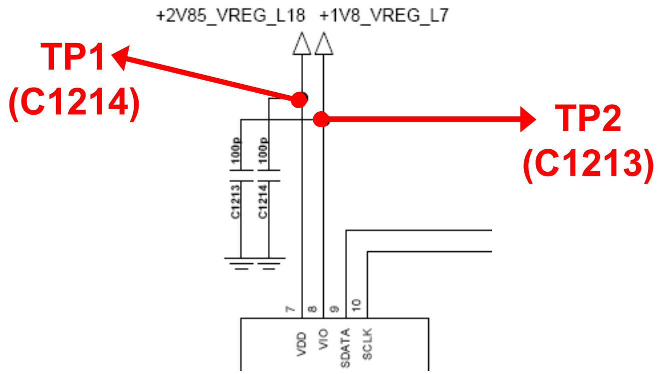


Image

Main Top



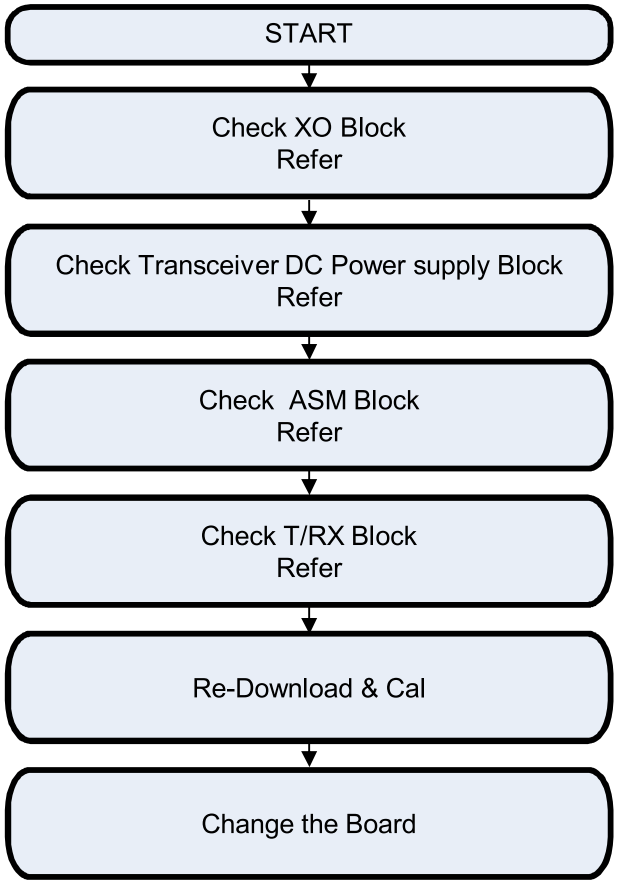
Circuit Diagram



3.5 GSM RF PART

GSM RF Part support GSM850/900/1800/1900

Checking Flow



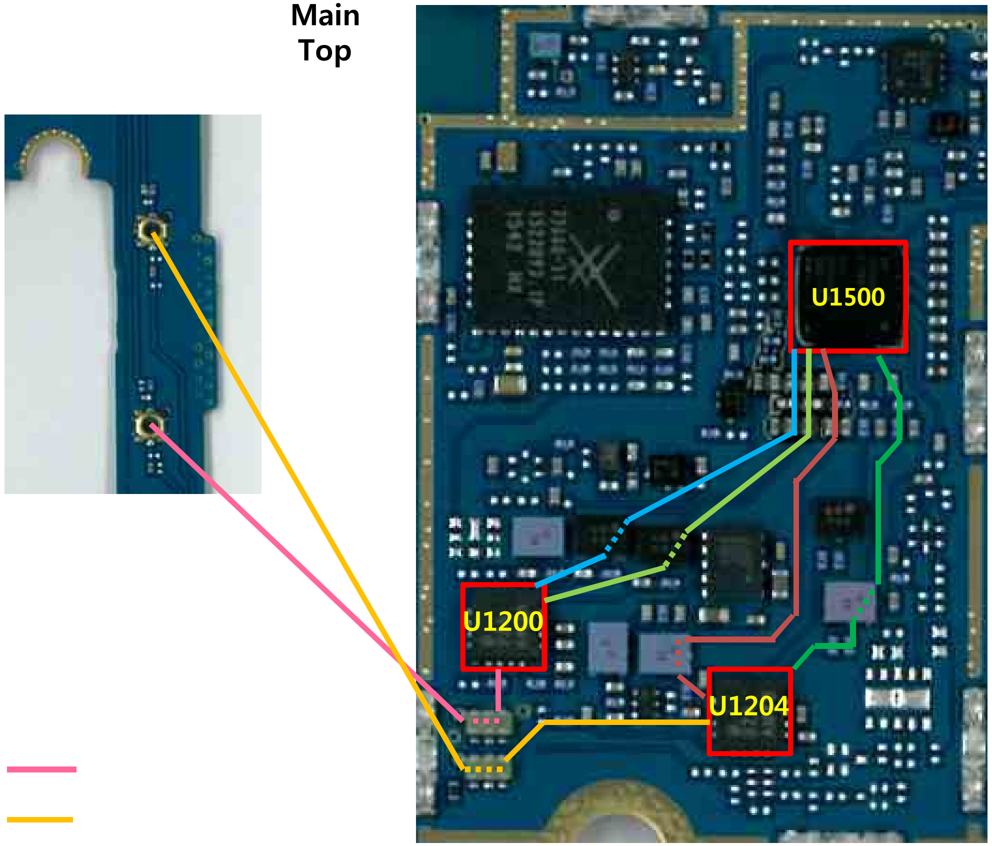
3.5 GSM RF PART

3.5.1 GSM RF Part RX RF PATH (GSM 850/900/1800/1900)

Image

Main BOT

Main Top



GSM 850 RX PATH
GSM 900 RX PATH
GSM 1800 RX PATH
GSM 1900 RX PATH
LOW BAND COMMON T/RX PATH
HIGH BAND COMMON T/RX PATH

3.5 GSM RF PART

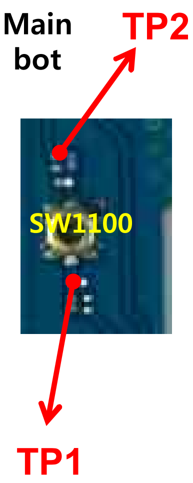
3.5.1.1 Checking RF Signal RX path(GSM 850)

Checking Flow

- START
- Check if there is over 3dB difference between TP1 and TP2
  - YES → Replace SW1100
  - NO ↓
- Check if there is over 3dB difference between TP2 and TP3
  - YES → Replace FL1116
  - NO ↓
- Check if there is over 3dB difference between TP3 and TP4
  - YES → Replace U1200
  - NO ↓
- Check if there is over 3dB difference between TP4 and TP5
  - YES → Replace FL1211
  - NO ↓
- Check Next step

Image

Main bot



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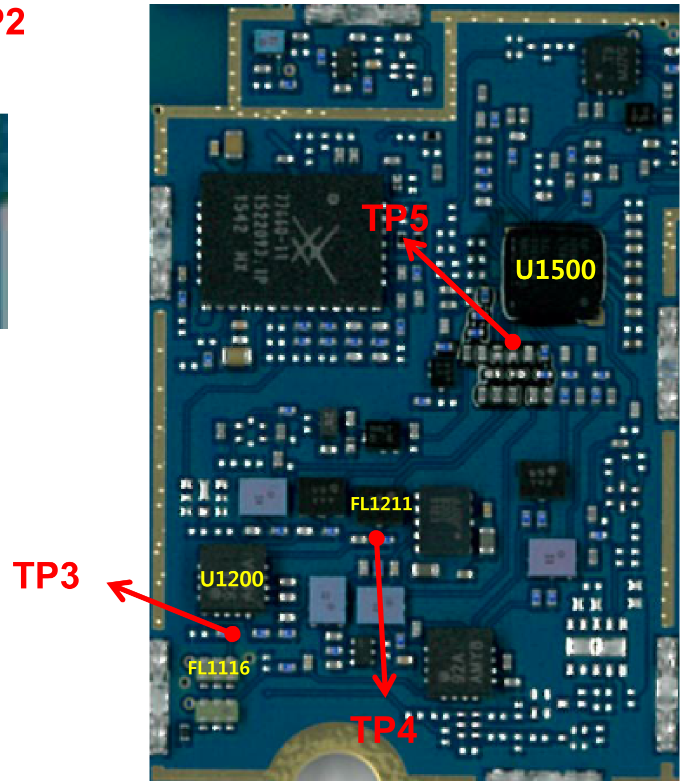
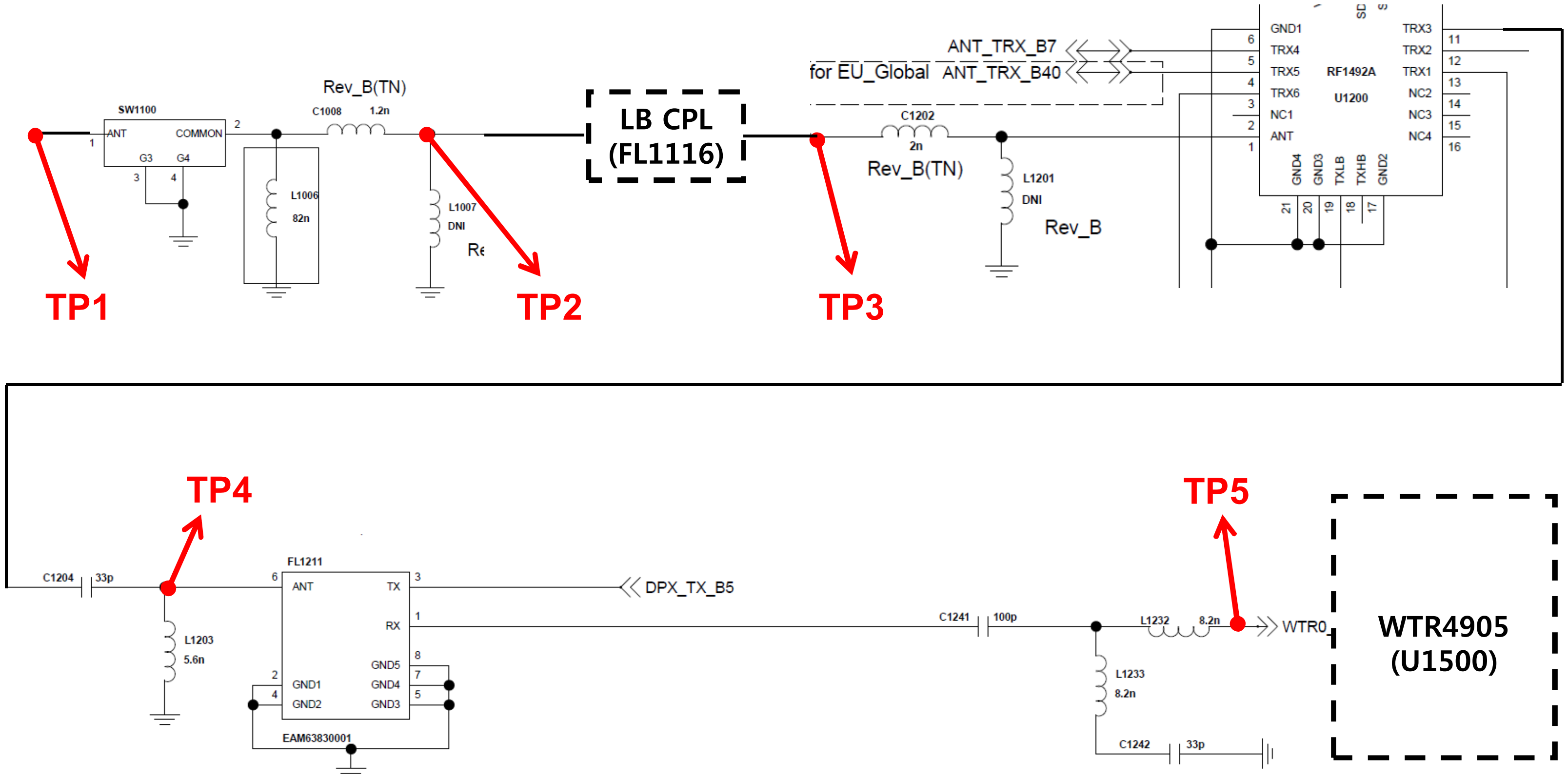


3.5 GSM RF PART

3.5.1.1 Checking RF Signal RX path(GSM 850)

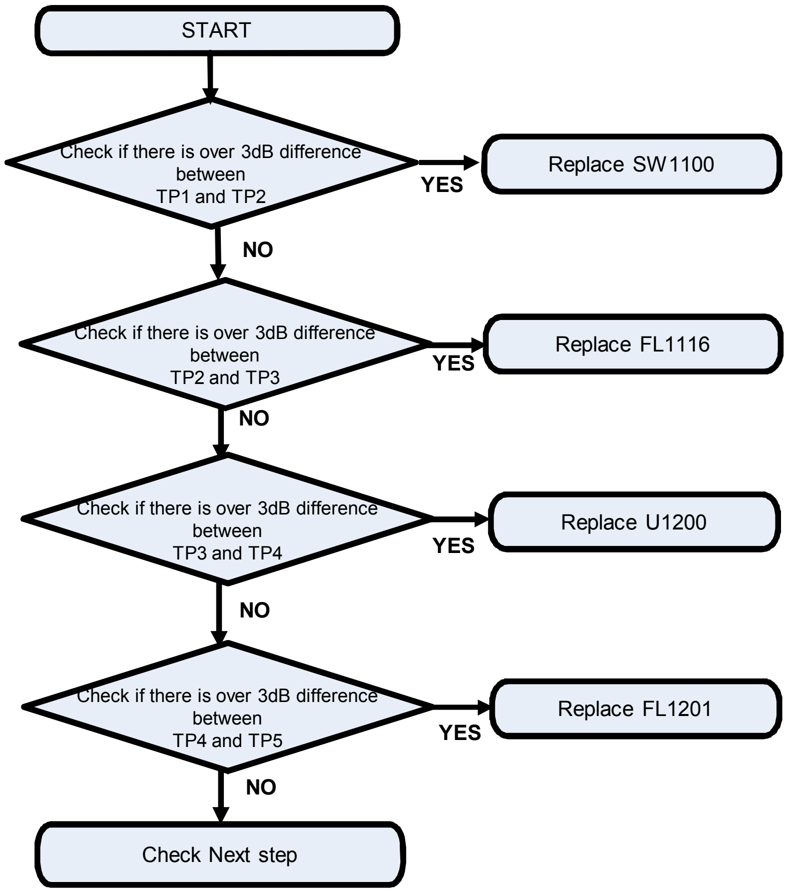
Circuit Diagram



3.5 GSM RF PART

3.5.1.2 Checking RF Signal RX path(GSM 900)

Checking Flow



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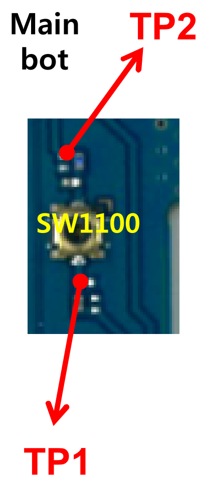
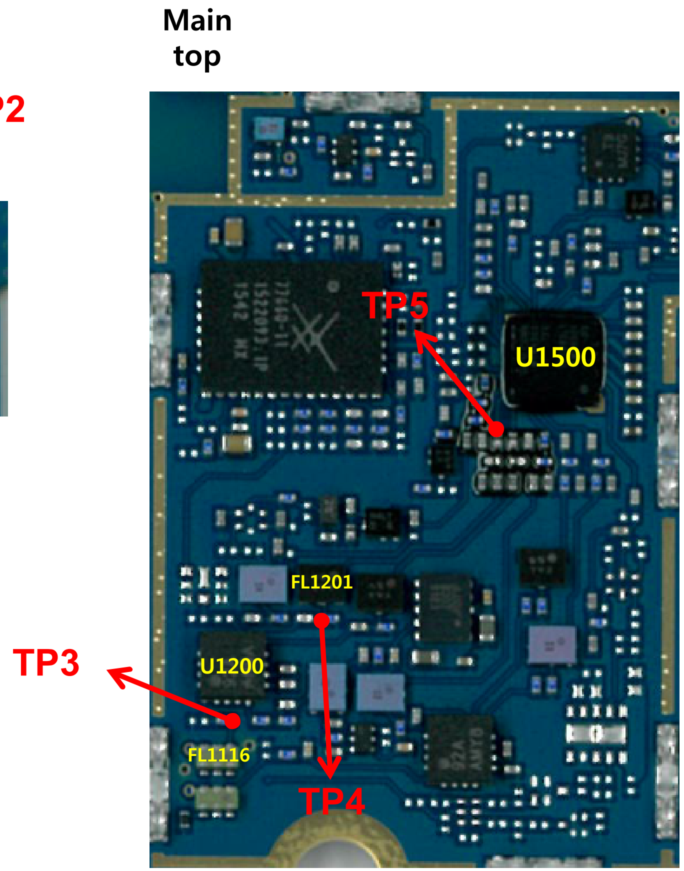
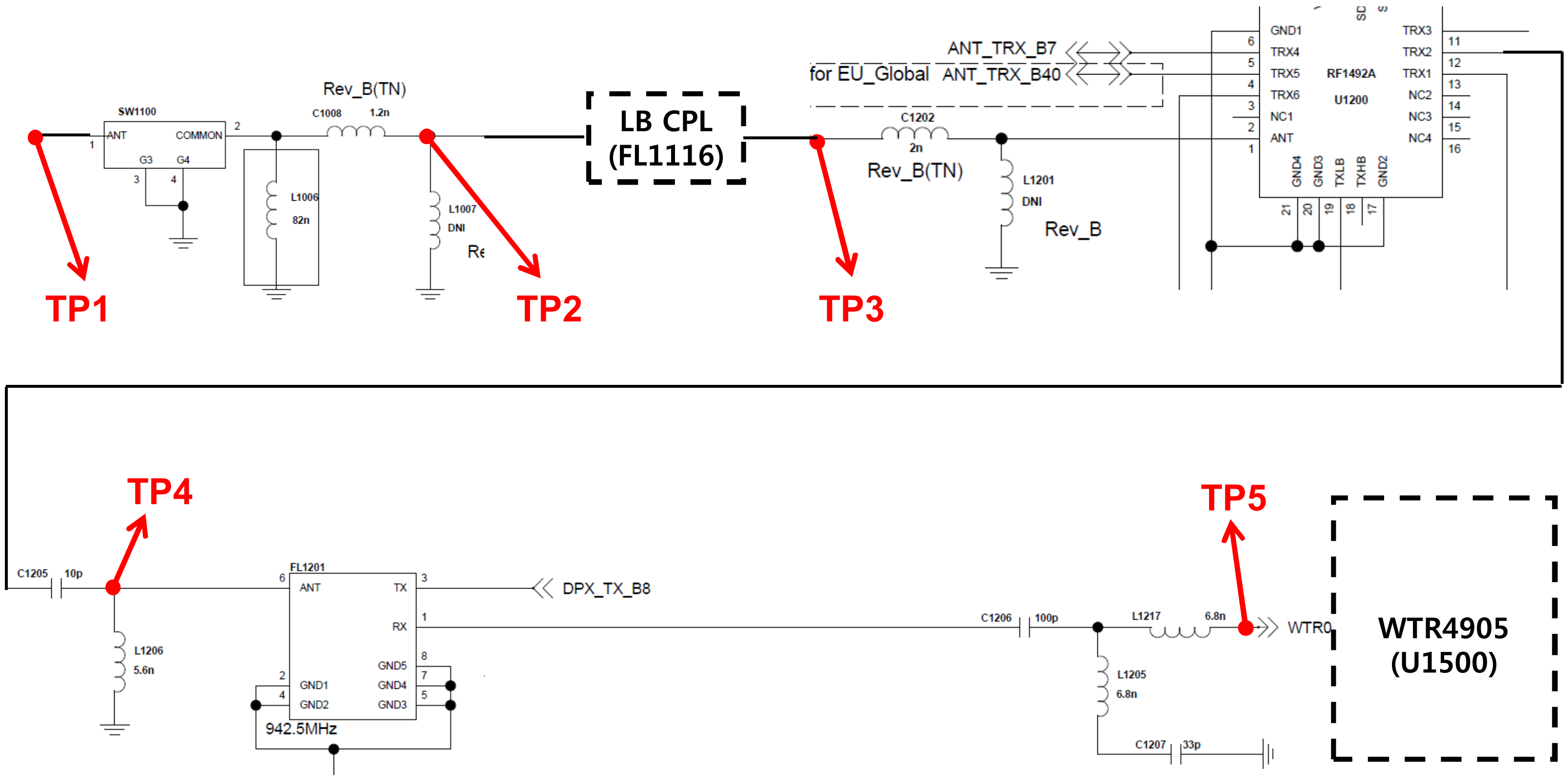




3.5 GSM RF PART

3.5.1.2 Checking RF Signal RX path(GSM 900)

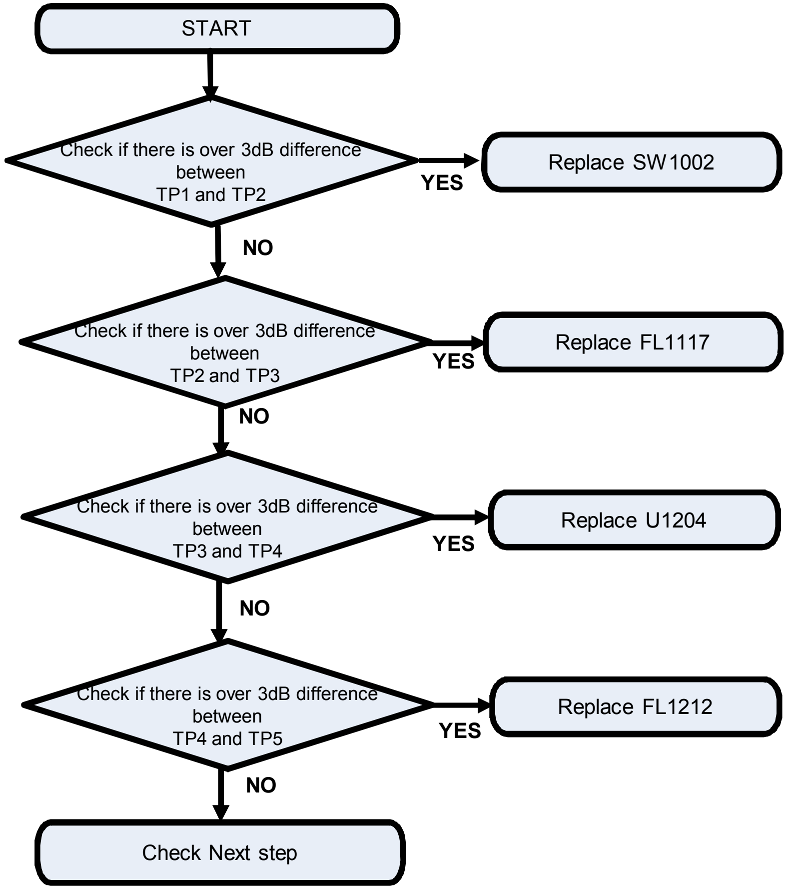
Circuit Diagram



3.5 GSM RF PART

3.5.1.3 Checking RF Signal RX path(GSM 1800)

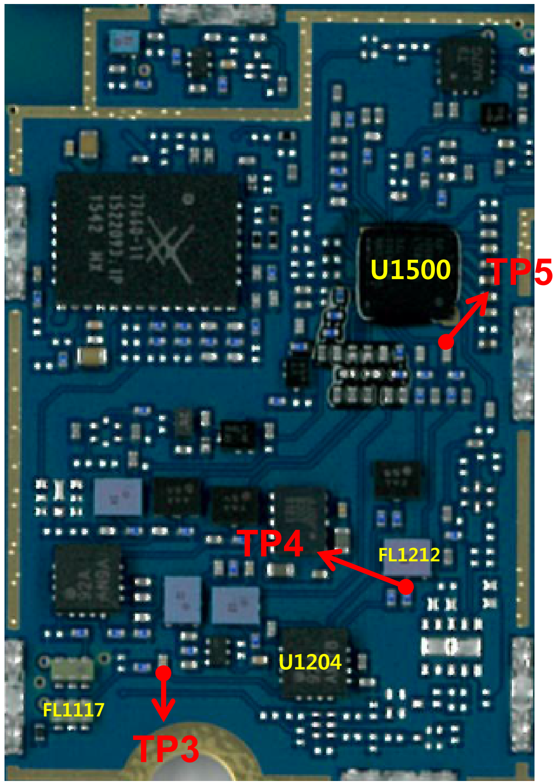
Checking Flow



Image

Main bot

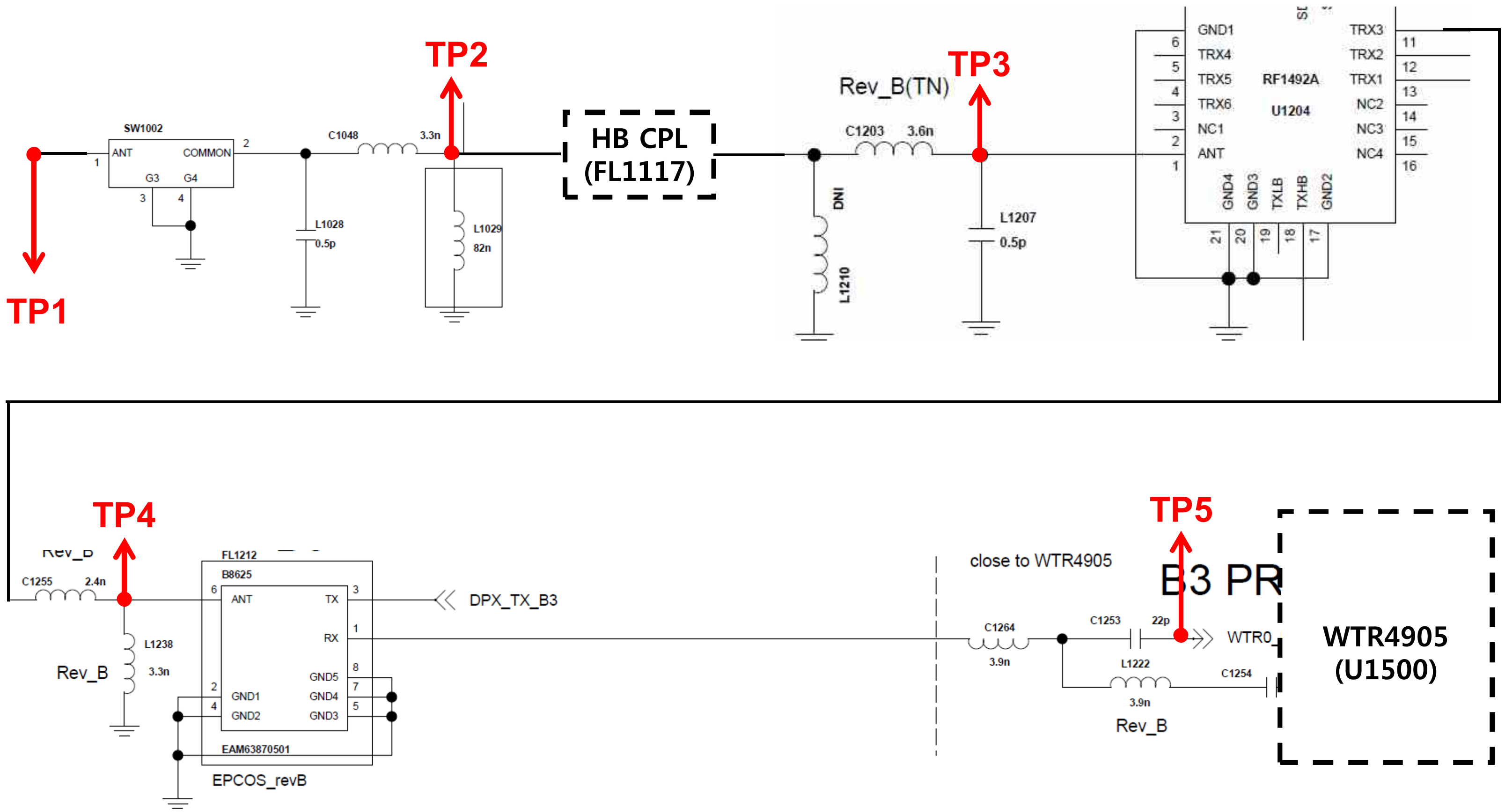
Main top



3.5 GSM RF PART

3.5.1.3 Checking RF Signal RX path(GSM 1800)

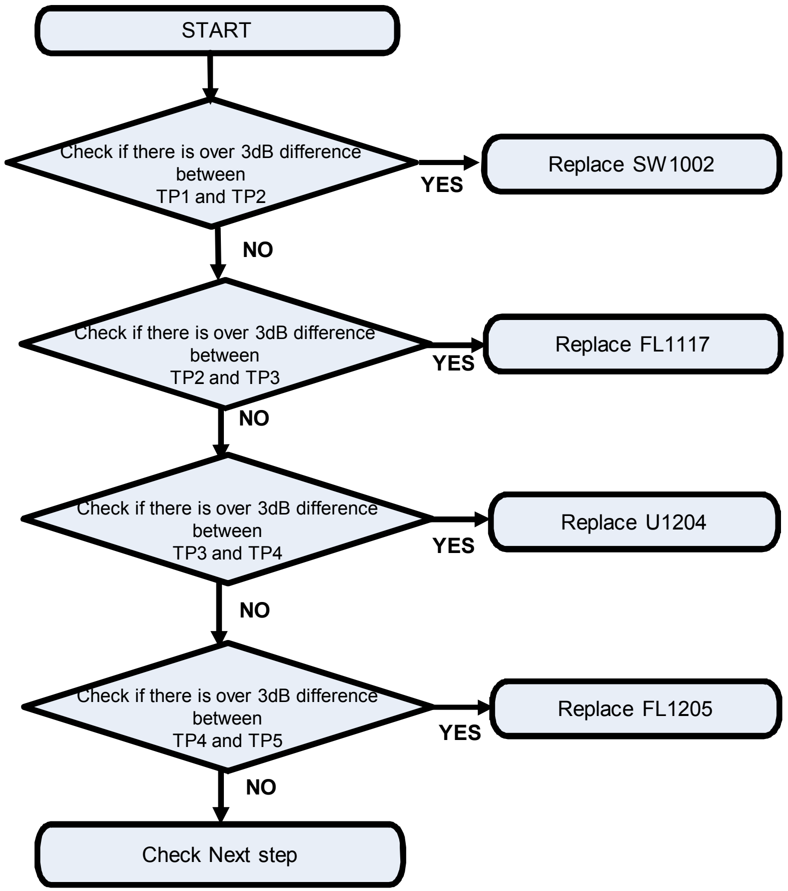
Circuit Diagram



3.5 GSM RF PART

3.5.1.4 Checking RF Signal RX path(GSM 1900)

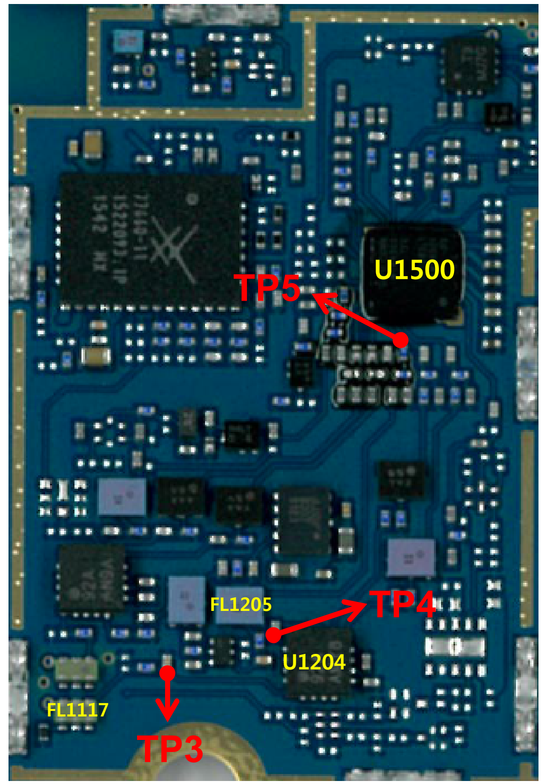
Checking Flow



Image

Main bot

Main top



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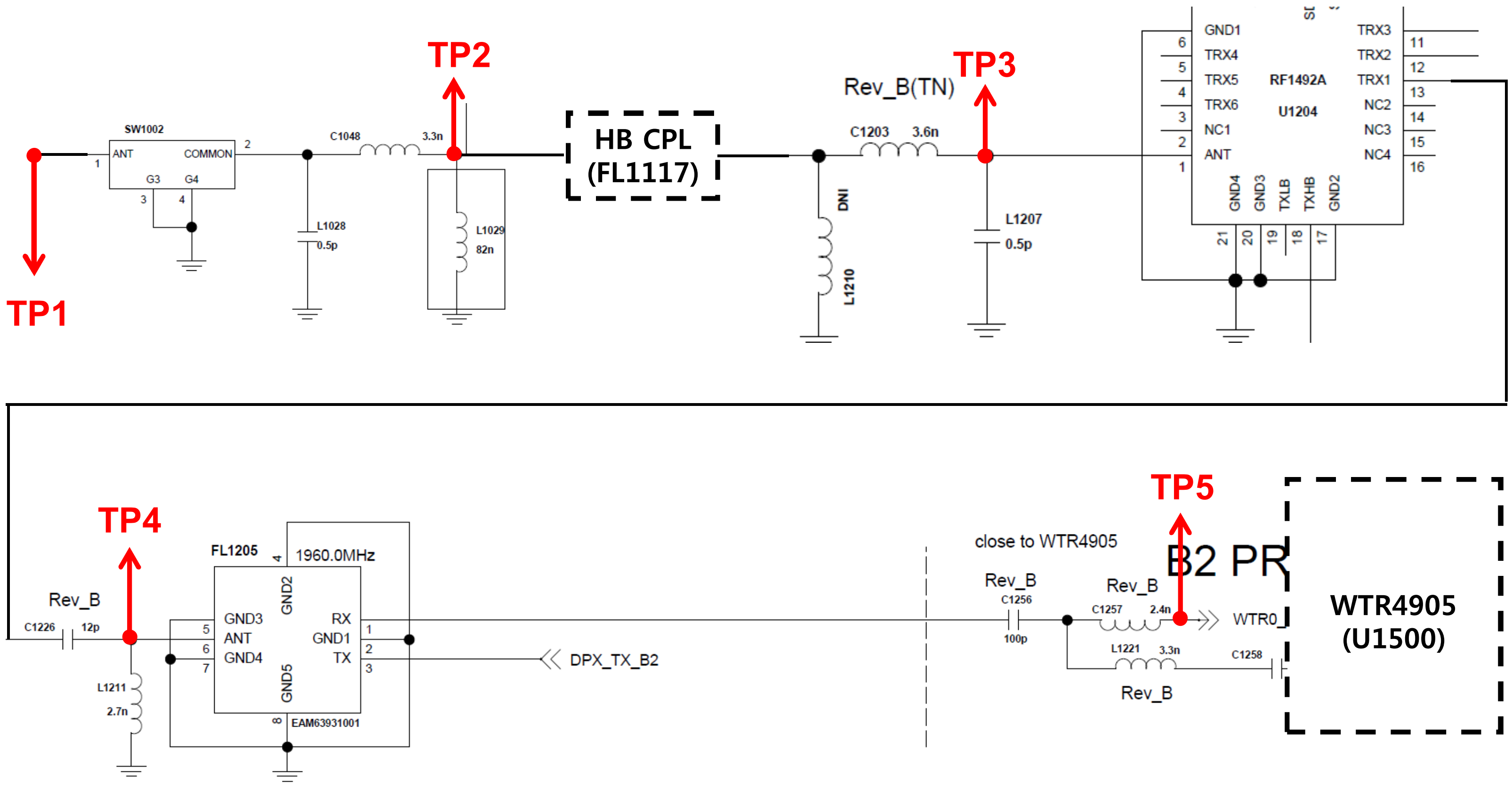
3.5 GSM RF PART

3.5.1.4 Checking RF Signal RX path(GSM 1900)

Circuit Diagram

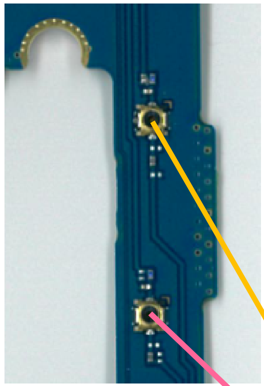


3.5 GSM RF PART

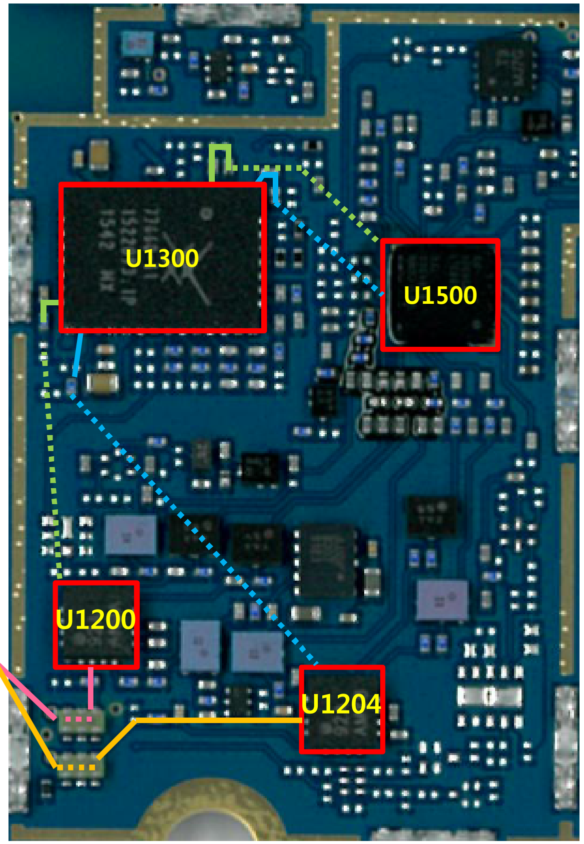
3.5.2 GSM RF Part TX RF PATH (GSM 850/900/1800/1900)

Image

Main BOT



Main Top



GSM 850/900 TX PATH

GSM 1800/1900 TX PATH

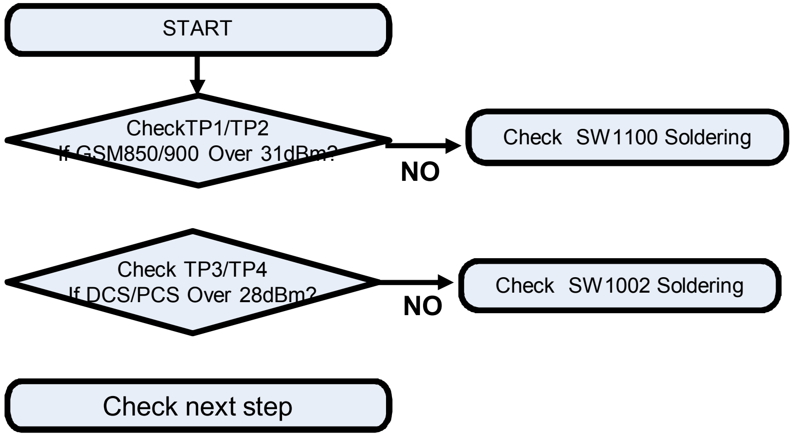
LOW BAND COMMON T/RX PATH

HIGH BAND COMMON T/RX PATH

3.5 GSM RF PART

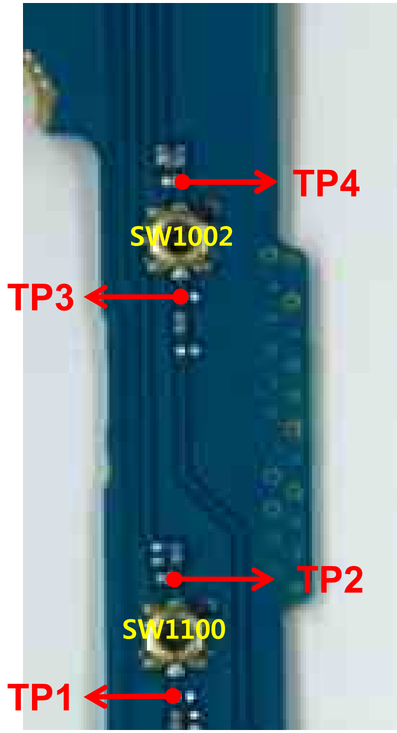
3.5.2.1 Checking RF Signal TX path (GSM 850/900/1800/1900)

Checking Flow

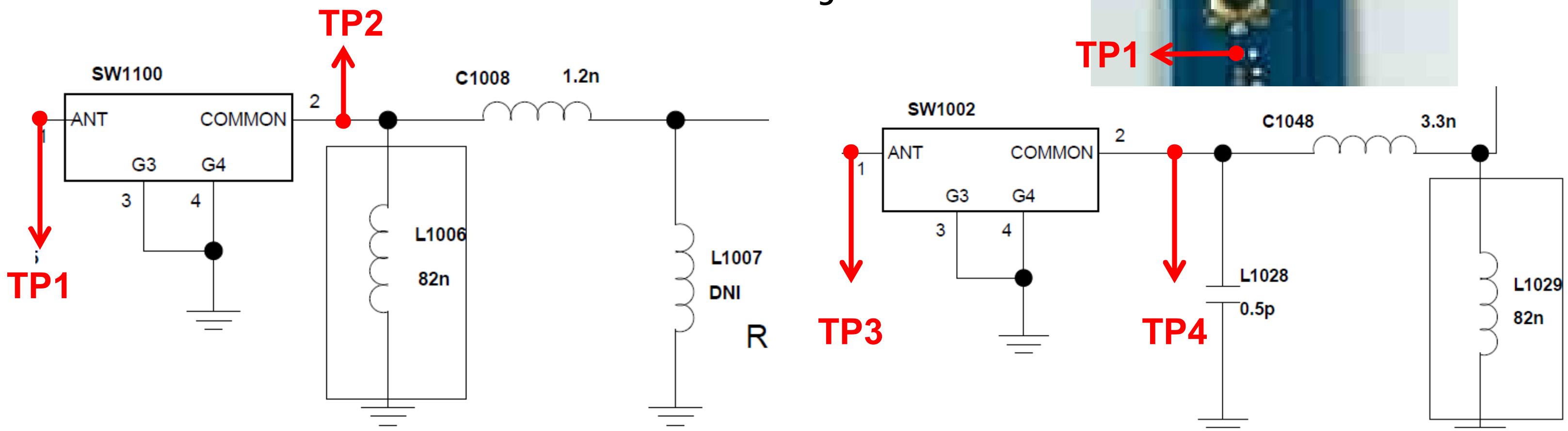


Image

Main bot



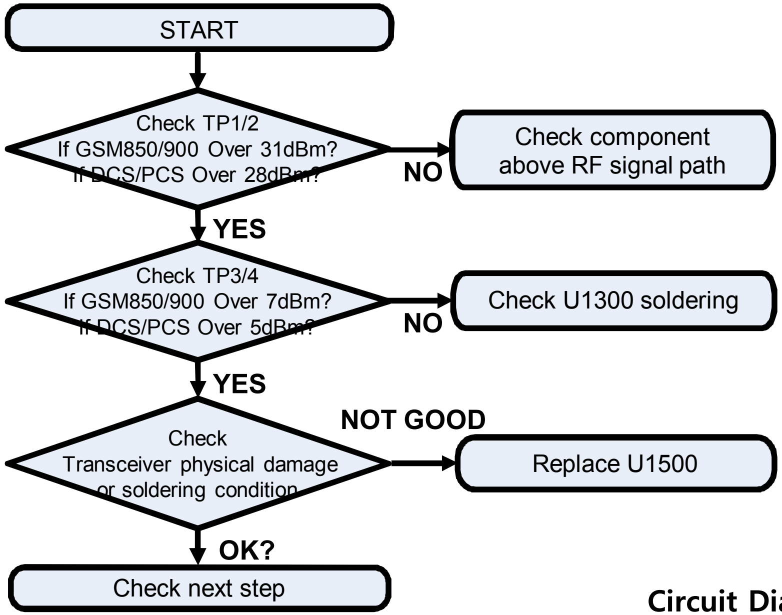
Circuit Diagram



3.5 GSM RF PART

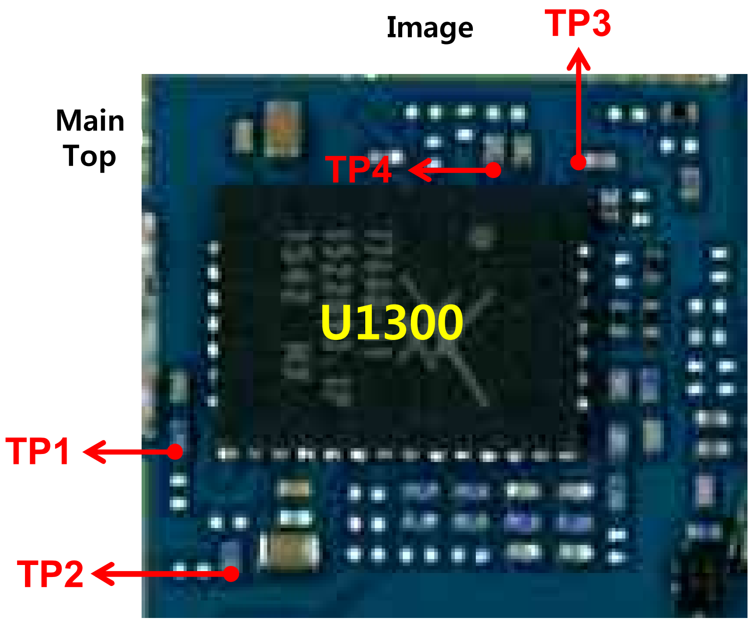
3.5.2.2 Checking RF Signal TX path (GSM 850/900/1800/1900)

Checking Flow

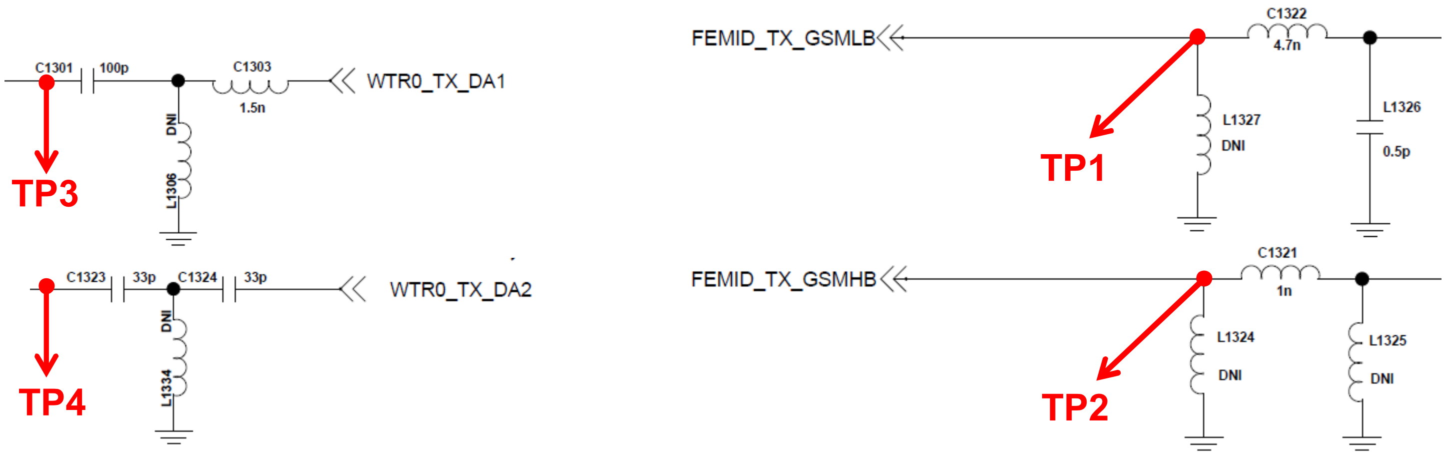


Image

Main
Top



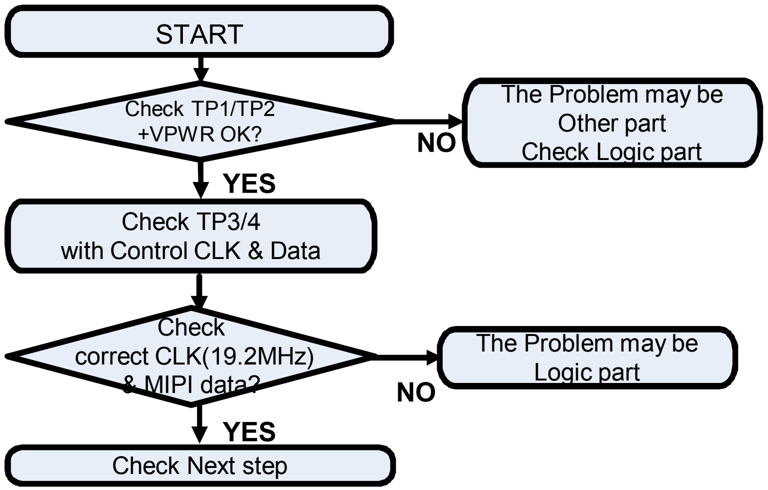
Circuit Diagram



3.5 GSM RF PART

3.5.3 Checking GSM PAM DC Power Circuit

Checking Flow



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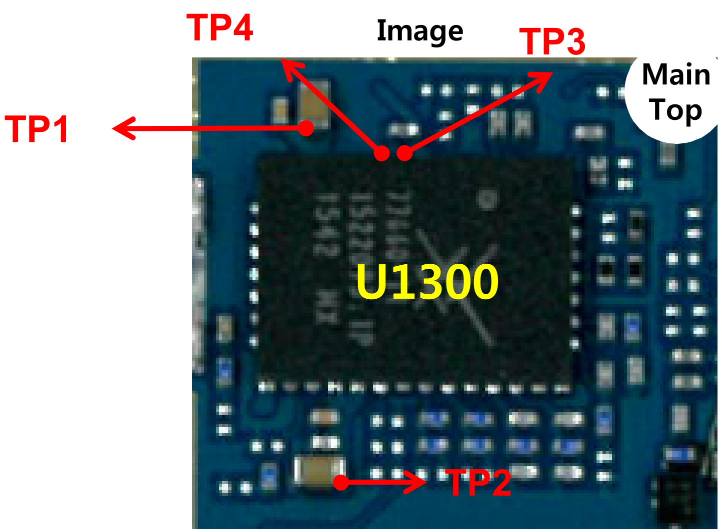
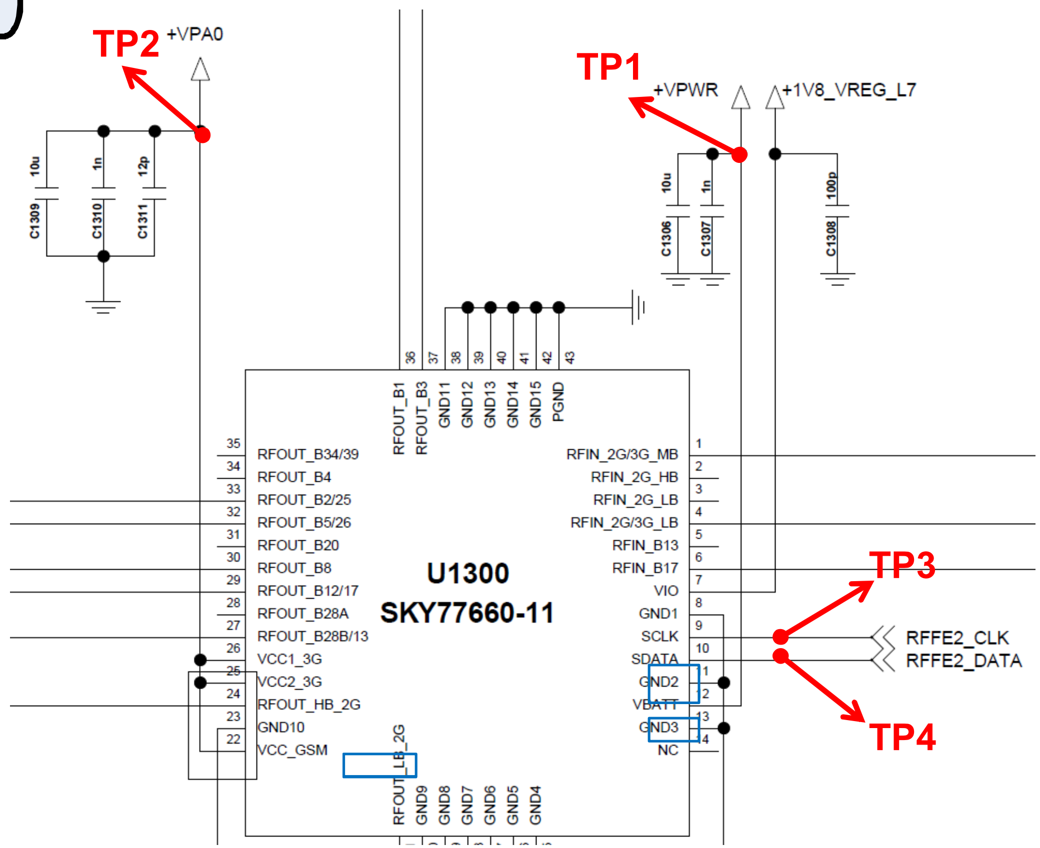


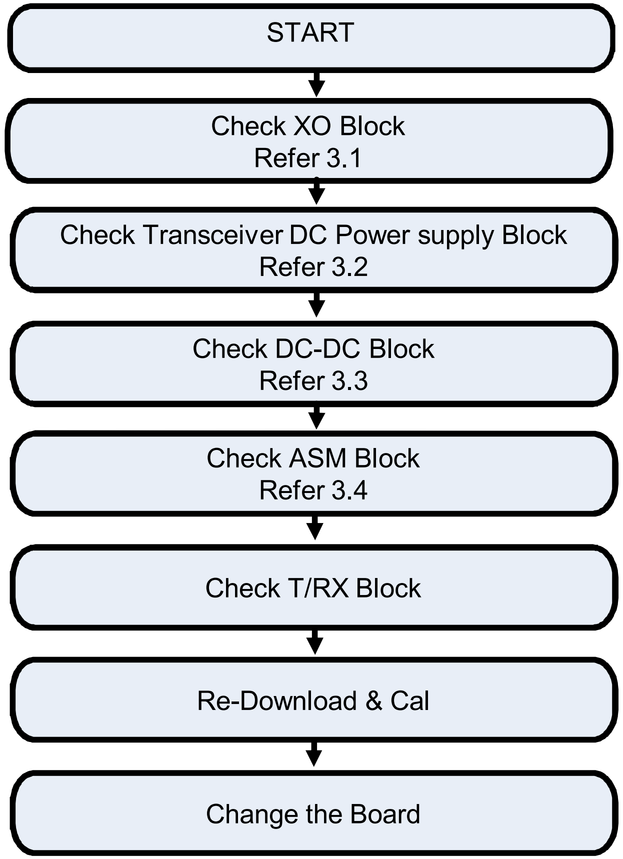
Circuit Diagram



3.6 WCDMA RF PART

WCDMA RF Part support WCDMA B1/2/5/8 with FEMiD, PAM, Transceiver component

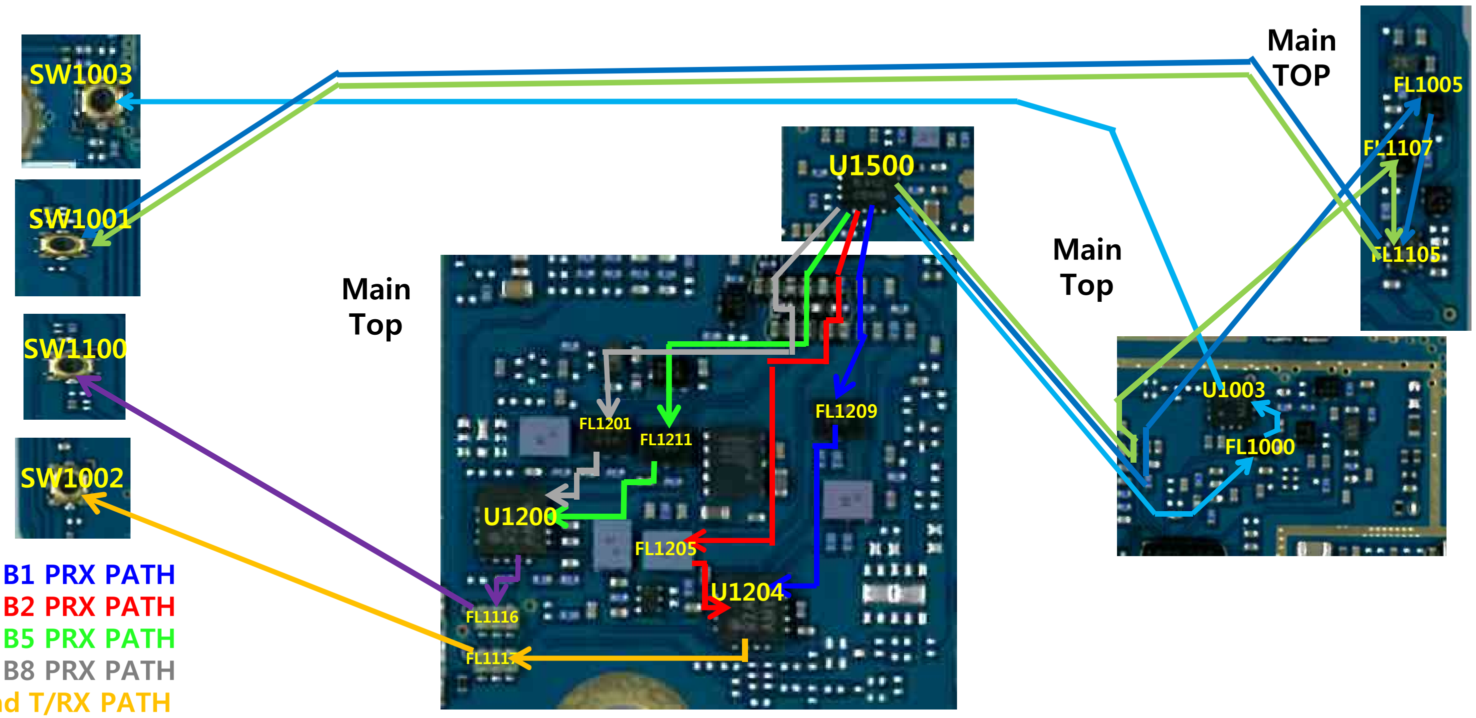
Checking Flow



3.6 WCDMA RF PART

3.6.1 WCDMA B1/2/5/8 RX RF PATH

Image

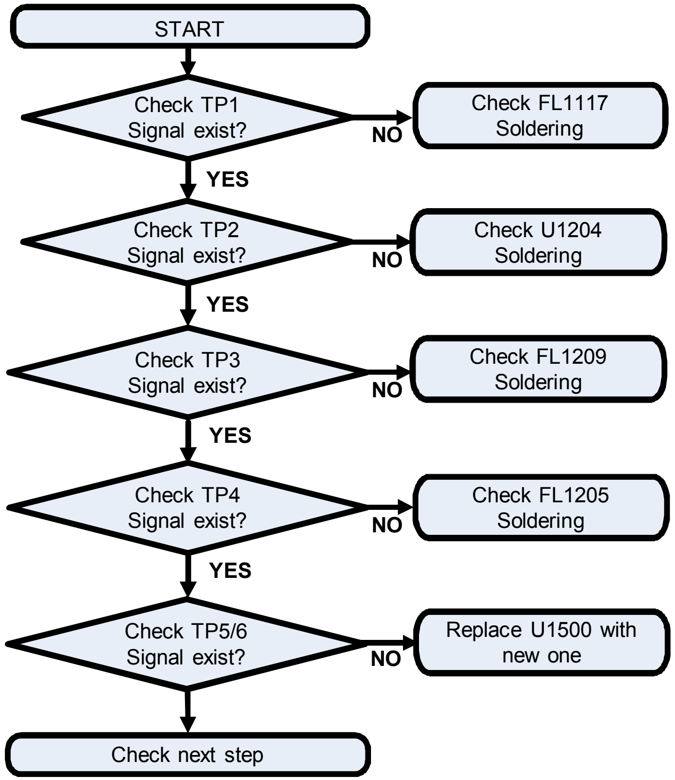


1. WCDMA B1 PRX PATH
2. WCDMA B2 PRX PATH
3. WCDMA B5 PRX PATH
4. WCDMA B8 PRX PATH
5. High Band T/RX PATH
6. Low Band T/RX PATH
7. WCDMA B1 DRX PATH
8. WCDMA B5 DRX PATH
9. WCDMA B8 DRX PATH

3.6 WCDMA RF PART

3.6.1.1 Checking RF signal PRX path(WCDMA B1/B2)

Checking Flow



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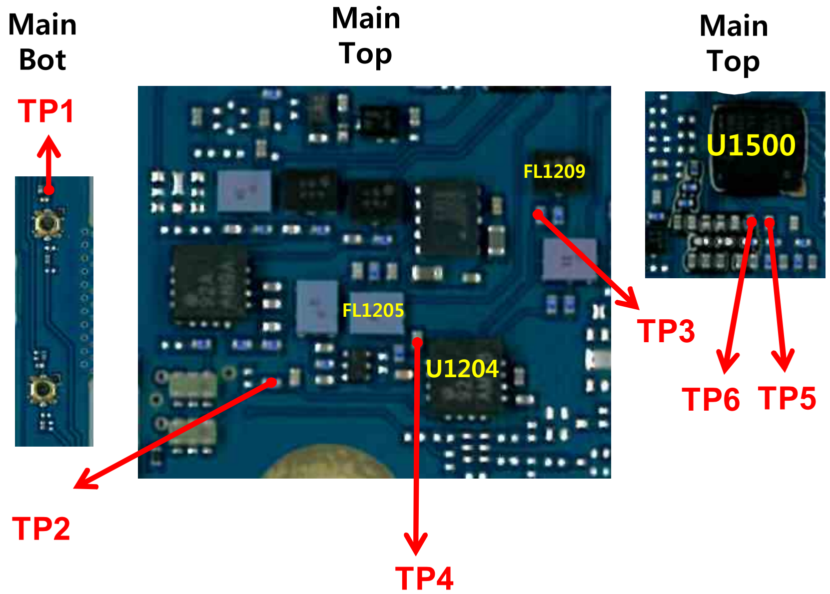
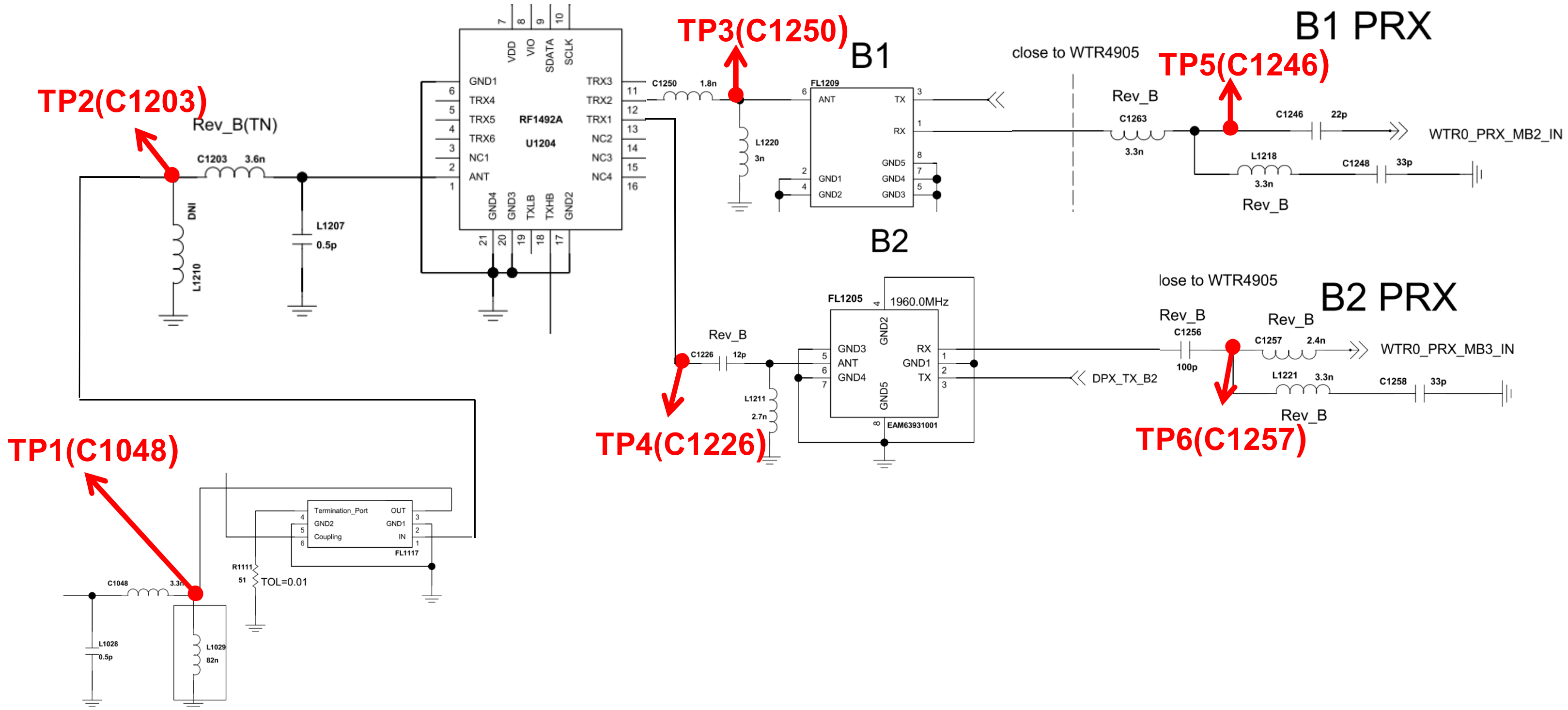


3.6 WCDMA RF PART

3.6.1.1 Checking RF signal PRX path(WCDMA B1/B2)

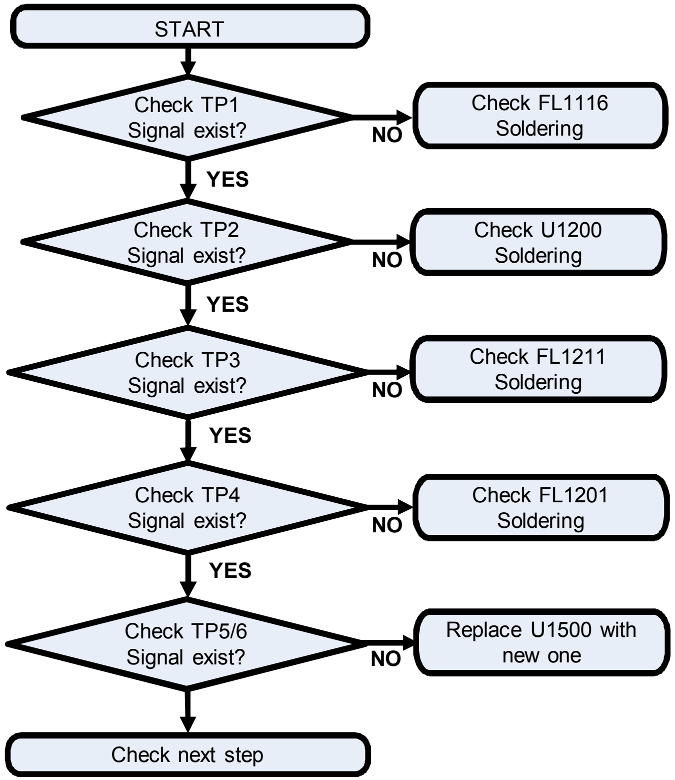
Circuit Diagram



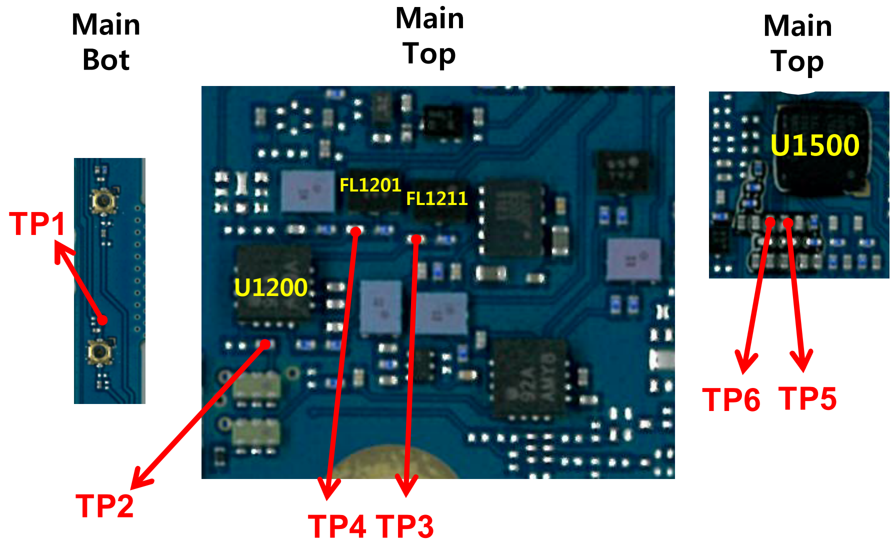
3.6 WCDMA RF PART

3.6.1.2 Checking RF Signal PRX path(WCDMA B5/B8)

Checking Flow



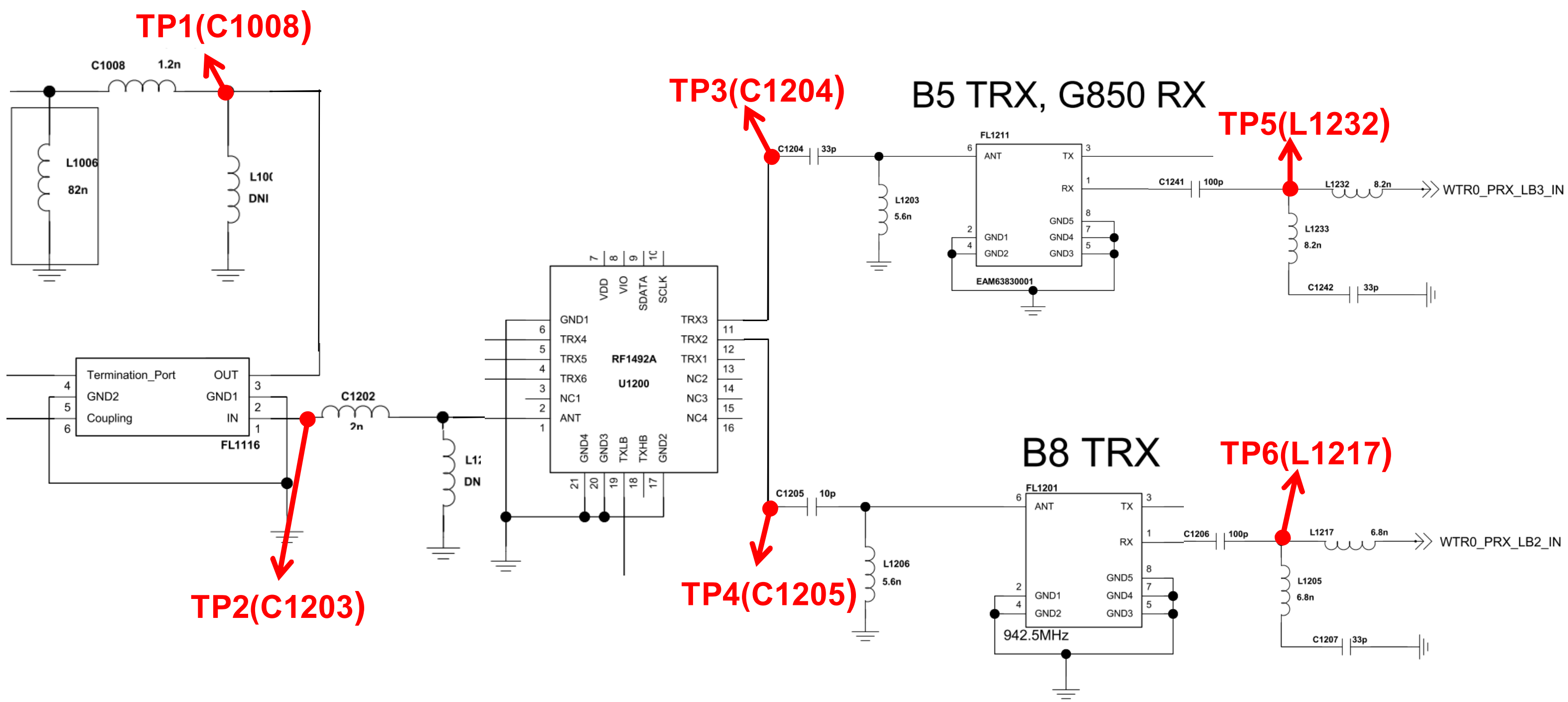
Image



3.6 WCDMA RF PART

3.6.1.2 Checking RF Signal PRX path(WCDMA B5/B8)

Circuit Diagram



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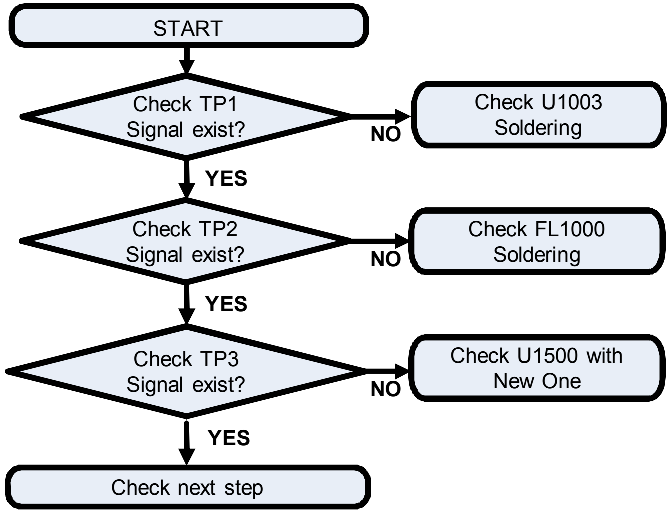
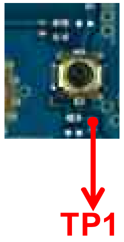
3.6 WCDMA RF PART

3.6.1.3 Checking RF signal DRX path(WCDMA B1)

Checking Flow



Main Bot



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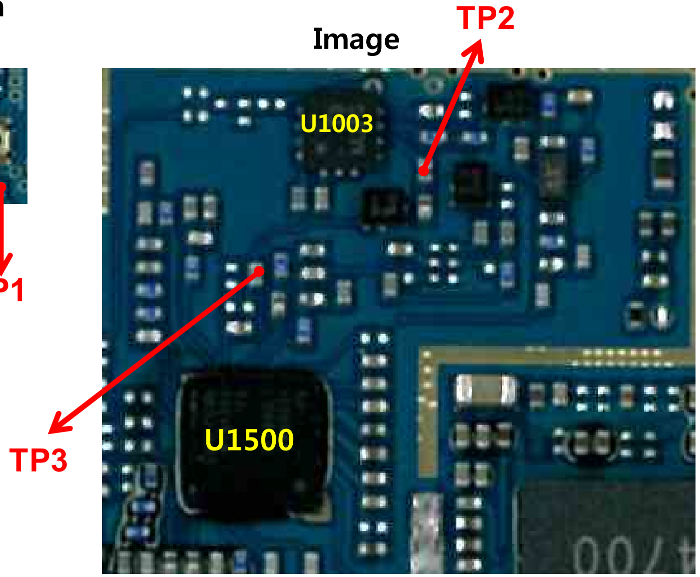
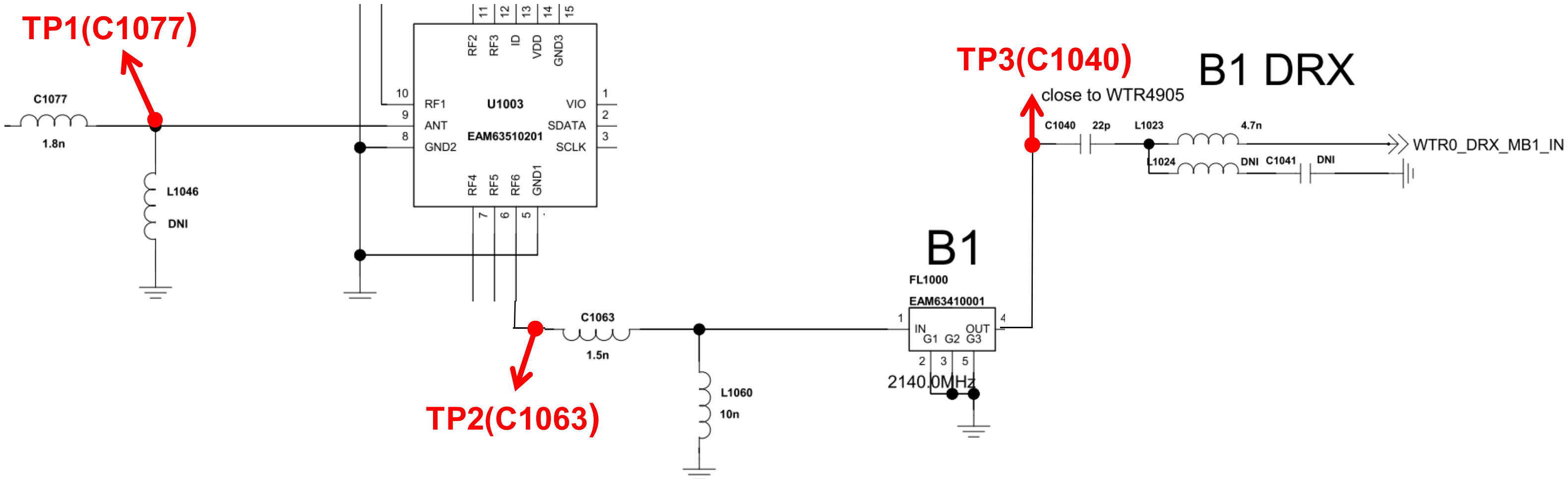


Circuit Diagram

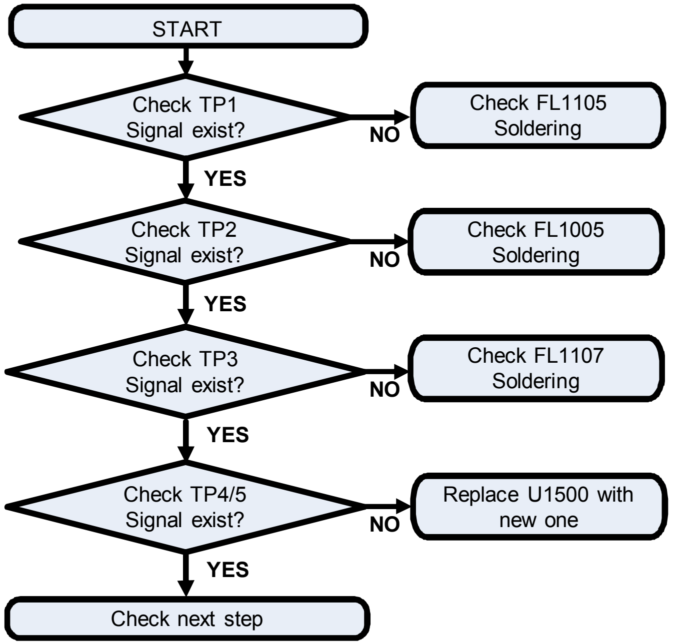


LGE Internal Use Only

3.6 WCDMA RF PART

3.6.1.4 Checking RF signal DRX path(WCDMA B5/B8)

Checking Flow



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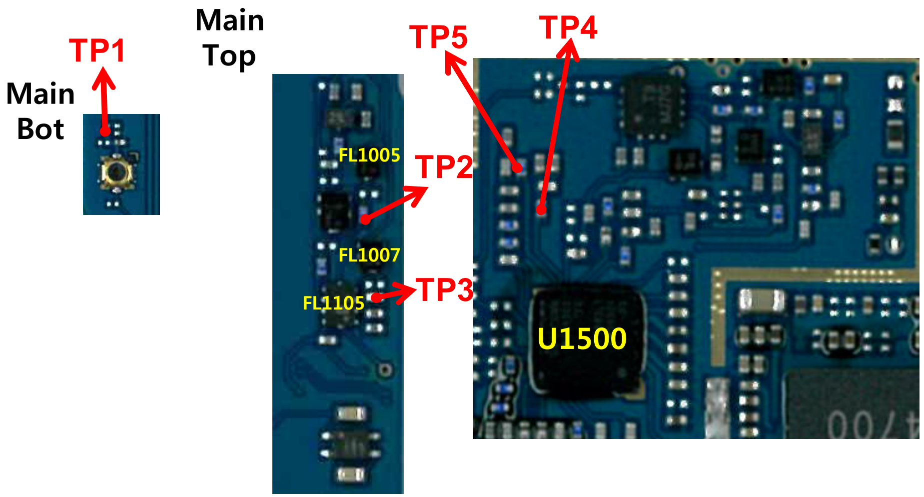
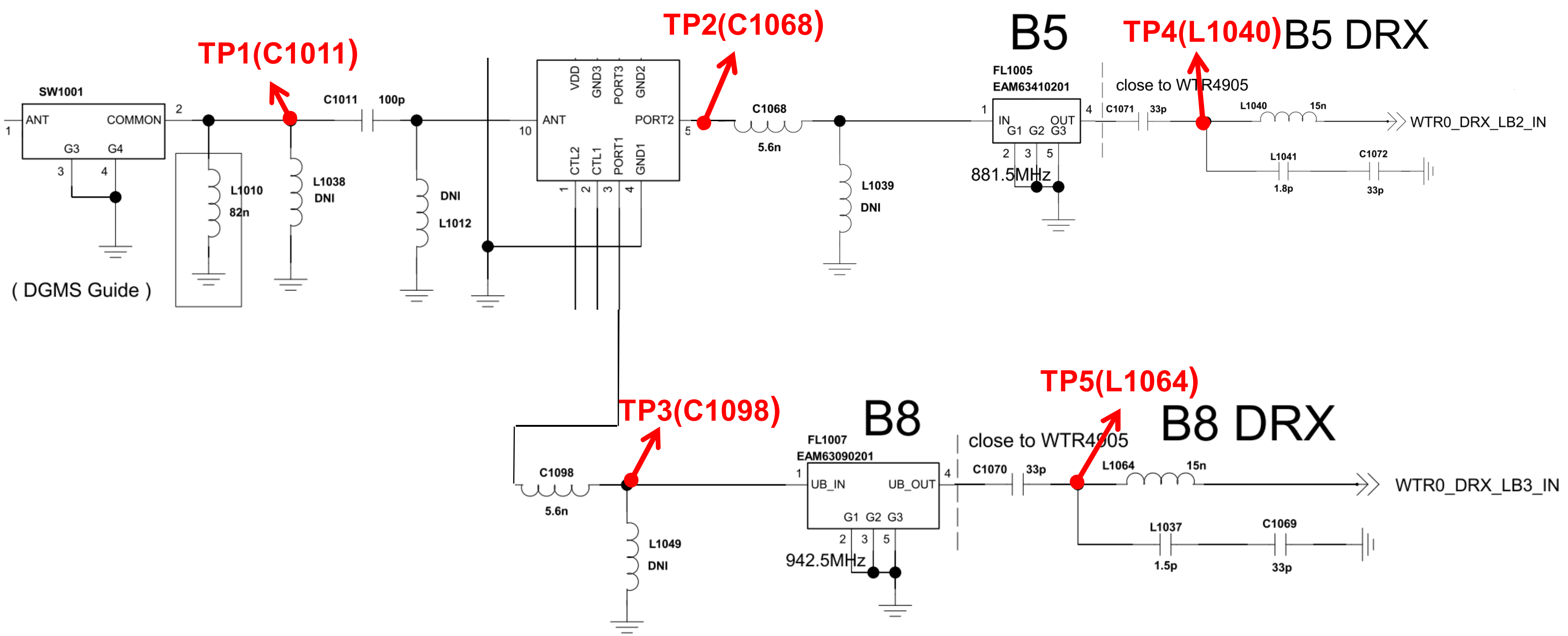


LGE Internal Use Only

3.6 WCDMA RF PART

3.6.1.4 Checking RF signal DRX path(WCDMA B5/B8)

Circuit Diagram



LGE Internal Use Only

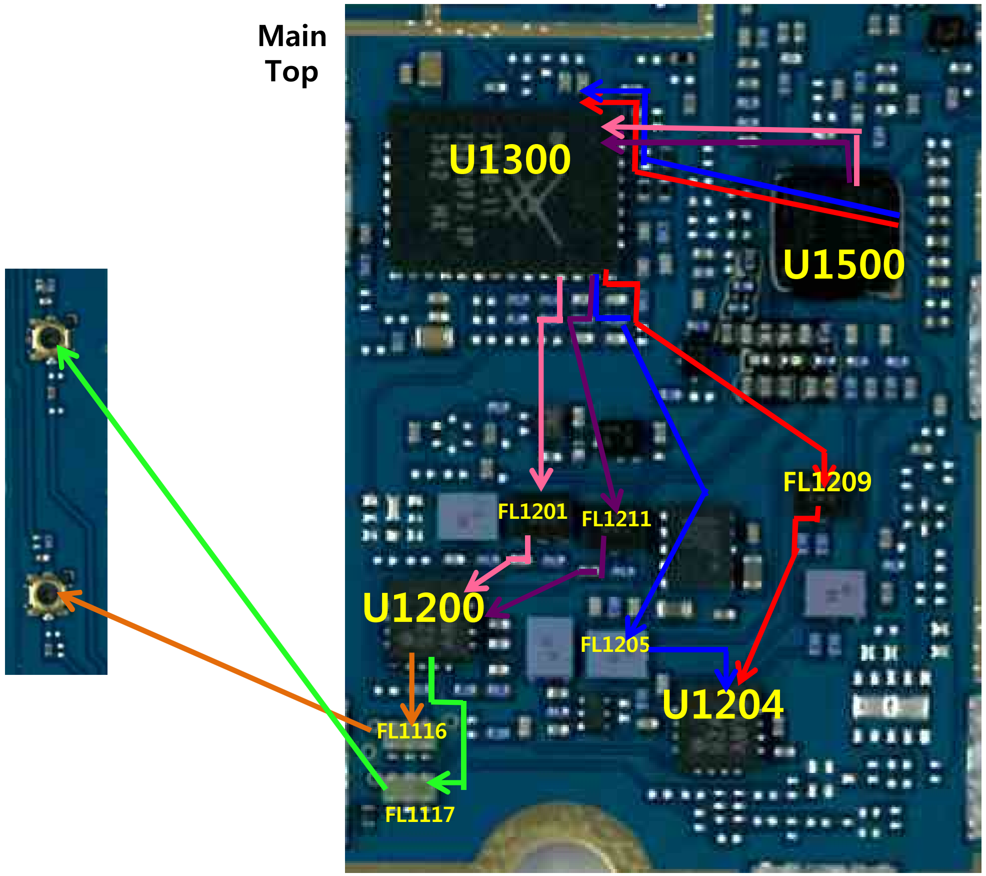
3.6 WCDMA RF PART

3.6.2 WCDMA B1/2/5/8 TX RF PATH

Image

Main Top

Main BOT

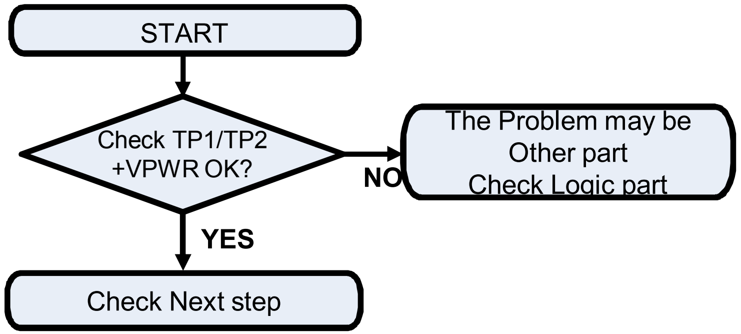


1. WCDMA B1 TX PATH
2. WCDMA B2 TX PATH
3. WCDMA B5 TX PATH
4. WCDMA B8 TX PATH
5. High Band T/RX PATH
6. Low Band T/RX PATH

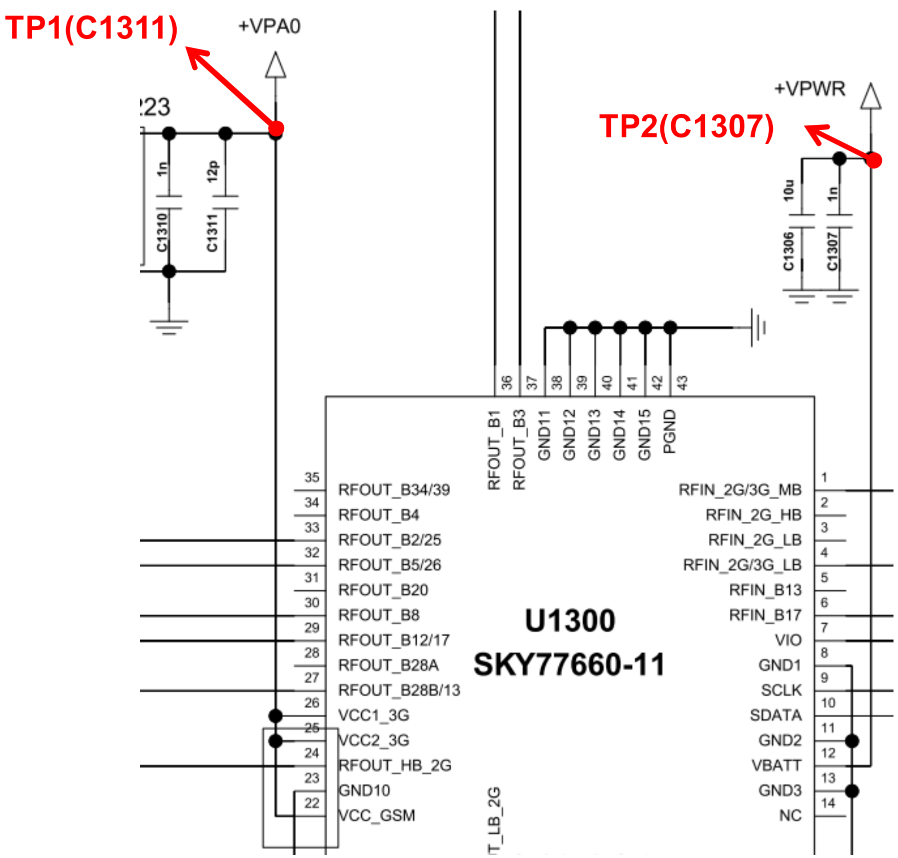
3.6 WCDMA RF PART

3.6.2.1 Checking WCDMA PAM DC Power Circuit

Checking Flow



Circuit Diagram



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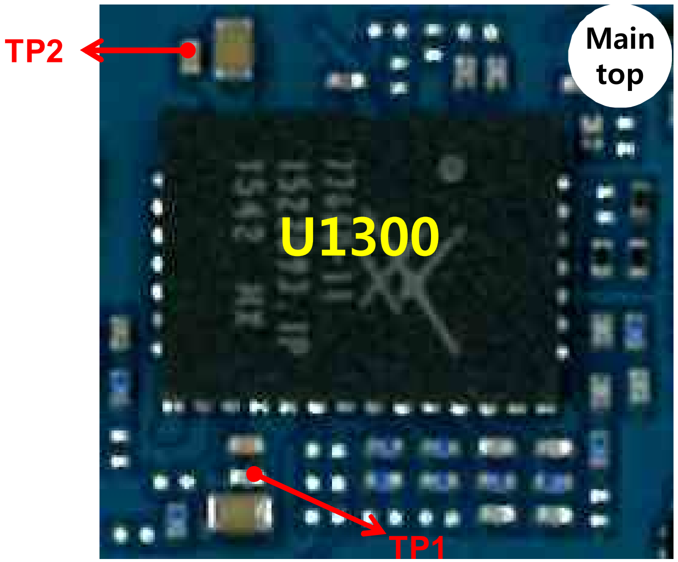
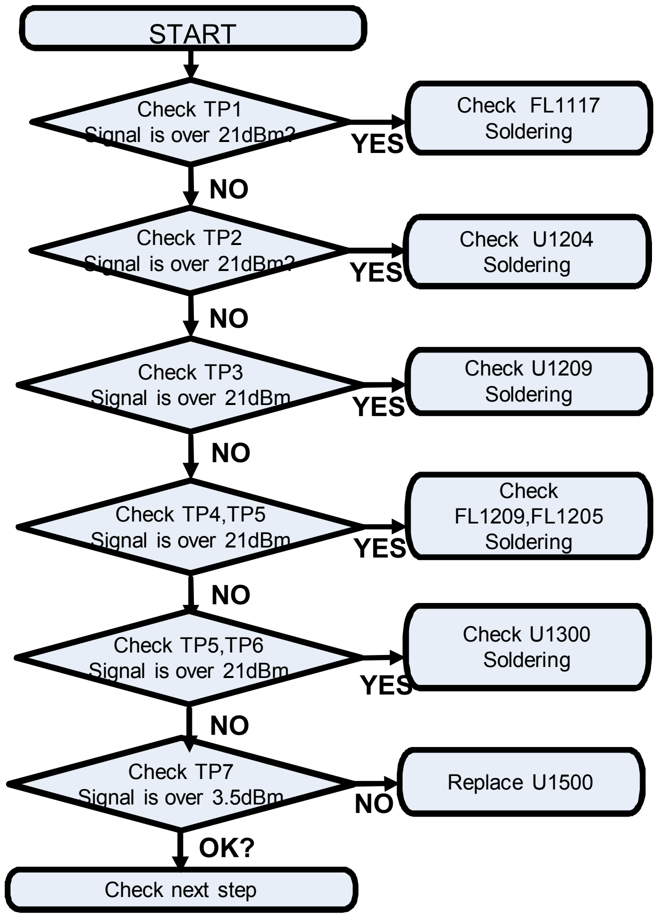


3.6 WCDMA RF PART

3.6.2.2 Checking RF Signal path(B1/B2)

Checking Flow



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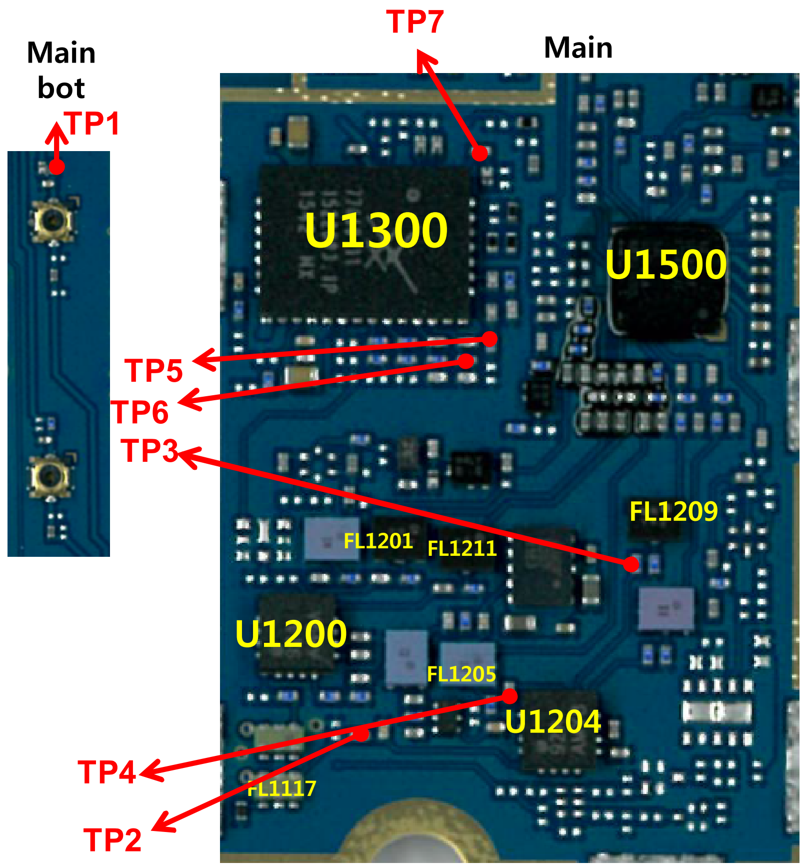
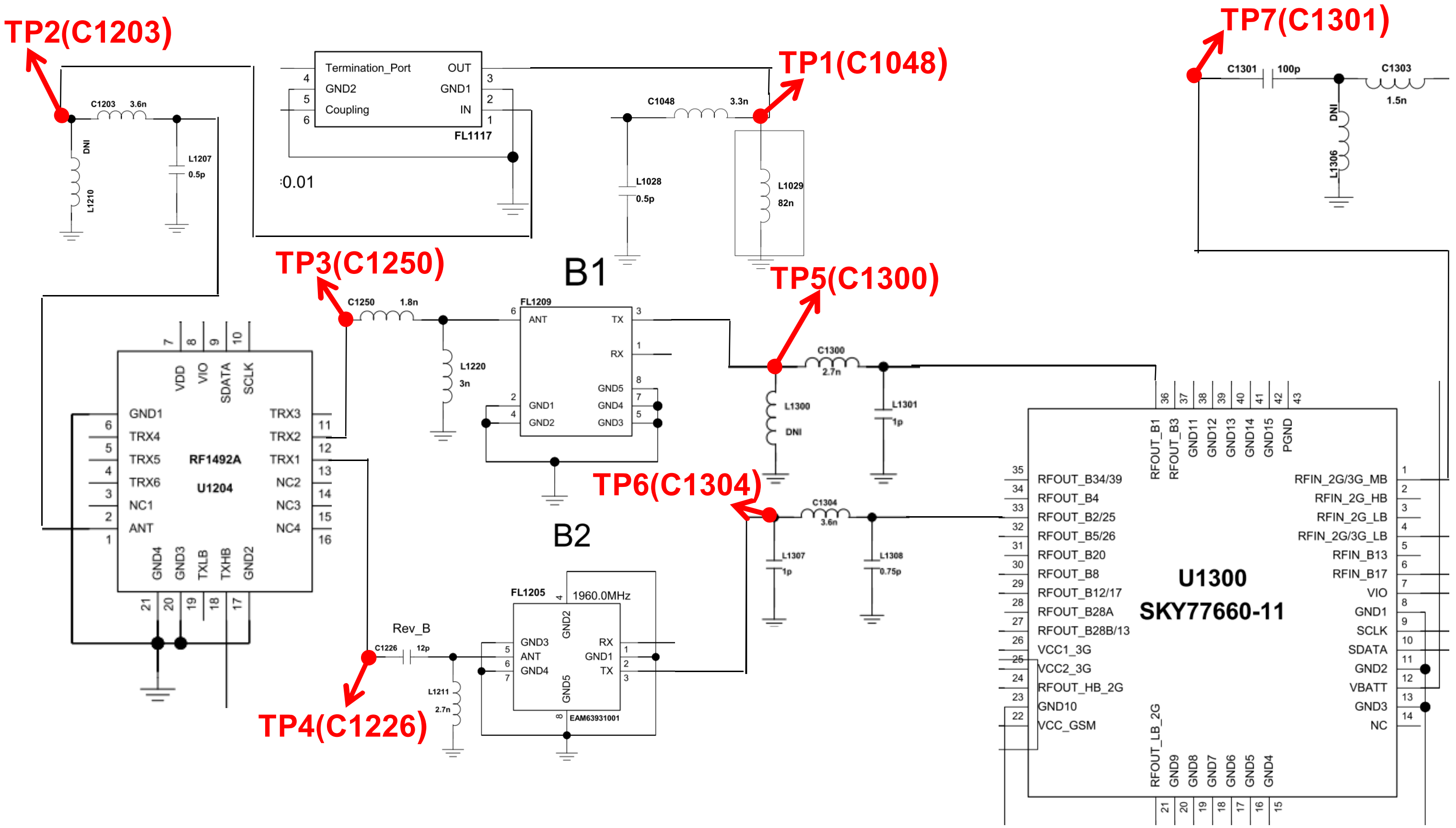


3.6 WCDMA RF PART

3.6.2.2 Checking RF Signal path(B1/B2)

Circuit Diagram



3.6 WCDMA RF PART

3.6.2.2 Checking RF Signal path(B5/B8)

Checking Flow

START

Check TP1 Signal is over 21dBm? — YES → Check FL1116 Soldering

NO ↓

Check TP2 Signal is over 21dBm? — YES → Check U1200 Soldering

NO ↓

Check TP3 Signal is over 21dBm — YES → Check FL1211,FL1201 Soldering

NO ↓

Check TP4,TP5 Signal is over 21dBm — YES → Check FL1209,FL1205 Soldering

NO ↓

Check TP5,TP6 Signal is over 21dBm — YES → Check U1300 Soldering

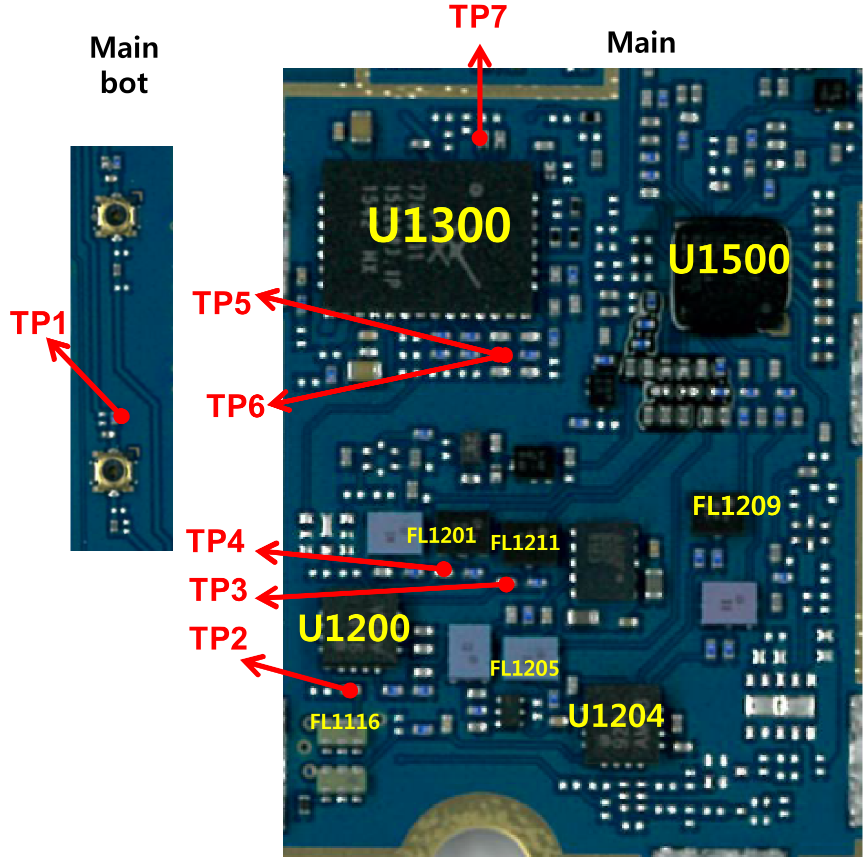
NO ↓

Check TP7 Signal is over 3.5dBm — NO → Replace U1500

OK? ↓

Check next step

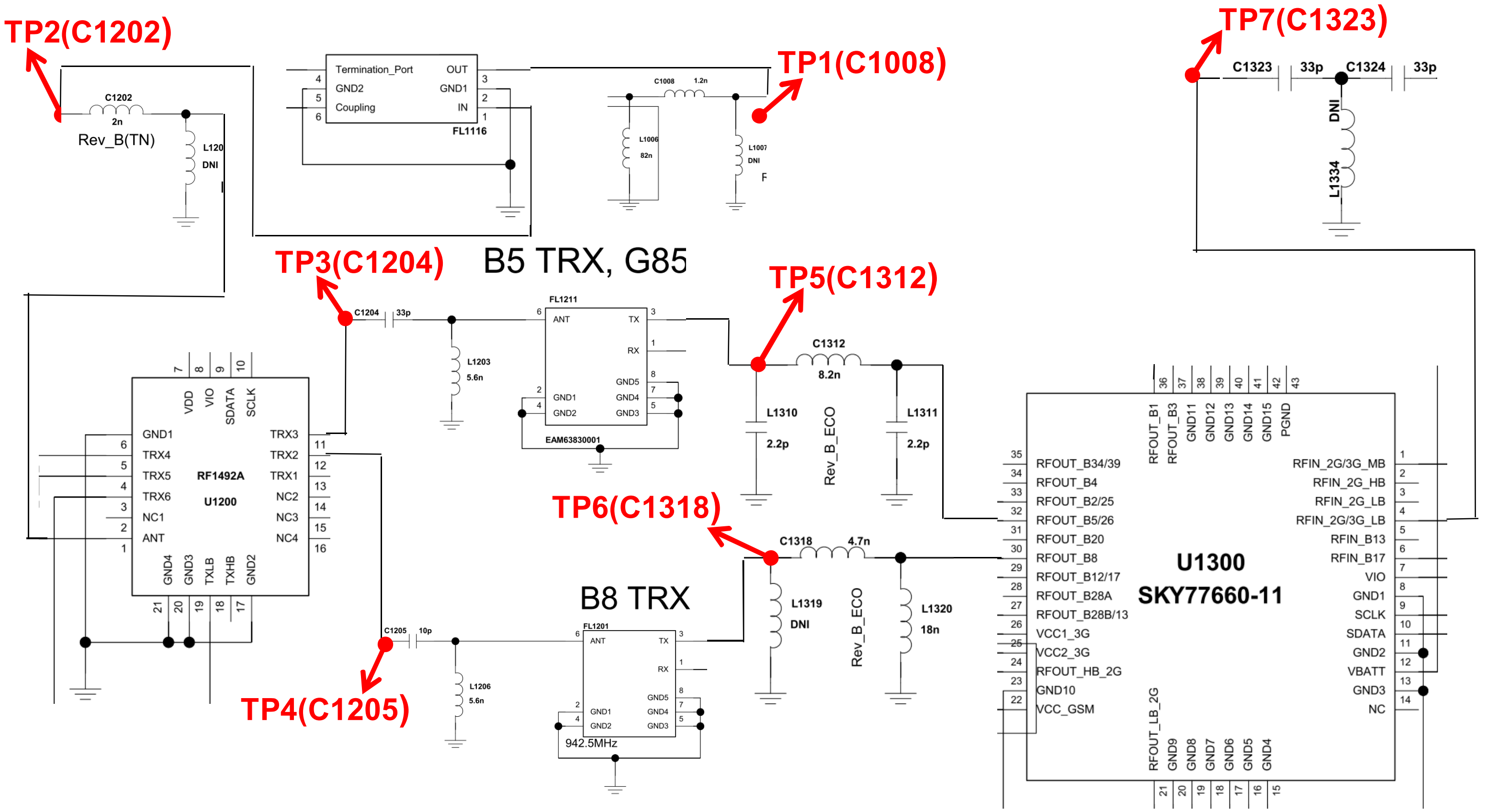
Image



3.6 WCDMA RF PART

3.6.2.2 Checking RF Signal path(B5/B8)

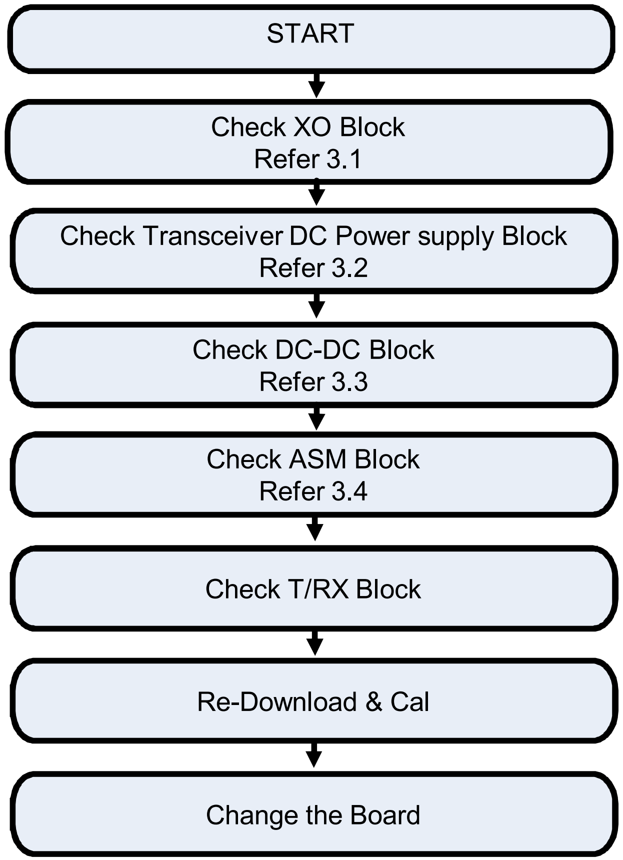
Circuit Diagram



3.7 LTE RF PART

LTE RF Part support LTE B1/B3/B7/B8/B20 with ASM, DPX, PAM, Transceiver component

Checking Flow



3.7 LTE RF PART

3.7.1 Checking RF signal path / 3.7.1.1 LTE B1/B8 TX,PRX,DRX / 3.7.1.2 Checking LTE PAM DC Power Circuit

Checking Flow	Image
For B1/B8(PRX,DRX) **Refer to 3.6.1** **For B1/B8(TX)** **Refer to 3.6.2**	

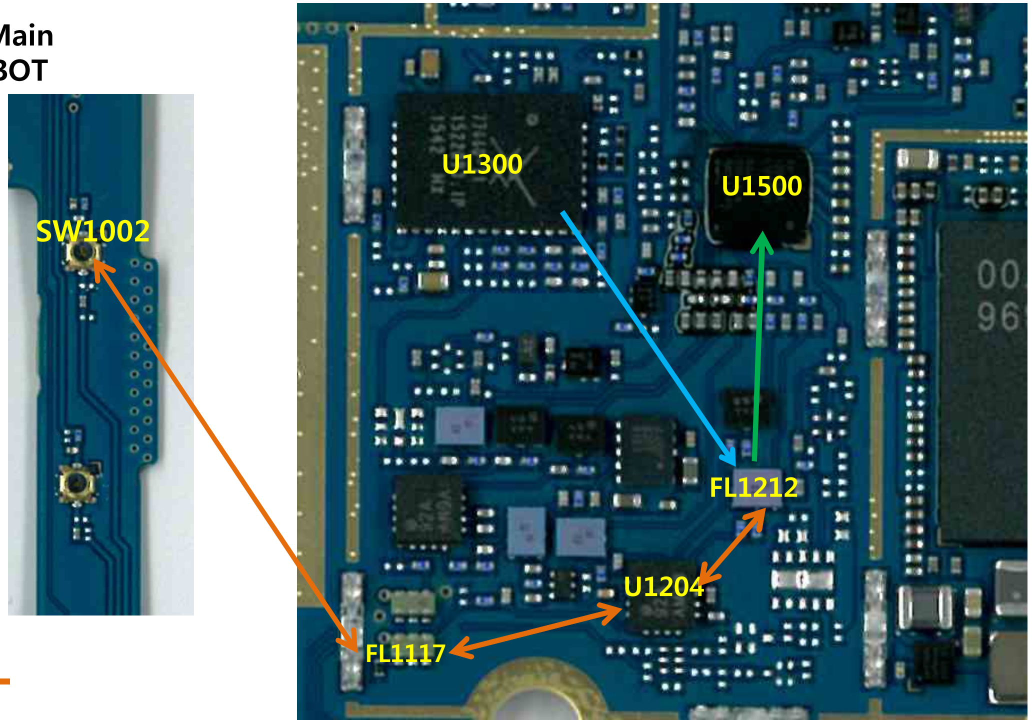
3.7 LTE RF PART

3.7.2 LTE B3 RF Part TX/PRX RF PATH

Image

Main Top

Main BOT



LTE B3 PRX PATH

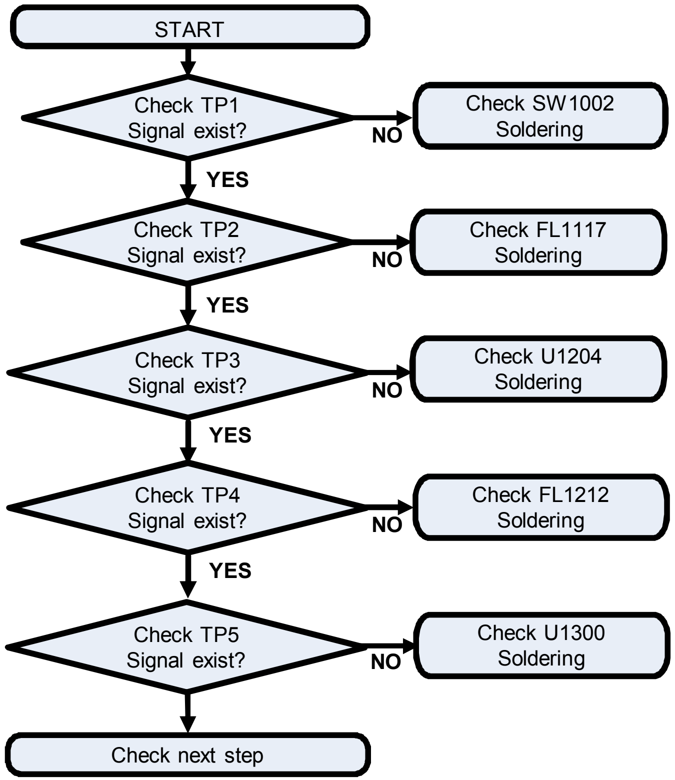
LTE B3 TX PATH

LTE B3 COMMON TX/PRX PATH

3.7 LTE RF PART

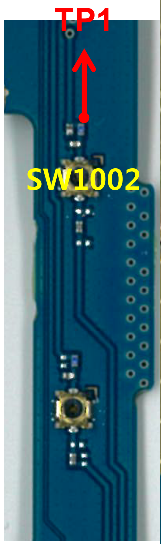
3.7.2.1 LTE B3 RF Part TX/PRX RF PATH

Checking Flow

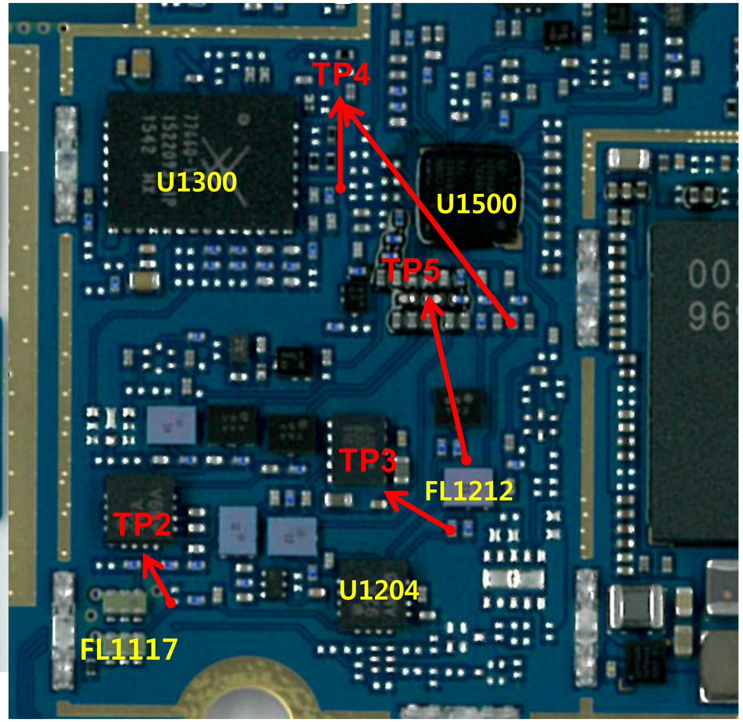


Image

Main Bot



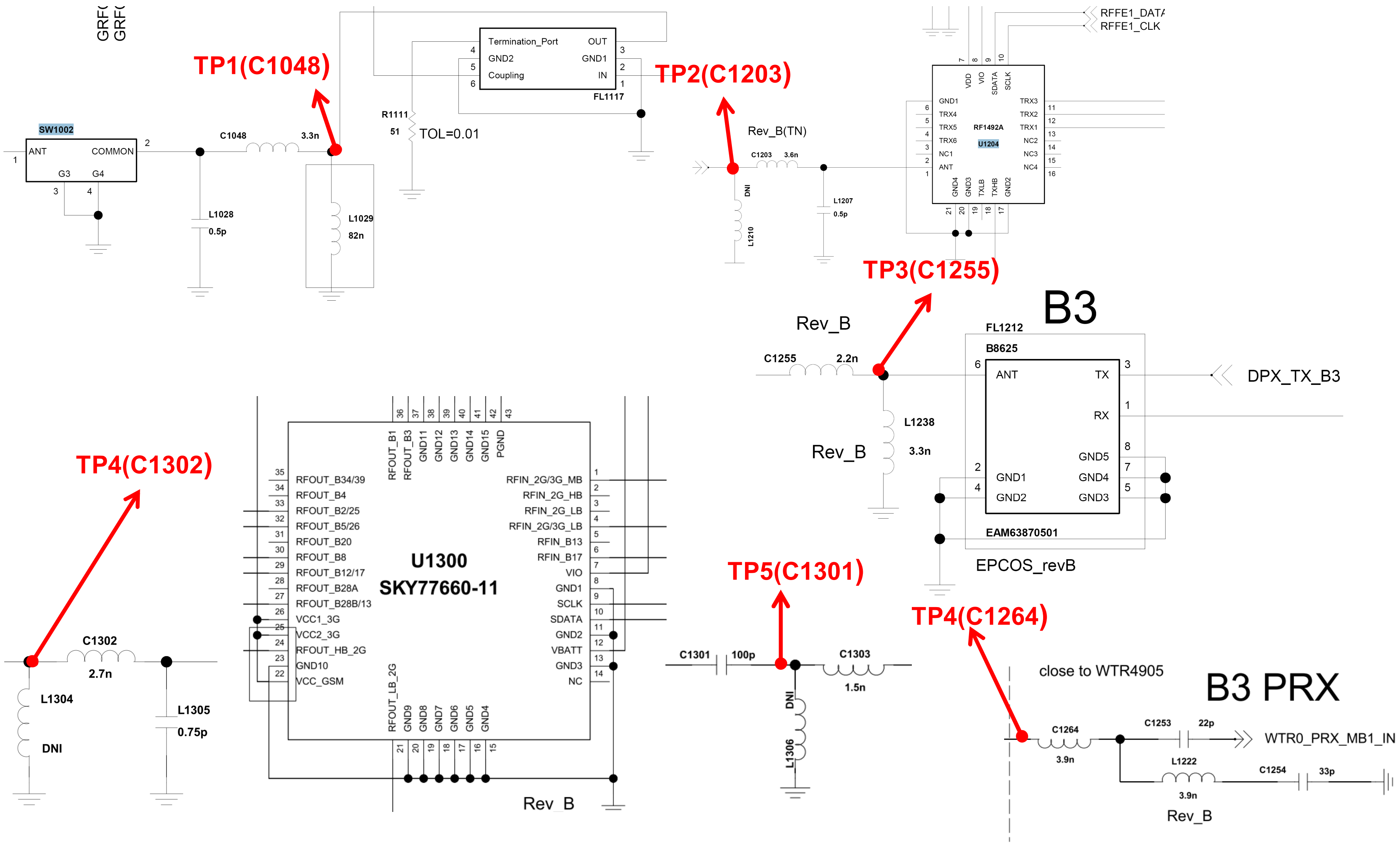
Main Top



3.7 LTE RF PART

3.7.2.1 LTE B3 RF Part TX/PRX RF PATH

Circuit Diagram



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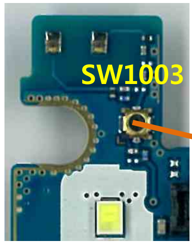
3.7 LTE RF PART

3.7.2.2 LTE B3 RF Part DRX RF PATH

Image

Main Bot



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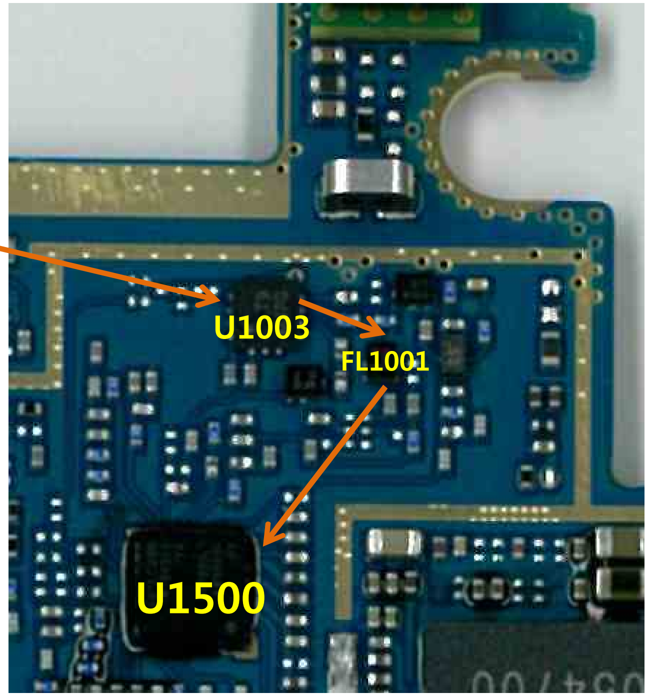
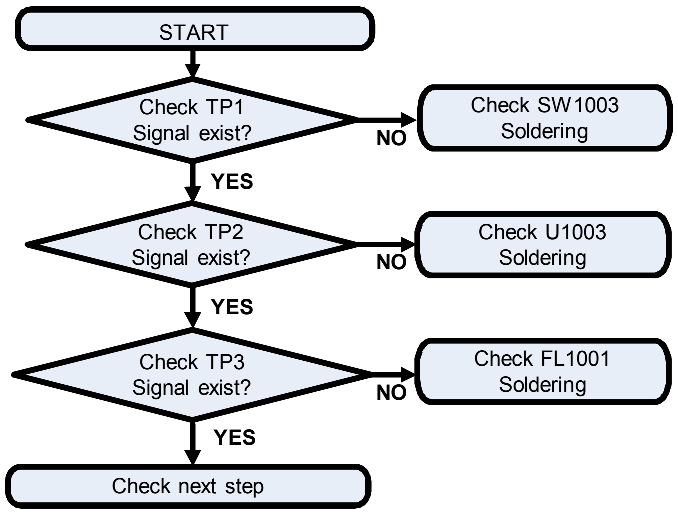


LTE B3 DRX PATH

3.7 LTE RF PART

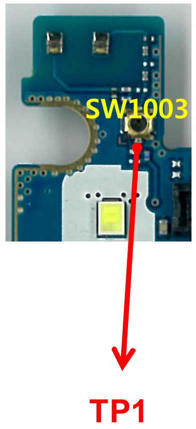
3.7.2.2 LTE B3 RF Part DRX RF PATH

Checking Flow



Image

Main Bot



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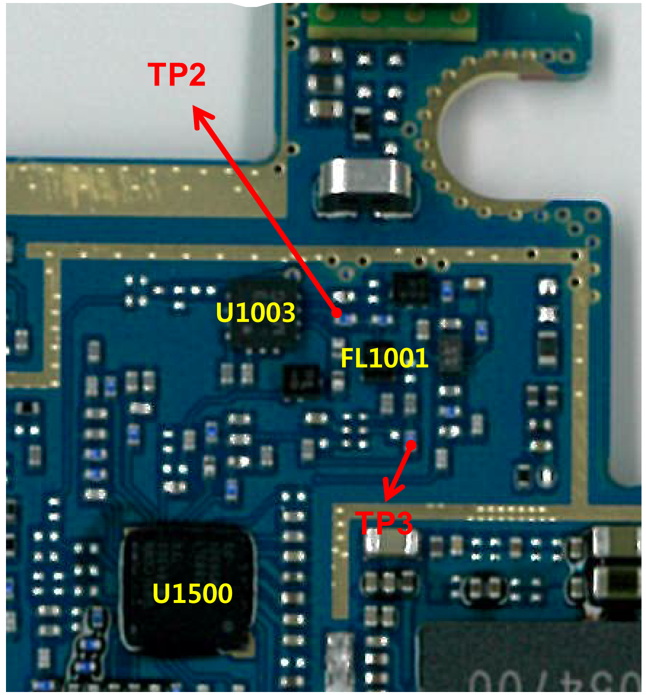
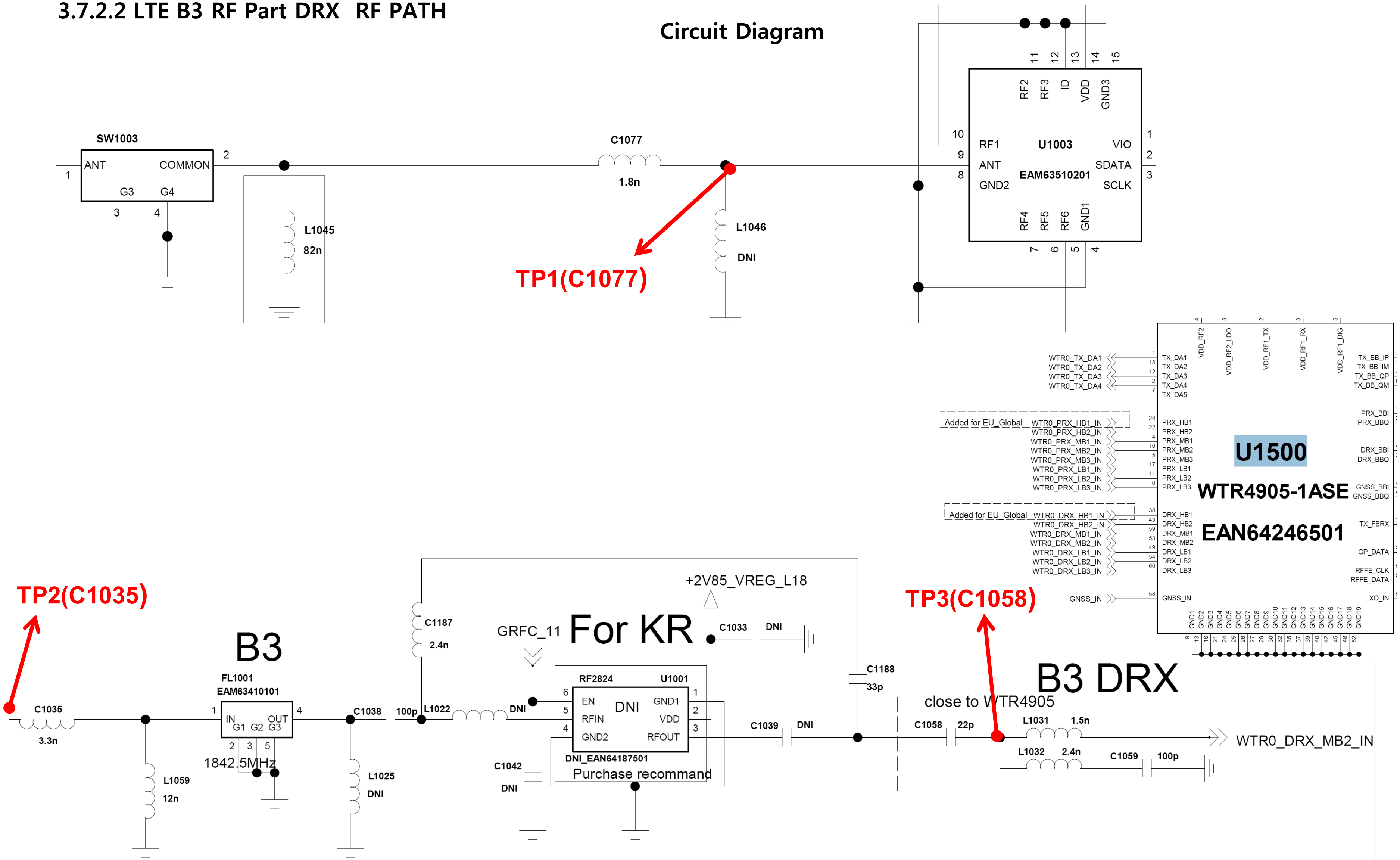


3.7 LTE RF PART

3.7.2.2 LTE B3 RF Part DRX RF PATH

Circuit Diagram



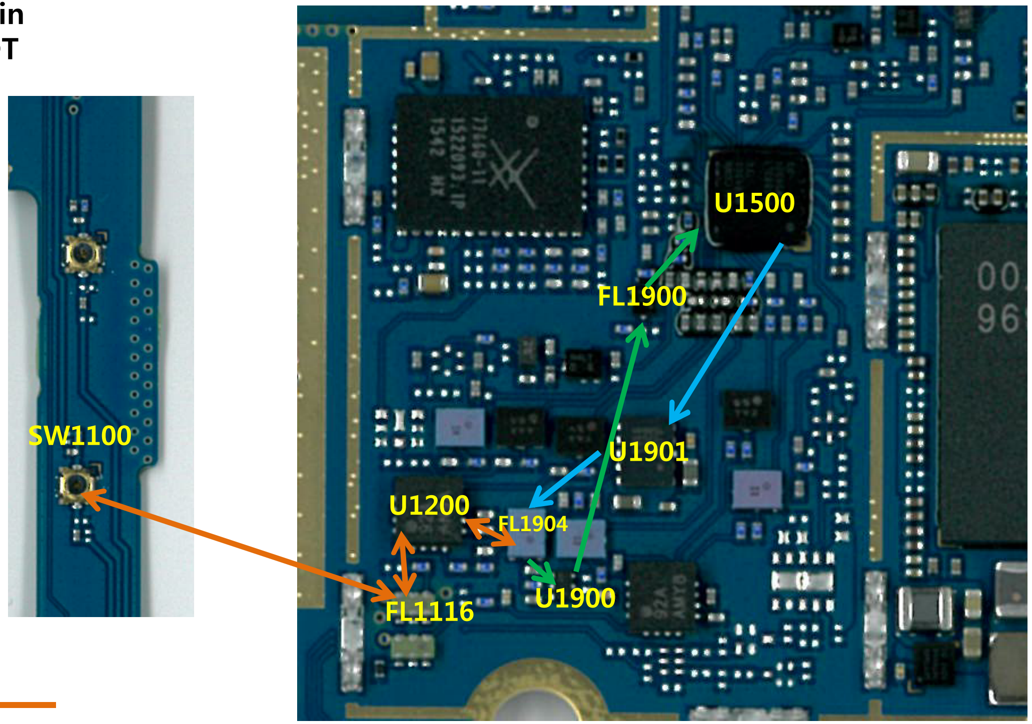
3.7 LTE RF PART

3.7.3 LTE B7 RF Part TX/PRX RF PATH

Image

Main
BOT

Main
Top



LTE B7 PRX PATH

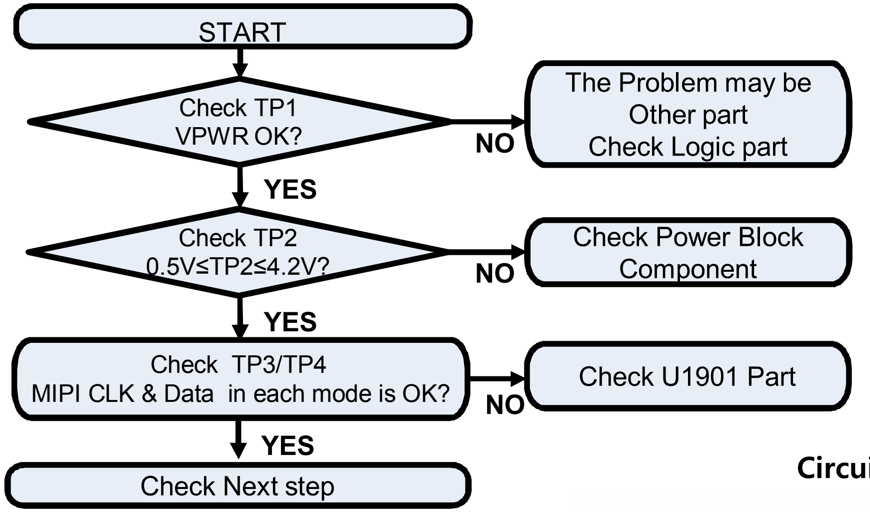
LTE B7 TX PATH

LTE B7 COMMON TX/PRX PATH

3.7 LTE RF PART

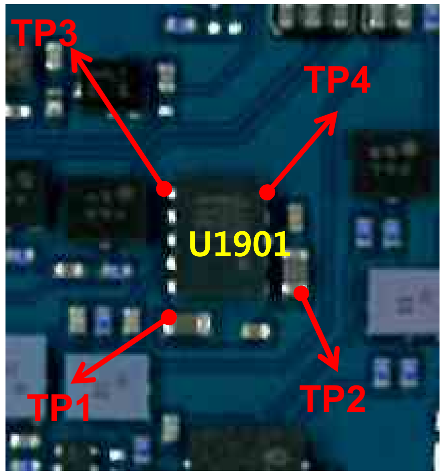
3.7.3.1 Checking LTE B7 PAM DC Power Circuit

Checking Flow

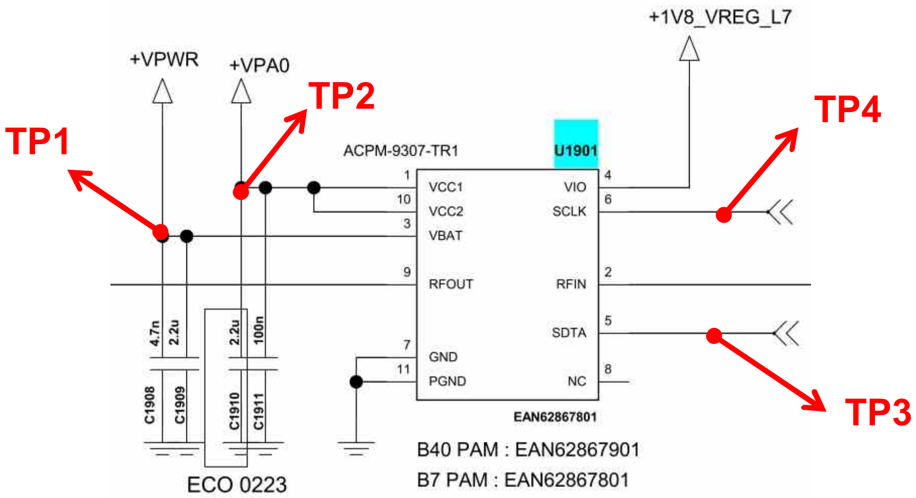


Image

Main Top



Circuit Diagram



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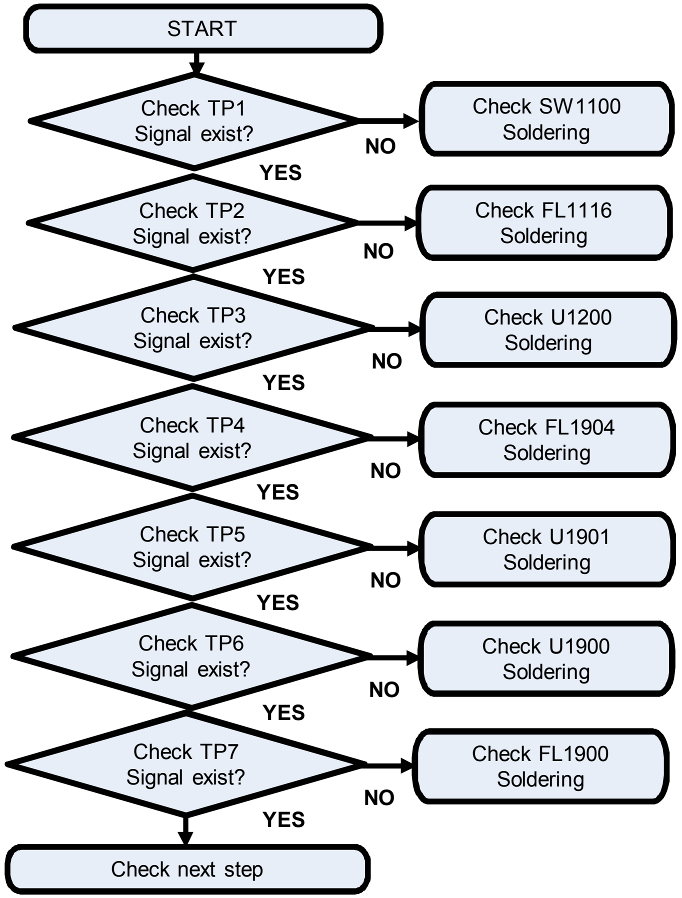
3.7 LTE RF PART

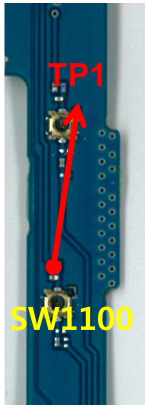
3.7.3.2 Checking LTE B7 TX/PRX RF common signal path

Checking Flow

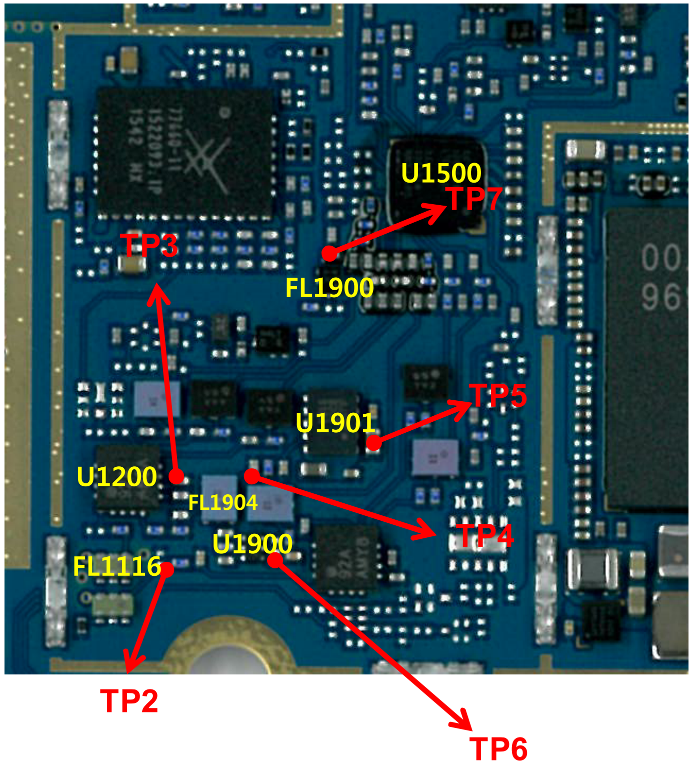


Image

Main
Bot



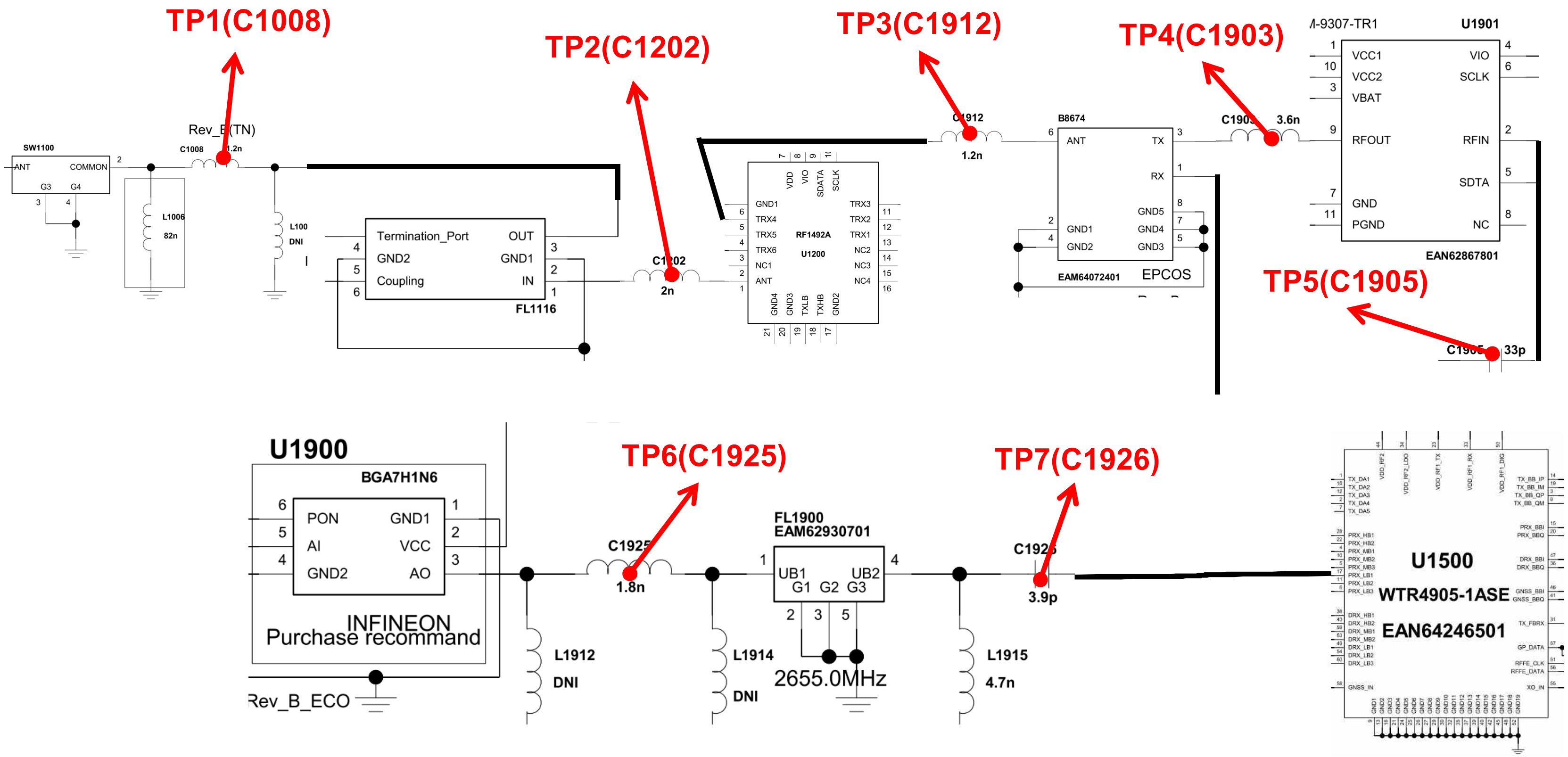
Main
Top



3.7 LTE RF PART

3.7.3.2 Checking LTE B7 TX/PRX RF common signal path

Circuit Diagram



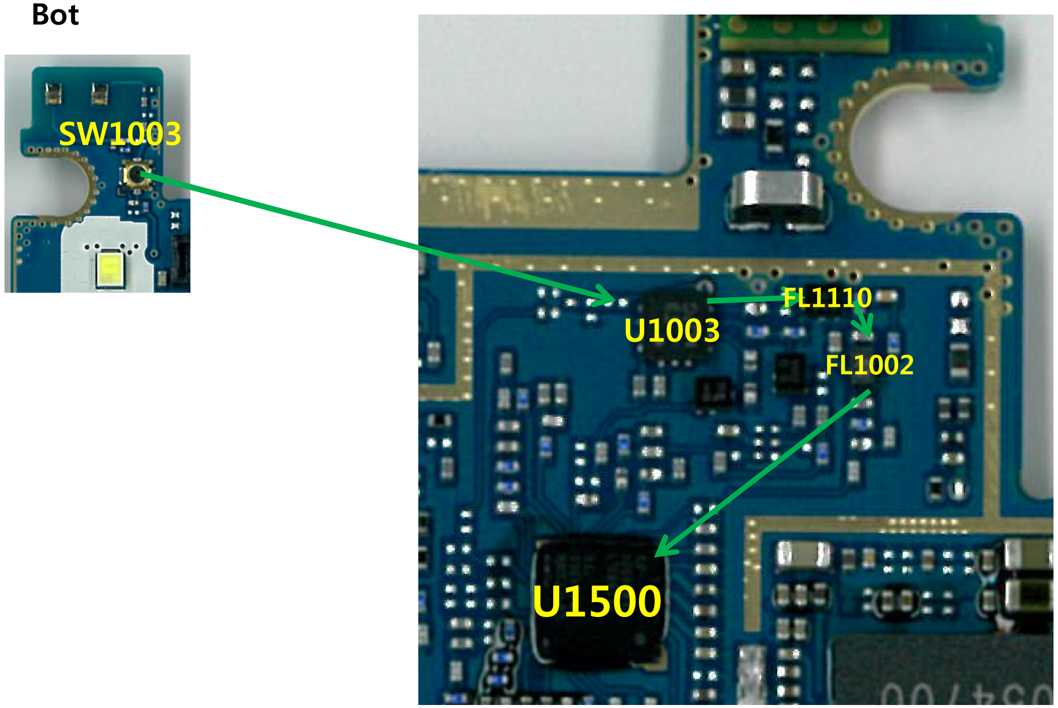
3.7 LTE RF PART

3.7.3.3 LTE B7 RF Part DRX RF PATH

Image

Main Bot

Main Top

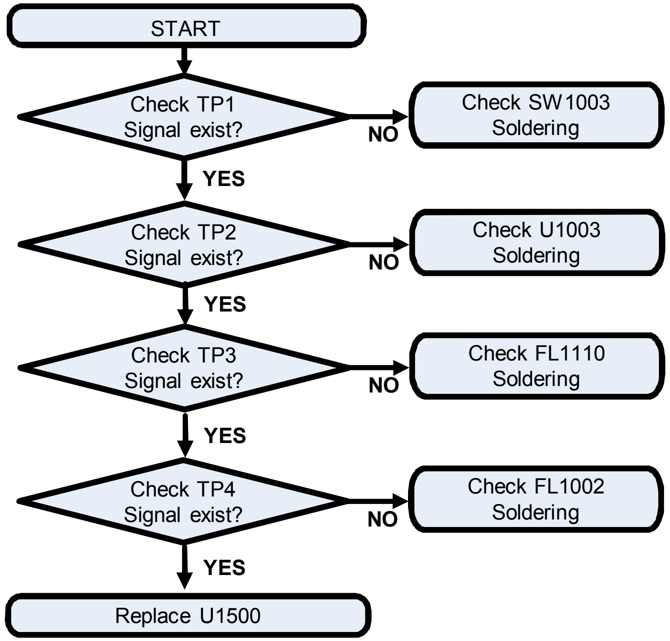


LTE B7 DRX PATH

3.7 LTE RF PART

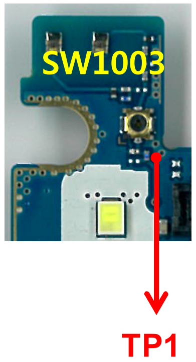
3.7.3.3 LTE B7 RF Part DRX RF PATH

Checking Flow

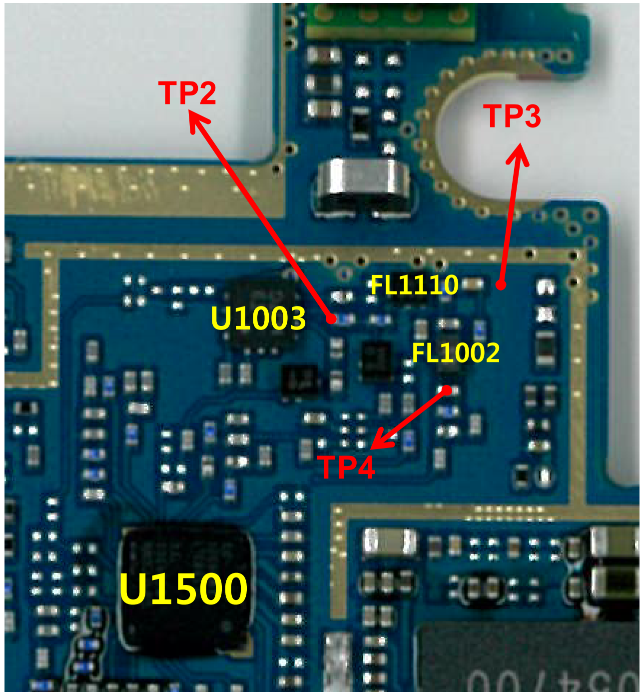


Image

Main
Bot



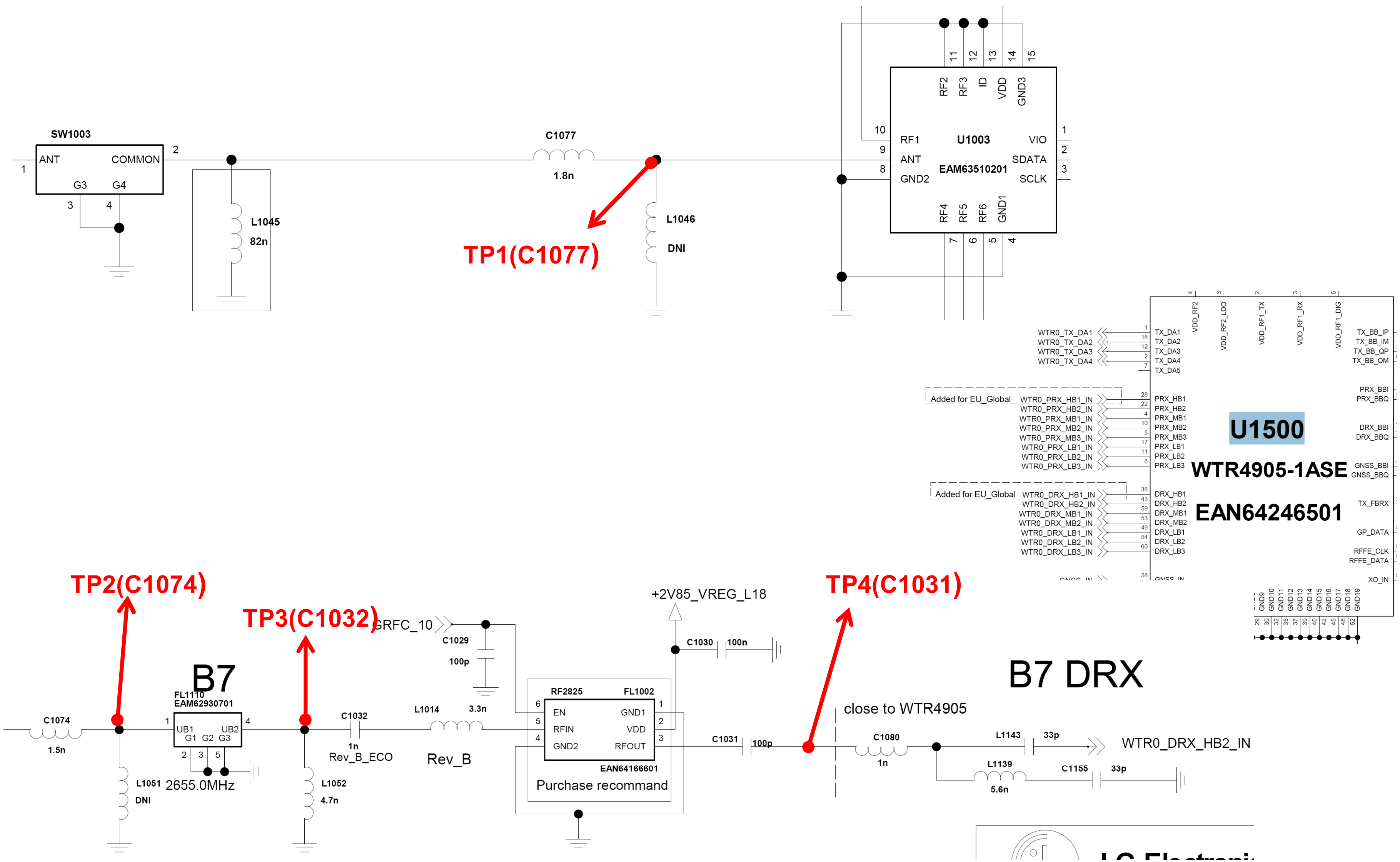
Main
Top



3.7 LTE RF PART

3.7.3.3 LTE B7 RF Part DRX RF PATH

Circuit Diagram



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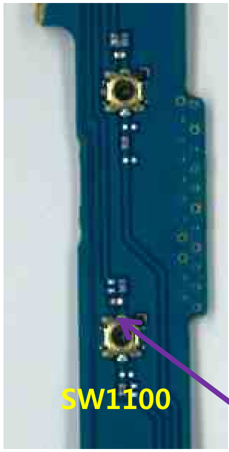
3.7 LTE RF PART

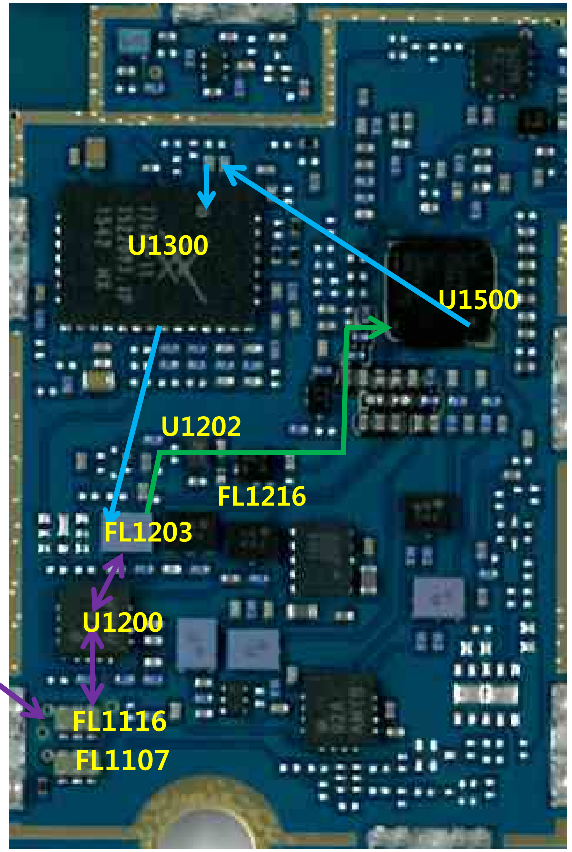
3.7.4 LTE B20 RF Part TX/PRX RF PATH

Image

Main
Top

Main
BOT



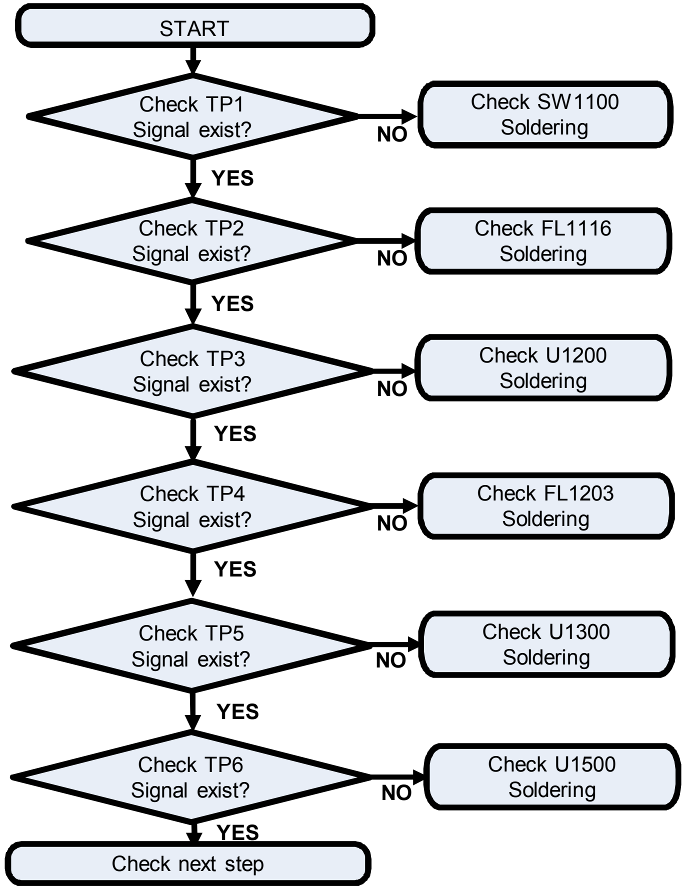


LTE B20 PRX PATH
LTE B20 TX PATH
LTE B20 COMMON TX/PRX PATH

3.7 LTE RF PART

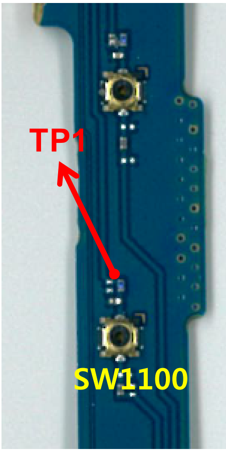
3.7.4.1 LTE B20 RF Part TX RF PATH

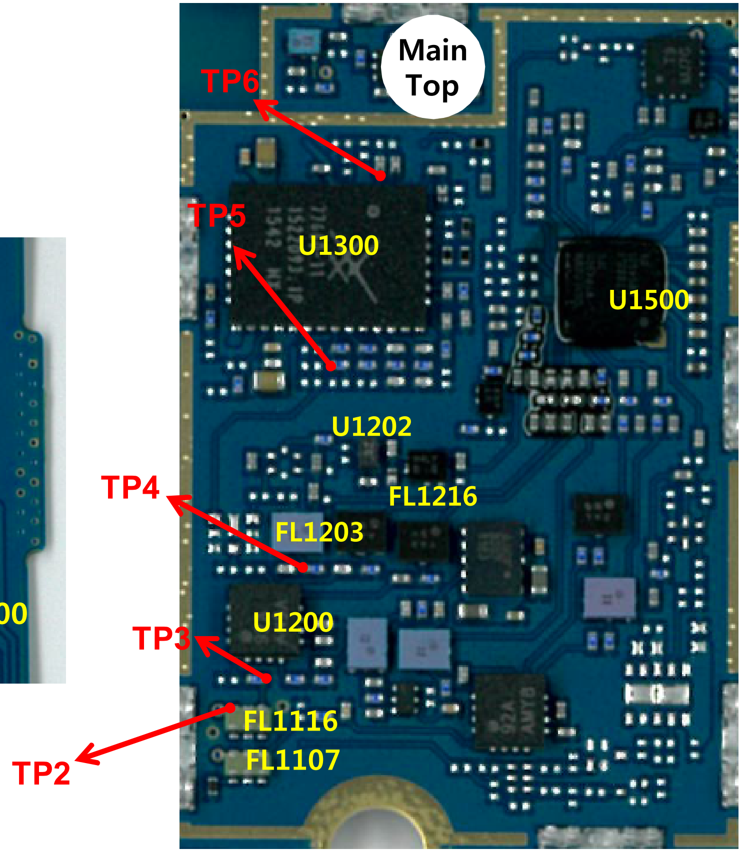
Checking Flow



Image

Main
Bot

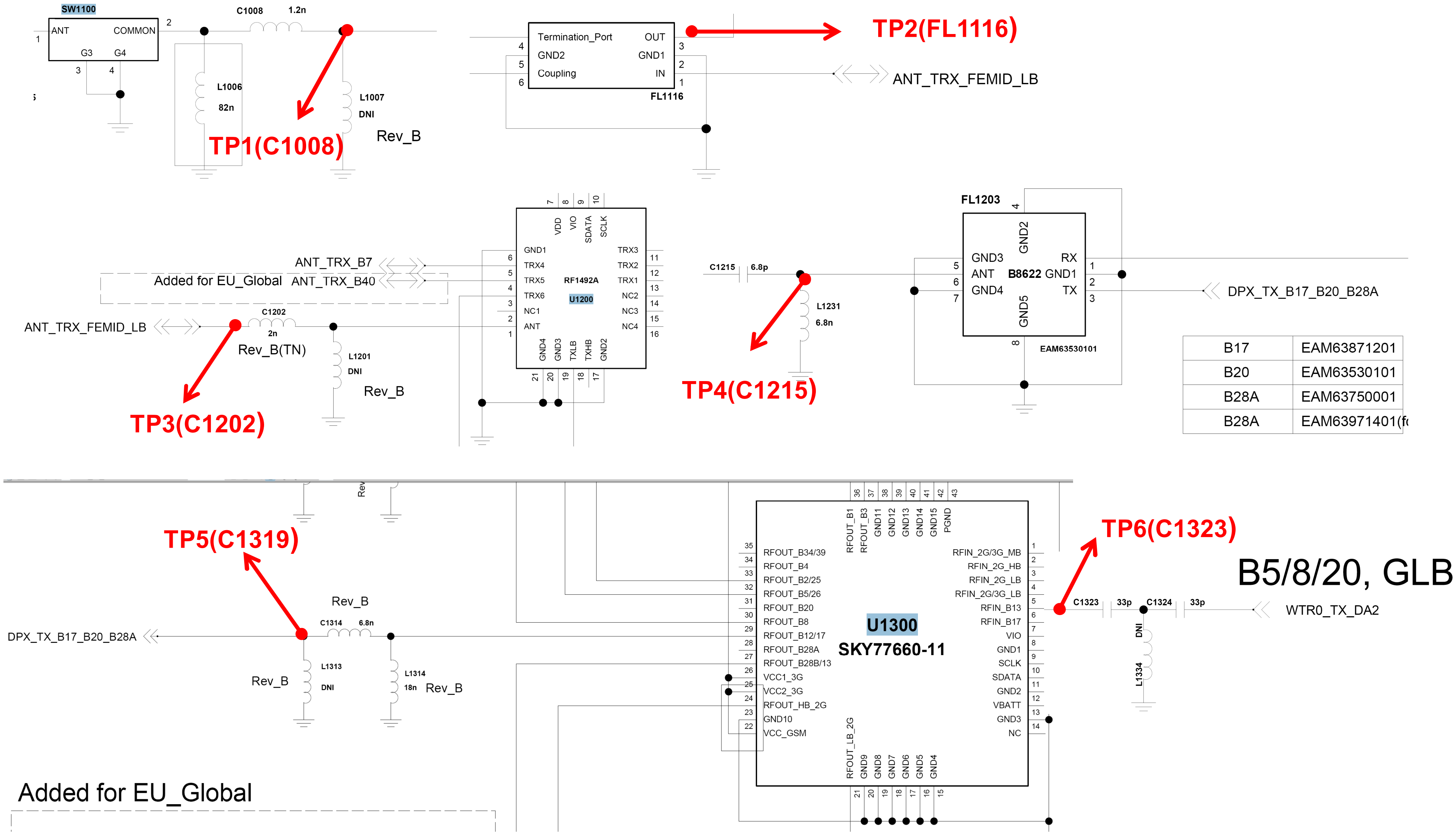




3.7 LTE RF PART

3.7.4.1 LTE B20 RF Part TX/PRX RF PATH

Circuit Diagram

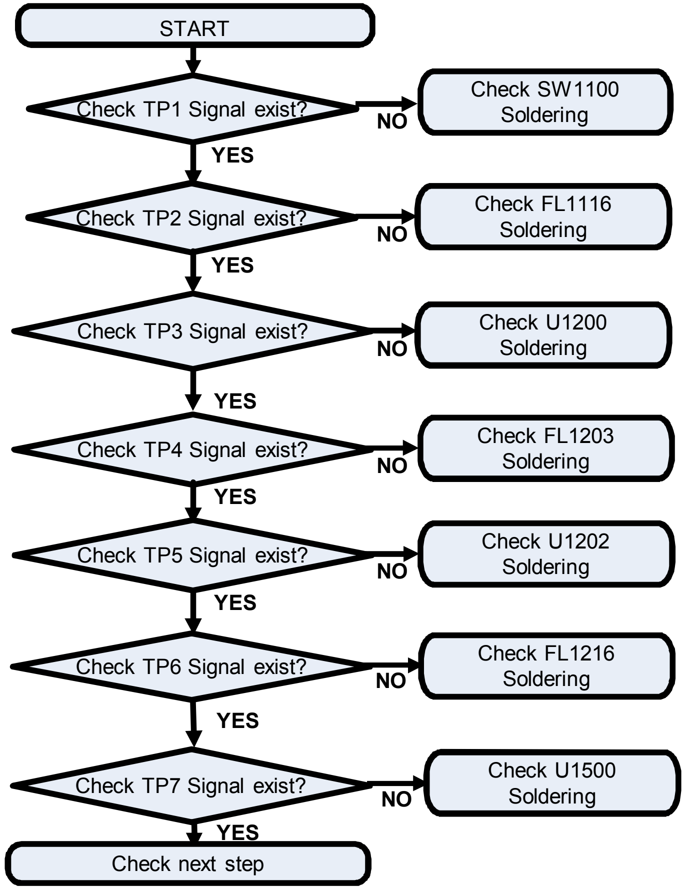


B17	EAM63871201
B20	EAM63530101
B28A	EAM63750001
B28A	EAM63971401(f

3.7 LTE RF PART

3.7.4.1 LTE B20 RF Part PRX RF PATH

Checking Flow



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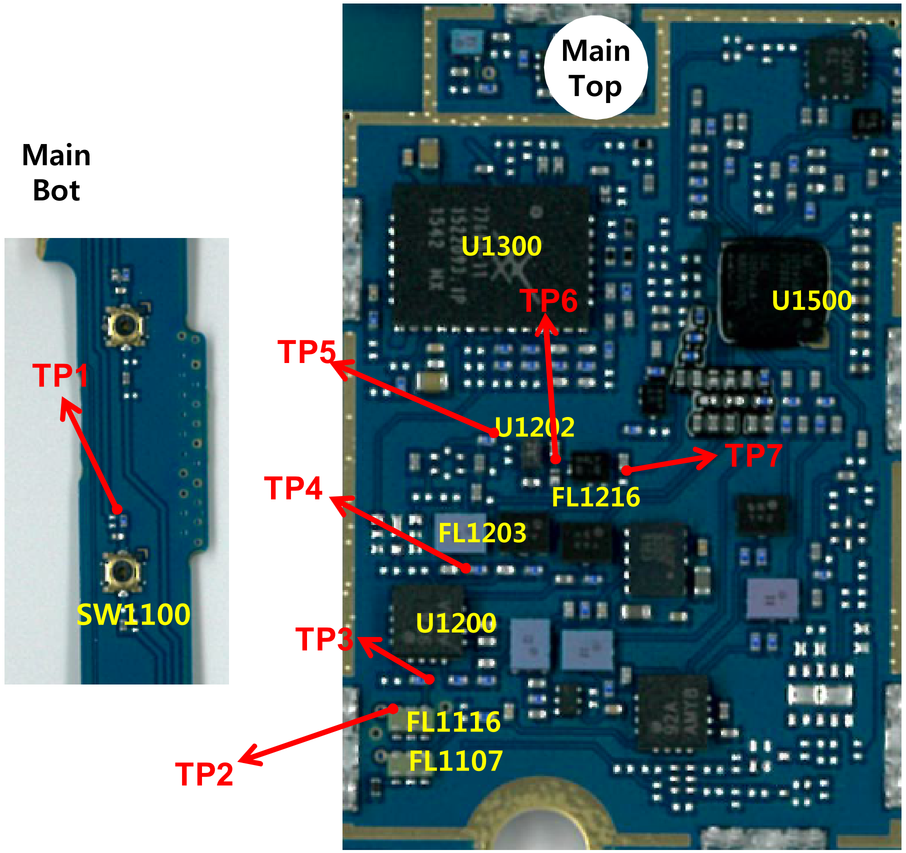

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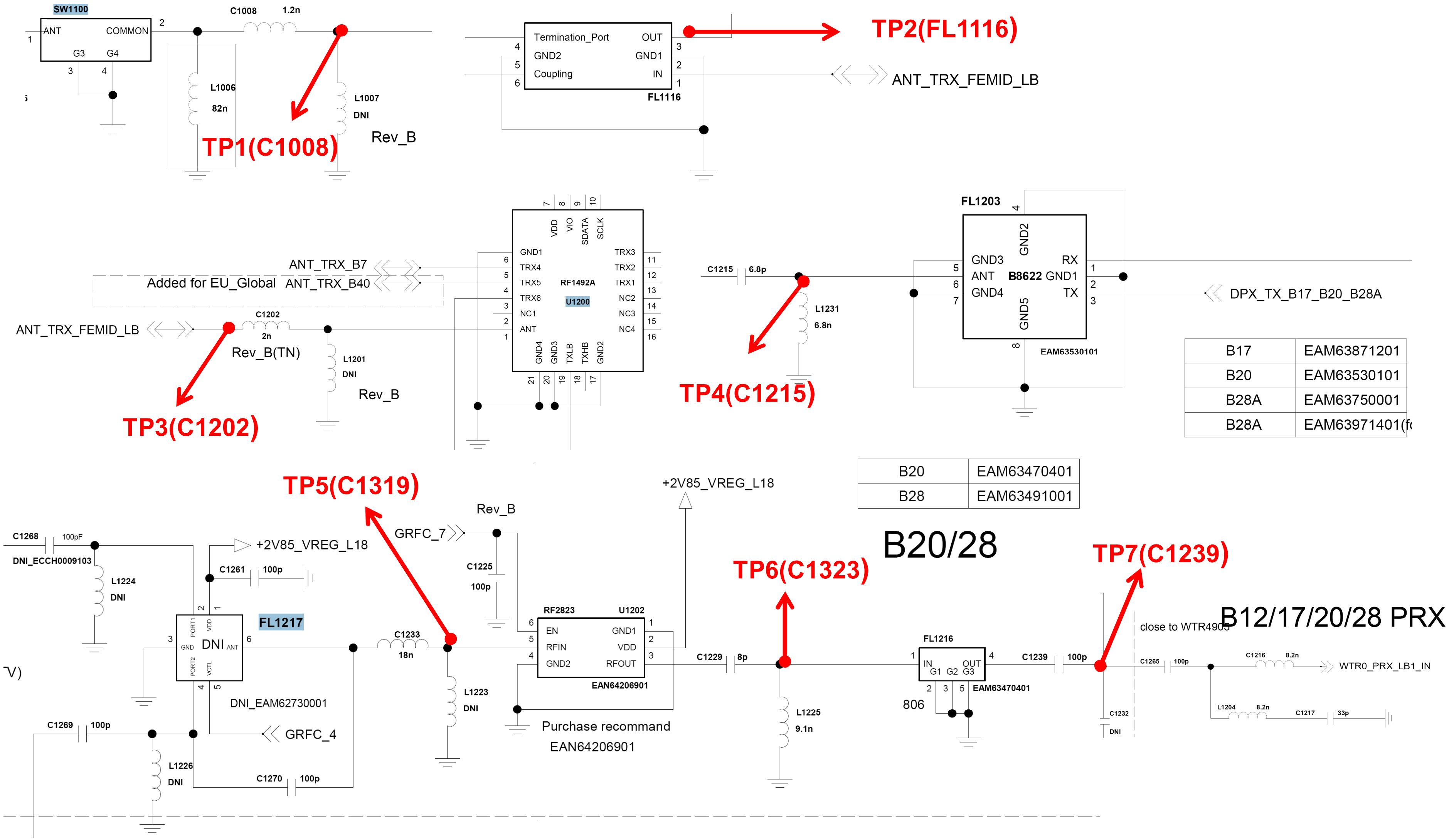
3.7 LTE RF PART

3.7.4.1 LTE B20 RF Part PRX RF PATH

Circuit Diagram



B17	EAM63871201
B20	EAM63530101
B28A	EAM63750001
B28A	EAM63971401(f

B20	EAM63470401
B28	EAM63491001

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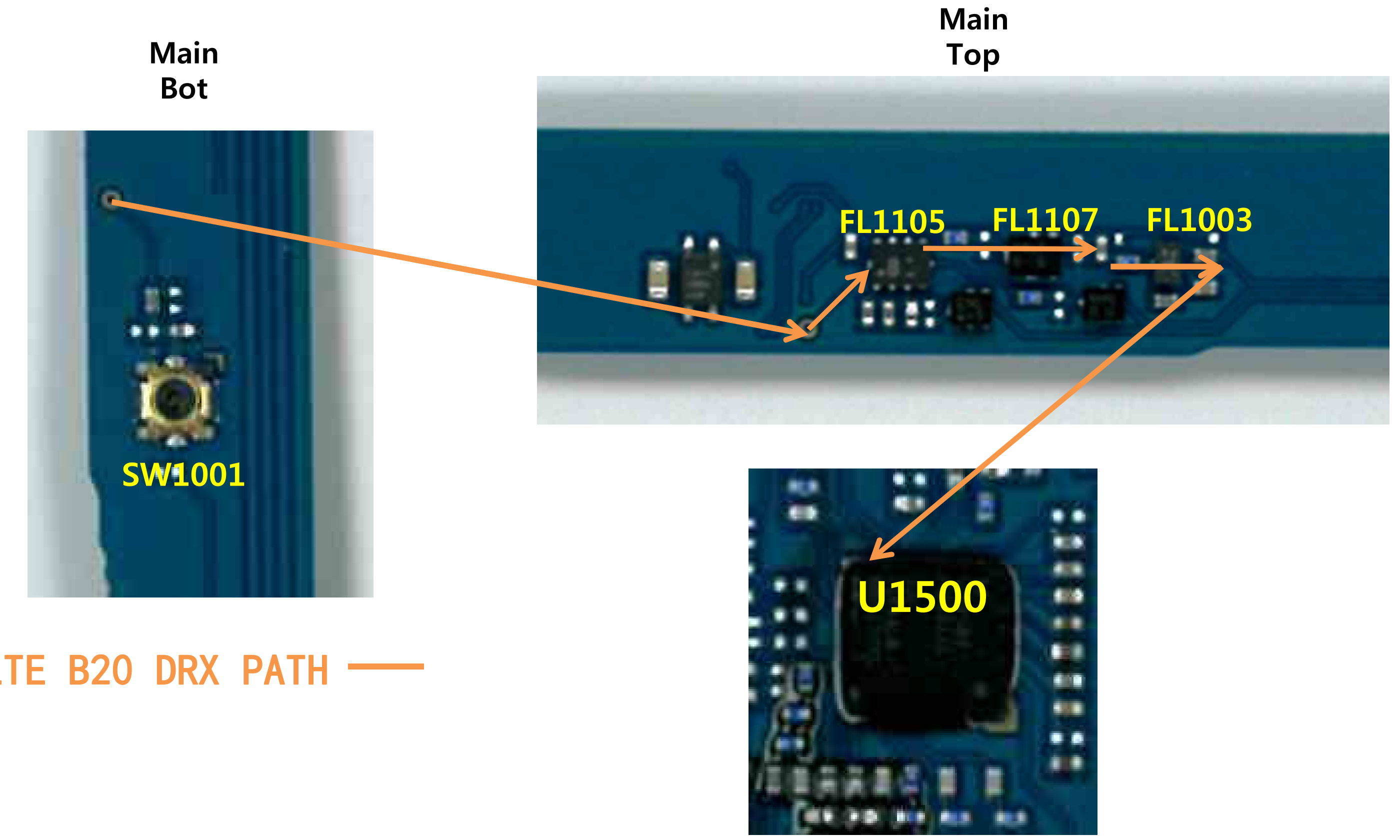
3.7 LTE RF PART

3.7.4.2 LTE B20 RF Part DRX RF PATH

Image

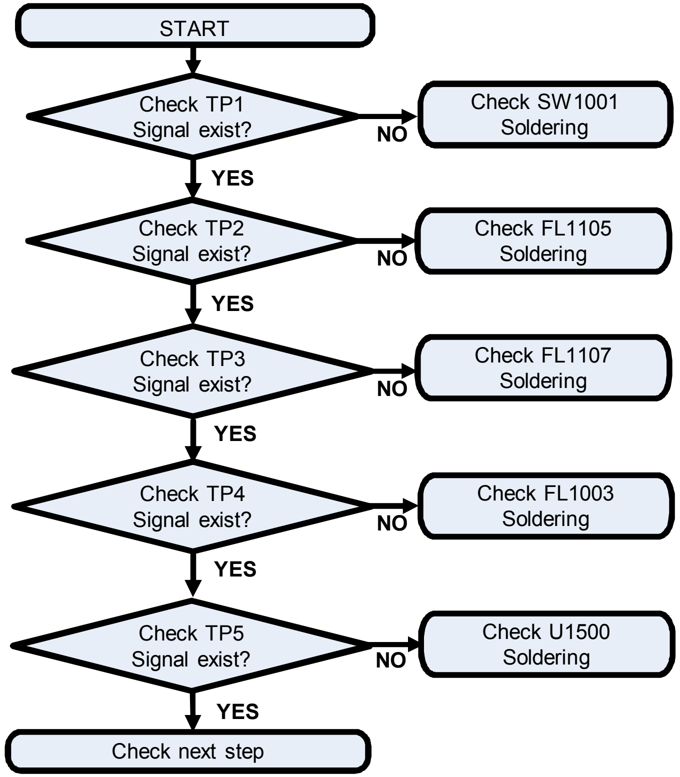


LTE B20 DRX PATH

3.7 LTE RF PART

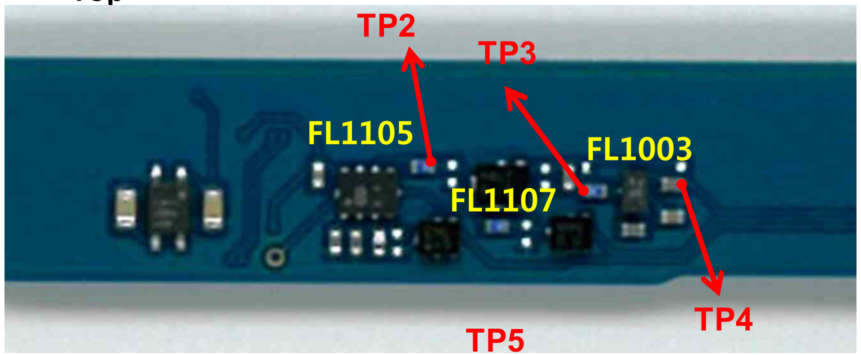
3.7.4.2 LTE B20 RF Part DRX RF PATH

Checking Flow

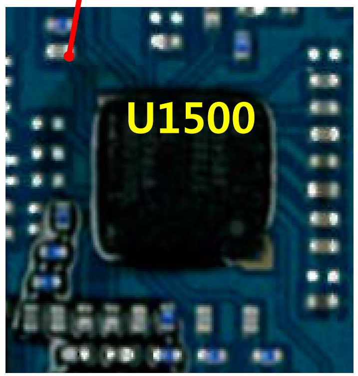


Image

Main Top



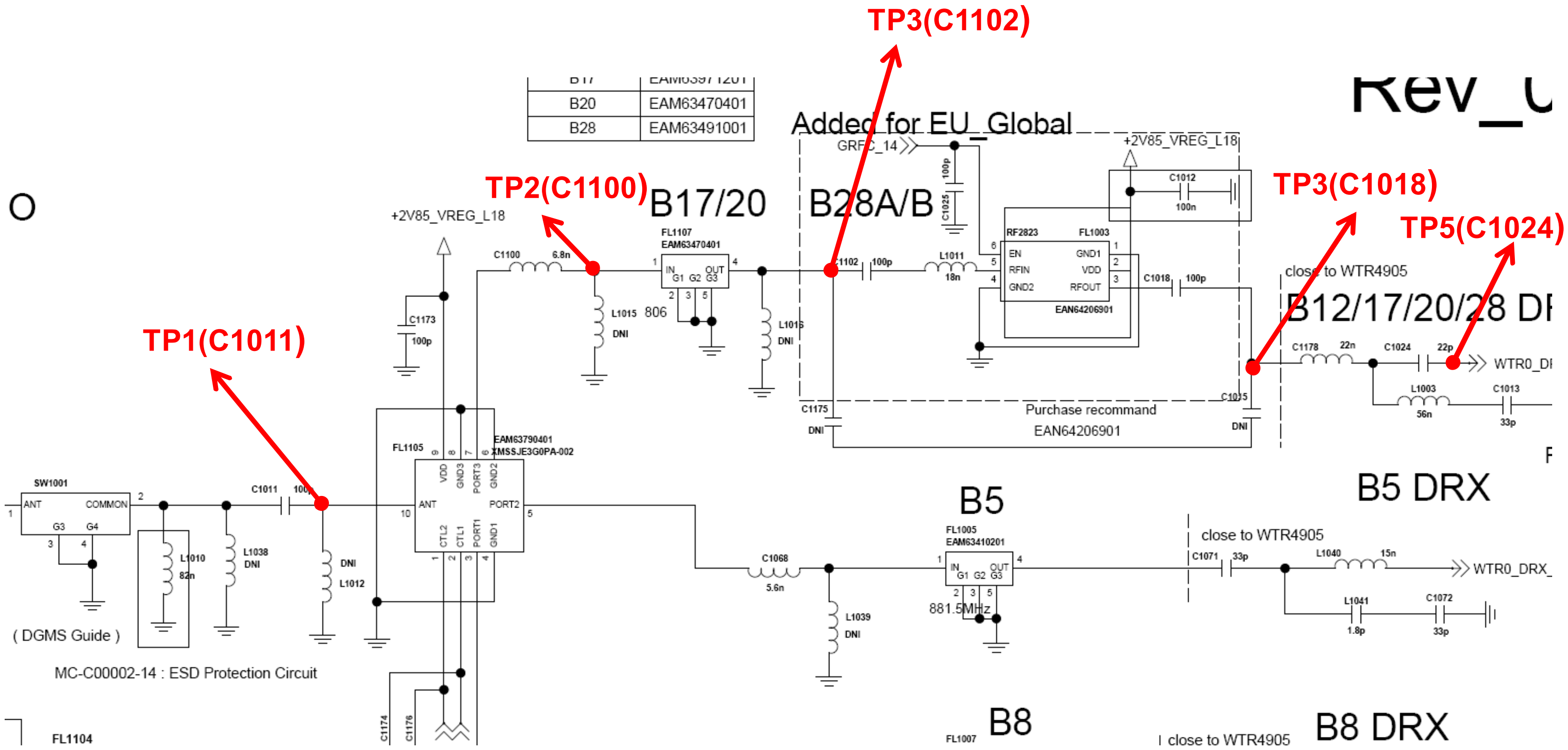
Main Bot



3.7 LTE RF PART

3.7.4.2 LTE B20 RF Part DRX RF PATH

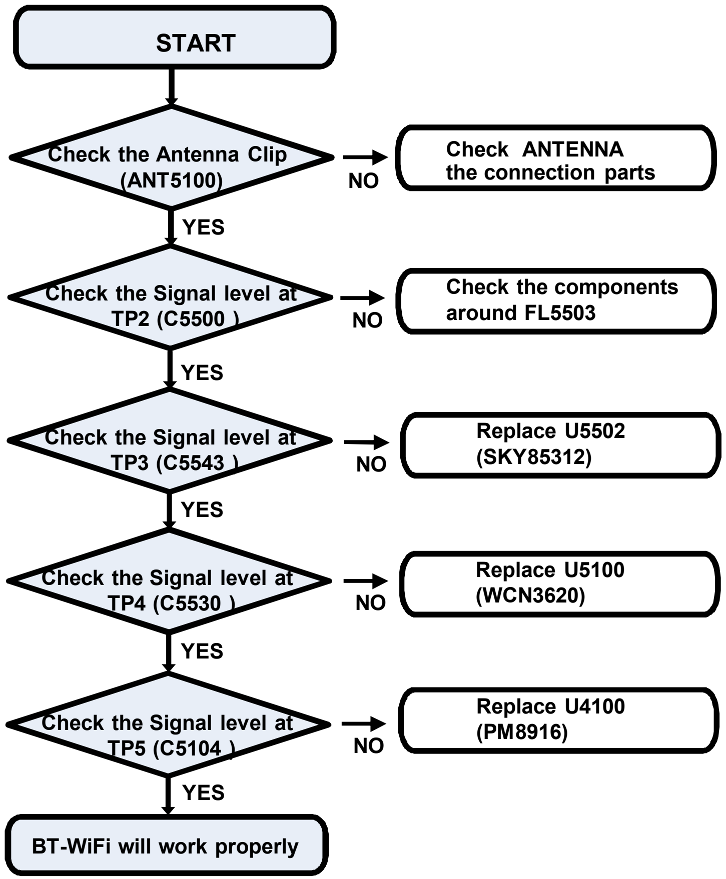
Circuit Diagram



3.8 Checking Wifi/BT Block

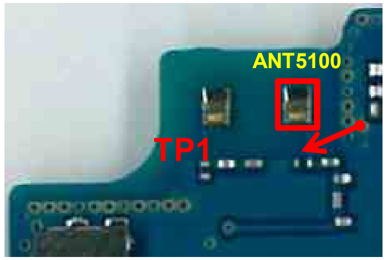
In case of the abnormal Operation of Wifi/BT, check the SKY85312, WCN3620 Output and PM8916 CLK

Checking Flow

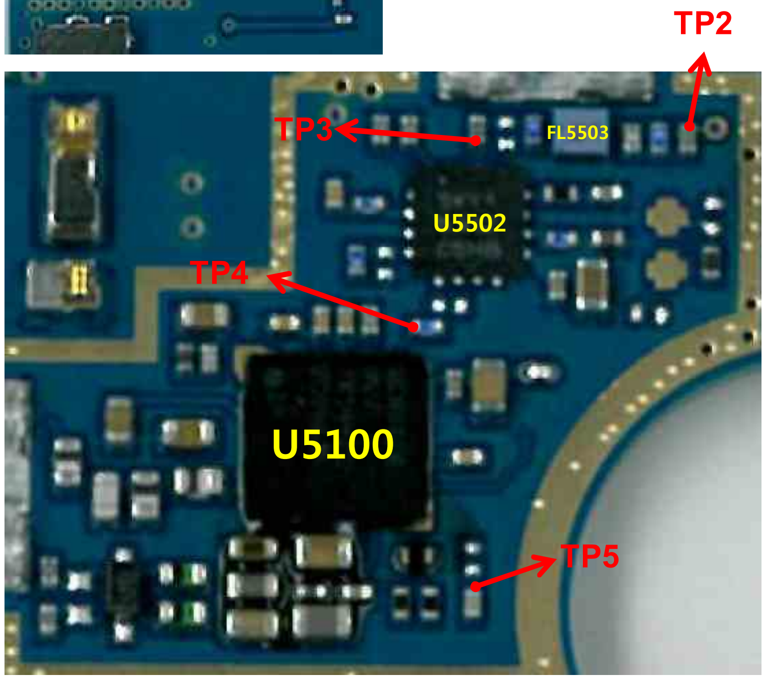


Image

Main Bot



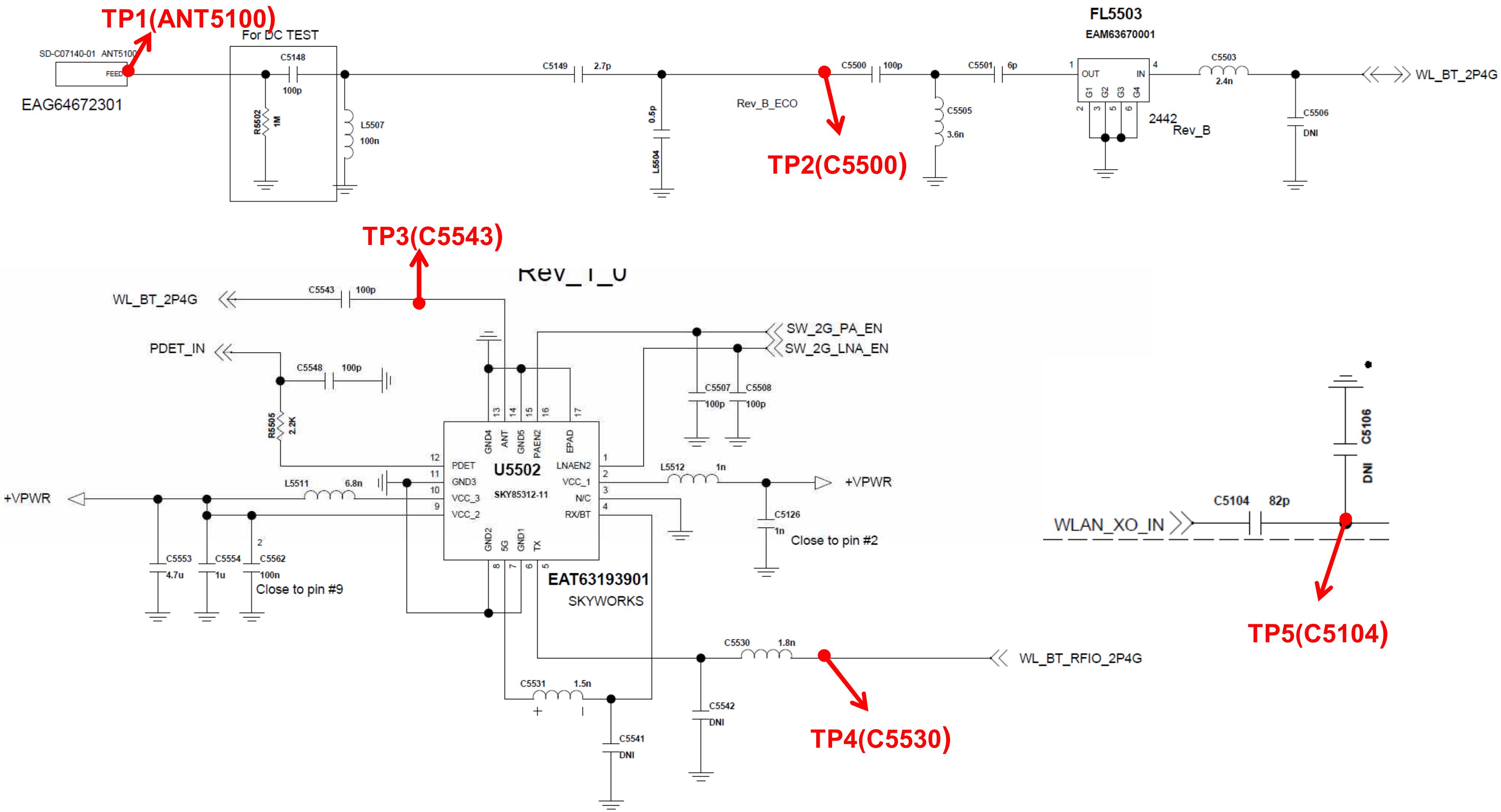
Main Top



3.8 Checking Wifi/BT Block

In case of the abnormal Operation of Wifi/BT, check the SKY85312, WCN3620 Output and PM8916 CLK

Circuit Diagram



LGE Internal Use Only

3.9 Checking GPS Block

In case of the abnormal Operation of GPS, check the WTR4905 Output and PM8916 CLK and LNA

Checking Flow

START

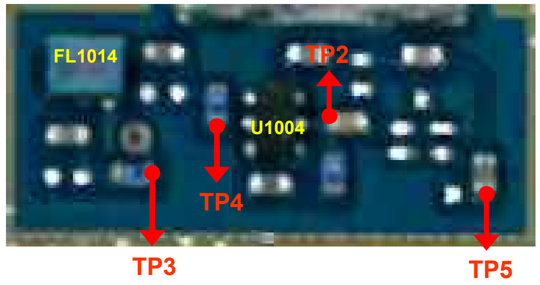
- Check the Signal level at TP1 (C1511) — NO → Replace U4100 (PM8916)
- YES → Check the voltage at, +2 V, TP2 (C1079) — NO → The Problem may be Logic part. Refer to Logic troubleshoot
- YES → Check the signal level at TP3 (C1084) — NO → Check ANTENNA the connection parts
- YES → Check the signal level at TP4 (L1148) — NO → Check the components around FL1014
- YES → Check the signal level at TP5 (L1048) — NO → Check the components around U1004

GPS will work properly

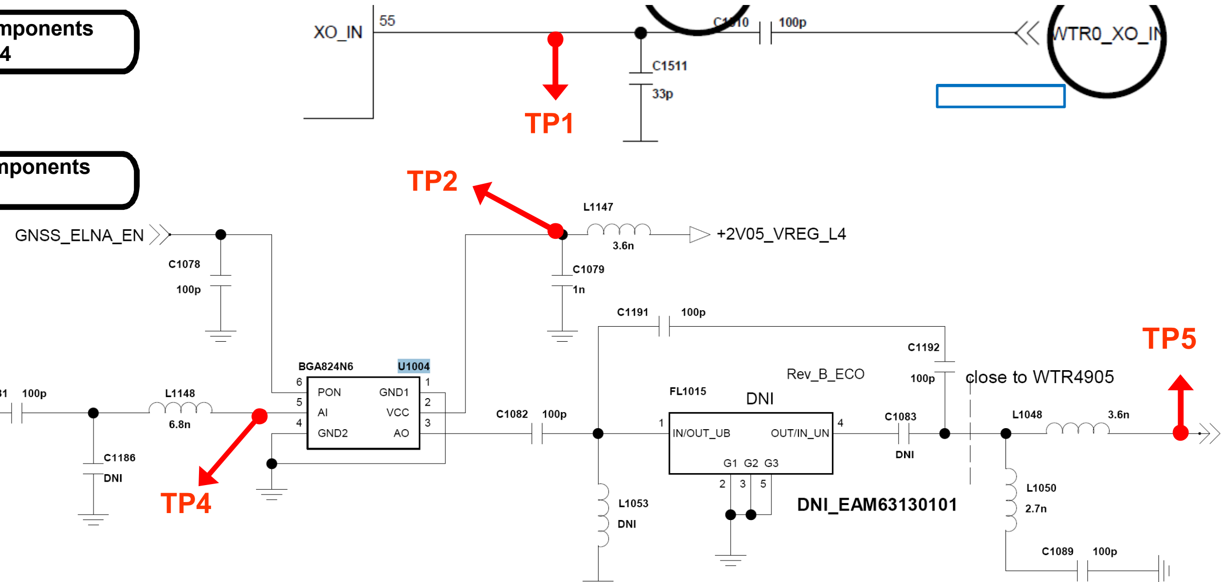
Image

TOP Side

BOT Side



Circuit Diagram

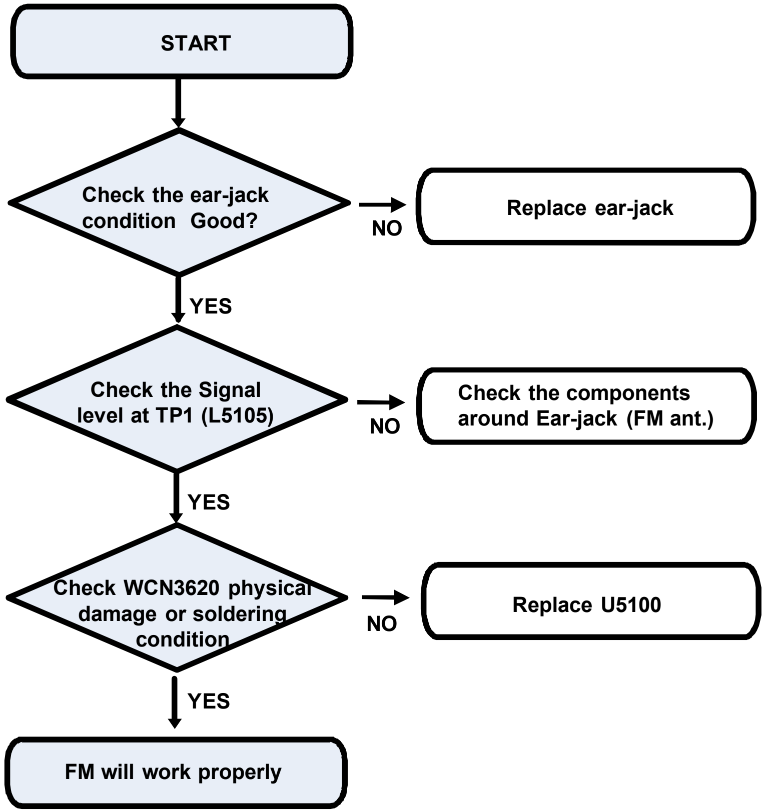


LGE Internal Use Only

3.10 Checking FM Block

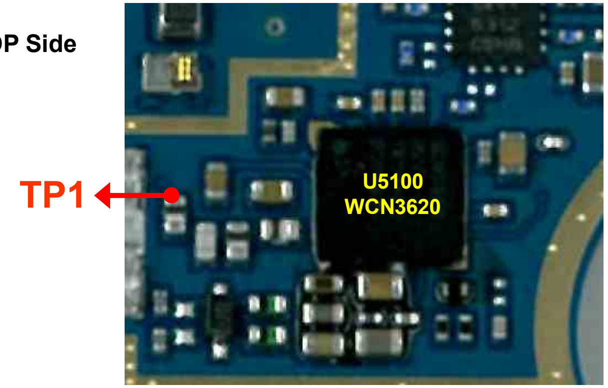
In case of the abnormal operation of FM Radio, check the WCN3620 Output and Ear-jack

Checking Flow

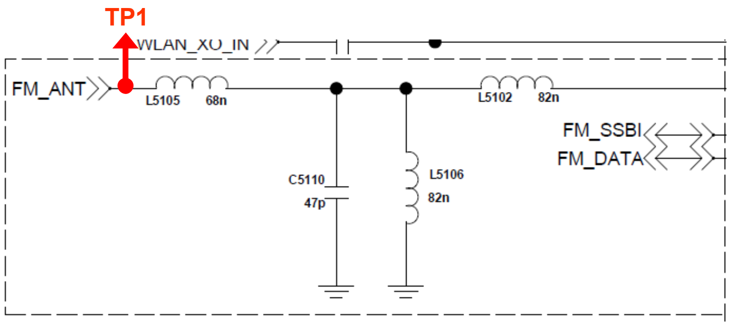


Image

TOP Side



Circuit Diagram

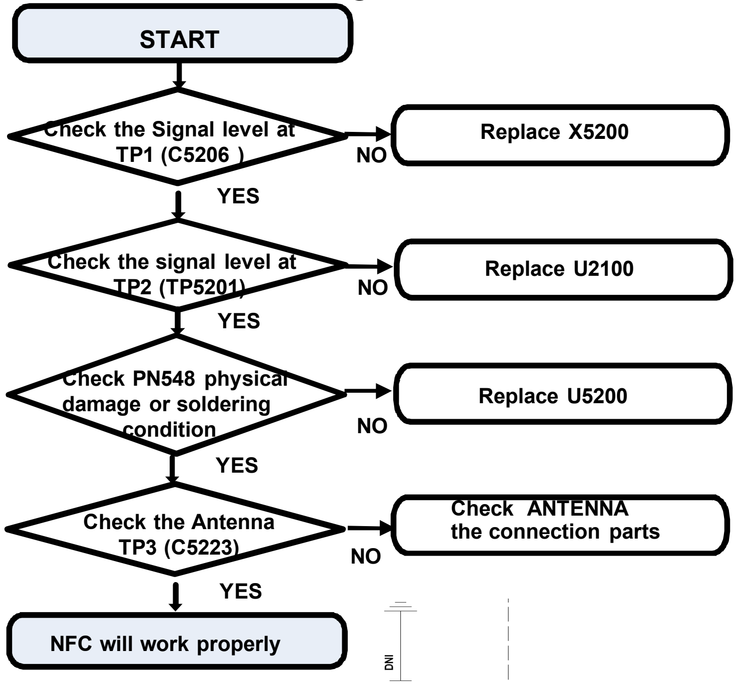


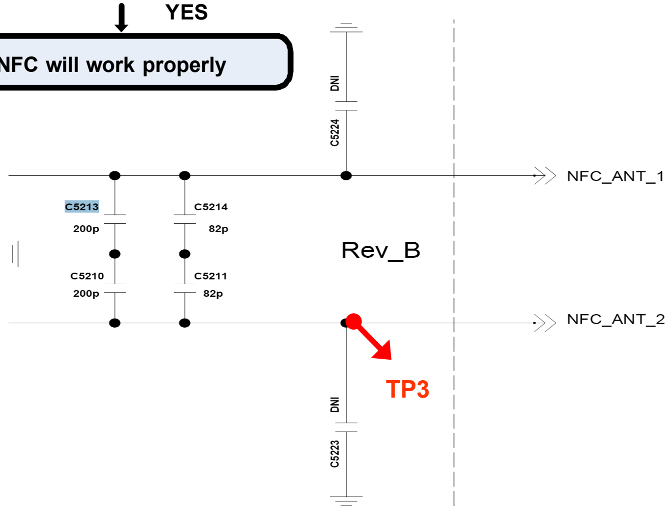
LGE Internal Use Only

3.11 Checking NFC Block

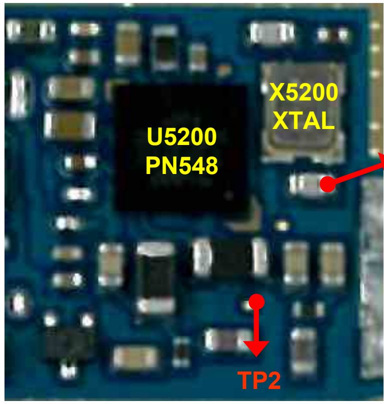
In case of the abnormal operation of NFC, check the PN548 Output and X5200 CLK

Checking Flow



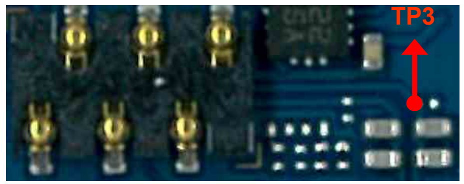


Image

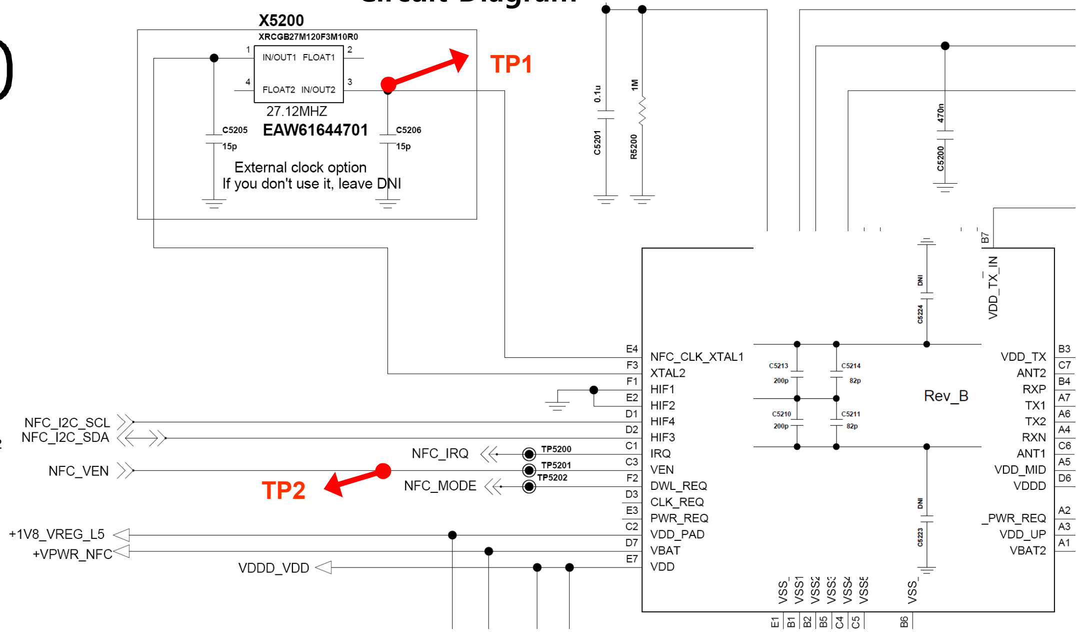


Top Side

BOT Side



Circuit Diagram



LGE Internal Use Only

3.12 Checking DAB/DMB Block

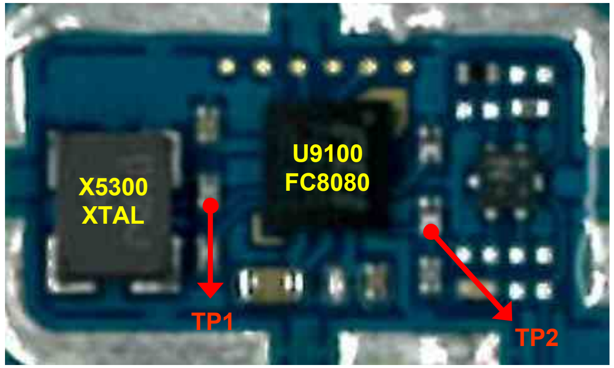
In case of the abnormal operation of DAB/DMB, check the FC8080 Output, Ear-jack and X5300 CLK

Checking Flow

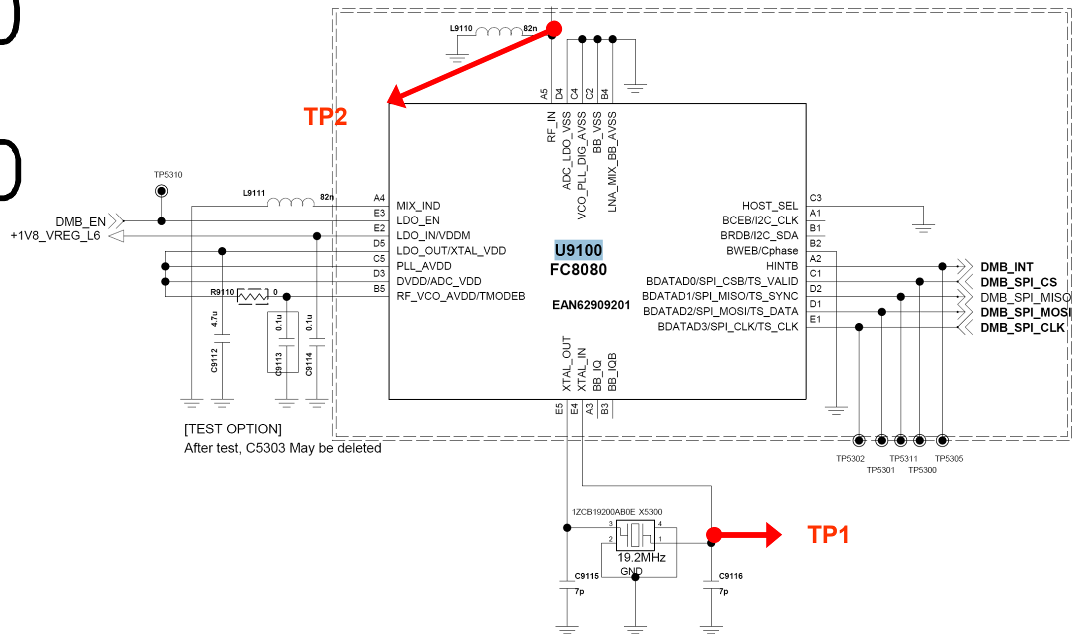
- START
- Check the Signal level at TP1 (C9116) → NO → Replace X5300
  - YES
- Check the ear-jack condition Good? → NO → Replace ear-jack
  - YES
- Check the Signal level at TP2 (L9110) → NO → Check the components around Ear-jack (DMB ant.)
  - YES
- Check FC8080 physical damage or soldering condition → NO → Replace U9100
  - YES
- NFC will work properly

Image

BOT Side



Circuit Diagram

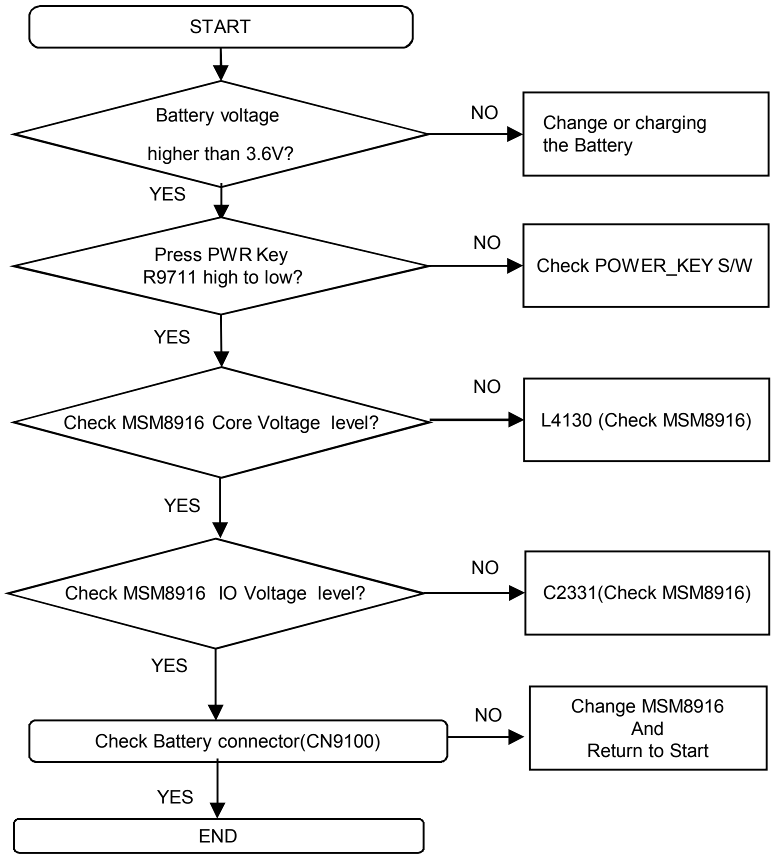


LGE Internal Use Only

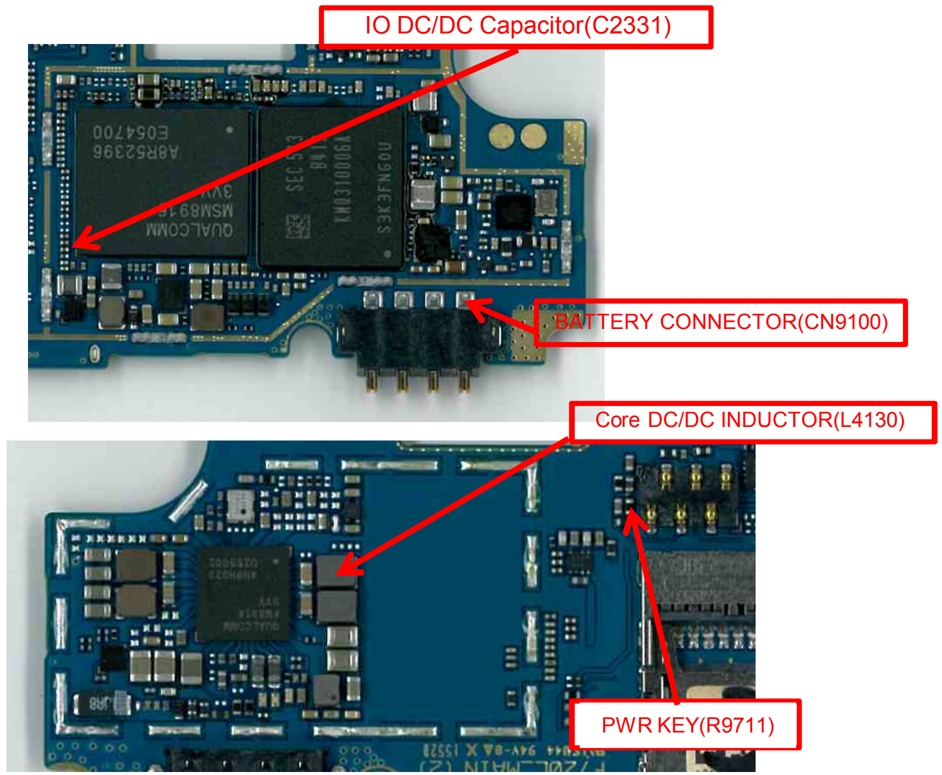
3.13 Power

Checking Power signal(Battery connector, Power Key, MSM Power)

Checking Flow



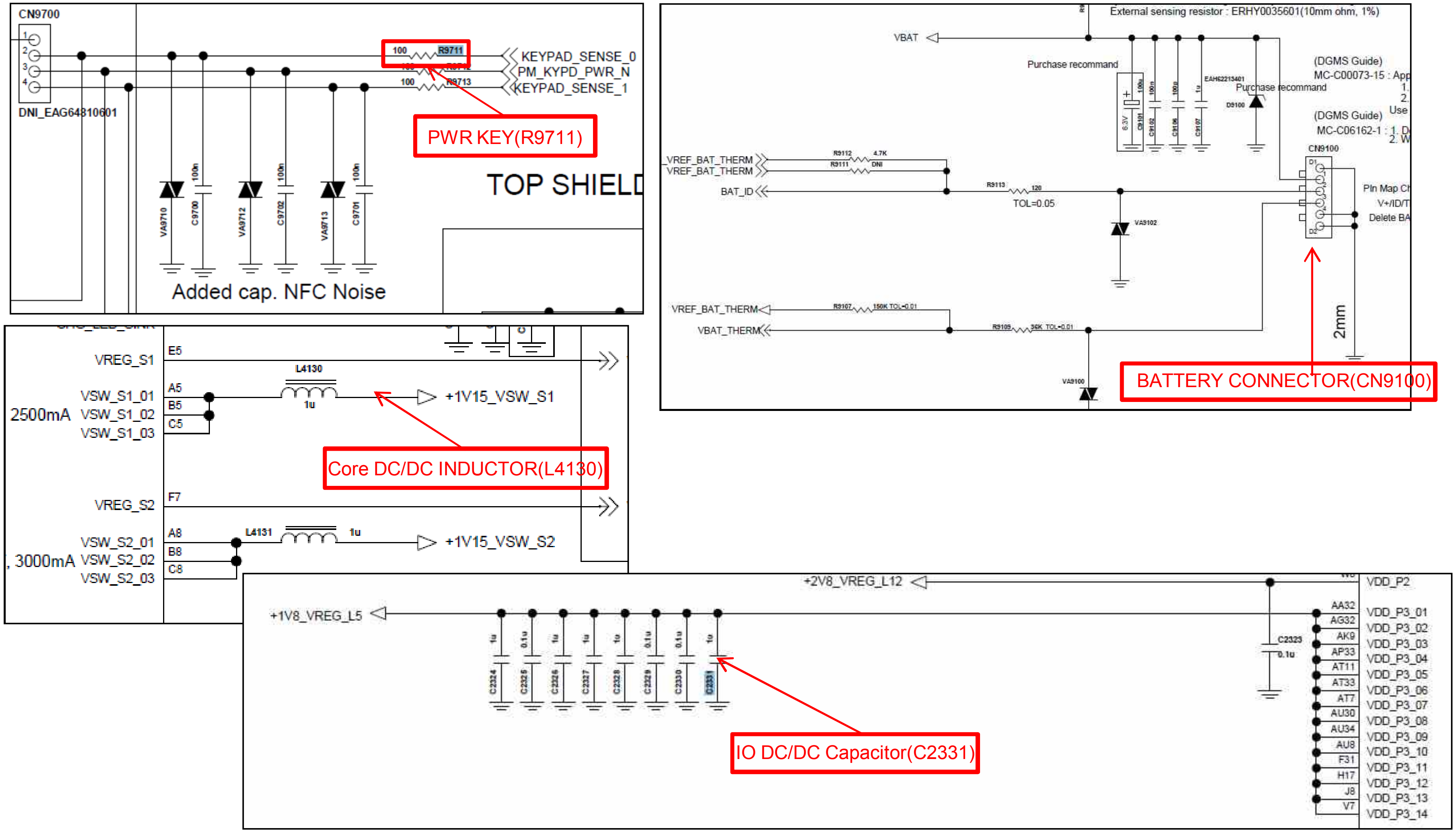
Image



3.13 Power

Checking Power signal(Battery connector, Power Key, MSM Power)

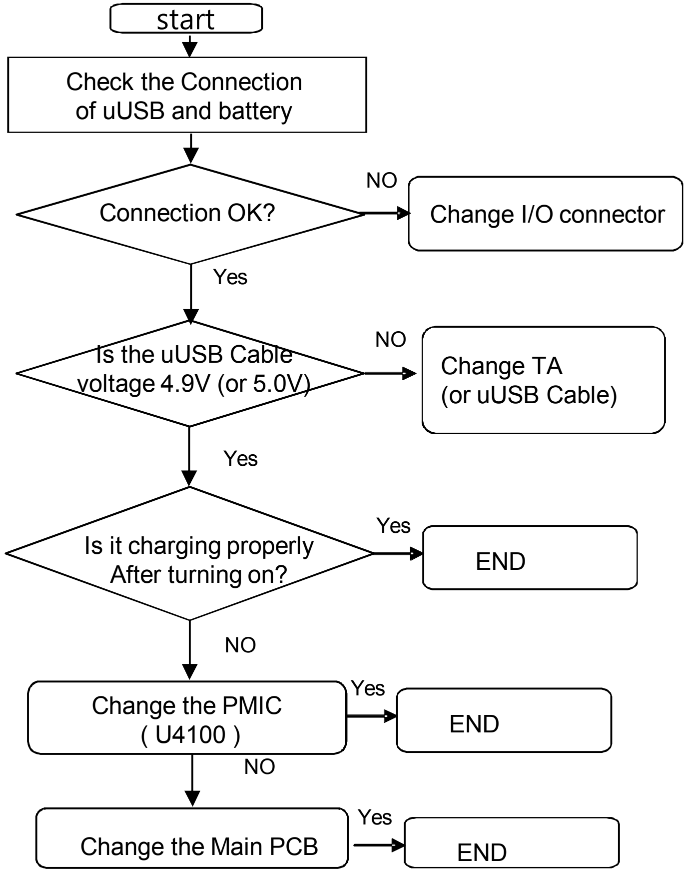
Circuit Diagram



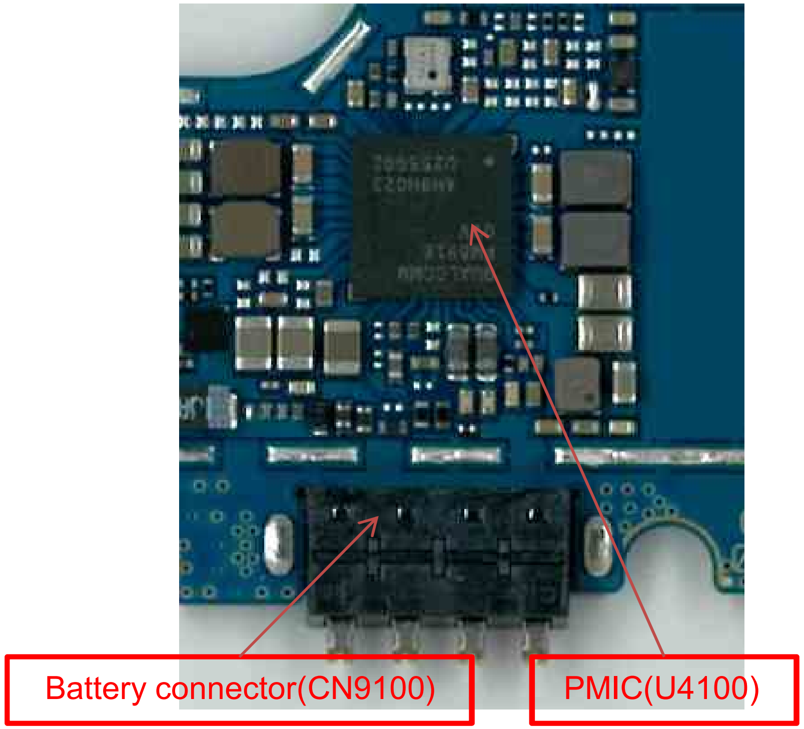
3.14 charger

The I/O connector and uUSB cable voltage(5.0V) is used as the reference one of PMIC for charging.

Checking Flow



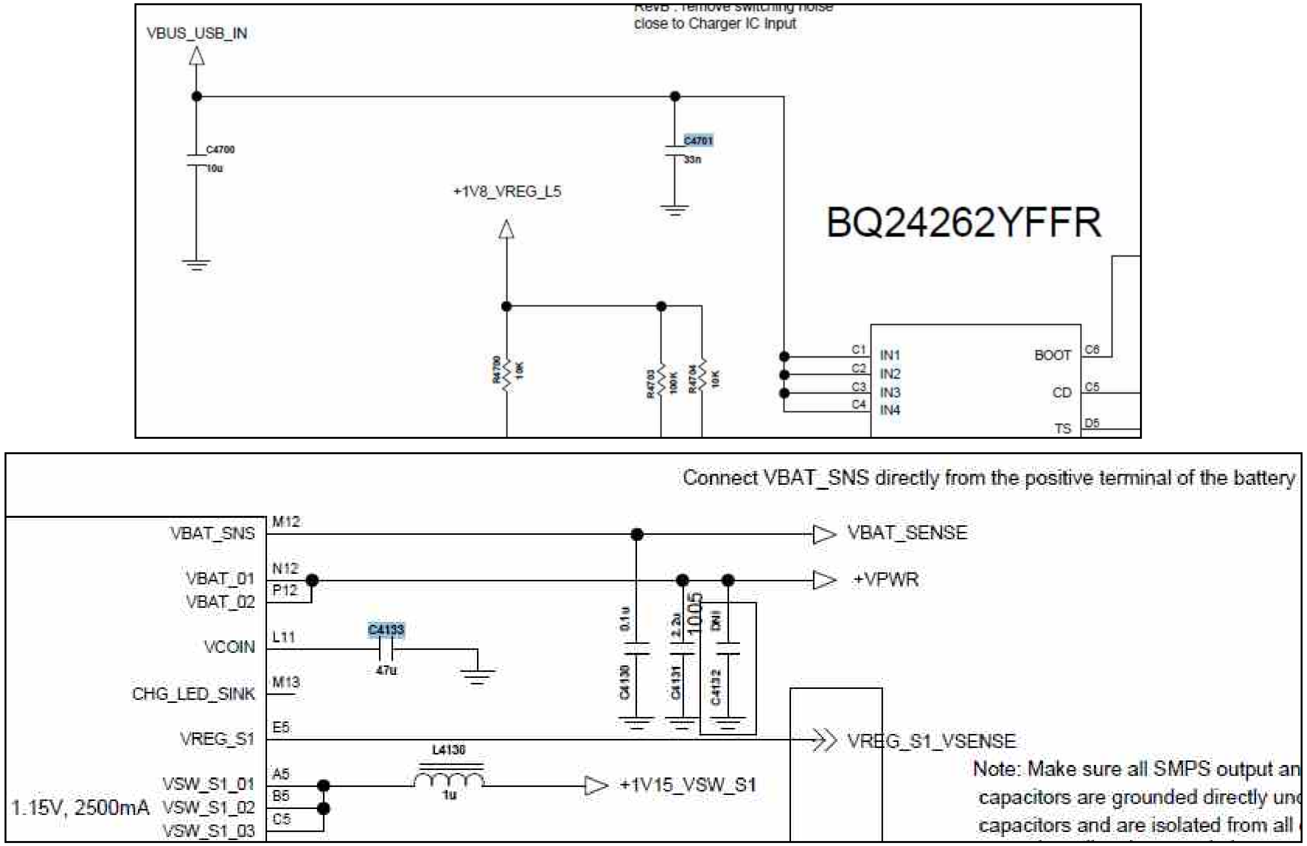
Image



3.14 charger

The I/O connector and uUSB cable voltage(5.0V) is used as the reference one of PMIC for charging.

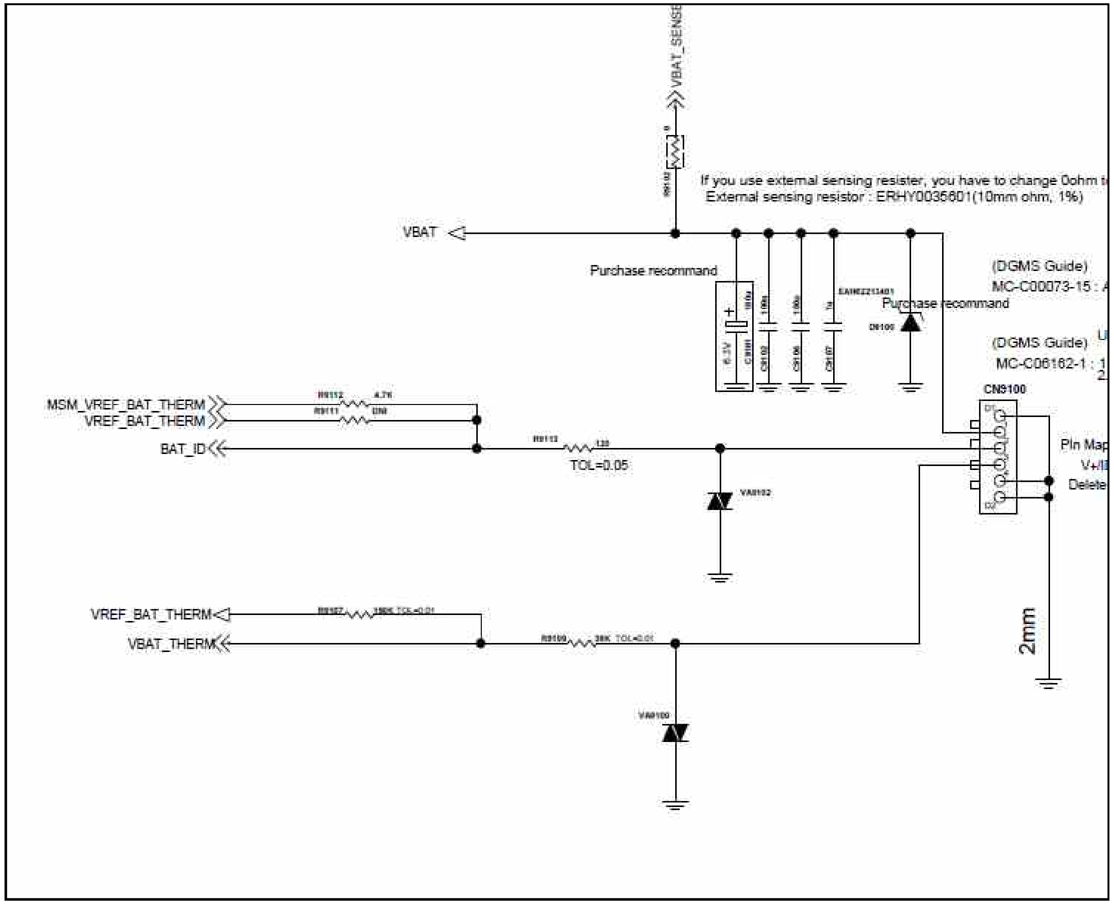
Circuit Diagram



3.14 charger

The I/O connector and uUSB cable voltage(5.0V) is used as the reference one of PMIC for charging.

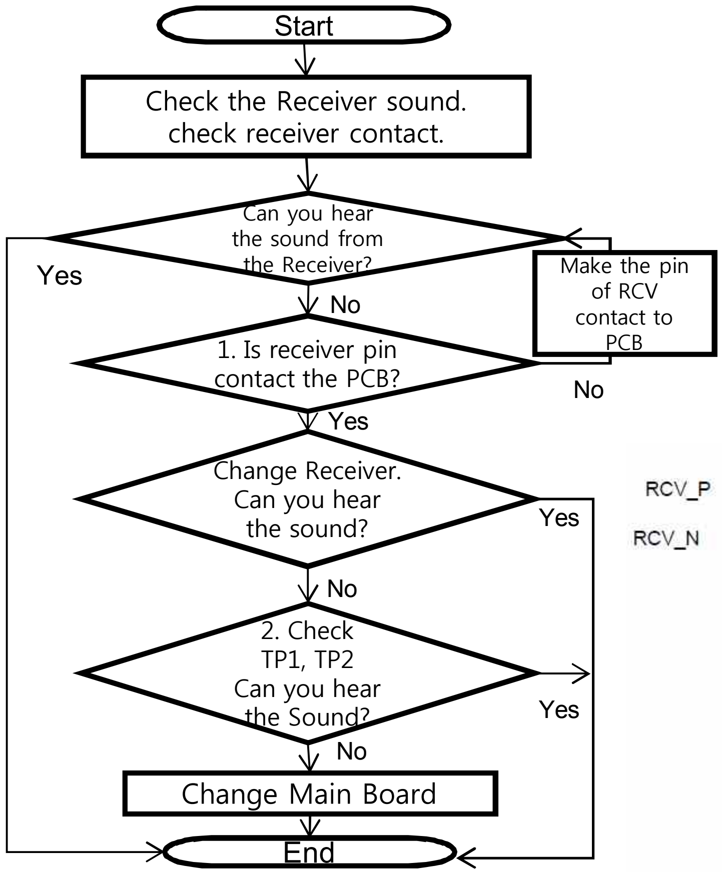
Circuit Diagram



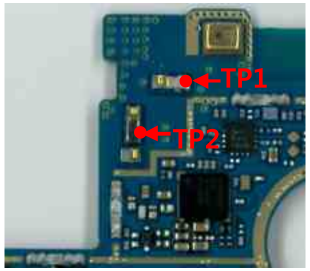
3.15.1 [Audio] receiver Block

The receiver control signals are generated by MSM8916(U2100), PM8916(U4100).
Voice Receiving path as below :MSM8916 → PM8916 → EARO_P/EARO_M → Receiver.

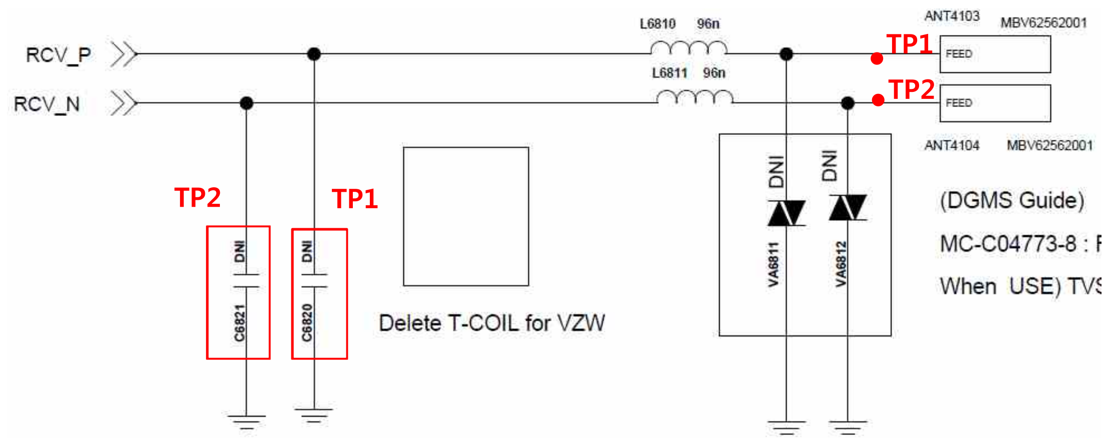
Checking Flow



Image



Circuit Diagram

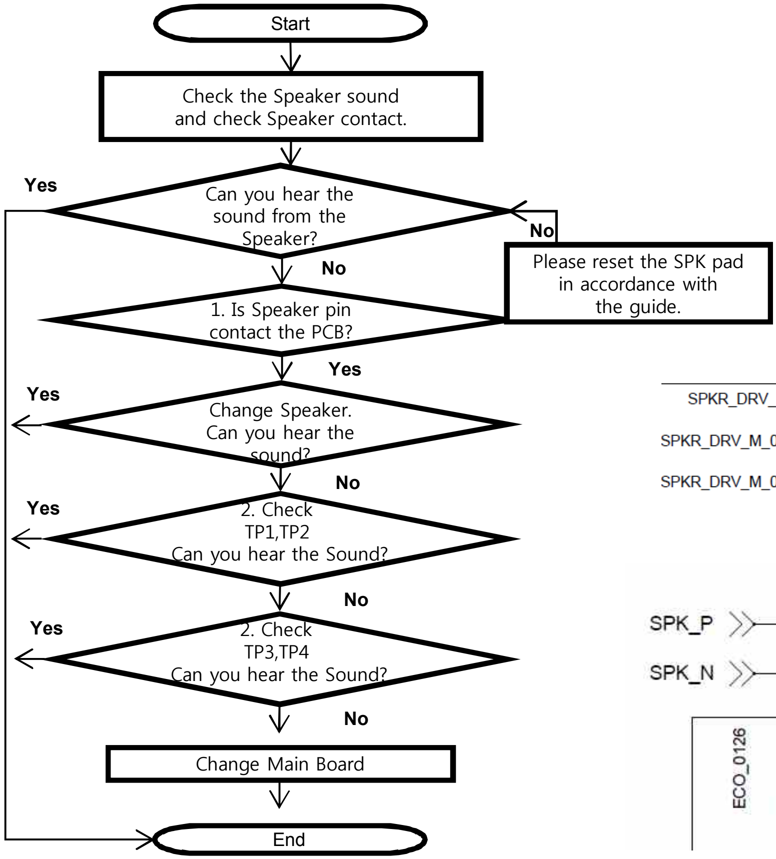


LGE Internal Use Only

3.15.2 [Audio] speaker

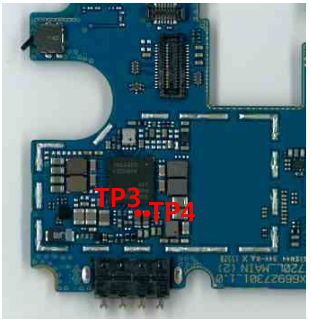
The Speaker control signals are generated by MSM8916(U2100), PM8916(U4100).
the MSM chip and the speaker are to be checked out.

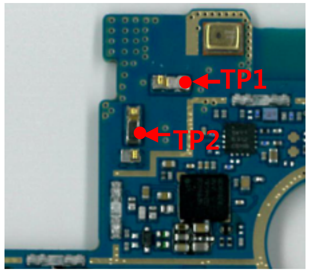
Checking Flow



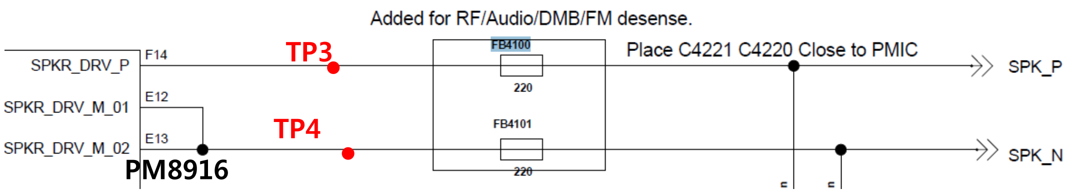
Image

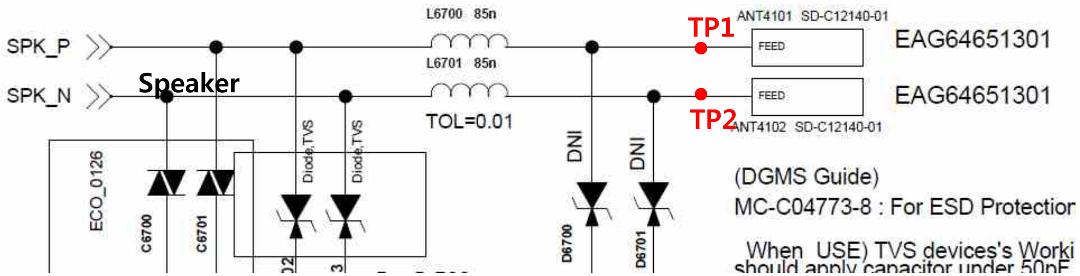
Speaker module





Circuit Diagram



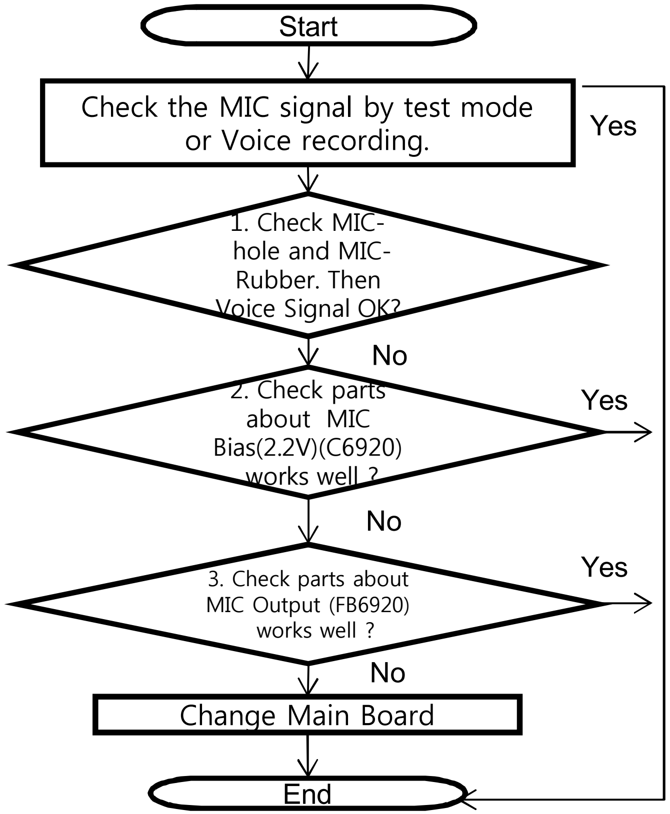


LGE Internal Use Only

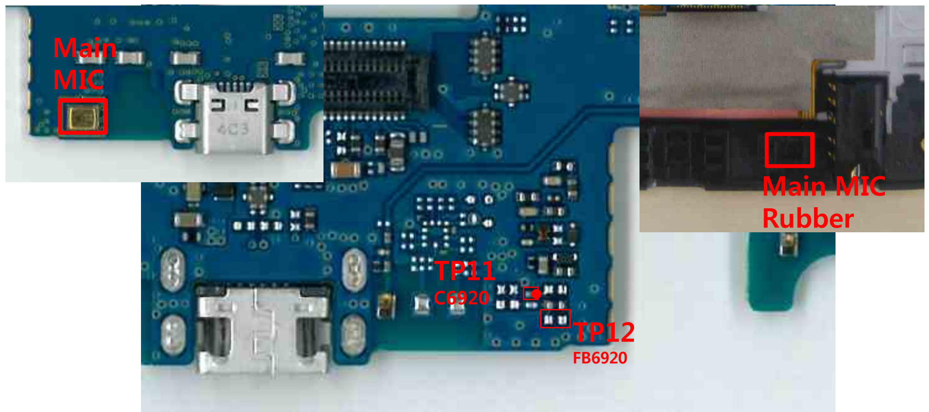
3.15.3 [Audio] Main MIC

A Main MIC is located at the bottom of PCB
It operates in case of voice call (Handset, Speaker phone), camcorder recording

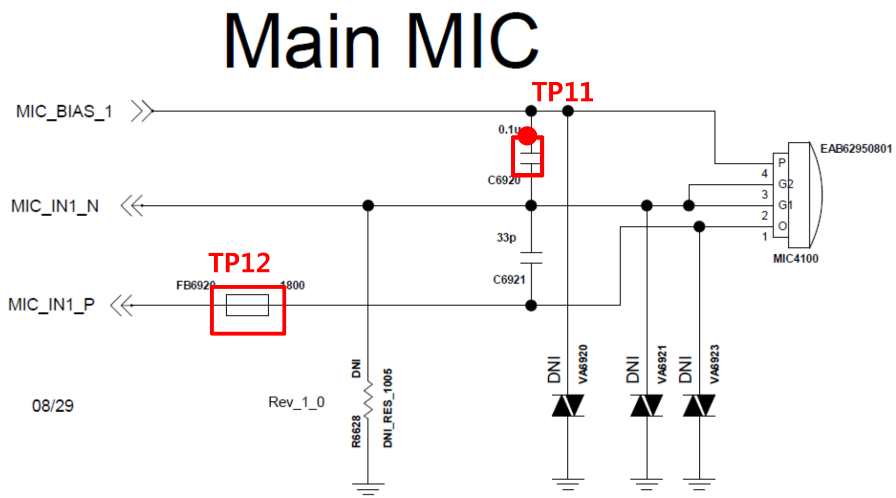
Checking Flow



Image

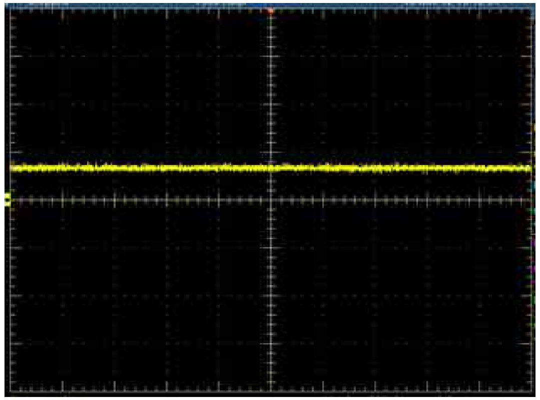
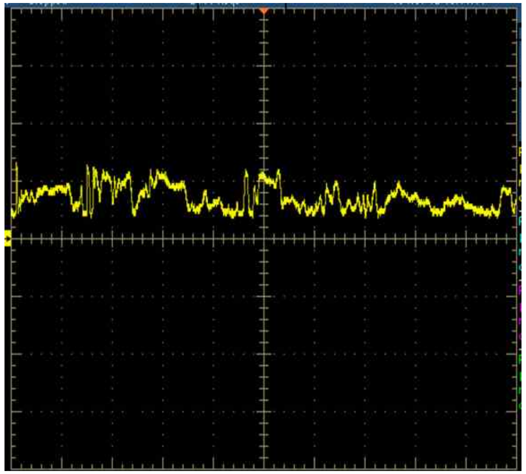


Circuit Diagram



3.15.3 [Audio] Main MIC

waveform with oscilloscope

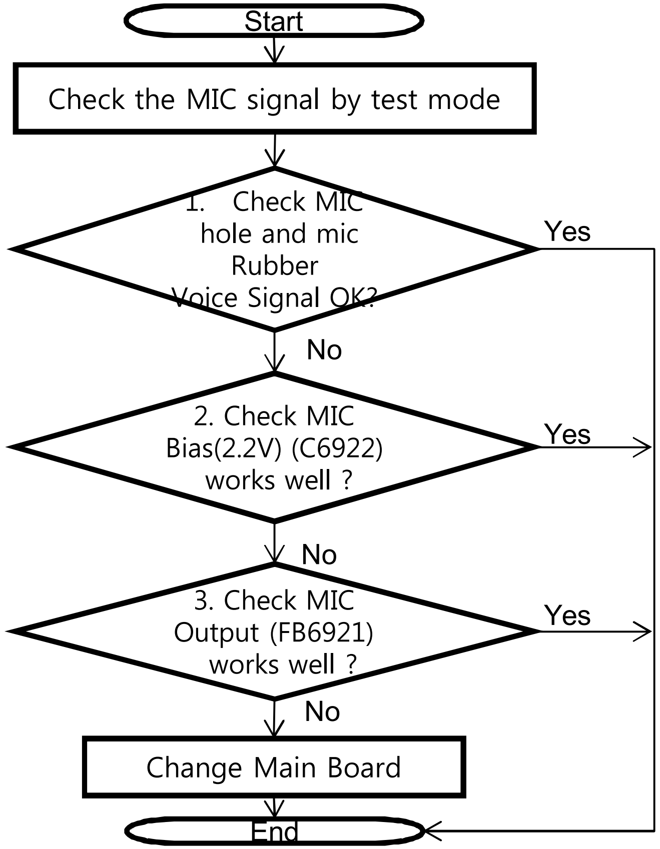
Normal MIC Out Signal (Idle)		About 0.7V Flat
Normal MIC Out Signal (Voice Input)		Moves (Dynamic)

LGE Internal Use Only

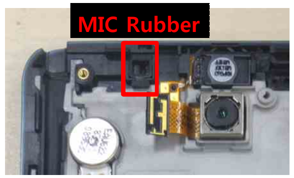
3.15.4 [Audio] SUB MIC

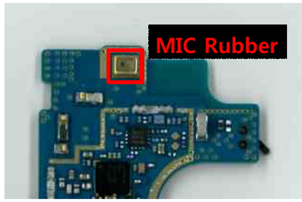
A SUB MIC is located at the TOP of PCB
It operates in case of voice call (Handset, Speaker phone).

Checking Flow



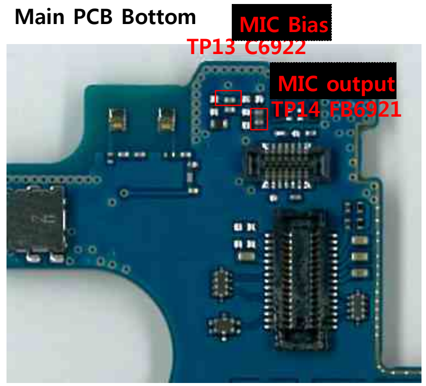
* Check SUB Mic hole.



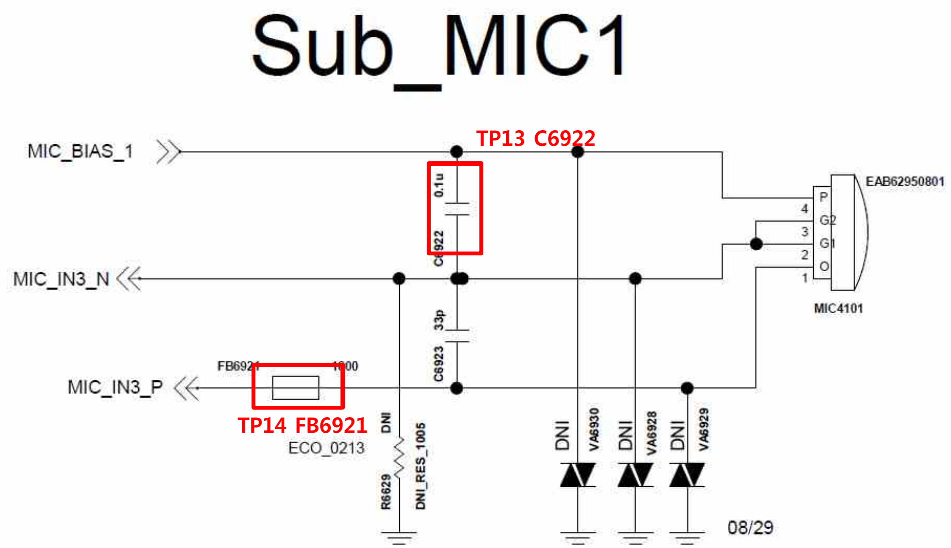


Image

Main PCB Bottom



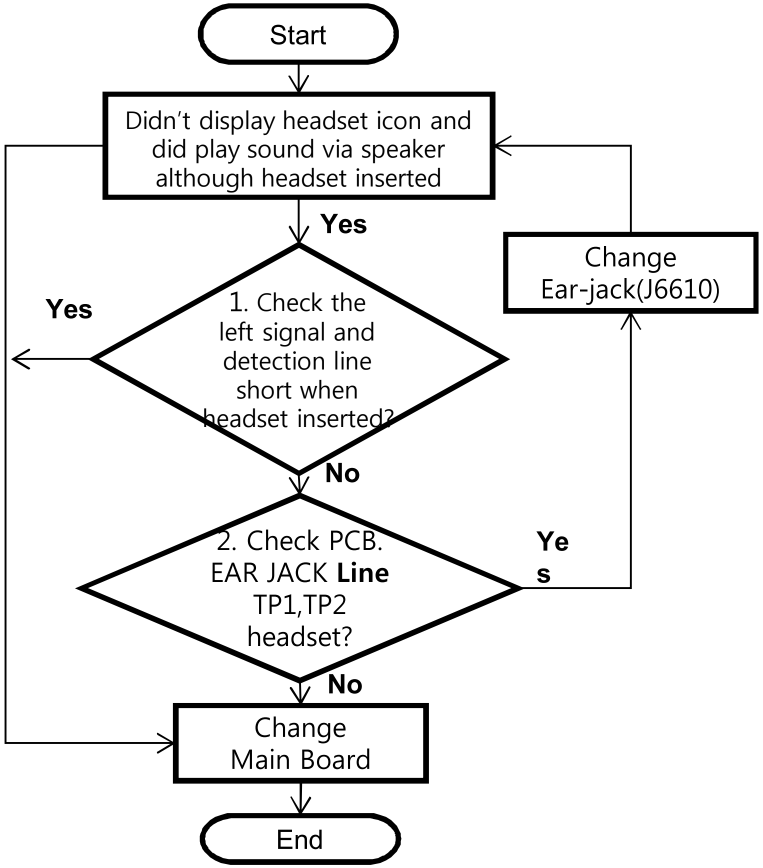
Circuit Diagram



3.15.5 [Audio] Ear MIC

The Ear MIC control signals are generate by MSM8916(U2100), PM8916(U4100)
Case A. Unable detection of headset insert.

Checking Flow

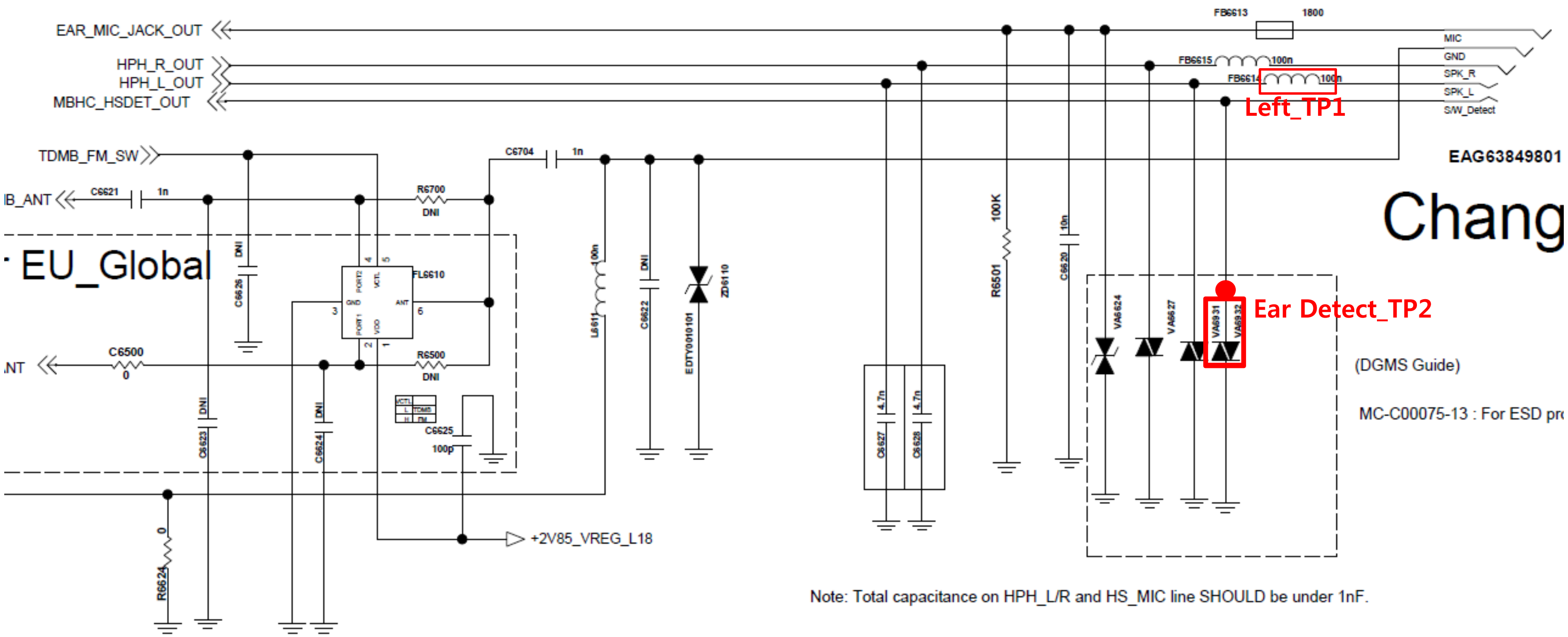


Image

3.15.5 [Audio] Ear MIC

The Ear MIC control signals are generate by MSM8916(U2100), PM8916(U4100)
Case A. Unable detection of headset insert.

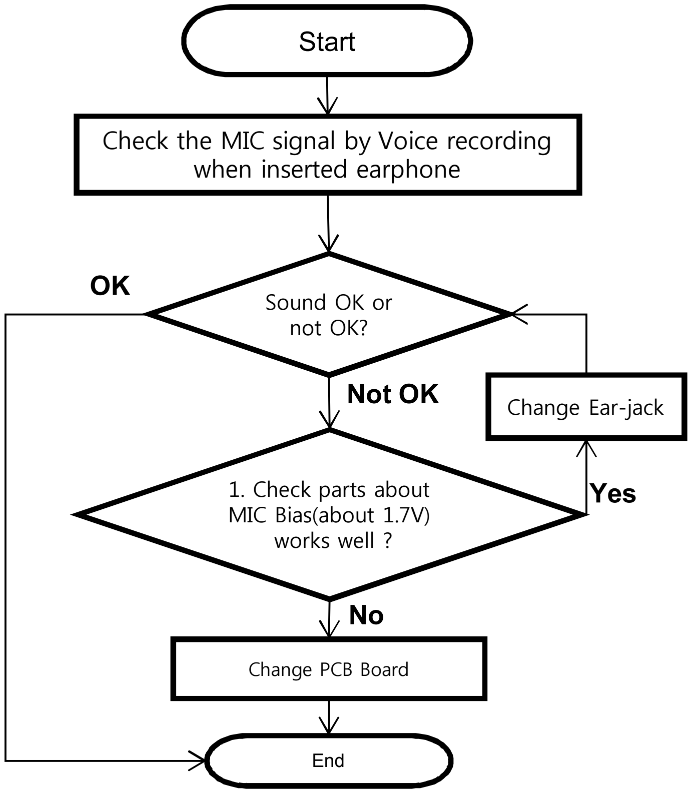
Circuit Diagram



3.15.5 [Audio] Ear MIC

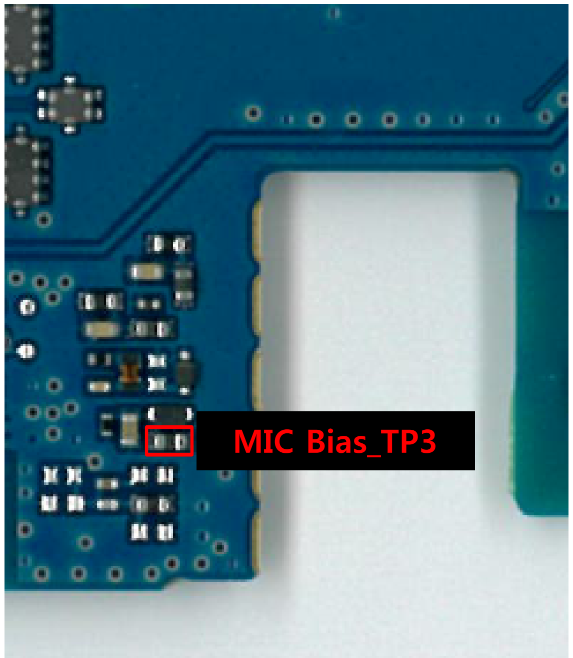
The Ear MIC control signals are generate by MSM8916(U2100), PM8916(U4100)
Case B. No Sound from MIC of earphone

Checking Flow



Image

Check parts about MIC Bias(about 1.7V) (FB6613)

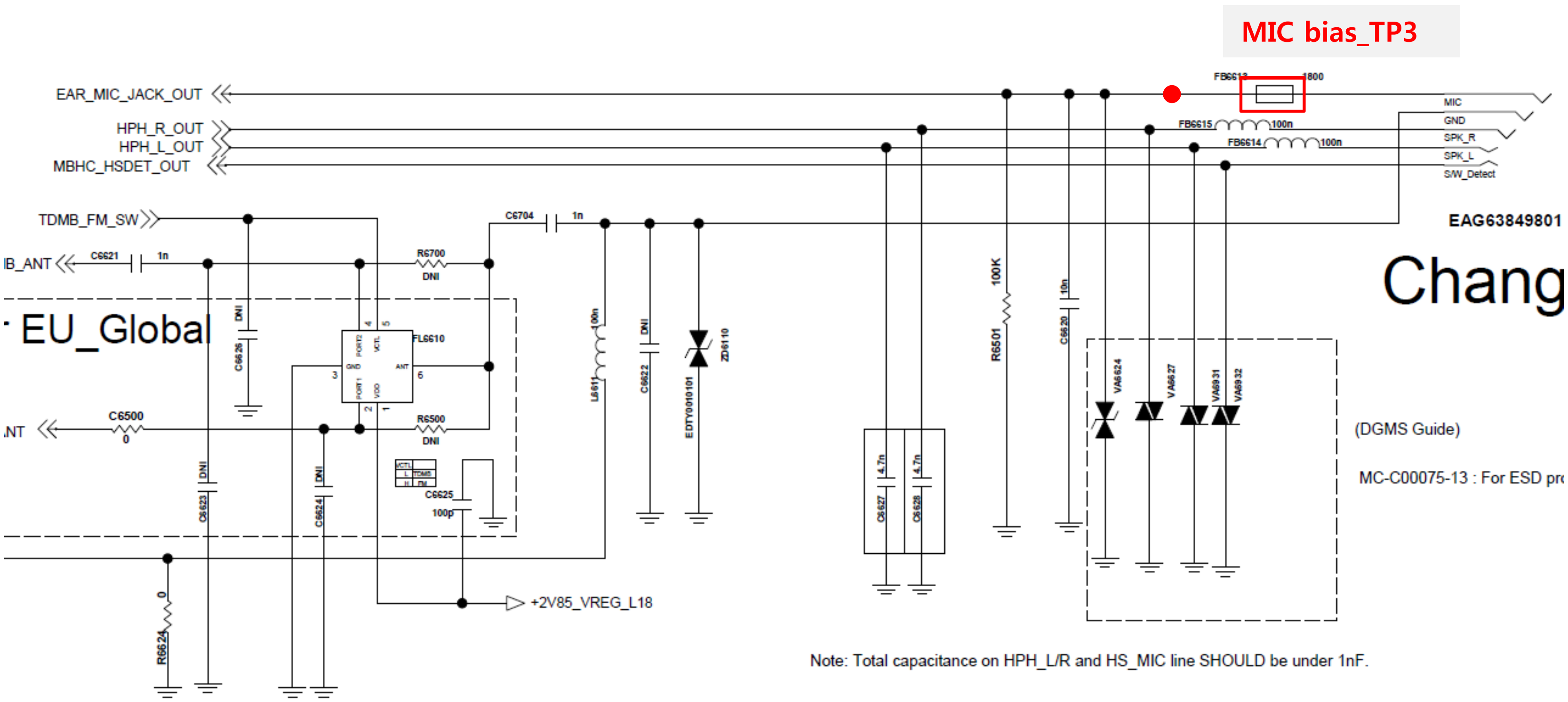


LGE Internal Use Only

3.15.5 [Audio] Ear MIC

The Ear MIC control signals are generate by MSM8916(U2100), PM8916(U4100)
Case B. No Sound from MIC of earphone

Circuit Diagram

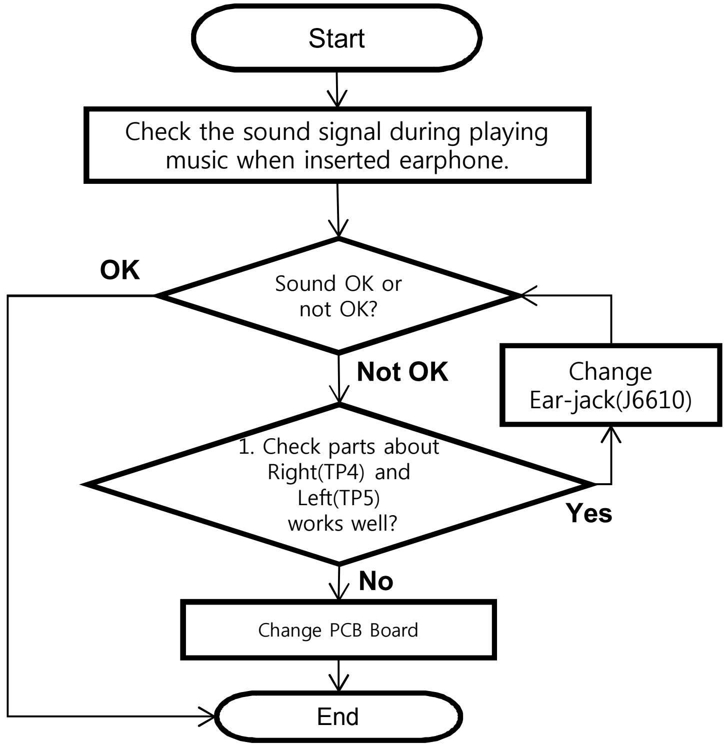


LGE Internal Use Only

3.15.5 [Audio] Ear MIC

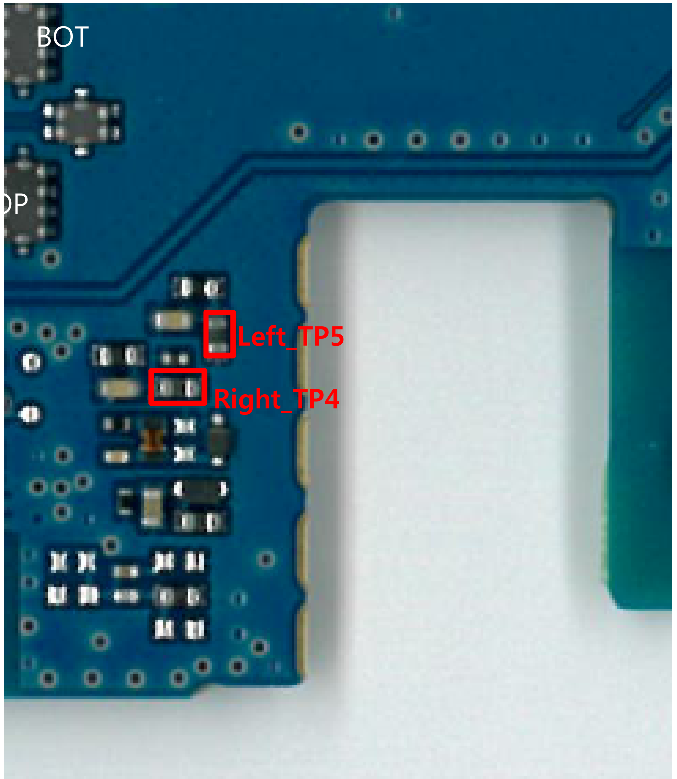
The Ear MIC control signals are generate by MSM8916(U2100), PM8916(U4100)
Case C. No Sound from speakers of earphone.

Checking Flow



Image

Check parts about Right(TP4) and Left(TP5)

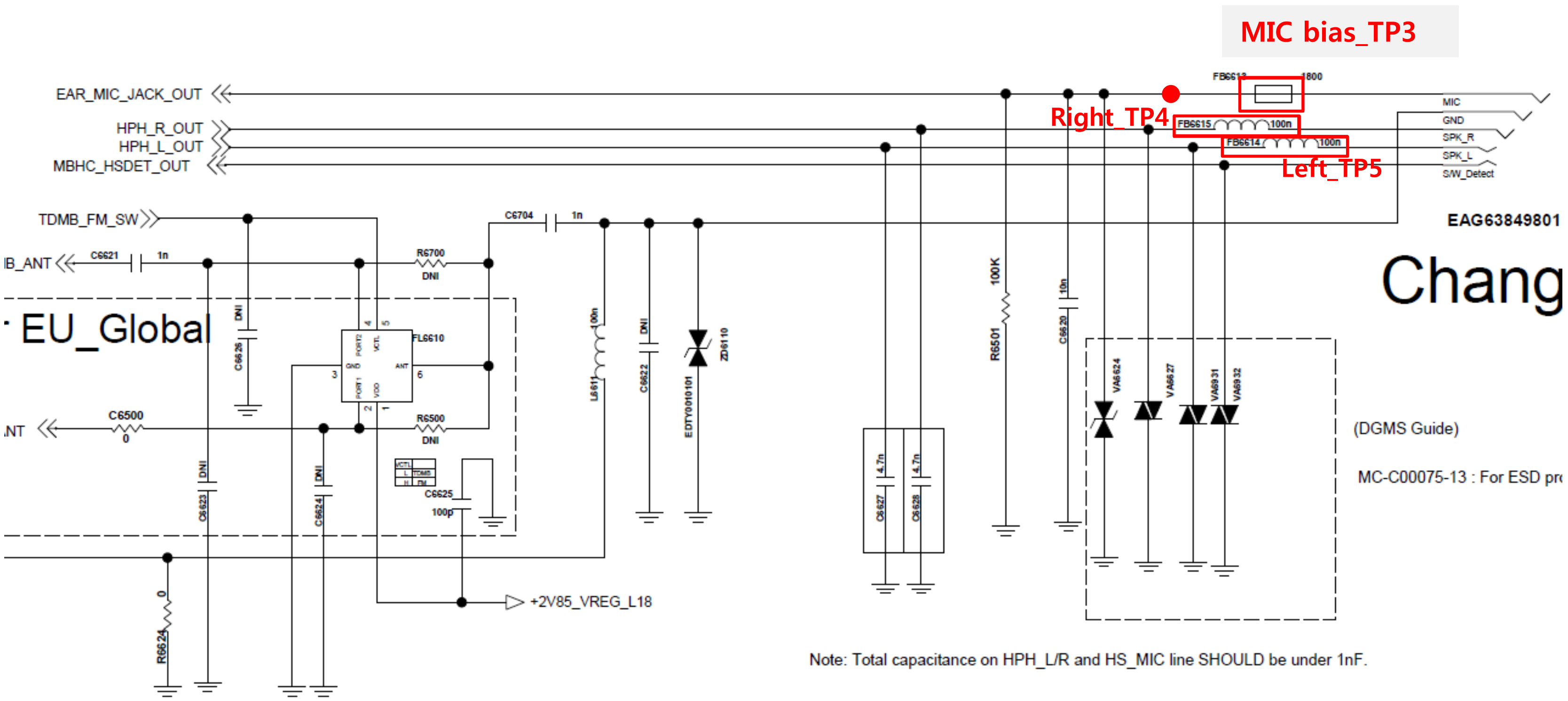


LGE Internal Use Only

3.15.5 [Audio] Ear MIC

The Ear MIC control signals are generate by MSM8916(U2100), PM8916(U4100)
Case C. No Sound from speakers of earphone.

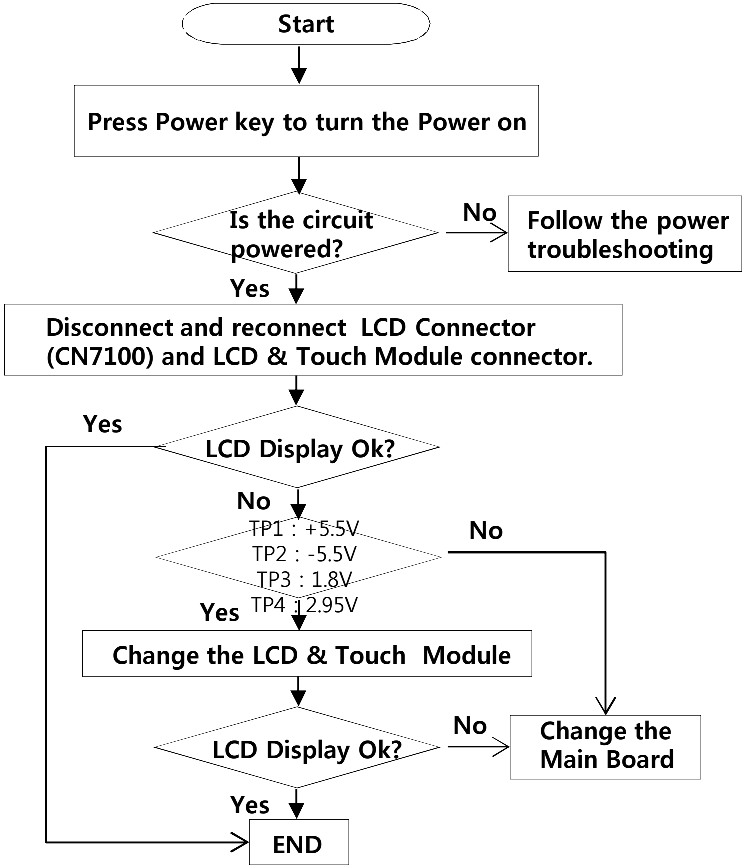
Circuit Diagram



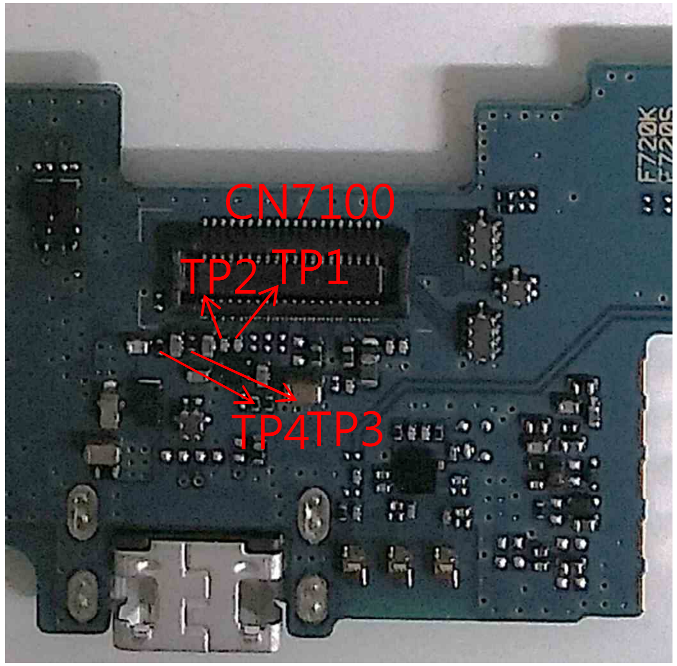
3.16 Checking LCD Block

The LCD control signals are generated by MSM8916. Its interface is MIPI having four data lanes and one clock lane.

Checking Flow



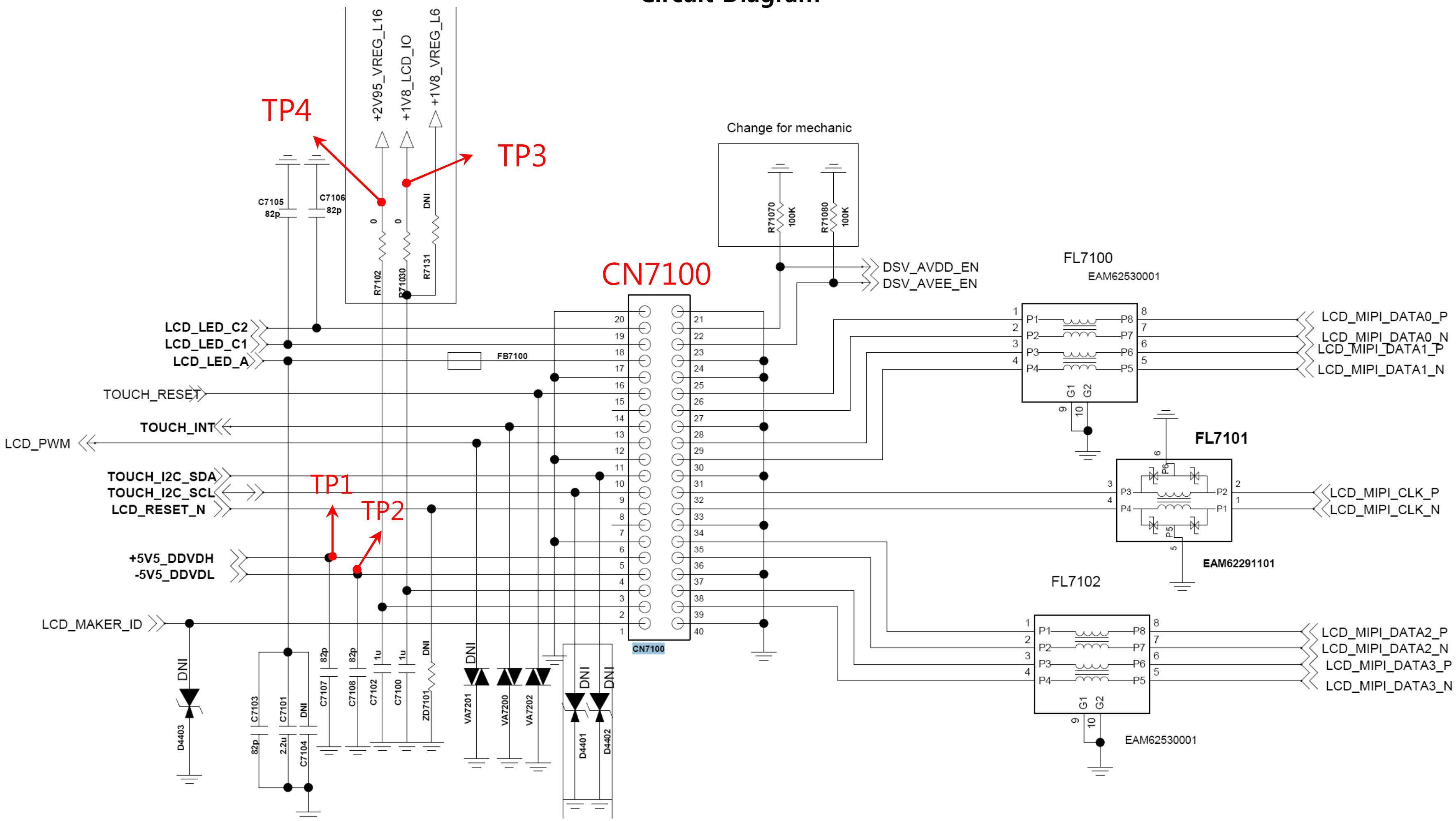
Image



3.16 Checking LCD Block

The LCD control signals are generated by MSM8916. Its interface is MIPI having four data lanes and one clock lane.

Circuit Diagram

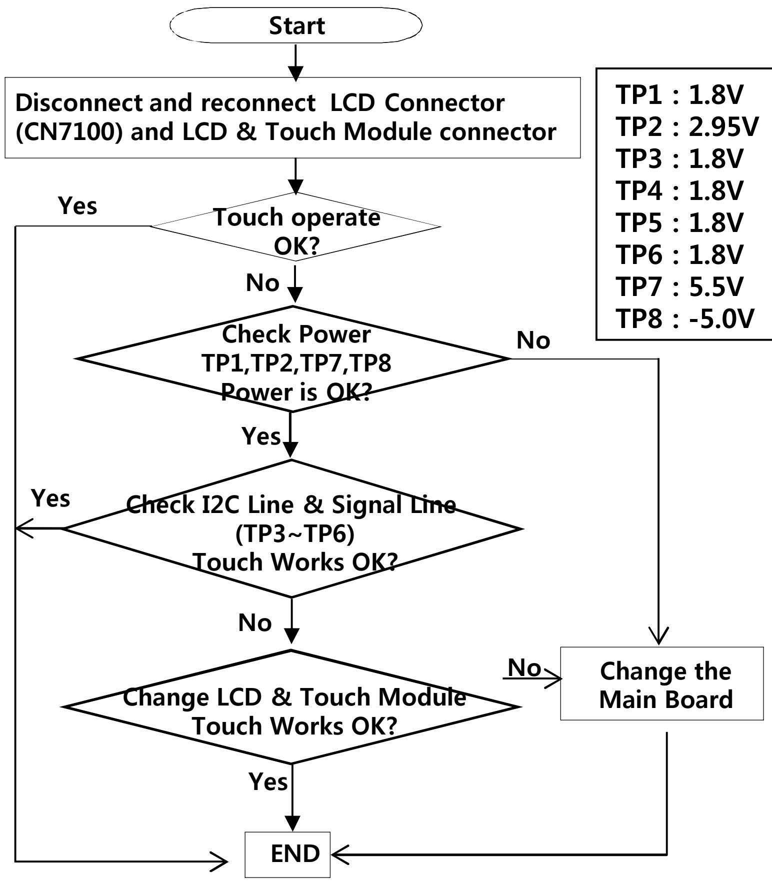


LGE Internal Use Only

3.17 Touch

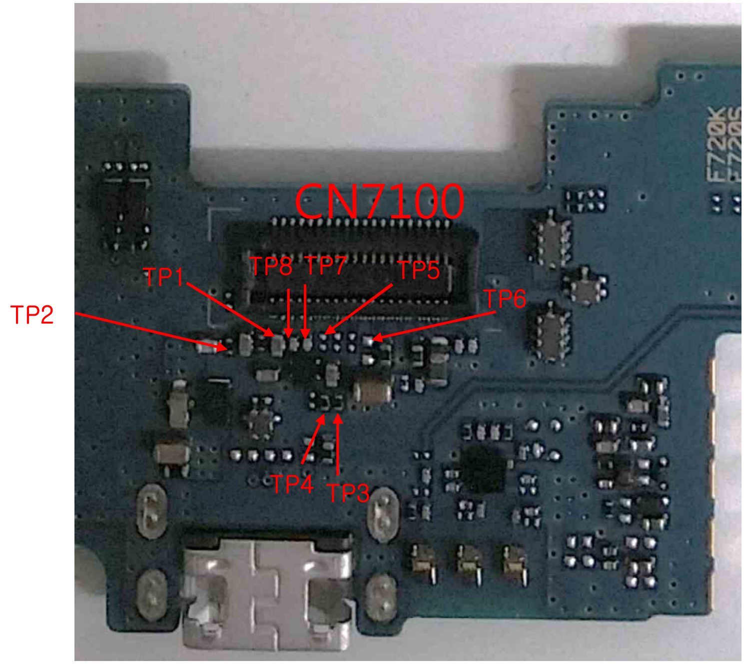
Touch control signals are generated by MSM8916. It is assembled with LCD.

Checking Flow



TP1 : 1.8V
TP2 : 2.95V
TP3 : 1.8V
TP4 : 1.8V
TP5 : 1.8V
TP6 : 1.8V
TP7 : 5.5V
TP8 : -5.0V

Image

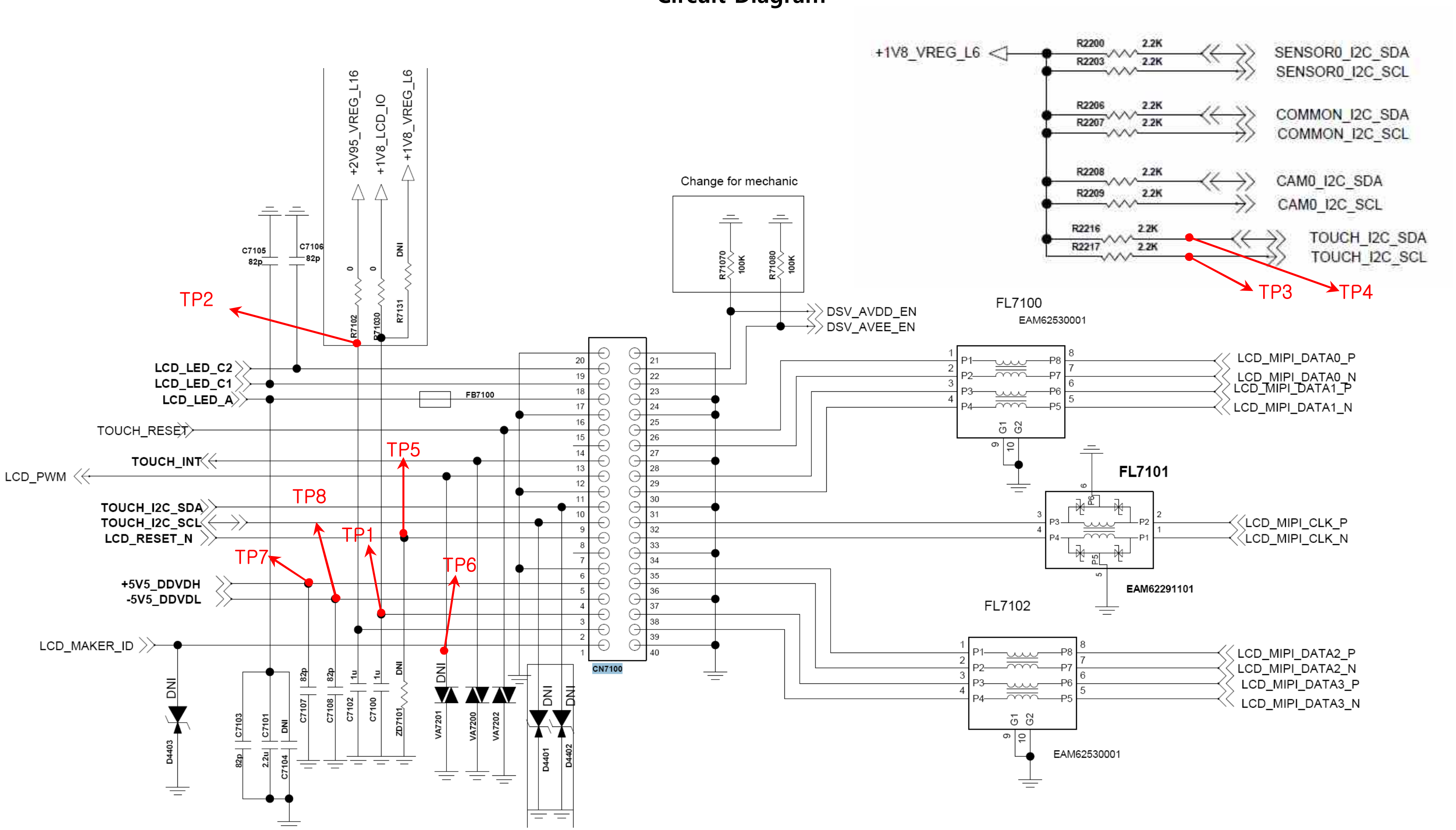


LGE Internal Use Only

3.17 Touch

The Touch control signals are generated by MSM8916. It is assembled with LCD.

Circuit Diagram

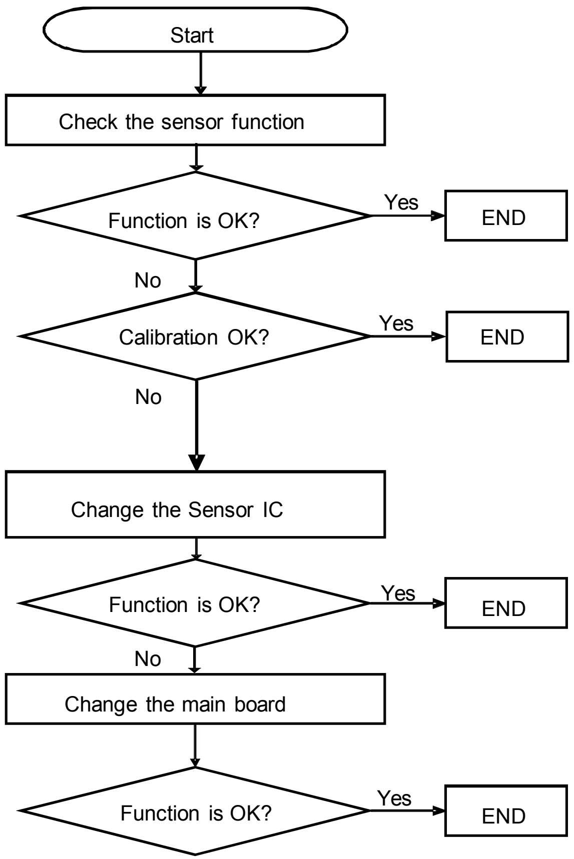


LGE Internal Use Only

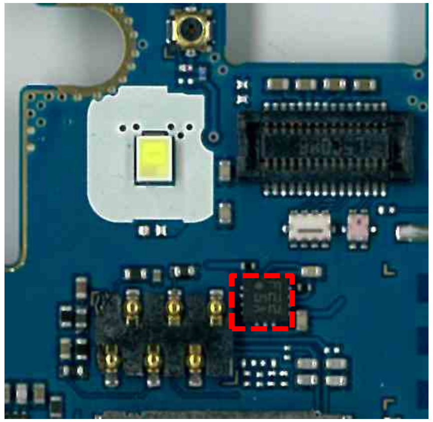
3.18.1 [Sensor] compass & accelerometer

The Accel. & compass sensors are calibrated by using SW algorithm.

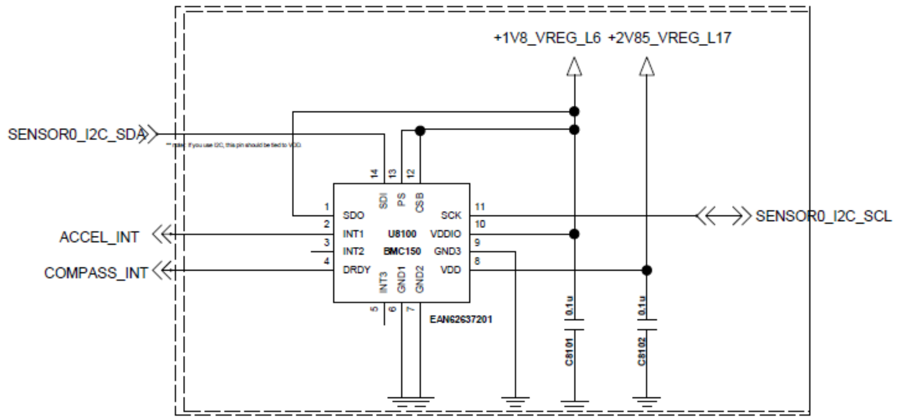
Checking Flow



Image



Circuit Diagram

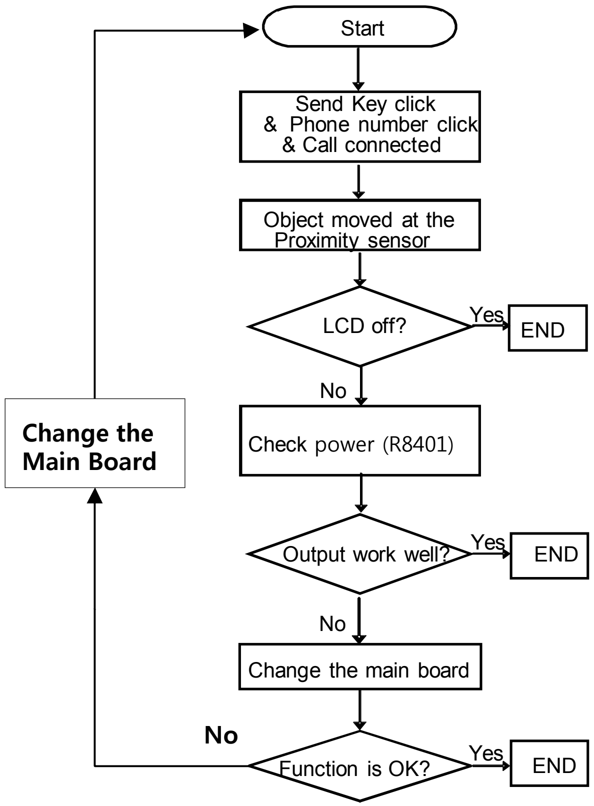


U8100

3.18.2 [Sensor] Proximity

Proximity Sensor is worked as below: Send Key click ➔ Phone number click ➔ Call connected ➔ Object moved at the sensor ➔ Control the screen's on/off operation automatically

Checking Flow



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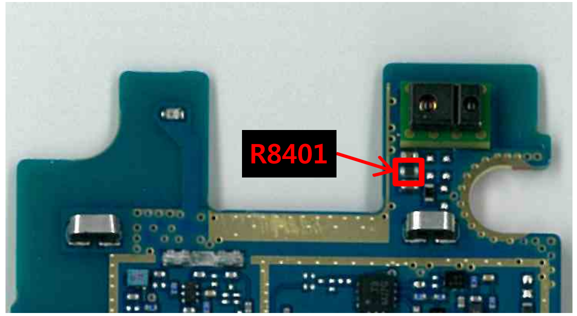
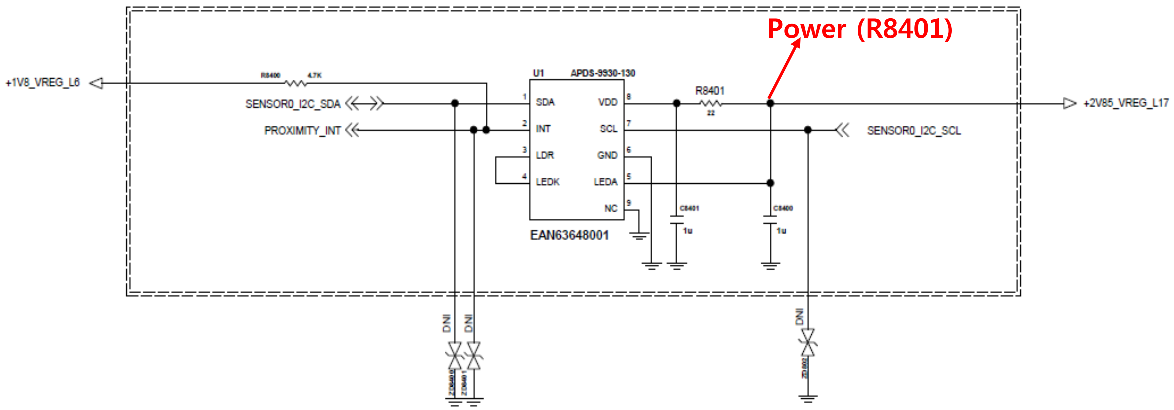


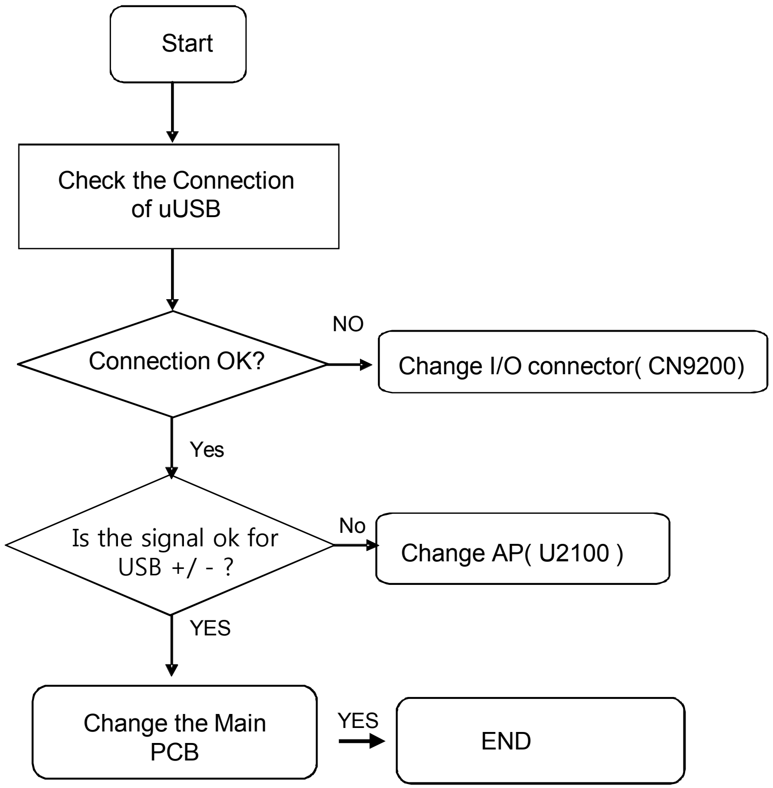
Circuit Diagram



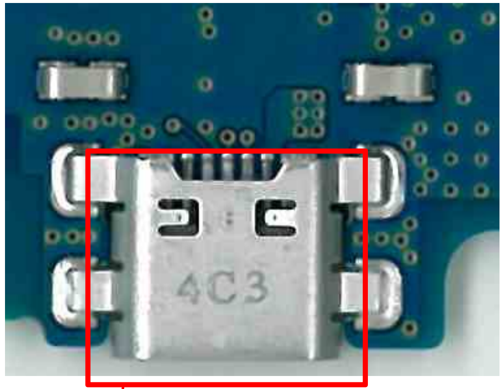
3.19 Checking USB Block

I/O connector is used as the USB port.

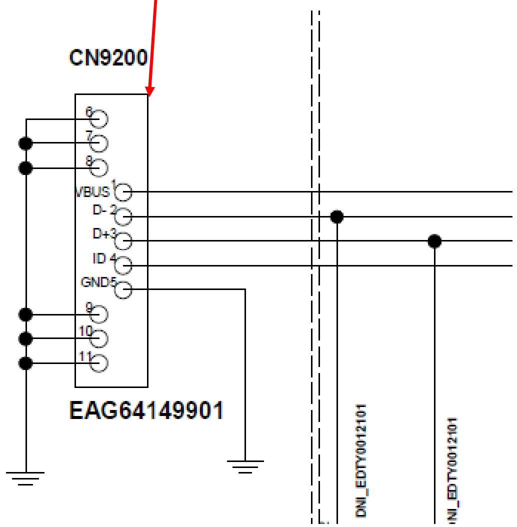
Checking Flow



Image



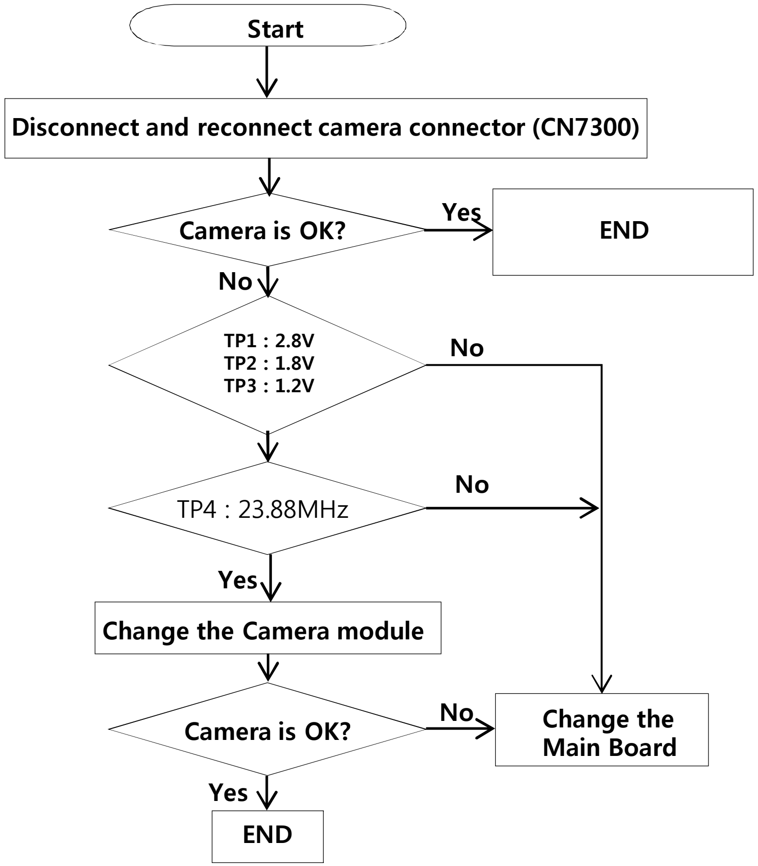
Circuit Diagram



3.20.1 [Camera] Main camera

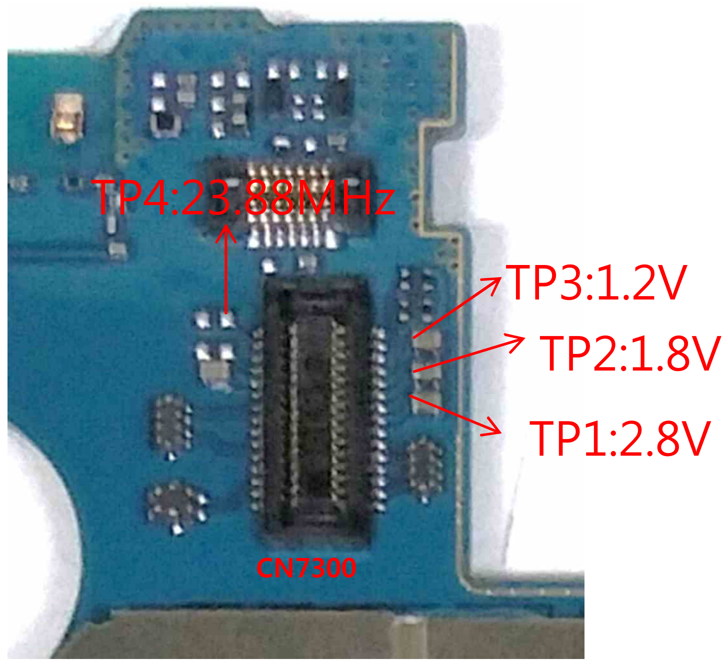
8M camera control signals are generated by MSM8916 (U2100 : Main Chipset). And powered by PMIC(U4100) & LDO(U7312).

Checking Flow



Image

BOT

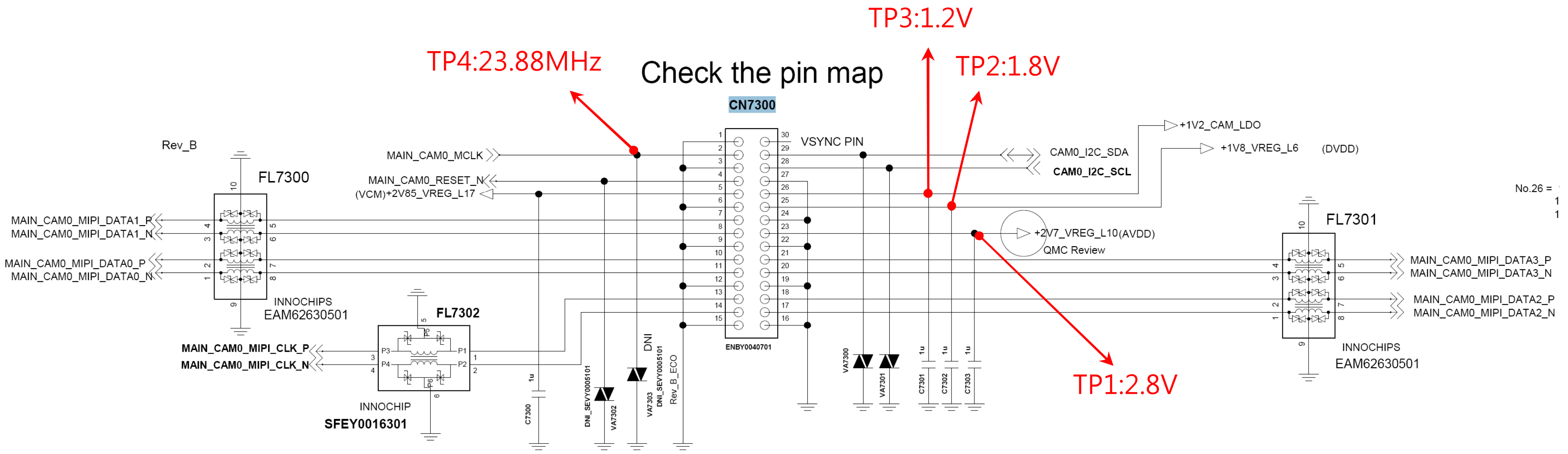


LGE Internal Use Only

3.20.1 [Camera] Main camera

8M camera control signals are generated by MSM8916 (U2100 : Main Chipset). And powered by PMIC(U4100) & LDO(U7312).

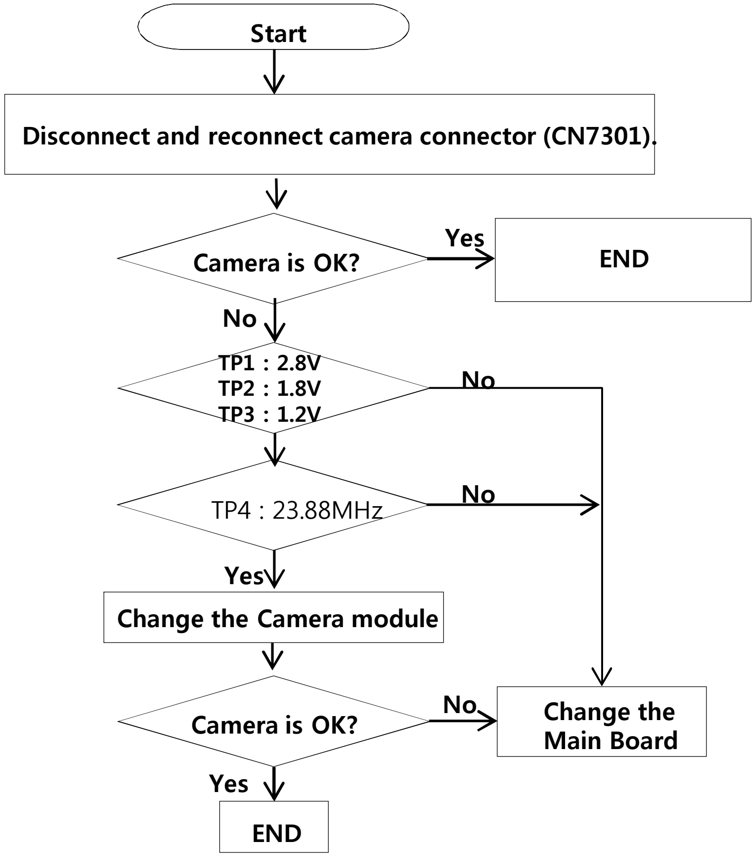
Circuit Diagram



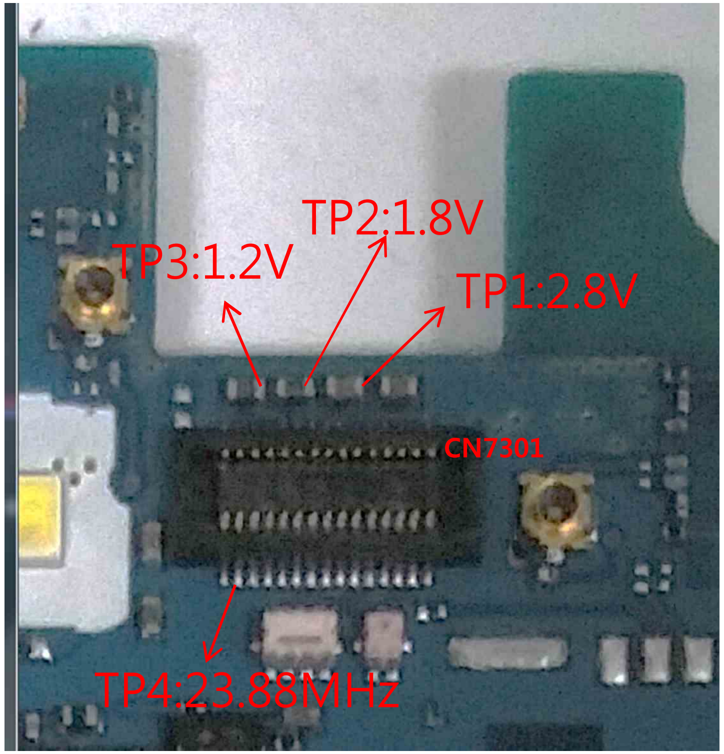
3.20.2 [Camera] VT camera

VT camera control signals are generated by MSM8916 (U2100 : Main Chipset). And powered by PMIC(U4100) & LDO(U7312).

Checking Flow



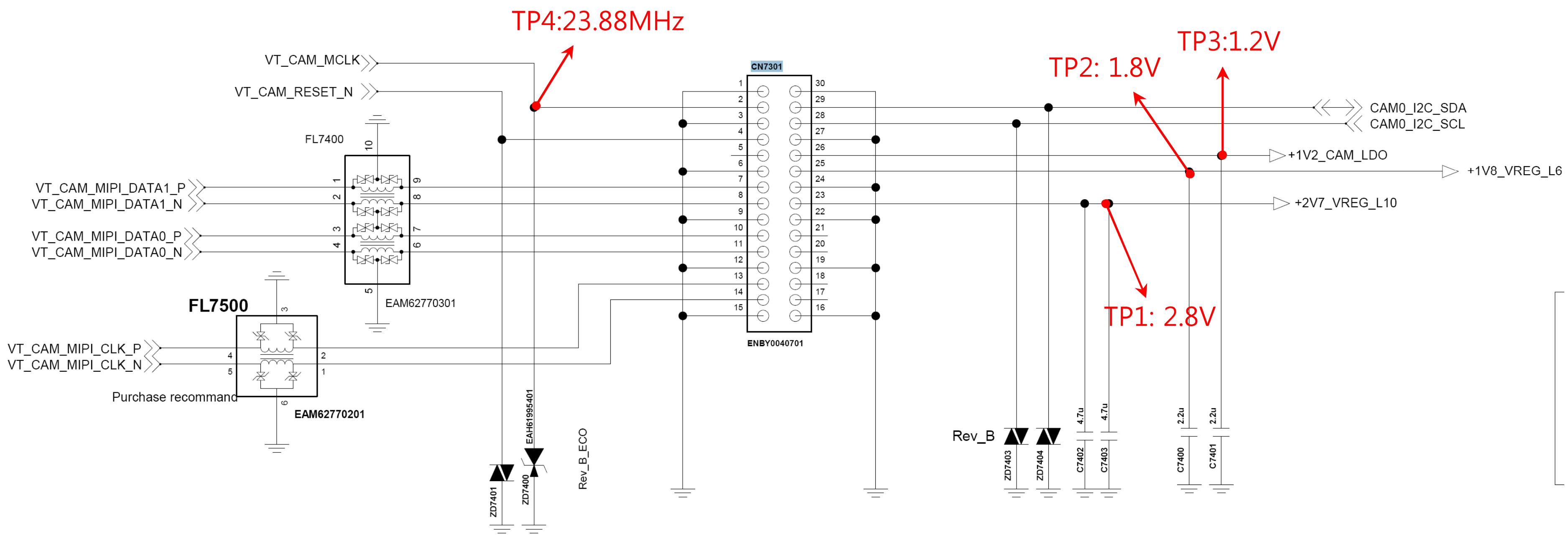
Image



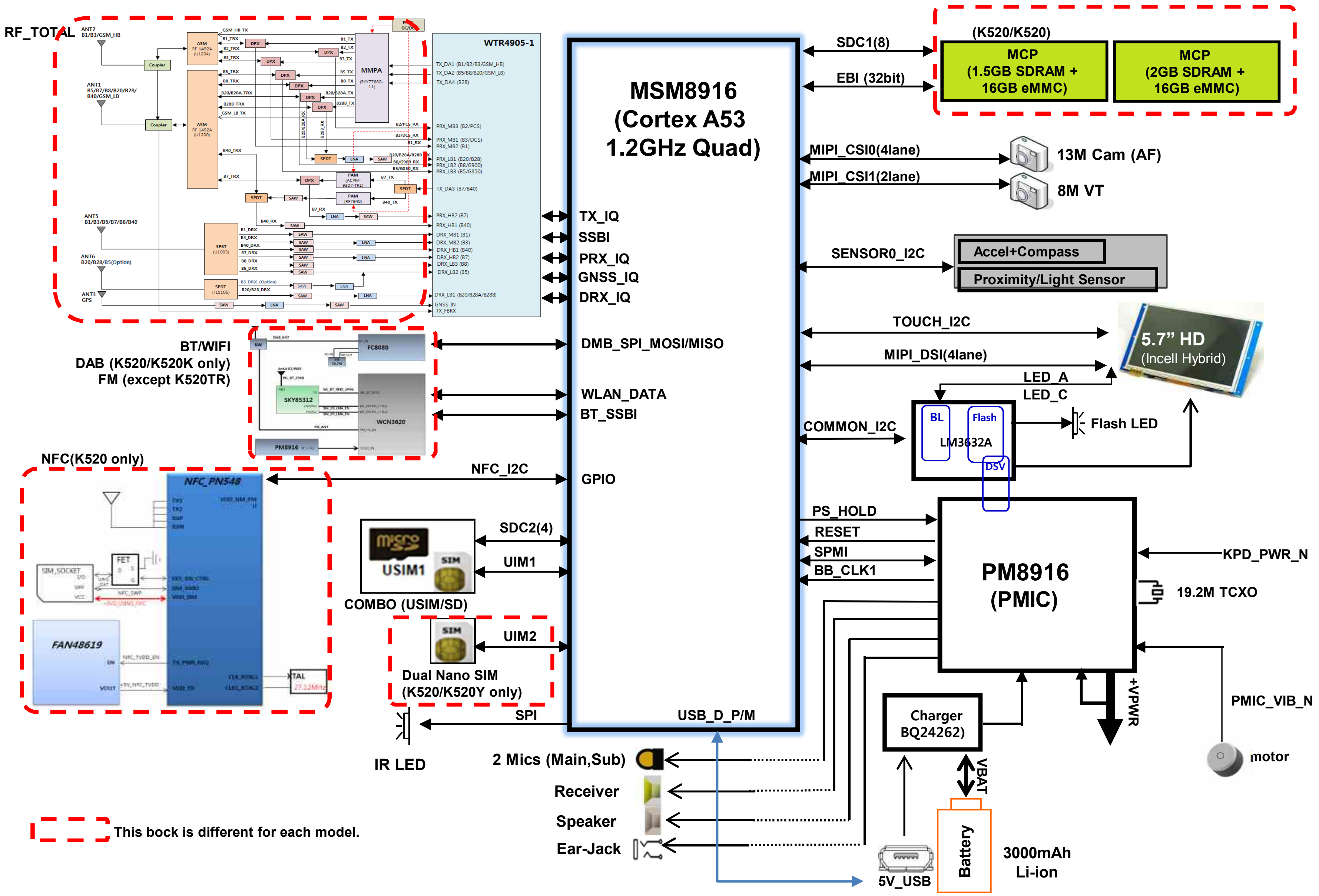
3.20.2 [Camera] VT camera

VT camera control signals are generated by MSM8916 (U2100 : Main Chipset). And powered by PMIC(U4100) & LDO(U7312).

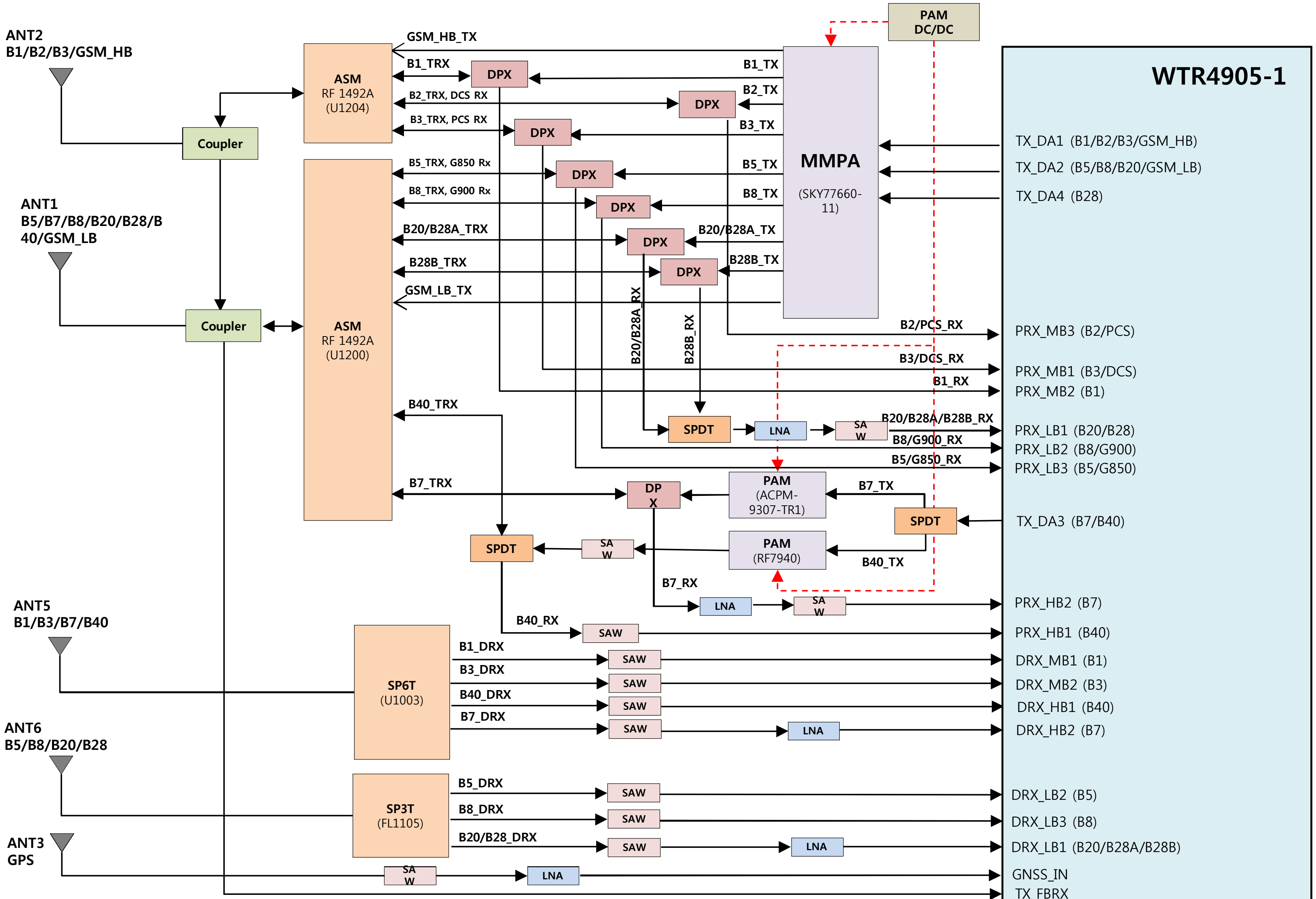
Circuit Diagram

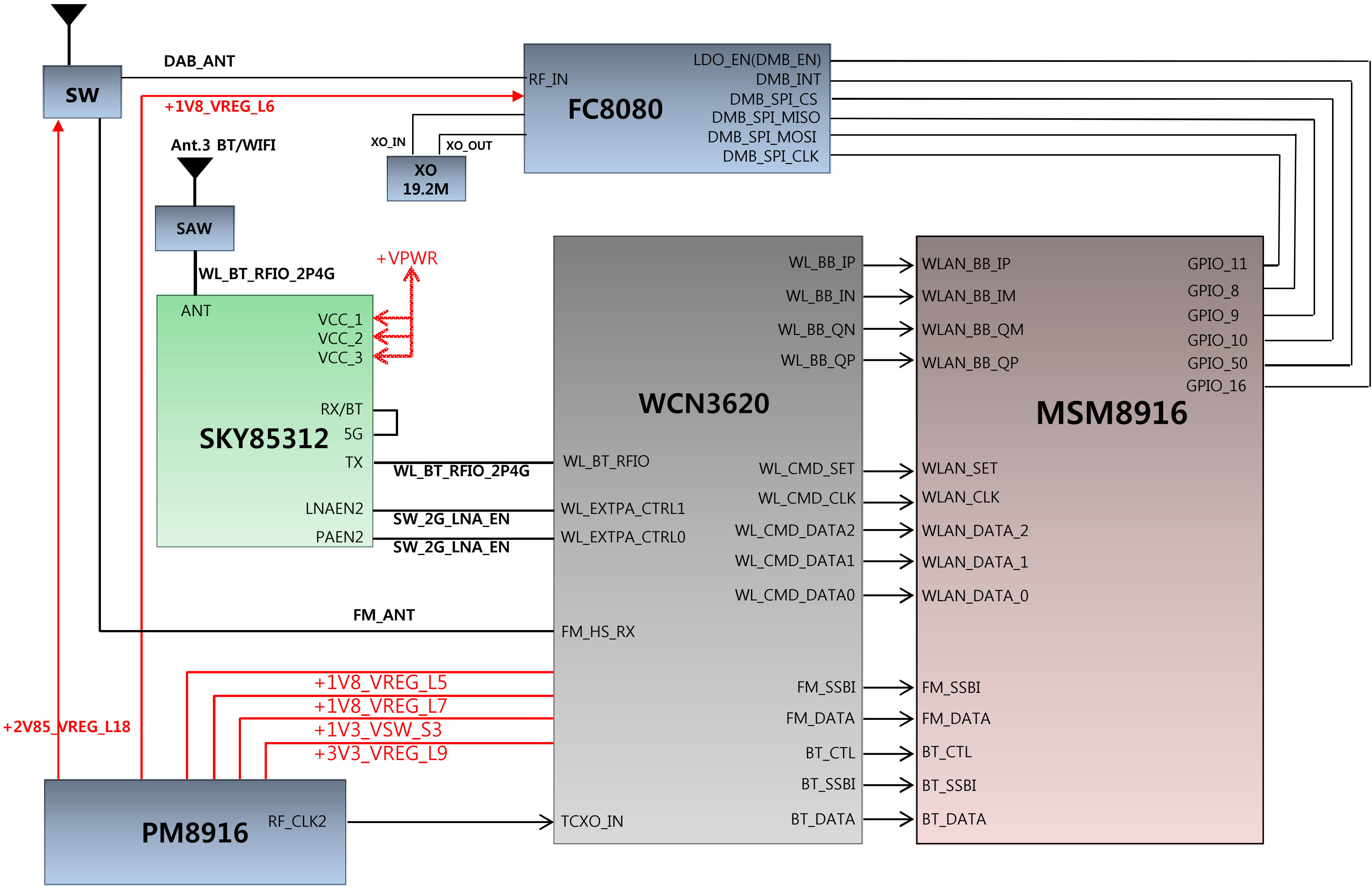


1. PH1 EU/GLOBAL TOTAL Block Diagram

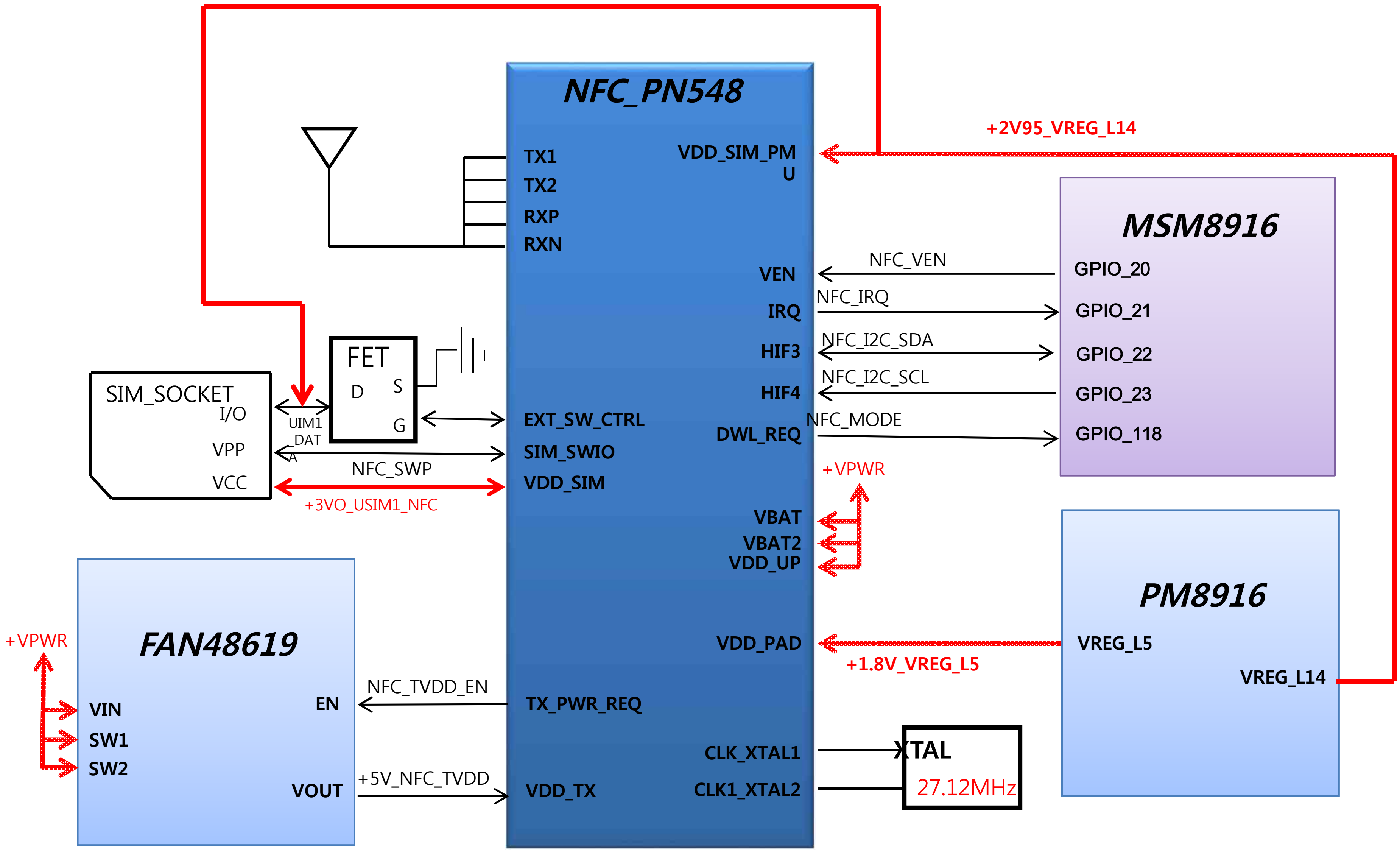


LGE Internal Use Only

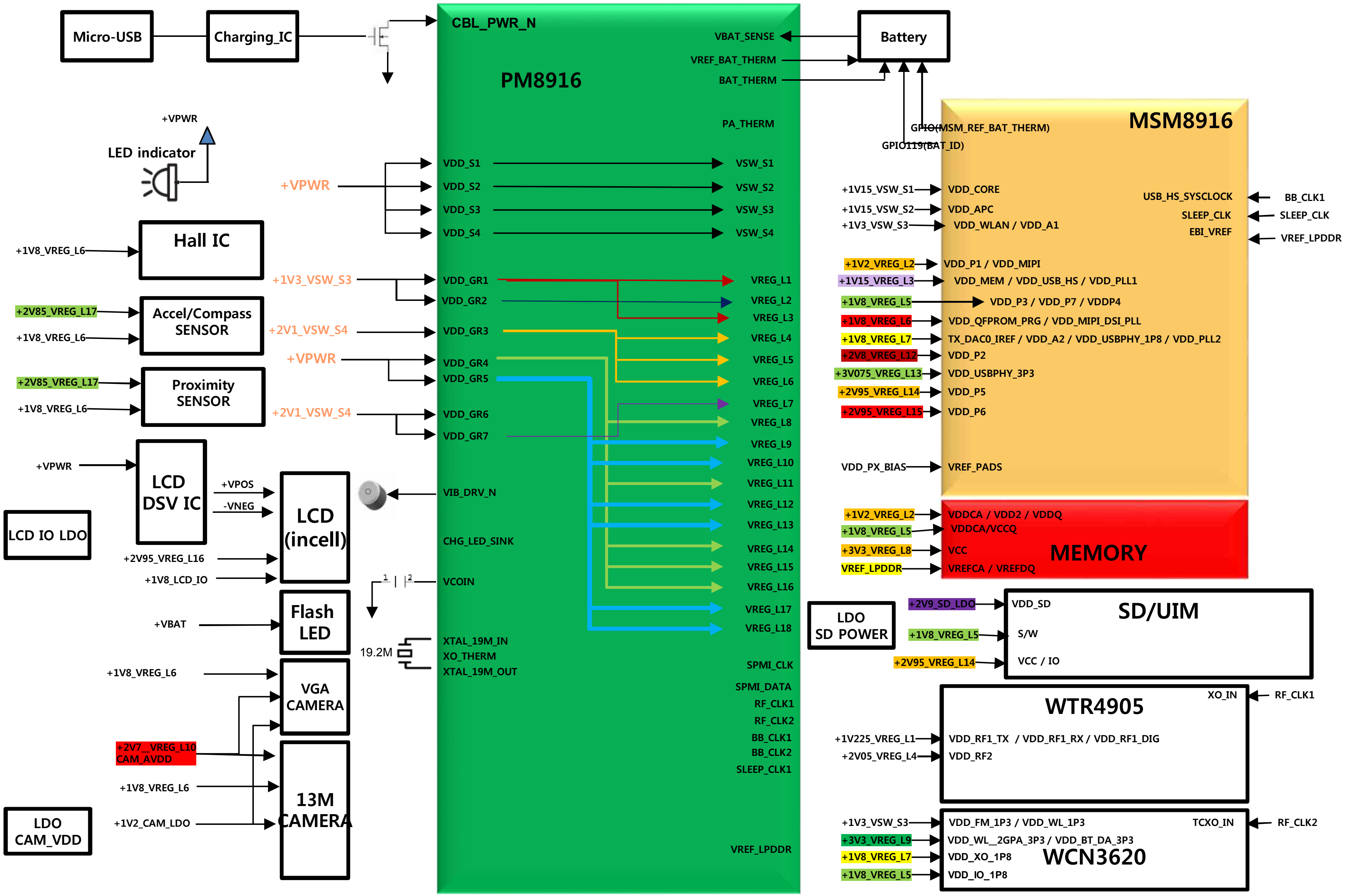

ANT2
B1/B2/B3/GSM_HB
Coupler
ASM
RF 1492A
(U1204)
GSM_HB_TX
B1_TRX
DPX
B1_TX
B2_TRX, DCS RX
DPX
B2_TX
B3_TRX, PCS RX
DPX
B3_TX
PAM
DC/DC
MMPA
(SKY77660-
11)
WTR4905-1
TX_DA1 (B1/B2/B3/GSM_HB)
TX_DA2 (B5/B8/B20/GSM_LB)
TX_DA4 (B28)
ANT1
B5/B7/B8/B20/B28/B
40/GSM_LB
ASM
RF 1492A
(U1200)
Coupler
B5_TRX, G850 Rx
DPX
B5_TX
B8_TRX, G900 Rx
DPX
B8_TX
B20/B28A_TRX
DPX
B20/B28A_TX
B28B_TRX
DPX
B28B_TX
GSM_LB_TX
B20/B28A_RX
B28B_RX
B2/PCS_RX
PRX_MB3 (B2/PCS)
B3/DCS_RX
PRX_MB1 (B3/DCS)
B1_RX
PRX_MB2 (B1)
B40_TRX
SPDT
LNA
SAW
B20/B28A/B28B_RX
PRX_LB1 (B20/B28)
B8/G900_RX
PRX_LB2 (B8/G900)
B5/G850_RX
PRX_LB3 (B5/G850)
B7_TRX
DPX
PAM
(ACPM-
9307-TR1)
B7_TX
SPDT
TX_DA3 (B7/B40)
SPDT
SAW
PAM
(RF7940)
B40_TX
B7_RX
LNA
SAW
PRX_HB2 (B7)
B40_RX
SAW
PRX_HB1 (B40)
ANT5
B1/B3/B7/B40
SP6T
(U1003)
B1_DRX
SAW
DRX_MB1 (B1)
B3_DRX
SAW
DRX_MB2 (B3)
B40_DRX
SAW
DRX_HB1 (B40)
B7_DRX
SAW
LNA
DRX_HB2 (B7)
ANT6
B5/B8/B20/B28
SP3T
(FL1105)
B5_DRX
SAW
DRX_LB2 (B5)
B8_DRX
SAW
DRX_LB3 (B8)
B20/B28_DRX
SAW
LNA
DRX_LB1 (B20/B28A/B28B)
ANT3
GPS
SAW
LNA
GNSS_IN
TX_FBRX

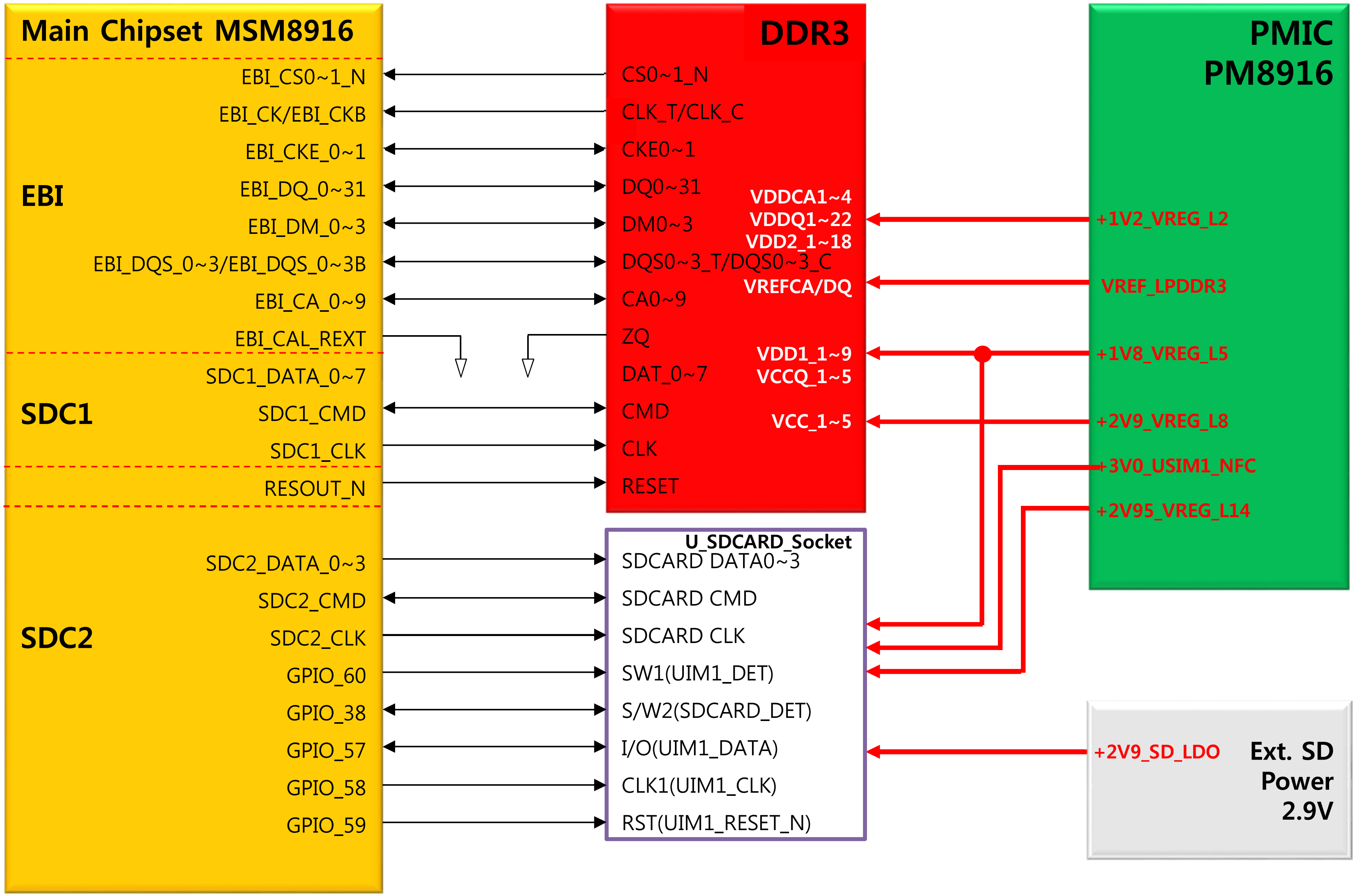


LGE Internal Use Only

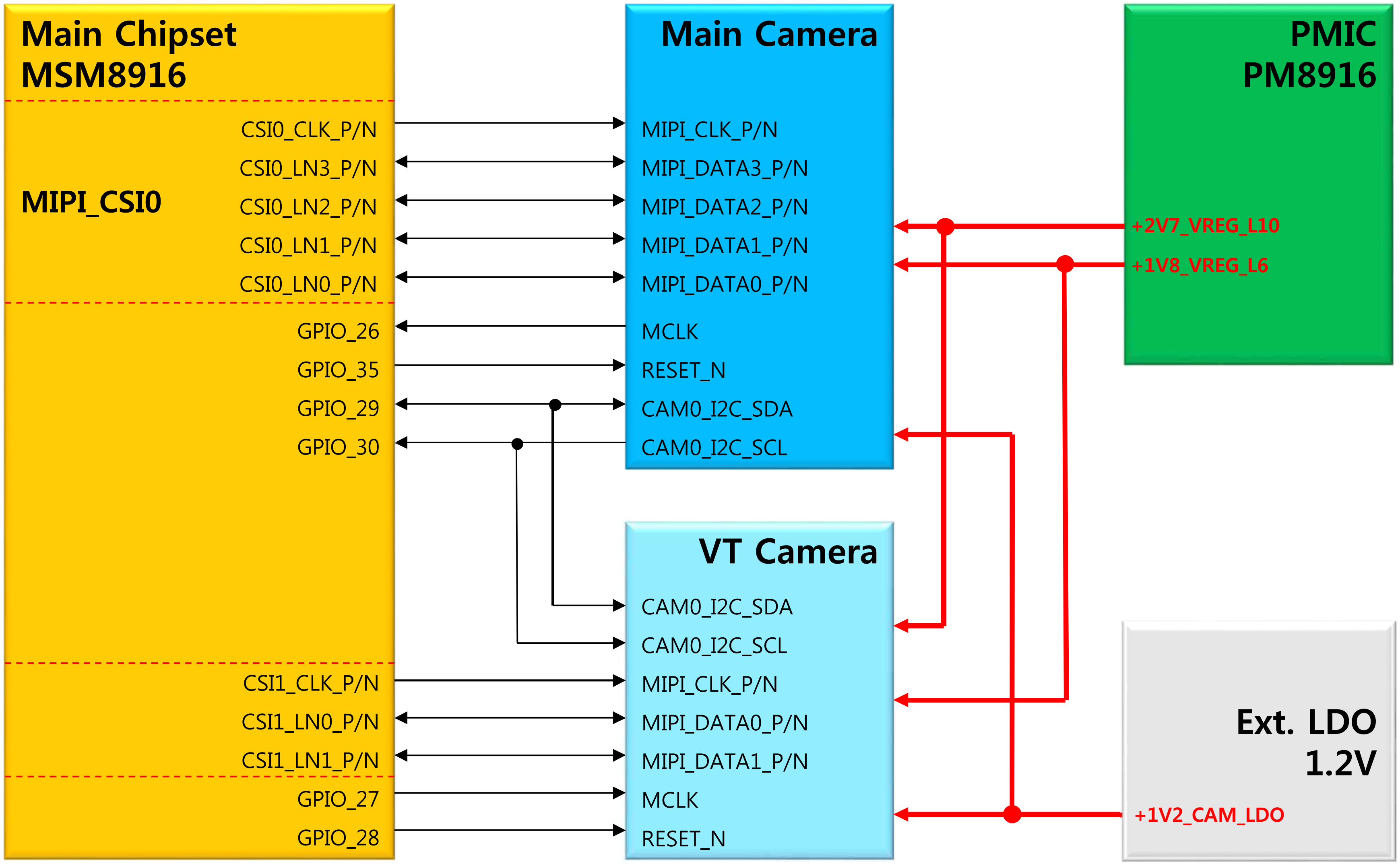

NFC_PN548
TX1
TX2
RXP
RXN
VDD_SIM_PMU
+2V95_VREG_L14
VEN
IRQ
HIF3
HIF4
DWL_REQ
EXT_SW_CTRL
SIM_SWIO
VDD_SIM
VBAT
VBAT2
VDD_UP
VDD_PAD
TX_PWR_REQ
CLK_XTAL1
CLK1_XTAL2
VDD_TX
MSM8916
GPIO_20
GPIO_21
GPIO_22
GPIO_23
GPIO_118
NFC_VEN
NFC_IRQ
NFC_I2C_SDA
NFC_I2C_SCL
NFC_MODE
+VPWR
PM8916
VREG_L5
VREG_L14
+1.8V_VREG_L5
XTAL
27.12MHz
FET
D
S
G
SIM_SOCKET
I/O
VPP
VCC
UIM1_DATA
NFC_SWP
+3VO_USIM1_NFC
FAN48619
+VPWR
VIN
SW1
SW2
EN
VOUT
NFC_TVDD_EN
+5V_NFC_TVDD

5. PH1 EU/GLOBAL POWER Block Diagram

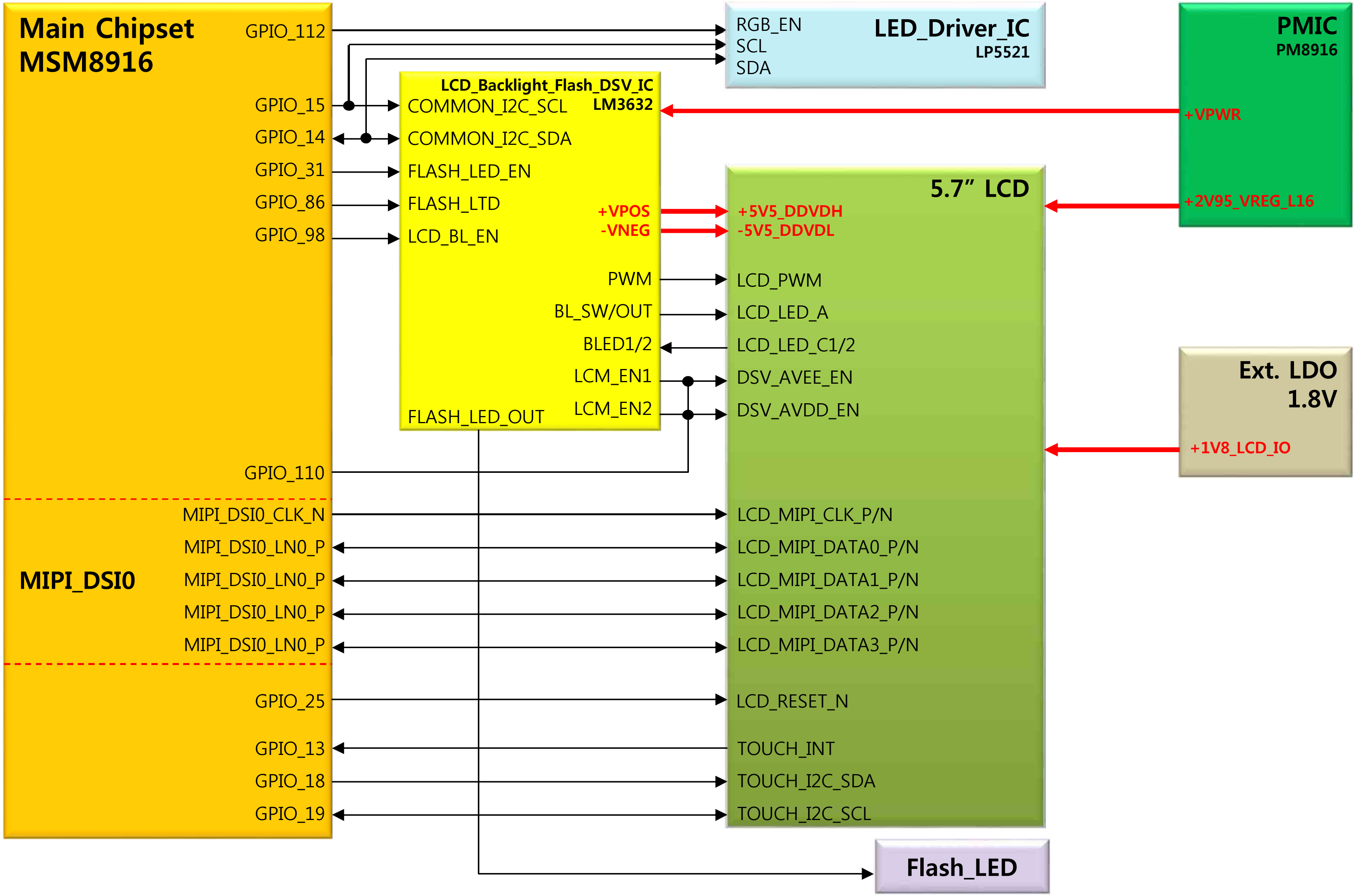




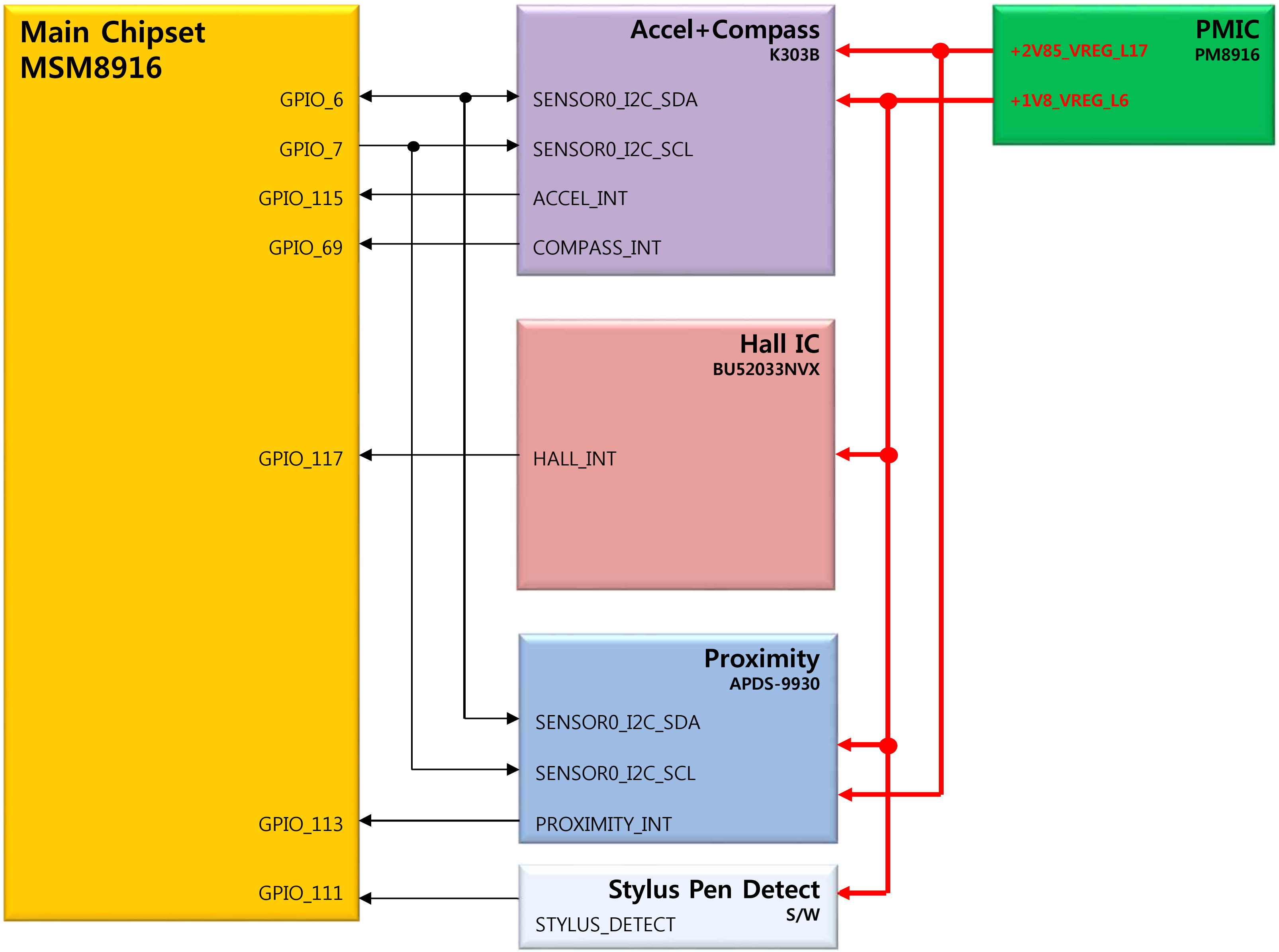
LGE Internal Use Only

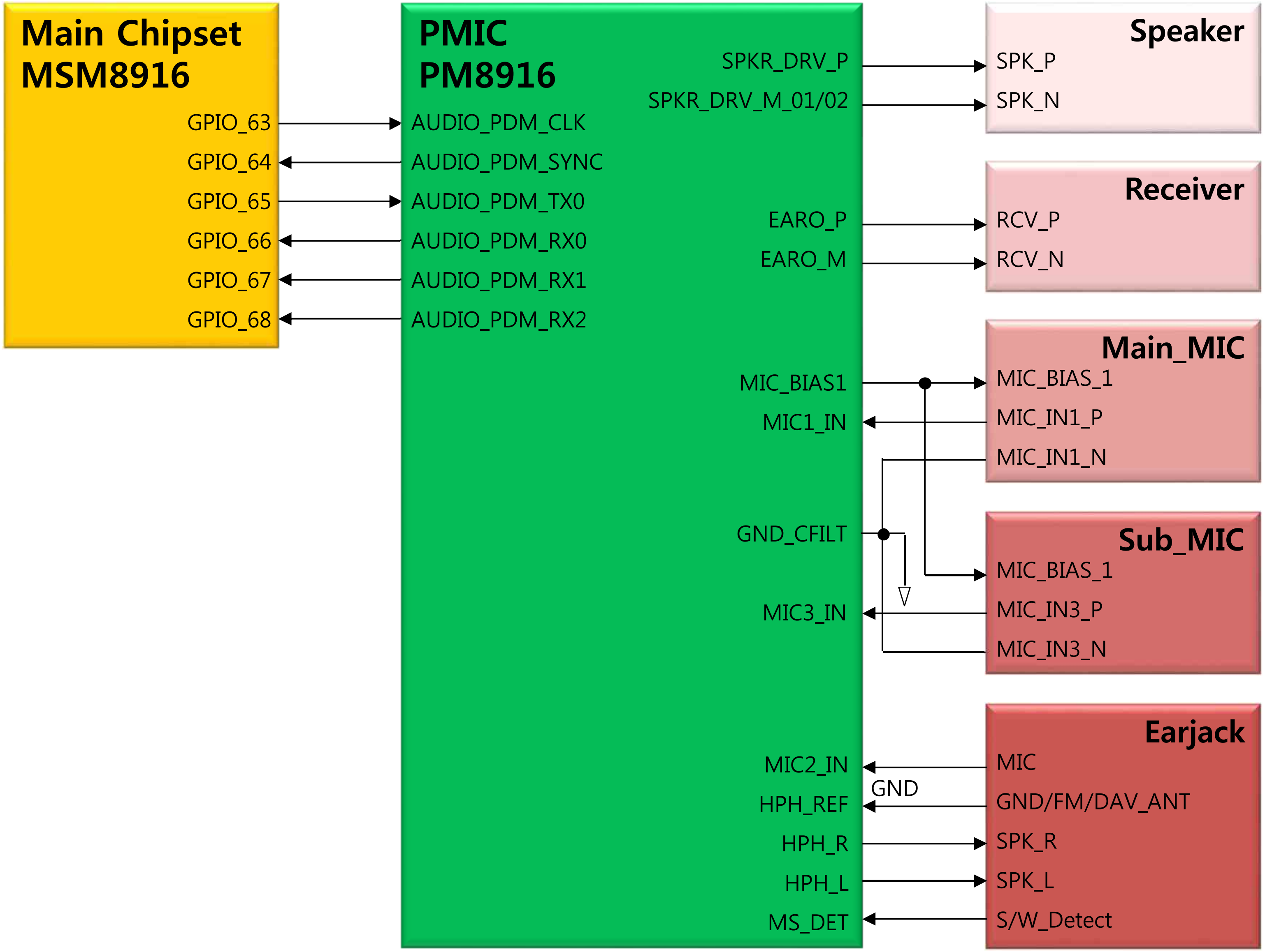


LGE Internal Use Only



LGE Internal Use Only


Main Chipset
MSM8916
GPIO_6
GPIO_7
GPIO_115
GPIO_69
GPIO_117
GPIO_113
GPIO_111
Accel+Compass
K303B
SENSOR0_I2C_SDA
SENSOR0_I2C_SCL
ACCEL_INT
COMPASS_INT
PMIC
PM8916
+2V85_VREG_L17
+1V8_VREG_L6
Hall IC
BU52033NVX
HALL_INT
Proximity
APDS-9930
SENSOR0_I2C_SDA
SENSOR0_I2C_SCL
PROXIMITY_INT
Stylus Pen Detect
S/W
STYLUS_DETECT

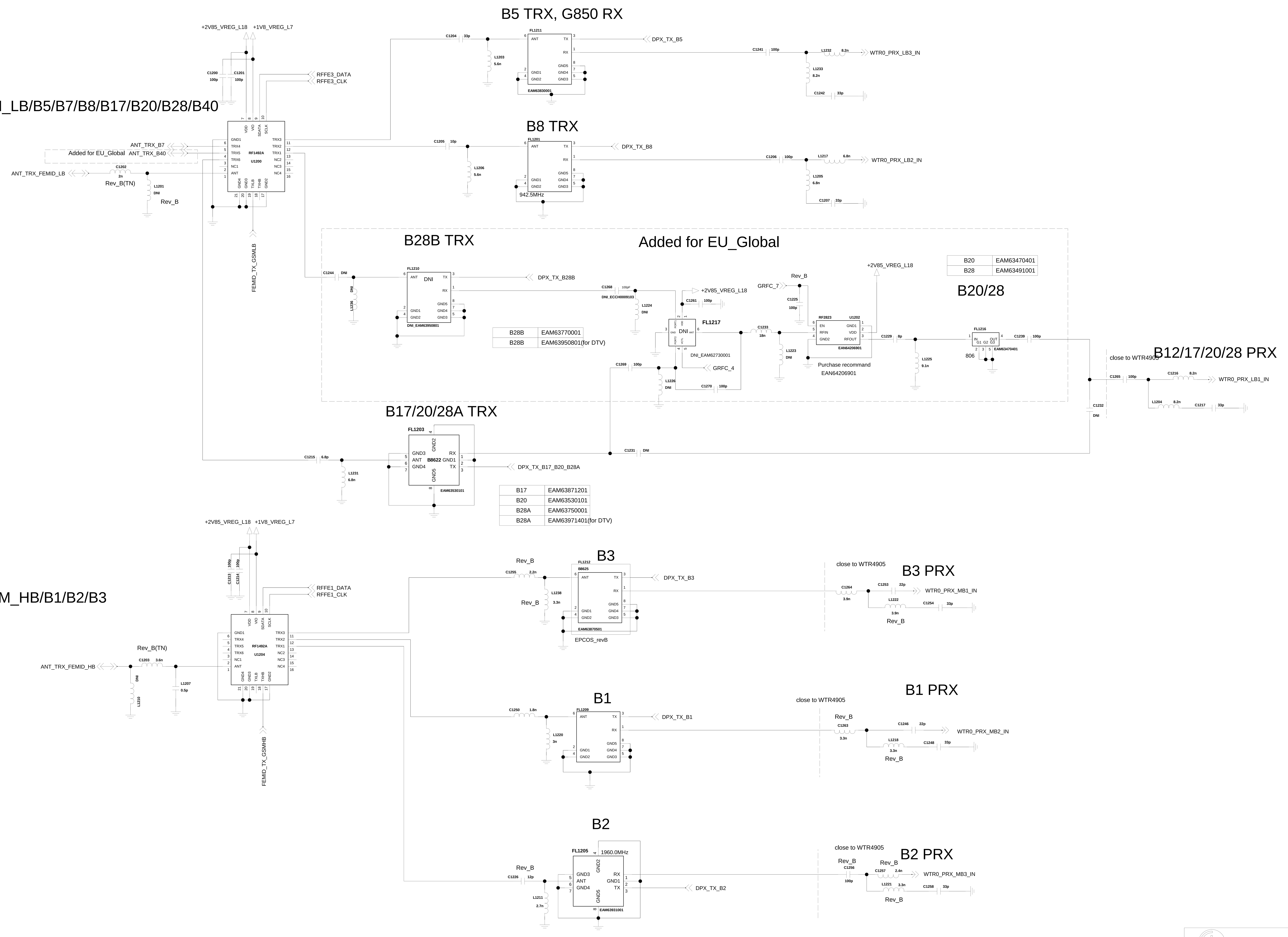


LGE Internal Use Only

CIRCUIT DIAGRAM

LGE Internal Use Only

< 1-10-1-2-1_SP8T_SP8T_nCA_CA> Rev_0.3



B20	EAM63470401
B28	EAM63491001

B28B	EAM63770001
B28B	EAM63950801(for DTV)

B17	EAM63871201
B20	EAM63530101
B28A	EAM63750001
B28A	EAM63971401(for DTV)

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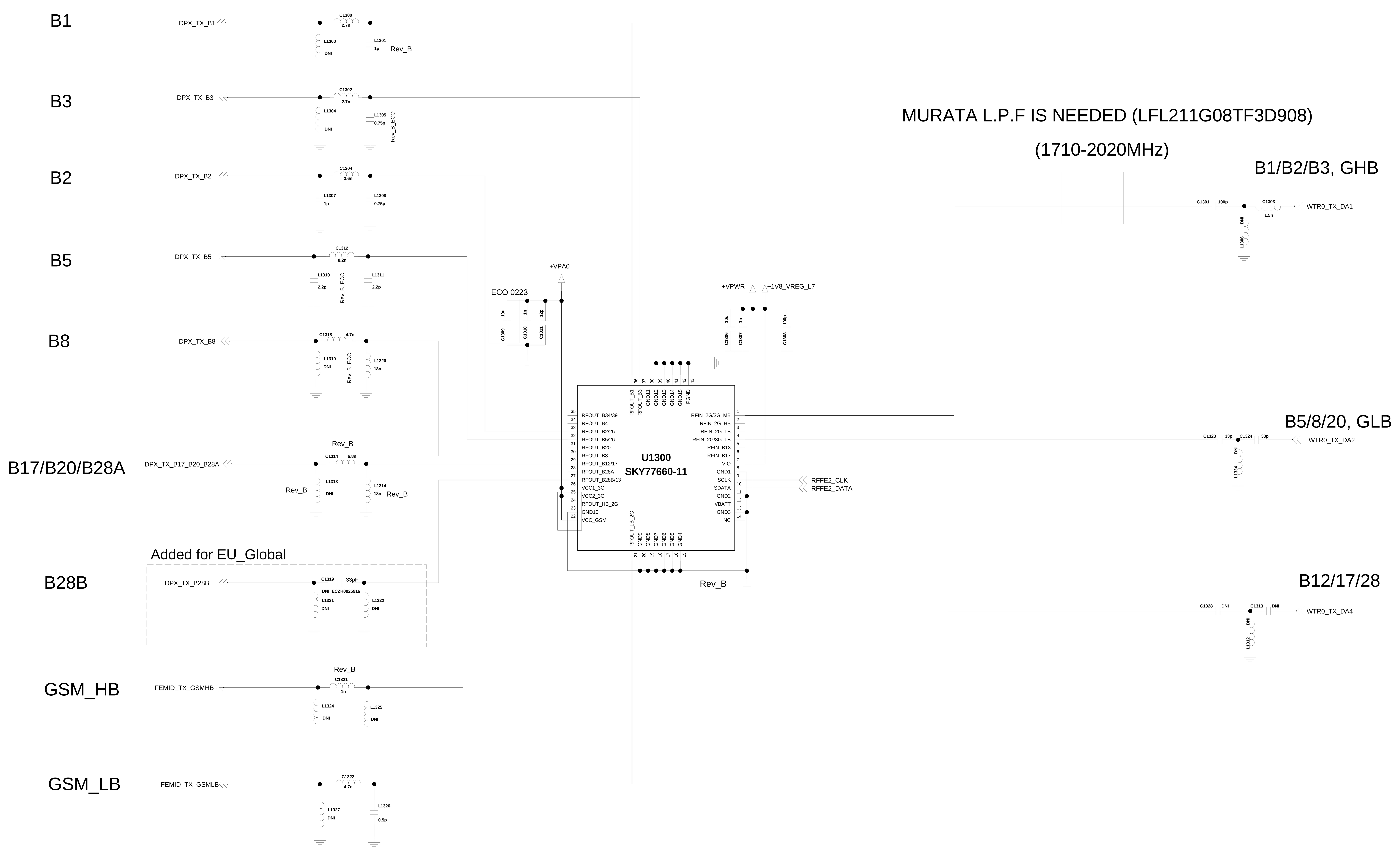
Schematic1

PDM NUMBER

< 1-11-1-3-1_uMMPA_APT_ACPM-7788 > Rev_0.1

MURATA L.P.F IS NEEDED (LFL211G08TF3D908)

(1710-2020MHz)



LGE Internal Use Only

LG Electronics

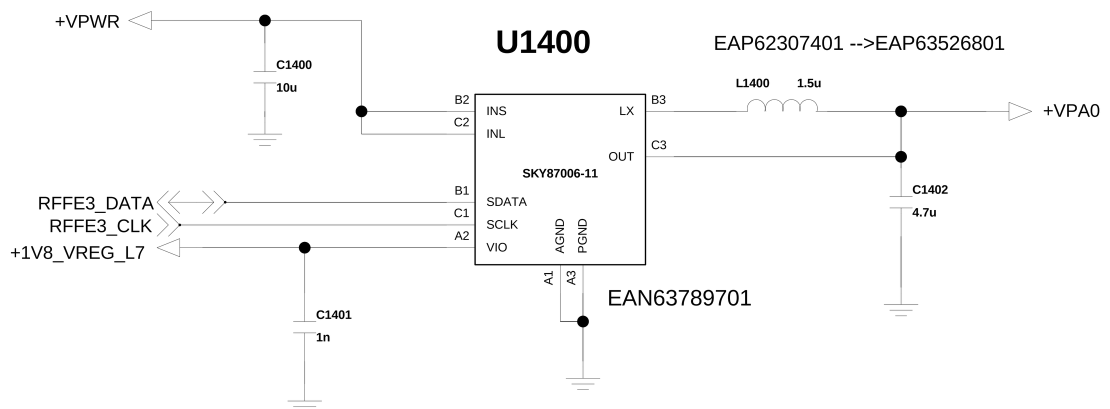
Schematic1

A2 PDM NUMBER Rev

User Name 25 25 Drawing Date

< 1-2-4-3_PT_APT_QFE2101 >

Rev_0.3 -> SKY87006



LGE Internal Use Only

LG Electronics

TITLE Schematic1

SIZE A2 | DWG NO PDM NUMBER | REV Rev

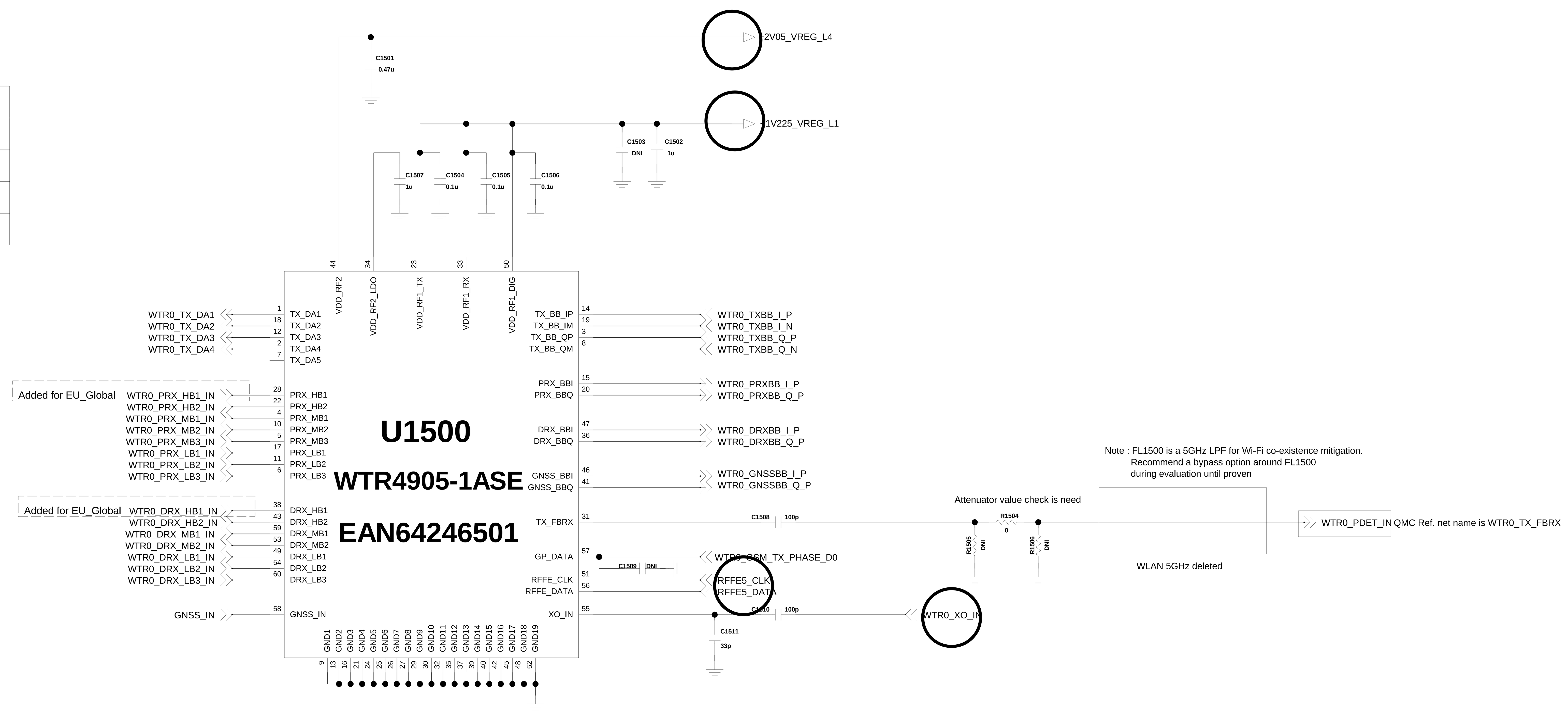
DRAWN BY User Name | SHEET 2 of 2 | Date Drawing Date

<1-8-5-1_WTR4905>

Rev_0.5

Tx Table

TX_DA1	B1/2/3, GSM_HB
TX_DA2	B5/8/20, GSM_LB
TX_DA3	B7, B40
TX_DA4	B17/28
TX_DA5	-



ECO_0213

LGE Internal Use Only

LG Electronics

TITLE Schematic1

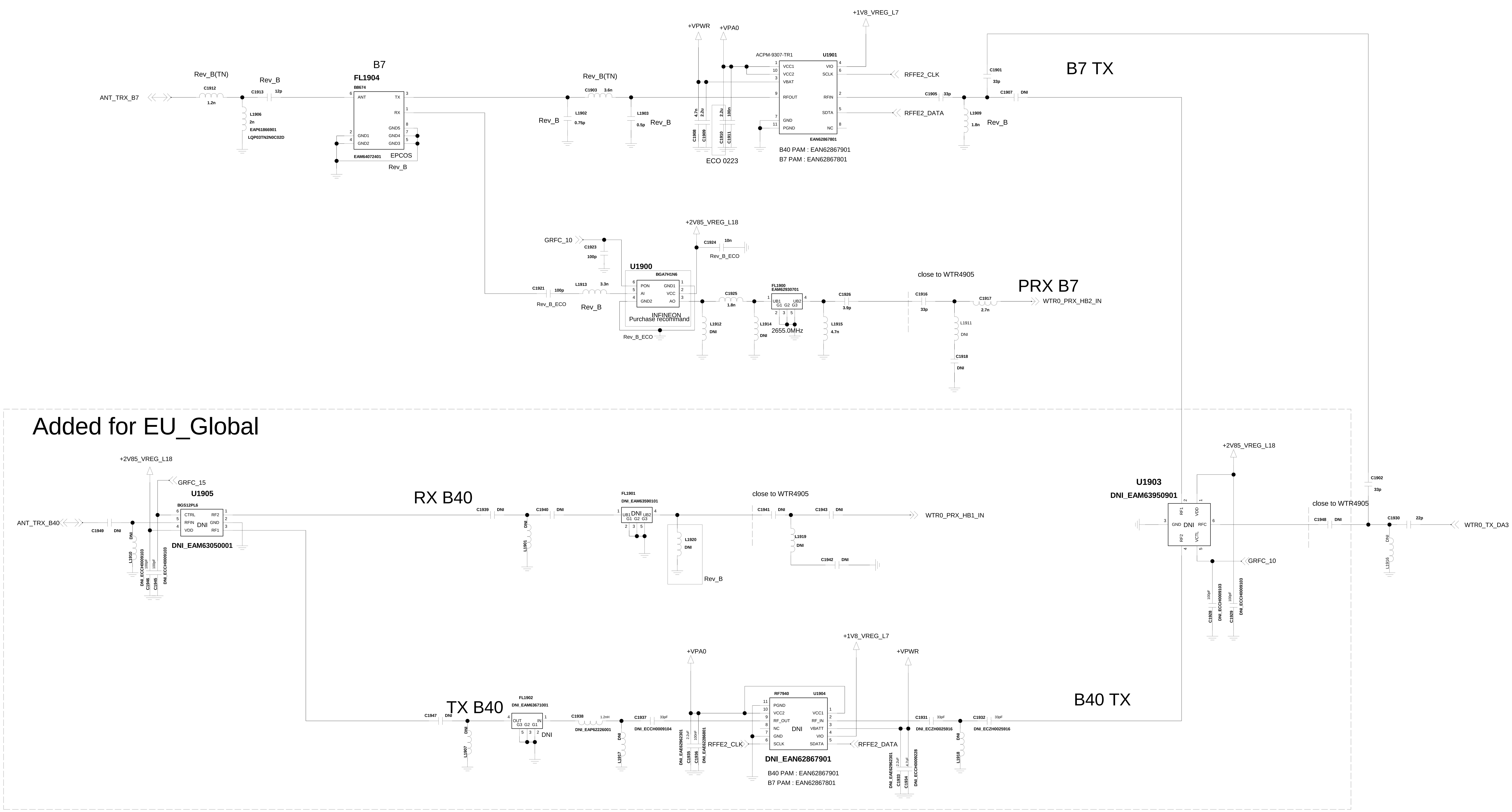
SIZE A2 PDM NUMBER Rev

SHEET 3 of 3 Drawing Date

DRAWN BY User Name

< 1-3-6-xx_TRX_B40_B7 >

Rev_0.4



LGE Internal Use Only

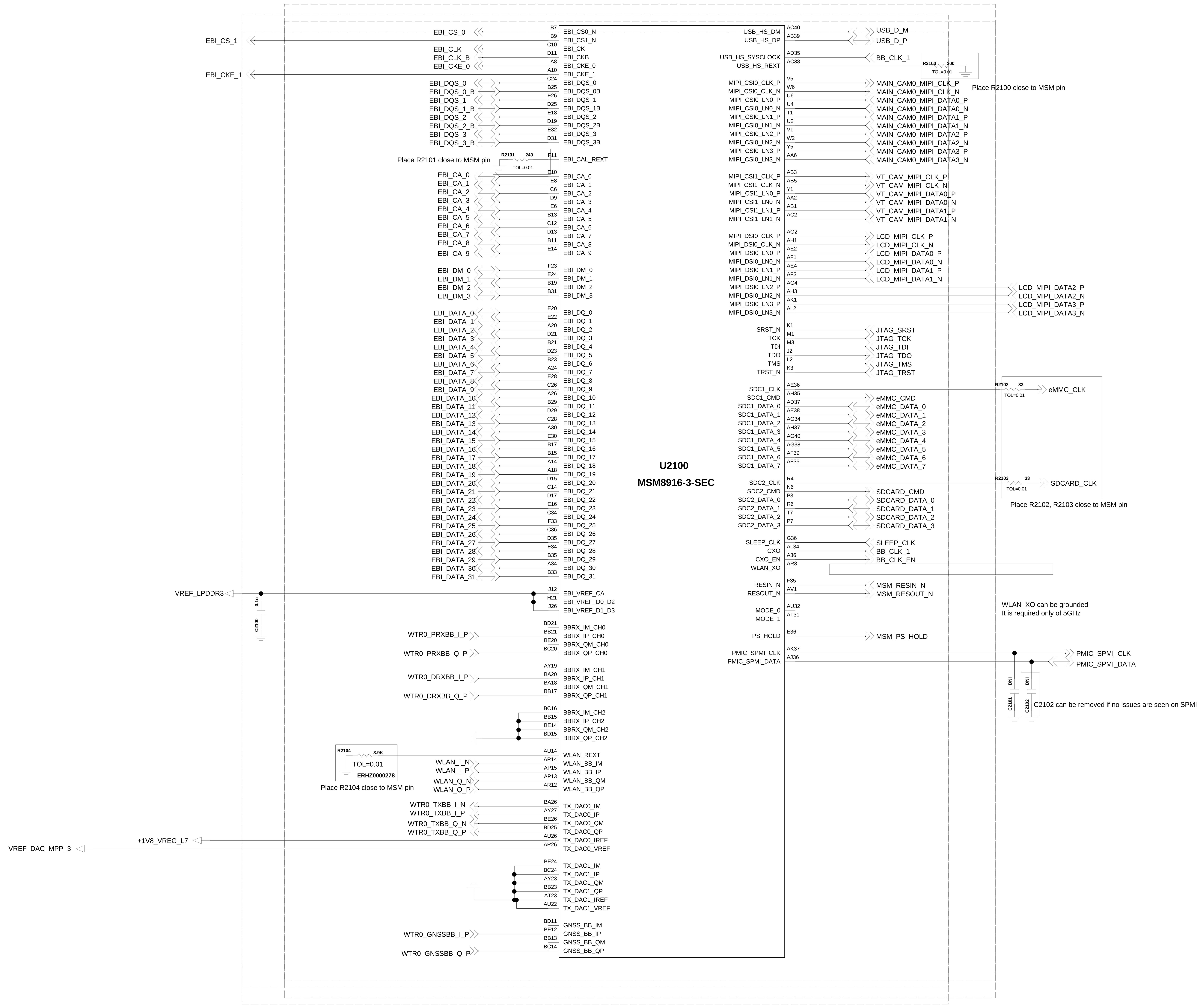
LG Electronics

Schematic1

A2	PDM NUMBER	Rev
User Name	4 of 3	Drawing Date

< 2-1-5-3-1_MSM8916-3-SEC_CONTROL > REV_0.3

CCDS CARD Information	
Release Date	June. 27 2014
Based on Reference Schematic	Rev. D
80-NK807-42_MSM8X16_+PM8916_Preliminary_Reference_Schematic	

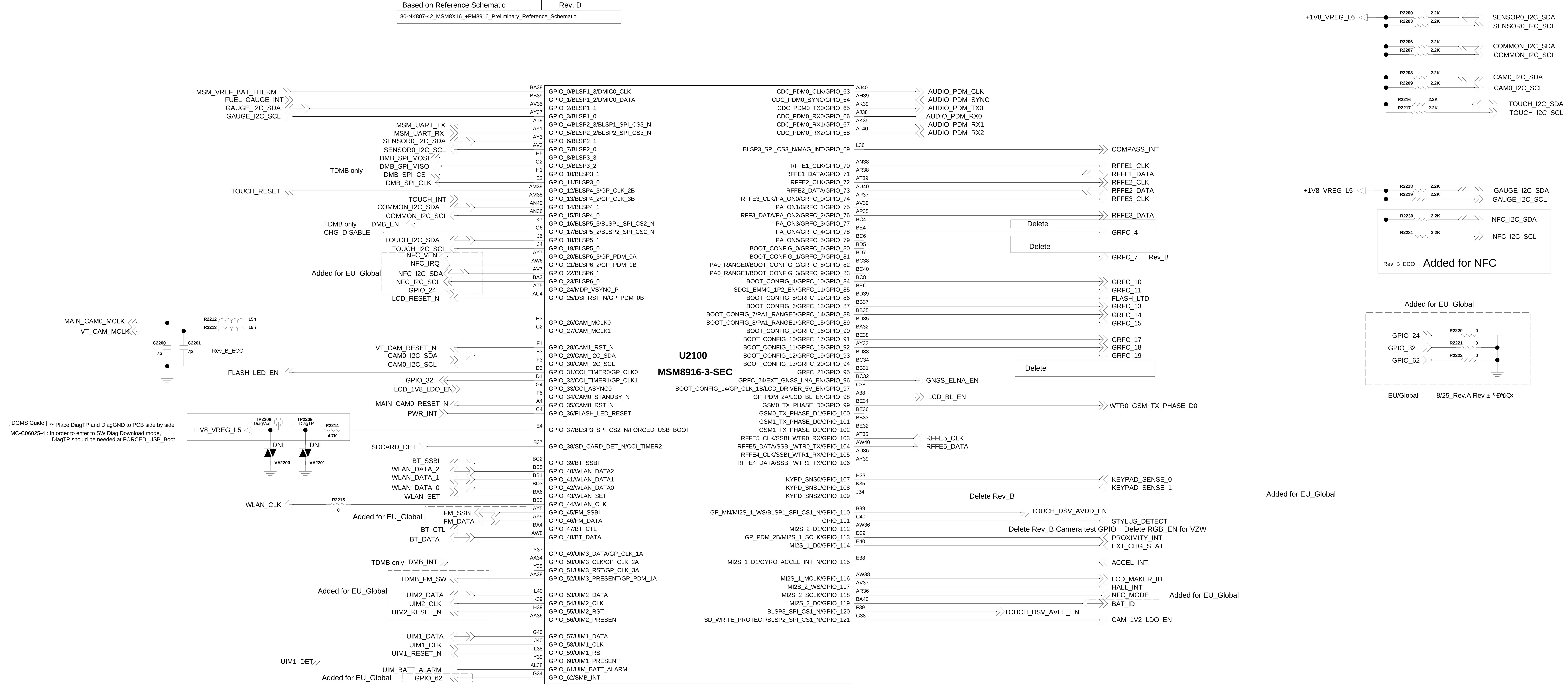


LGE Internal Use Only

LG Electronics	
TITLE	Schematic1
SIZE A2	PDM NUMBER	Rev
SHEET 2 of 2	Drawing Date
DRAWN BY	User Name

< 2-1-5-3-2_MSM8916-3-SEC_GPIO >   Rev_0.3

CCDS CARD Information	
Release Date	Sep. 24 2014
Based on Reference Schematic	Rev. D
80-NK807-42_MSM8X16_+PM8916_Preliminary_Reference_Schematic	



LGE Internal Use Only

LG Electronics

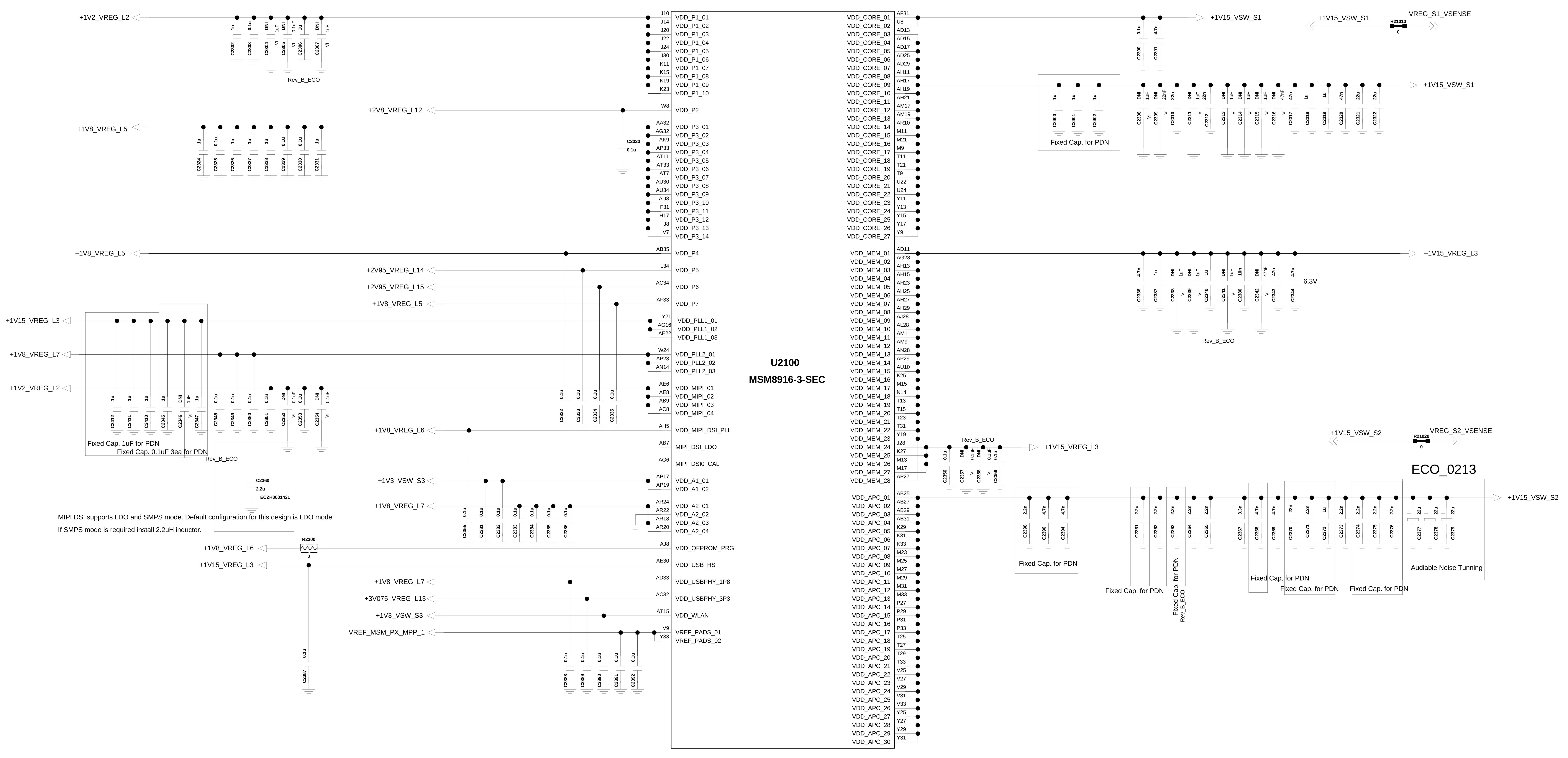
Schematic1

A2 PDM NUMBER Rev

User Name Drawing Date

< 2-1-5-3-3_MSM8916-3-SEC_POWER > Rev_0.3

CCDS CARD Information	
Release Date	Sep. 24 2014
Based on Reference Schematic	Rev. D
80-NK807-42_MSM8X16_+PM8916_Preliminary_Reference_Schematic	

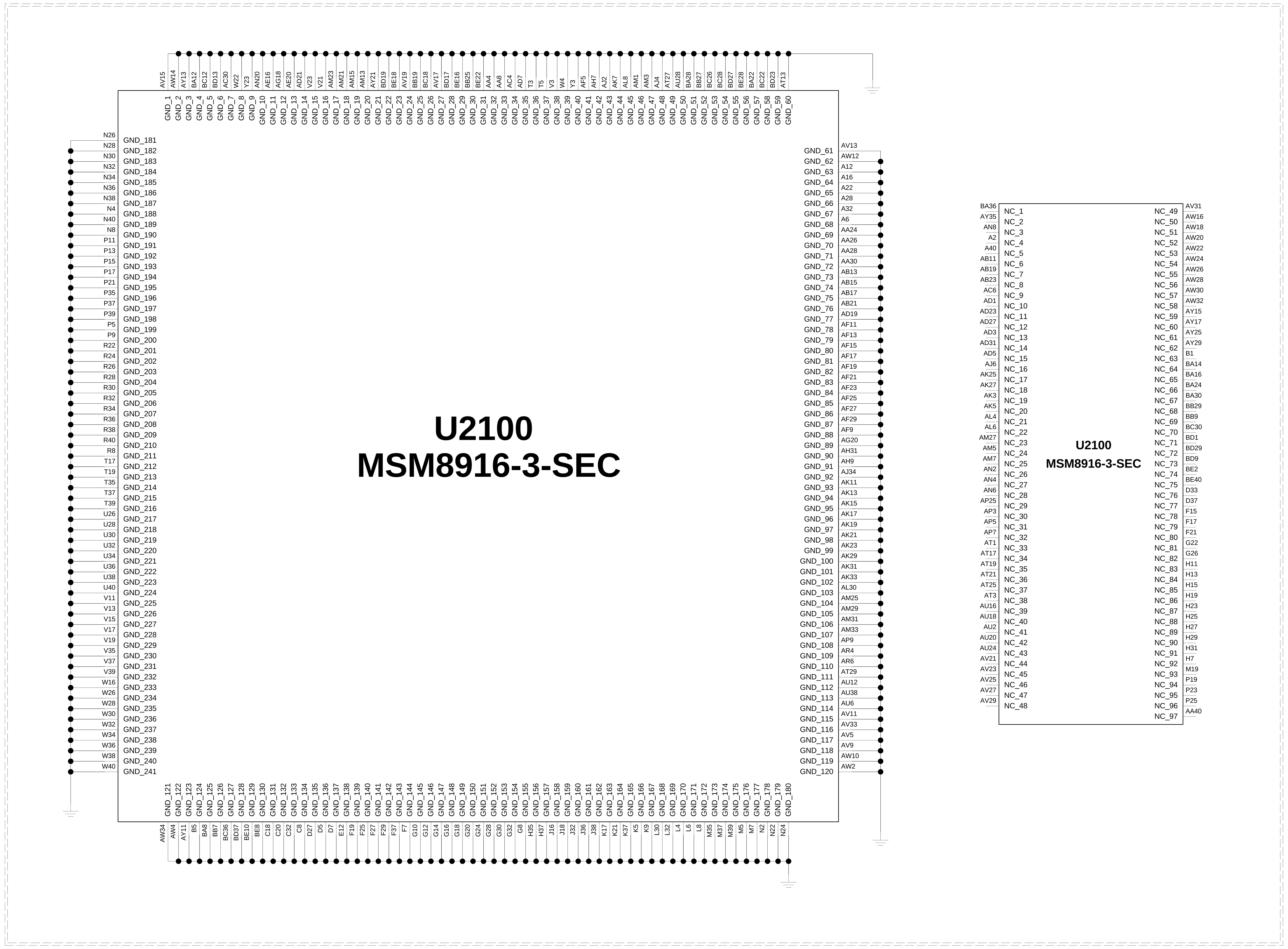


LGE Internal Use Only

LG Electronics
Schematic1
A2 PDM NUMBER Rev
User Name
Drawing Date

< 2_1_5_3_4_MSM8916_3_SEC_GND_NC > REV_0.3

CCDS CARD Information	
Release Date	Sep. 24 2014
Based on Reference Schematic	Rev. D
80-NK807-42_MSM8X16_+PM8916_Preliminary_Reference_Schematic	



LG Electronics

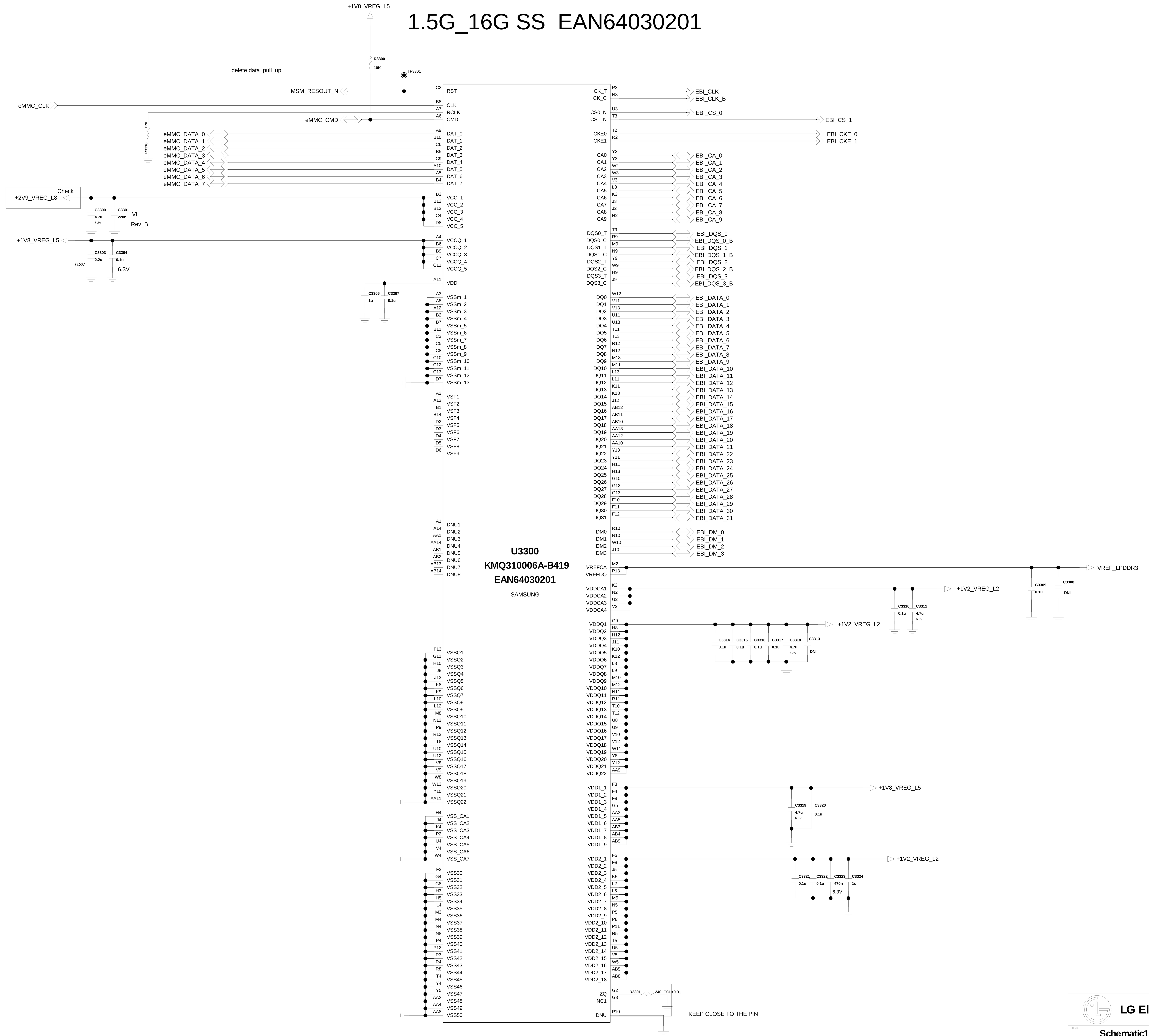
TITLE Schematic1

SIZE A2 | PDM NUMBER | Rev

User Name | Drawing Date

< 3-3-3-3-1_MCP_eMMC_5_0_8G_DDR3_12G_Samsung > Rev_0.3

1.5G_16G SS EAN64030201



LGE Internal Use Only

LG Electronics

Schematic1

A2 | PDM NUMBER | Rev

1 of 29

User Name

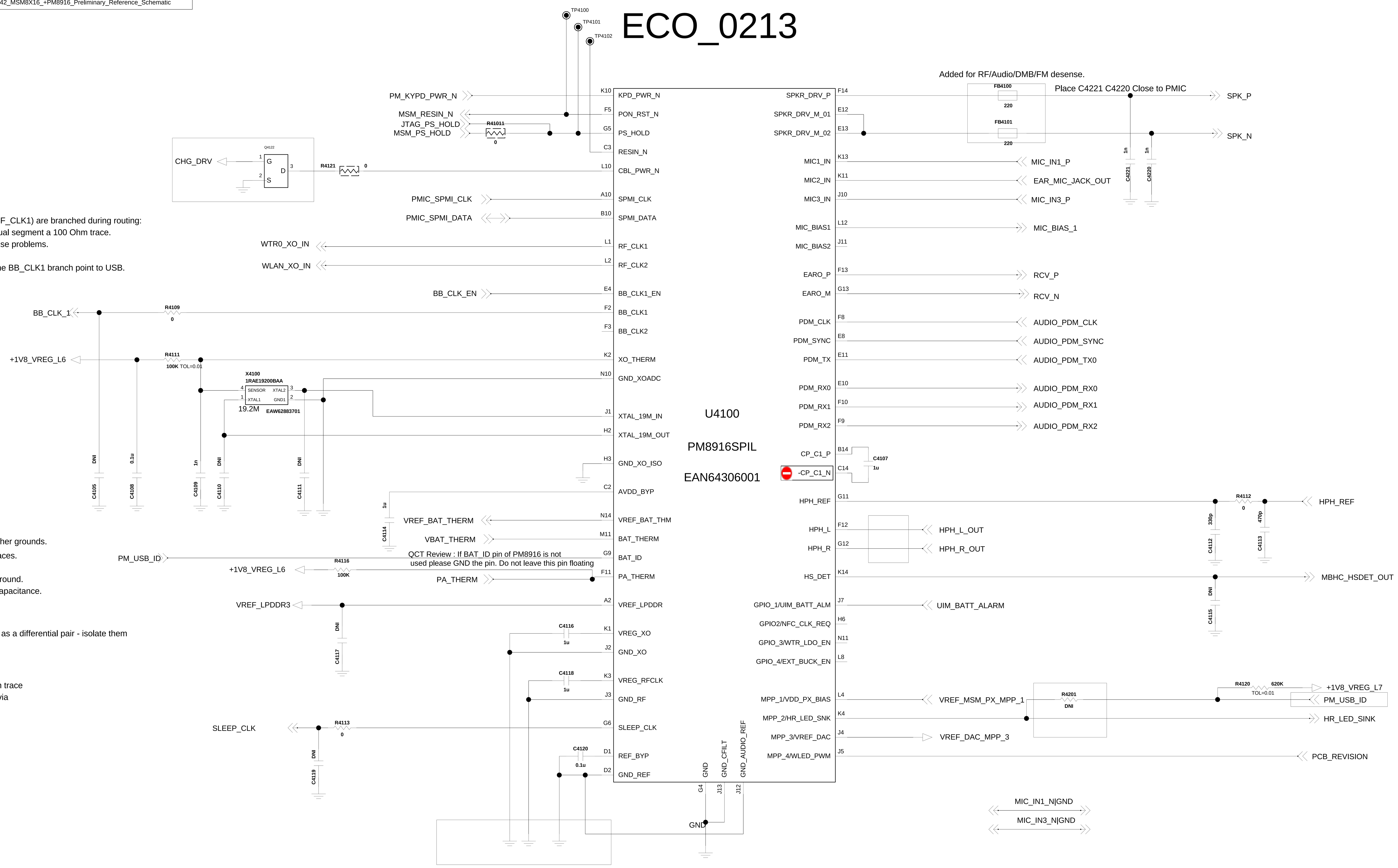
Drawing Date

<4-1-9-1_PM8916_DATA> Rev_1.0

CCDS CARD Information	
Release Date	June. 27 2014
Based on Reference Schematic	Rev. D
80-NK807-42_MSM8X16_+PM8916_Preliminary_Reference_Schematic	

ECO_0213

When the PMIC¡¯s 19.2 MHz clock signals (BB_CLK1 and RF_CLK1) are branched during routing:
¡¤Make the common route a 50 Ohm trace, and each individual segment a 100 Ohm trace.
¡¤Match the branch lengths; even a 5 mm difference can cause problems.
¡¤Keep branches as short as possible.
¡¤If these routing guidelines cannot be met, add a buffer at the BB_CLK1 branch point to USB.



A ¡®moat¡¯ cleared of metal that isolates XO ground from other grounds.
XO ground points connect to GND_XOADC pin using thin traces.
GND below in/out lines for improved isolation
Via to main ground plane under the XO_THERM capacitor ground.
Maintain clearance below in/out pads for reduced parasitic capacitance.

Locate the crystal 1 cm from the PMIC
Use thin traces for high thermal resistance
XTAL_19M_IN and XTAL_19M_OUT should NOT be routed as a differential pair - isolate them
Isolate from other signals (no parallel routing)
R and C close to crystal
Ground connections
Connect crystal GND to PMIC GND_XOADC pin using a thin trace
Connect GND_XOADC pin to main GND using a dedicated via

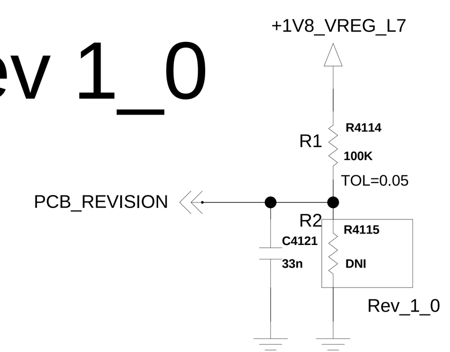
	VZW	Other
MPP_2	VZW_CHG	HR_LED_SINK
R4201	0	DNI
R4202	DNI	0

GND_REF must connect directly to C4120 and then to main GND
without any other intermediate connections to GND fills

GND_XO and GND_RF must connect directly to C4118 and C4116 respectively
and then connect to main GND without any other intermediate connections to GND fills

PCB REVISION

Rev 1_0



Rev.	R1	R2	Voltage
A	100K	20K	0.300
B	100K	27K	0.383
C	100K	36K	0.48
D	100K	51K	0.680
E	100K	75K	0.771
F	100K	100K	0.900
G	100K	130K	1.017
1.0	100K	DNI	1.800
1.1	DNI	100K	0.000
MKT	100K	200K	1.2

It is recommended that the handset microphone ground or negative
are routed differentially with microphone input trace
and ground to the same ground as GND_CFILT
80-NK808-121 PM8916 CODEC BANDGAP REFERENCE AND MICROPHONE PCB CONNECTIONS TECHNICAL MEMO

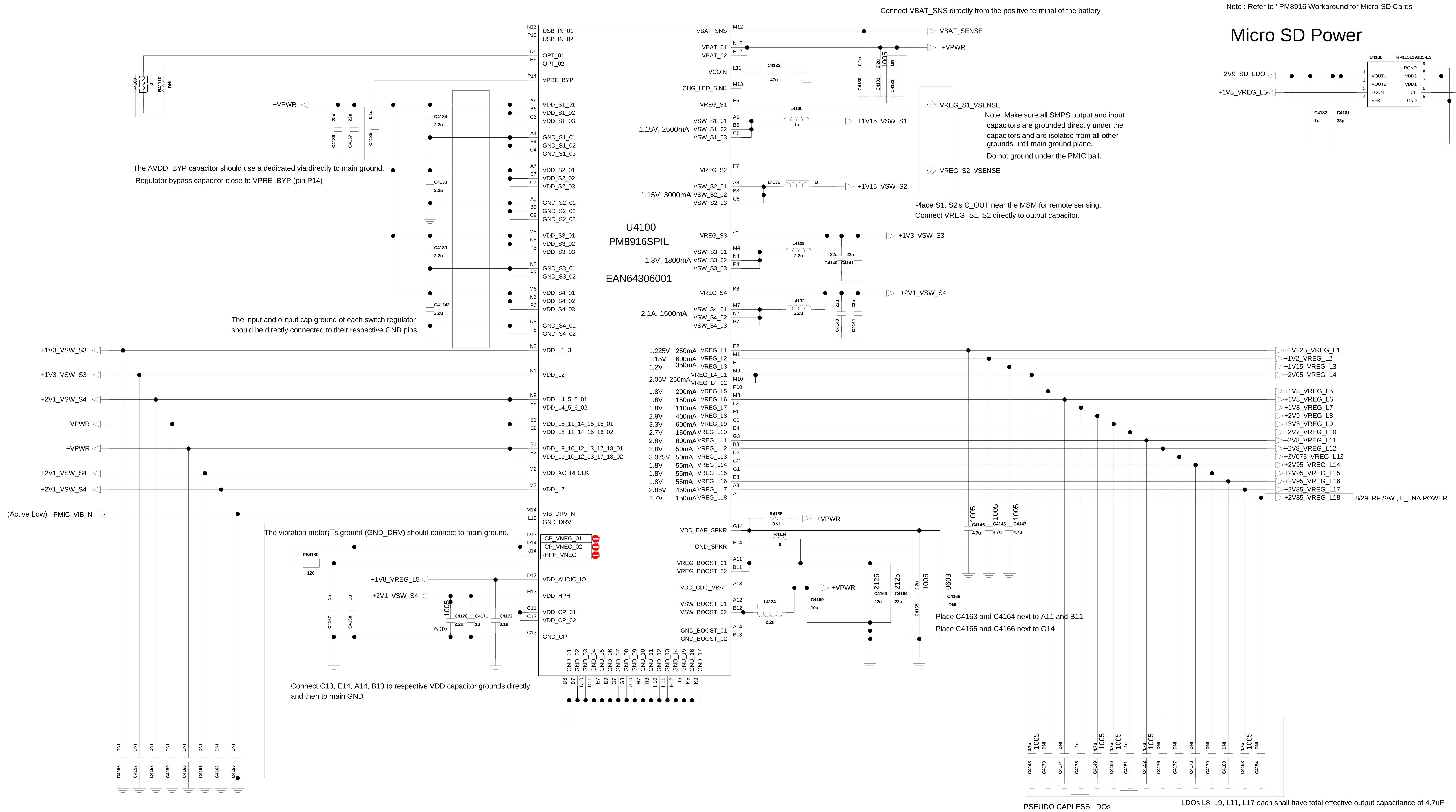
LG Electronics
Schematic1
A2 PDM NUMBER Rev
28 27 User Name Drawing Date

LGE Internal Use Only

<4-1-9-2_PM8916_POWER> REV_1.0

CCDS CARD Information	
Release Date	June. 27 2014
Based on Reference Schematic	Rev. D
80-NK807-42_MSM8X16_+PM8916_Preliminary_Reference_Schematic	

ECO_0213



LDOs L8, L9, L11, L17 each shall have total effective output capacitance of 4.7uF

Rest of the LDOs shall each have total effective output capacitance of 1.0uF

The Pseudo capless LDOs must comply to following tolerable routing requirement:
P50: R<400mOhms, L<45nH
P150: R<200mOhms, L<20nH
P300: R<150mOhms, L<18nH
P600: R<150mOhms, L<18nH

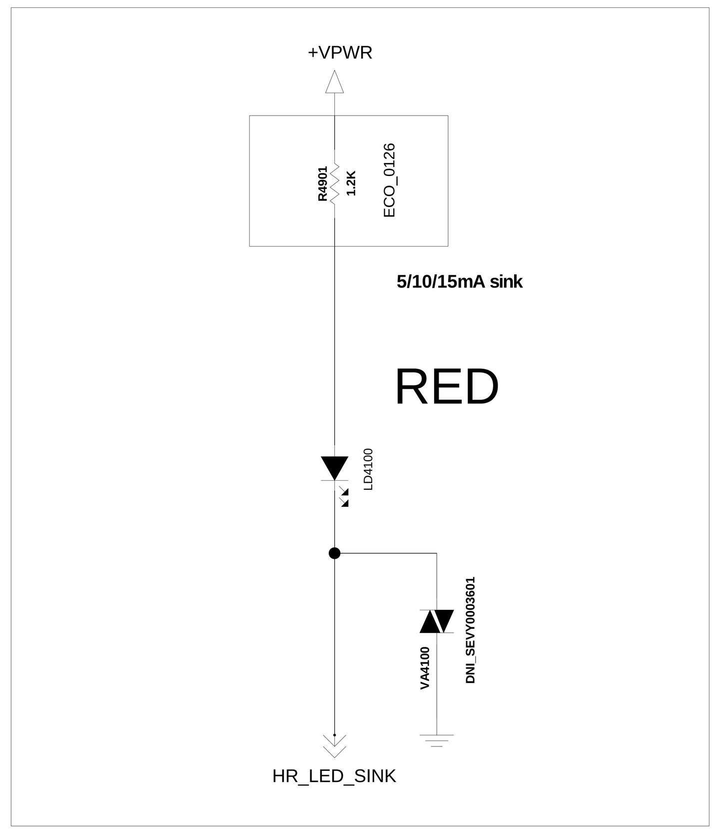
Function	Circuit Type	Default Voltage	Specified Range	Programmable Range	Rated Current	Default On
S1	HF_SMPS	1.15V				Y
S2	HF_SMPS	1.15V				N
S3	HF_SMPS	1.3V				Y
S4	HF_SMPS	2.1V				Y
L1	NMOS LDO	1.225V				N
L2	NMOS LDO	1.15V				Y
L3	NMOS LDO	1.2V				Y
L4	PMOS LDO	2.05V				N
L5	PMOS LDO	1.8V				Y
L6	PMOS LDO	1.8V				N
L7	PMOS LDO	1.8V				N
L8	PMOS LDO	2.9V				N
L9	PMOS LDO	3.3V				N
L10	PMOS LDO	2.7V				N
L11	PMOS LDO	2.8V				N
L12	PMOS LDO	2.8V				N
L13	PMOS LDO	3.075V				N
L14	PMOS LDO	1.8V				N
L15	PMOS LDO	1.8V				N
L16	PMOS LDO	1.8V				N
L17	PMOS LDO	2.85V				N
L18	PMOS LDO	2.7V				N
VREG_XO	PMOS LDO	2.9V				
VREG_RF_CLK	PMOS LDO	2.95V				

LGE Internal Use Only

LG Electronics
Schematic1
A2 PDM NUMBER Rev
User Name
29 28
Drawing Date

<4-6-1_LED_Driver_IC_LP5521>

REV.1.0



For VZW

<4-8-2_Fuel_Gauge_MAX17050>

Rev_1.0

Delete Fuel Gauge MAX17050 for VZW

For VZW

<4-3-4_OVP_R5560Z011A-TL-F>

15/02/13 Change for low cost

OVP Default 6.8V(C1 Pin Connet to GND)

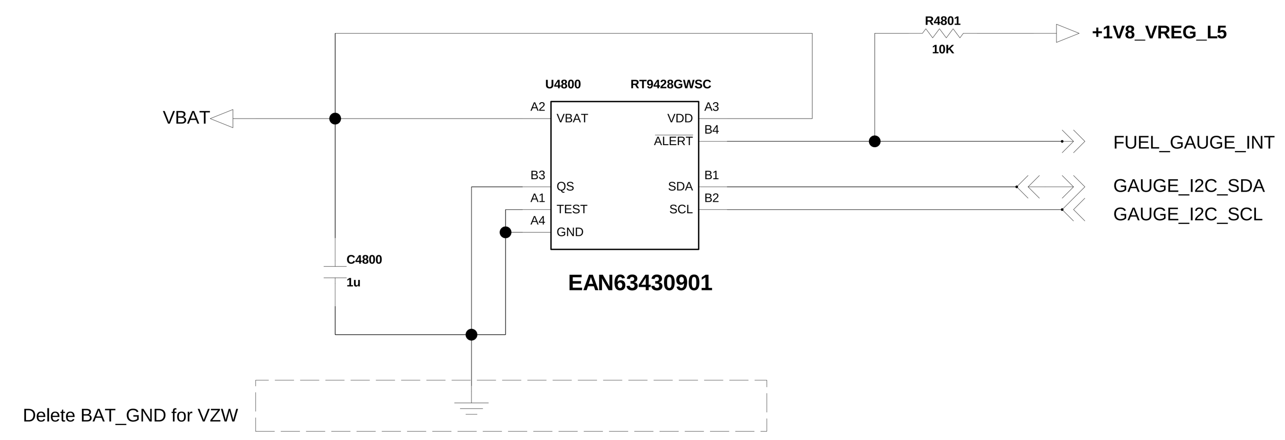
REV_1.0

Delete OVP for VZW

<4-8-1_Fuel_Gauge_RT9428>

Only use line VBAT1 from Battery Connector.

CLOSE TO Battery CONNECTOR



Don't use Board GND

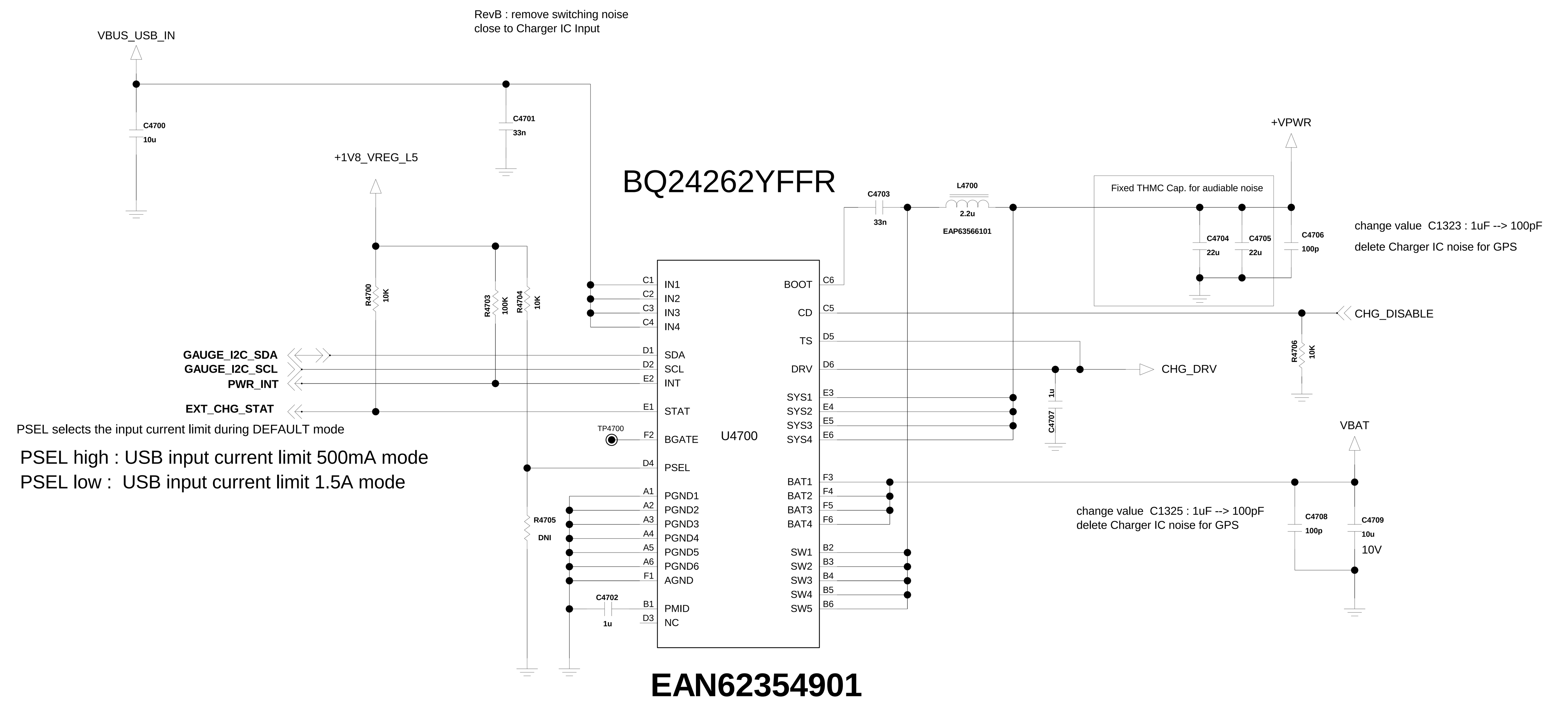
Use line GND From Battery GND Pin#4

For VZW

Delete BQ24292 For VZW

CHARGING IC

Add Charger IC _130808



PSEL selects the input current limit during DEFAULT mode

PSEL high : USB input current limit 500mA mode

PSEL low : USB input current limit 1.5A mode

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LG Electronics

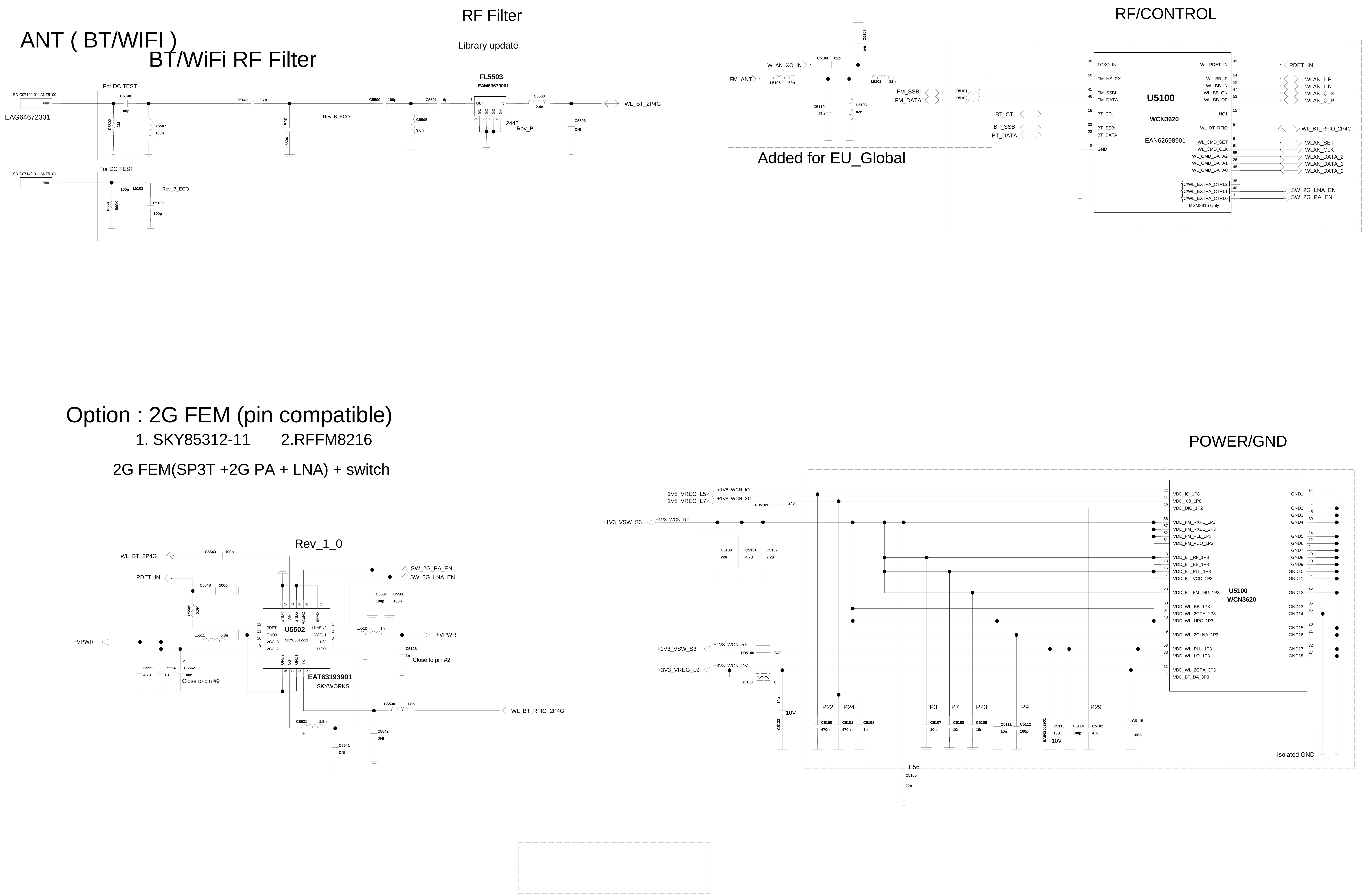
Schematic1

A2 PDM NUMBER Rev

33 of 30

User Name Drawing Date

< 5-1-1-17_BT_Wi-Fi_WCN3660B > Rev_1.0



LG Electronics

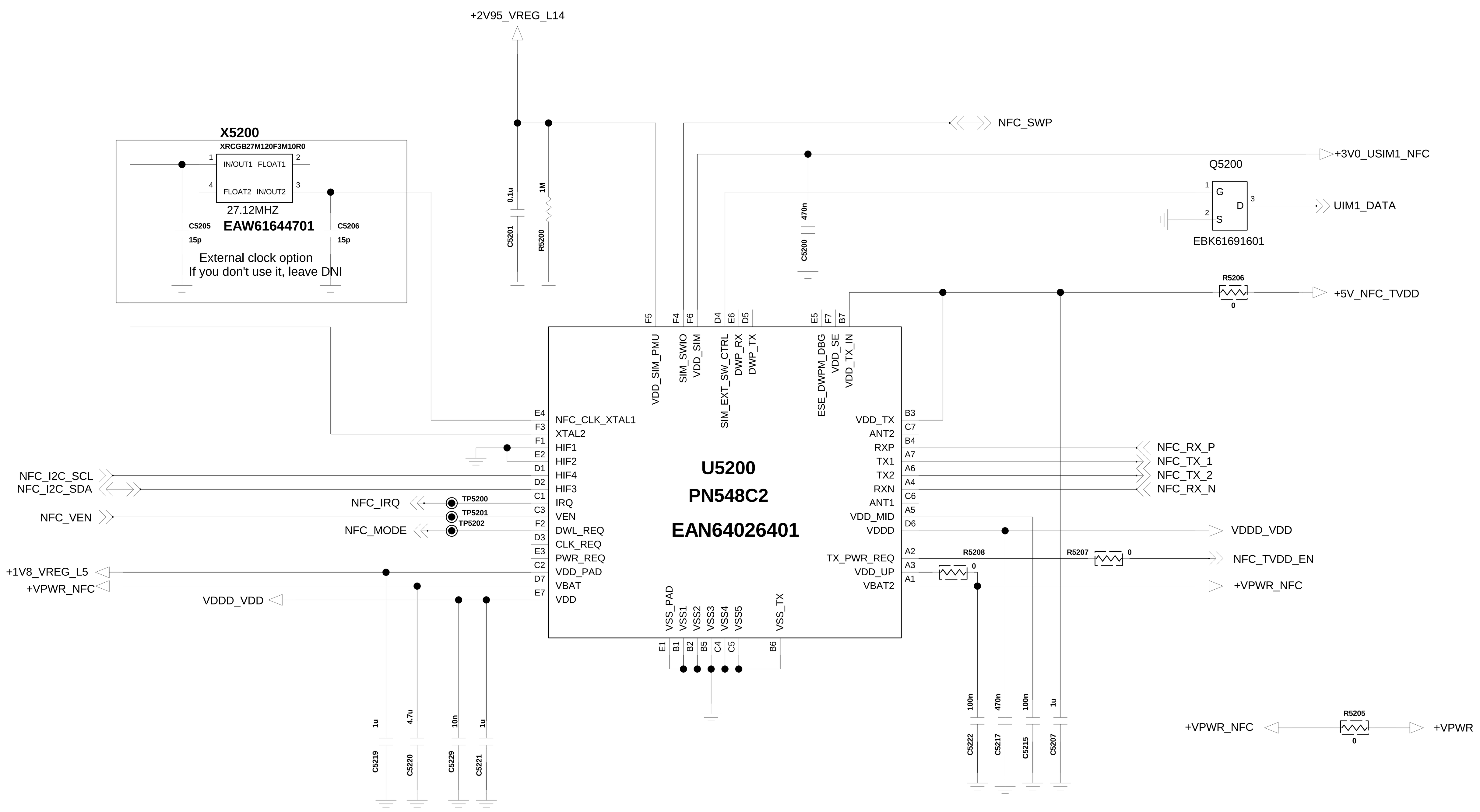
DRAWN BY: heesoo.kim

SIZE: A2

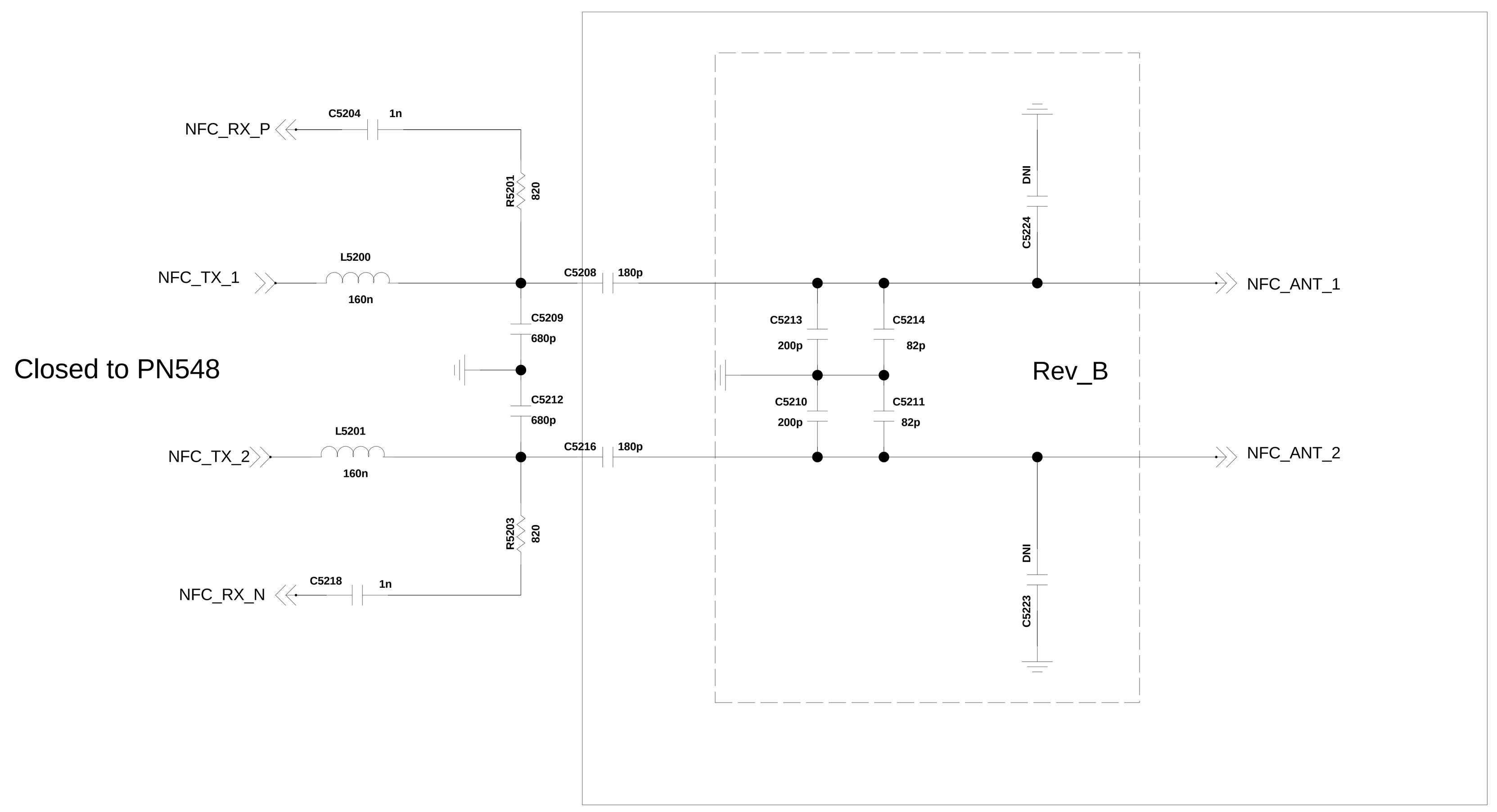
LGE Internal Use Only

Added for NFC Rev_B_ECO

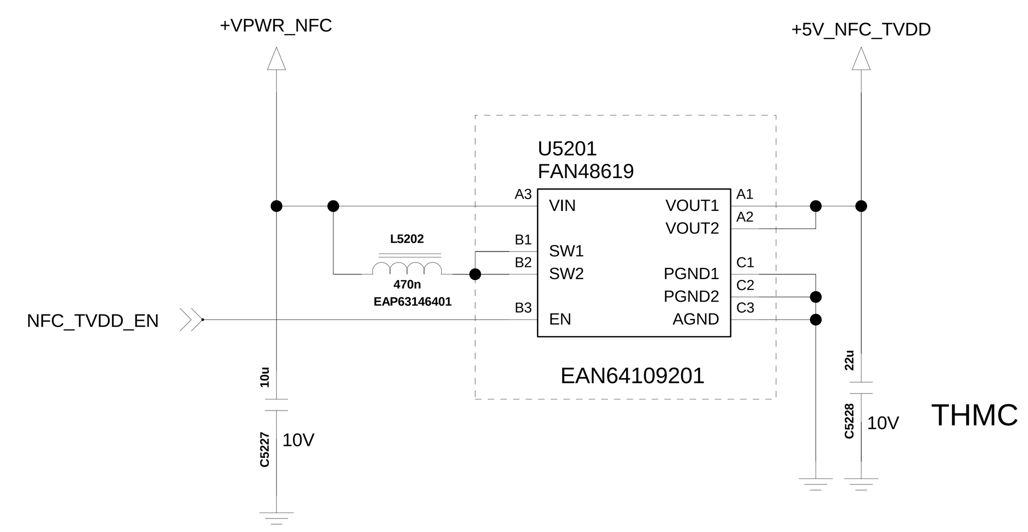
<NFC_PN548>



NFC Antenna



+5V Booster



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LG Electronics

Schematic1

A2 PDM NUMBER Rev

User Name

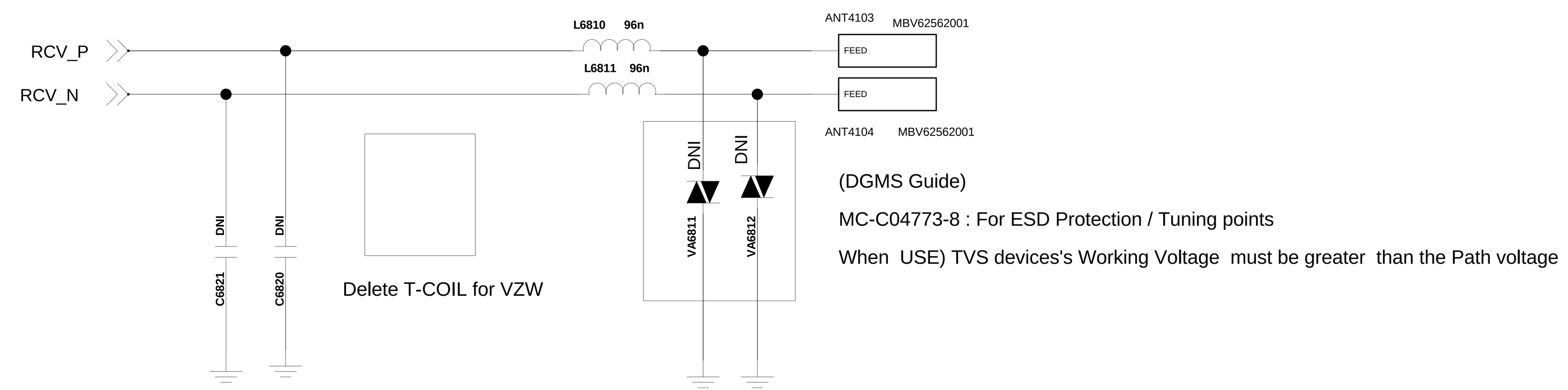
29 of 28

Drawing Date

< 6-8-2_Receiver_TCOIL >

Rev_0.4

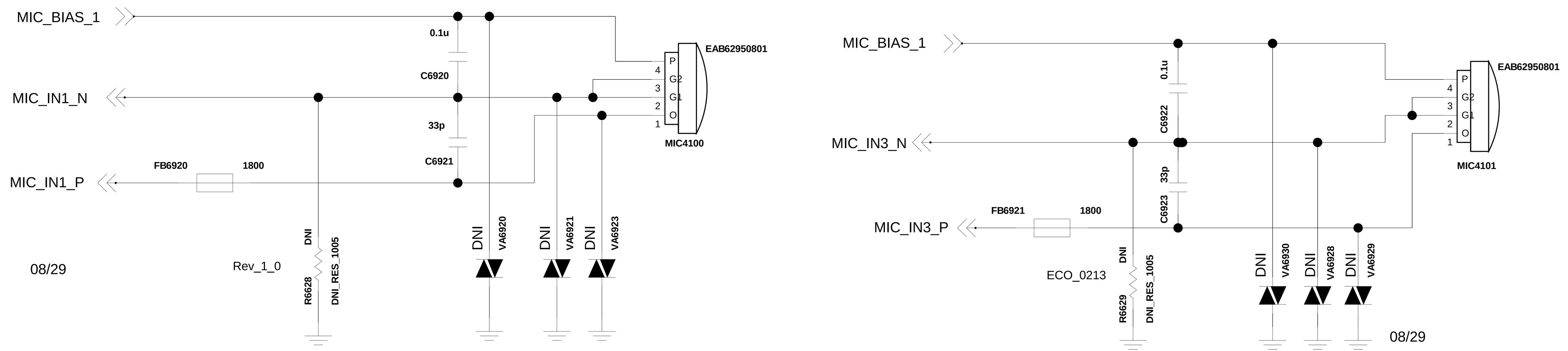
FPCB type : EAB64148501(no HAC, KIRIN)
FPCB type : EAB63688601(for TMUS)



< 6-9-2_2MIC >

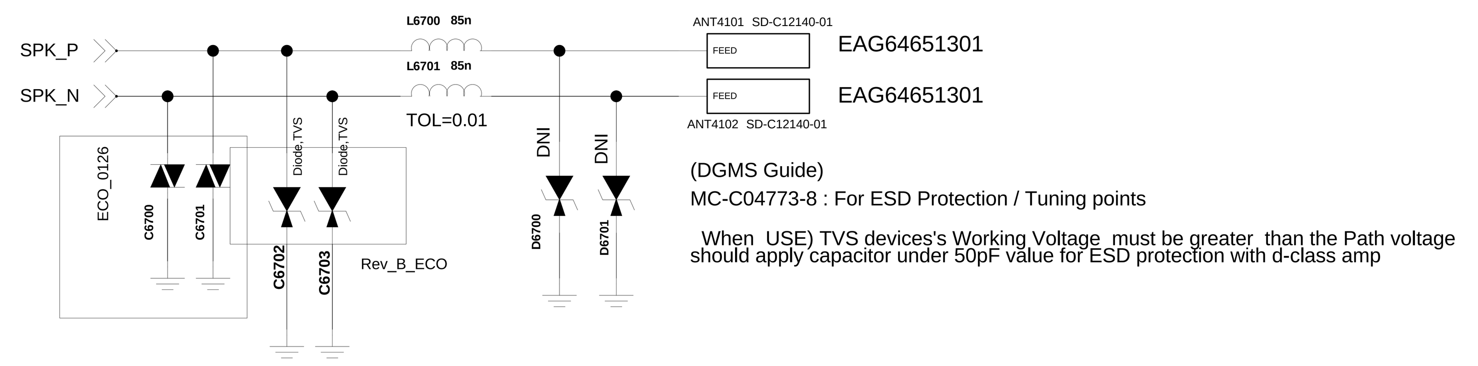
Main MIC

Sub_MIC1



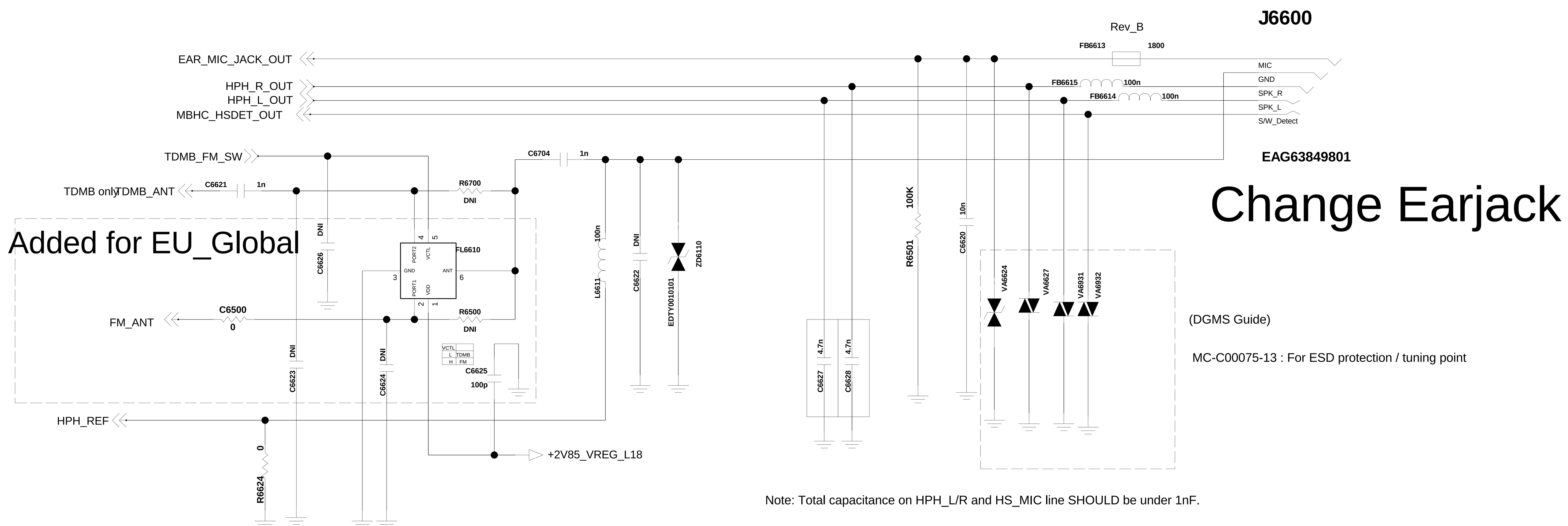
REV10_DNI varistor changed to 0 Ohm for camcording noise improvement. (03/28)

< 6-7-1_Speaker > Rev_0.4



< 6-6-1_Earjack > Rev_0.6

Circuit 2. Ear jack_3.5pi with MBHC

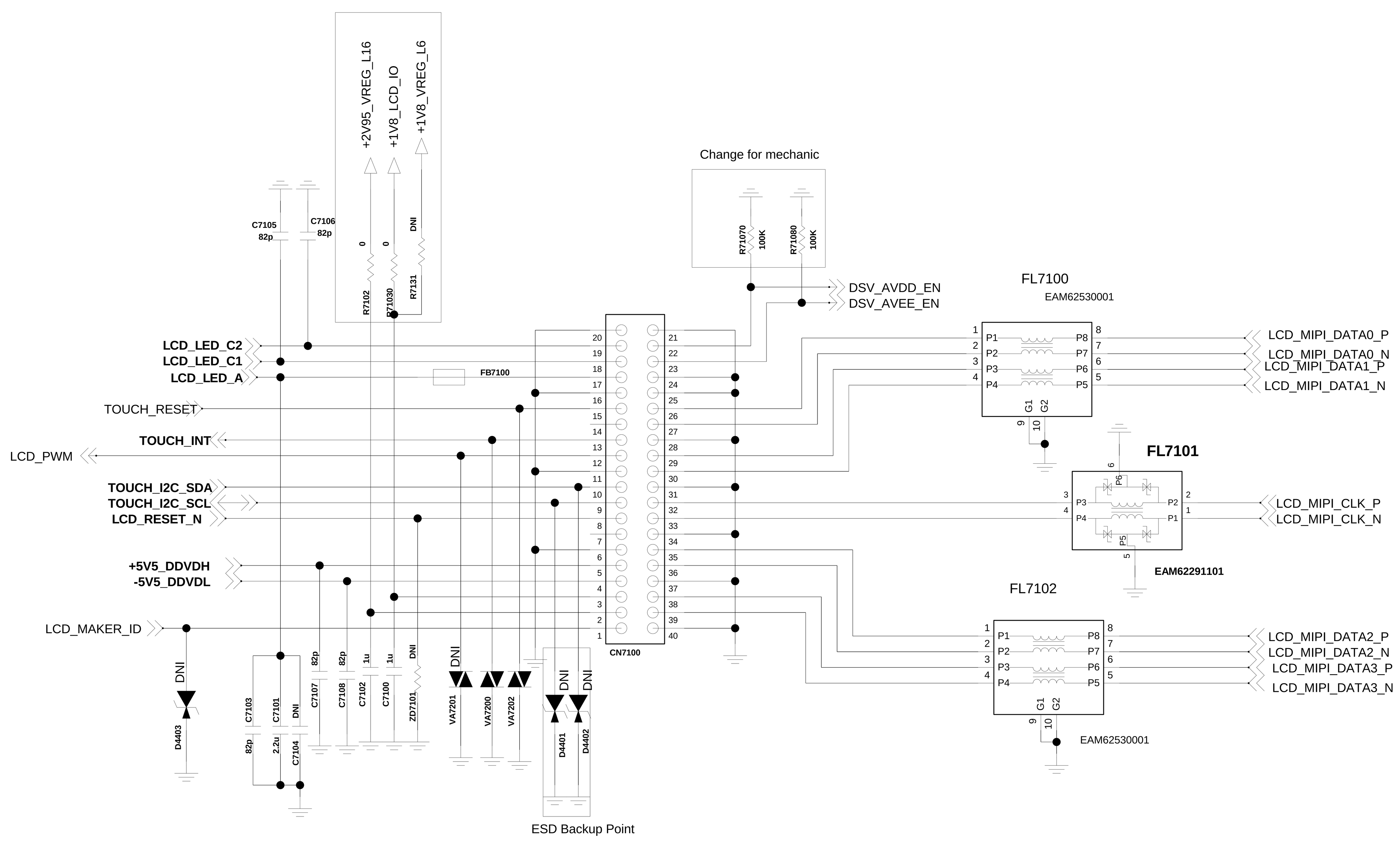


Note: Total capacitance on HPH_L/R and HS_MIC line SHOULD be under 1nF.

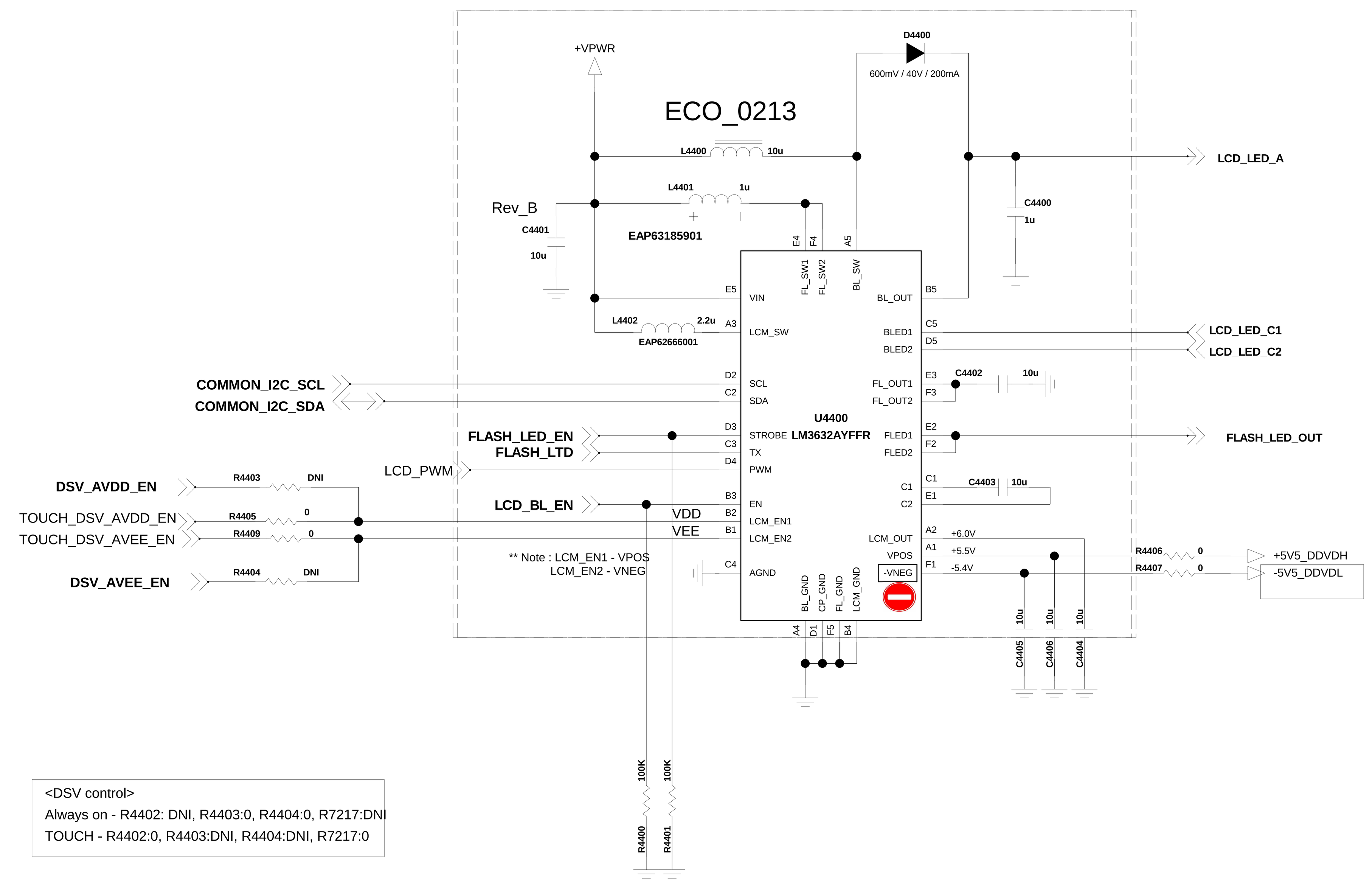
LGE Internal Use Only

LG Electronics
Schematic1
A2 PDM NUMBER Rev
34 of 31
User Name Drawing Date

PH1_5.7" LCD Connector



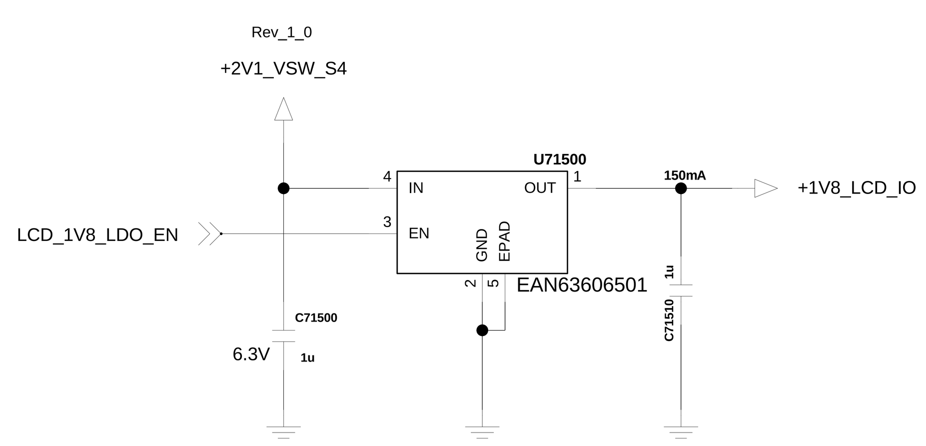
<4-4-7_LCD_Backlight_Flash_DSV_IC_LM3632> Rev_0.3



Delete LCD LDO spare

<Flash_LED>

LCD IO LDO



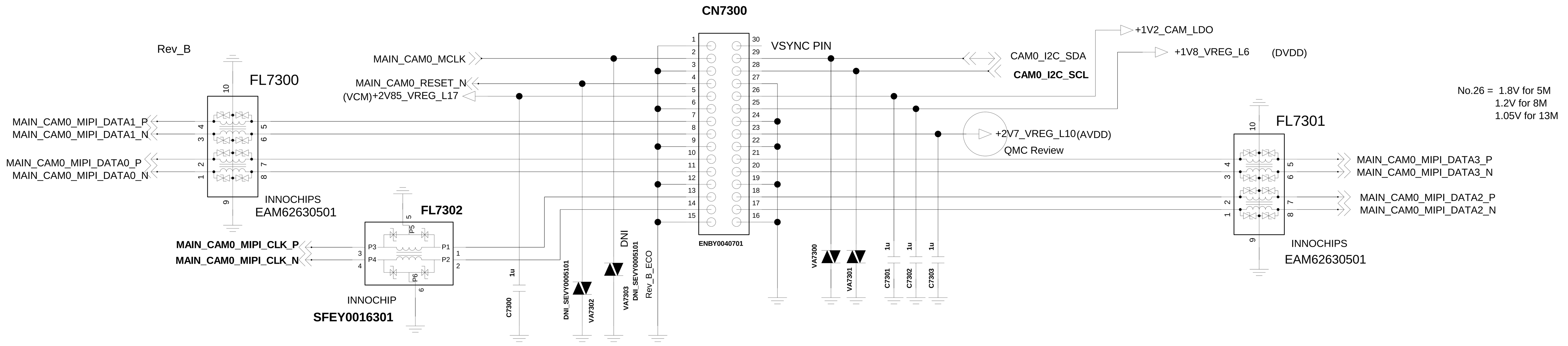
LGE Internal Use Only

LG Electronics
MODEL NAME
PDM NUMBER
A2
Rev
User Name
Drawing Date
32 of 29

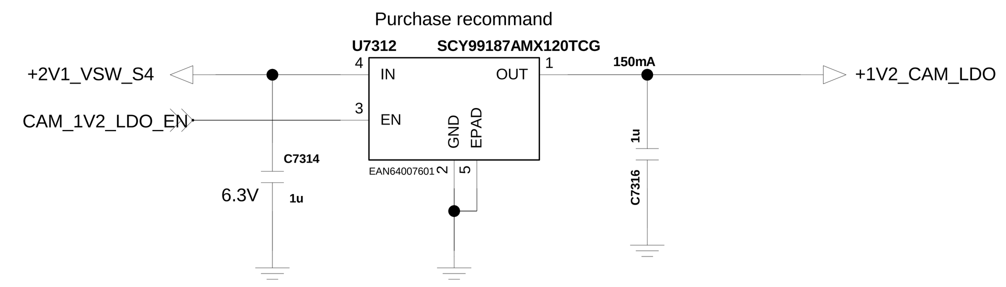
<7-3-5-1_30pin_5M_8M_13M_AF>

13 Mega camera module : EBP62702201 Rev_0.3

Check the pin map



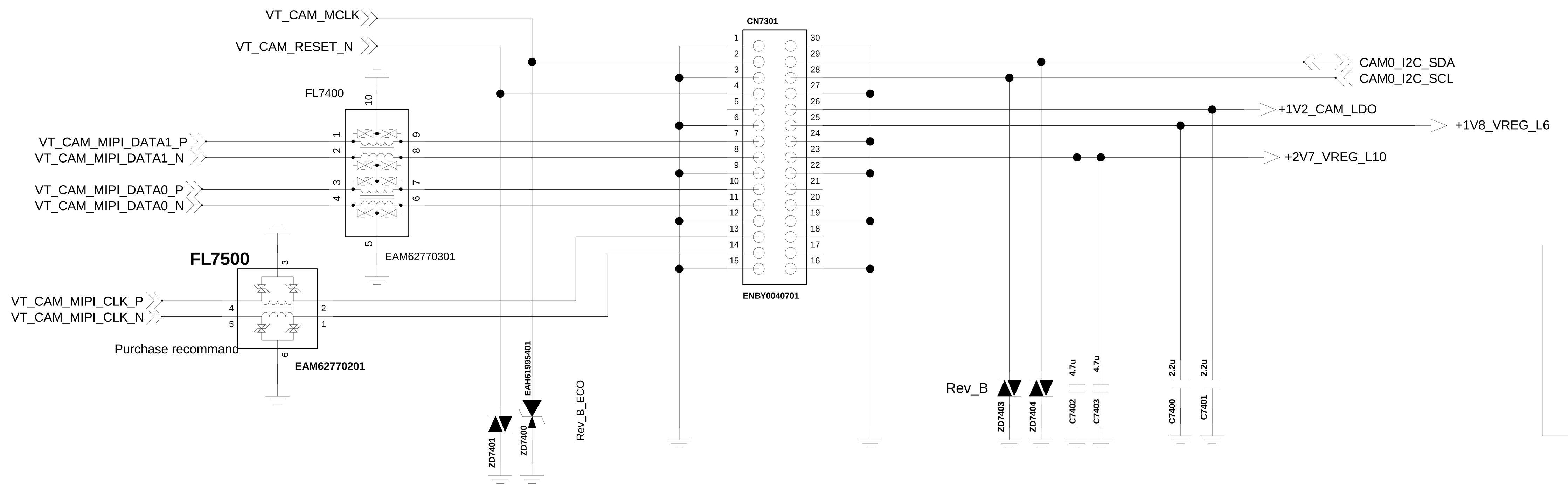
CAM LDO 13M&8M



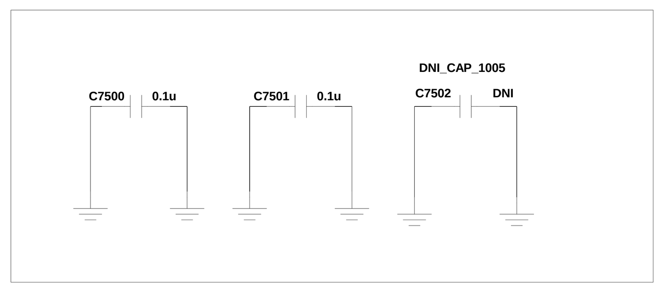
< 7-4-x_Front_Camera_30pin_5M_FF>

[DGMS Guide]

MC-C00081-17: 1. Add ESD Protection curcuit for Camera signal lines



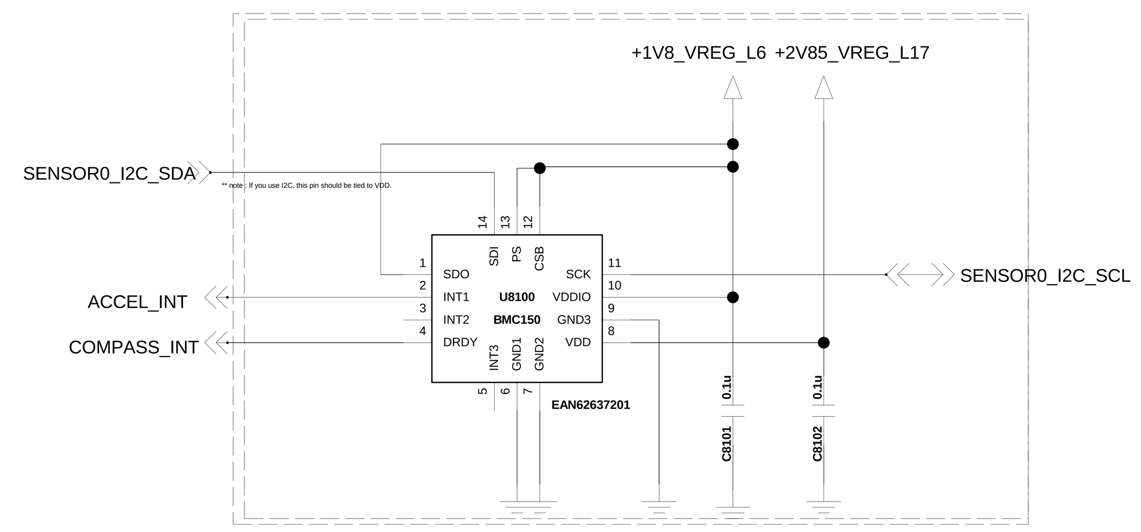
Added Cap. for ESD



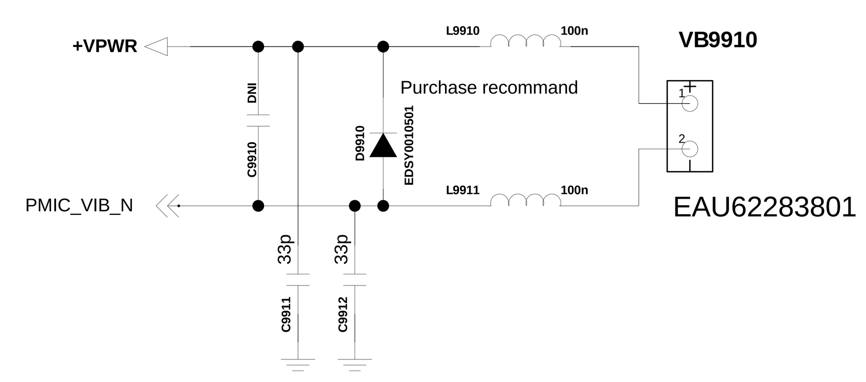
LGE Internal Use Only

LG Electronics	
TITLE	MODEL NAME
SIZE A2	PDM NUMBER
REV	Rev
DRAWN BY	User Name
SHEET	1 of 2
Date	Drawing Date

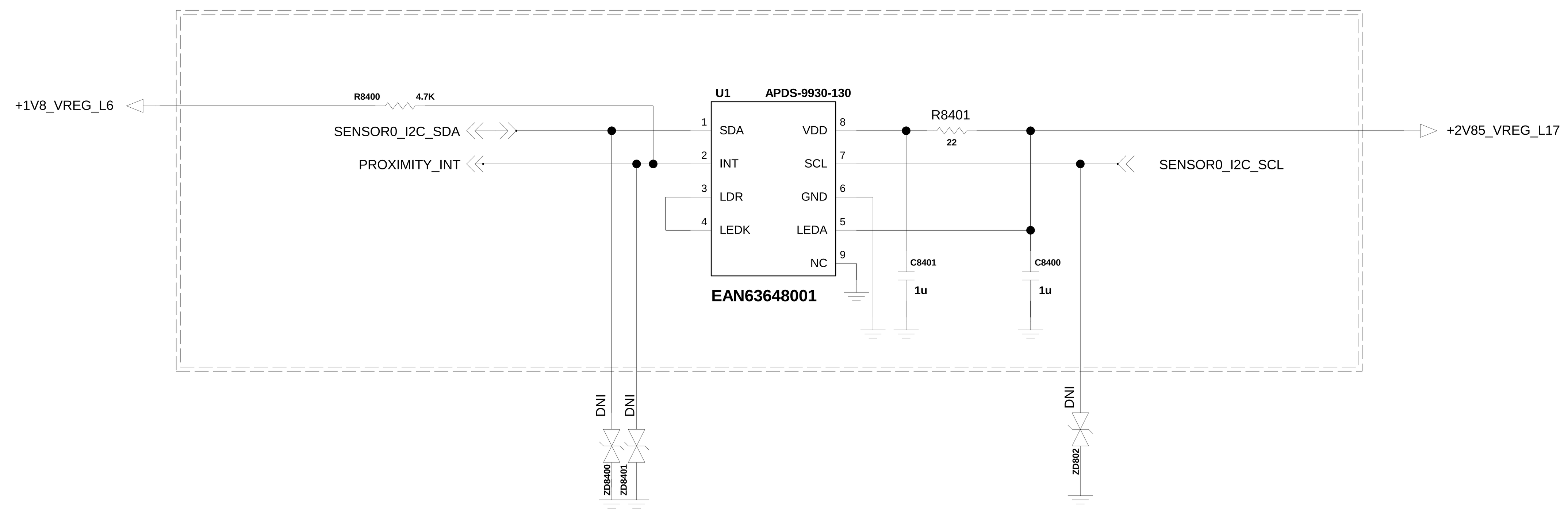
<8-1-2-2_Accel_Compass_BMC150> Rev_1.0



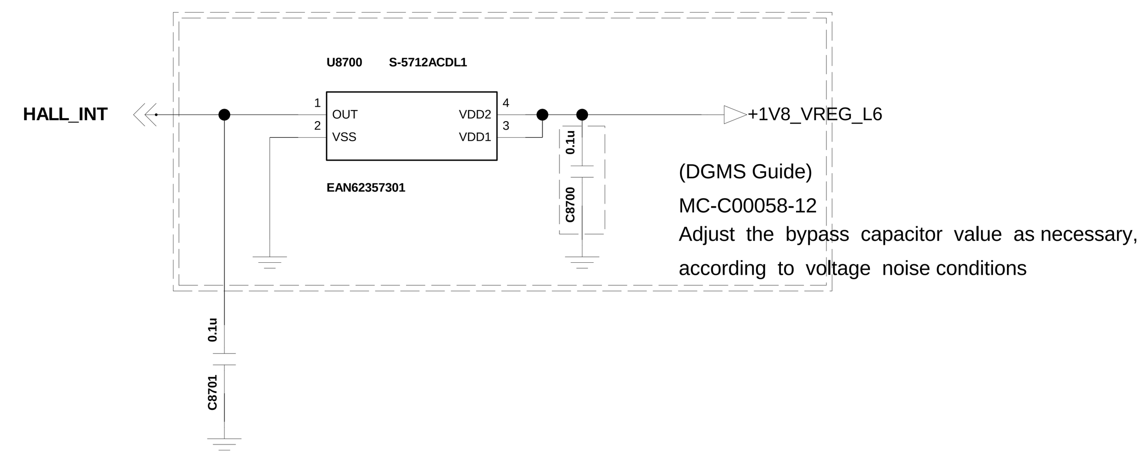
< 9-9-1-1_Motor > Rev_0.3



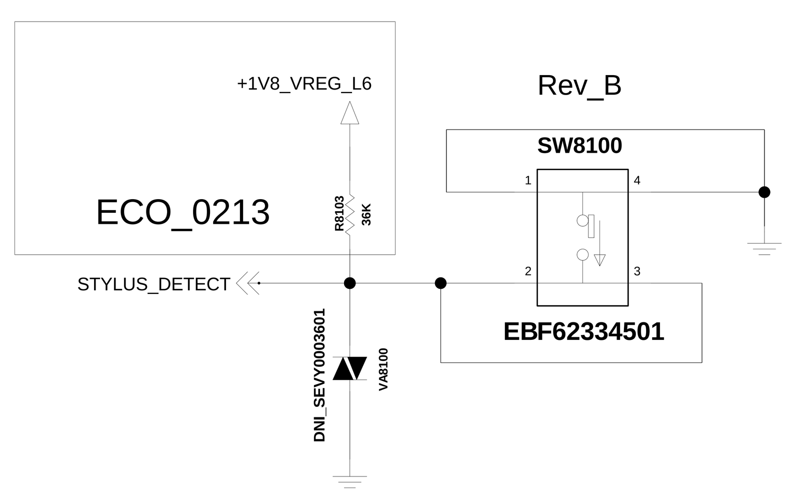
<8-4-1-1_Proximity_APDS-9930> REV.1.0



<8-7-1-6_Hall_IC_BU52033NVX> Rev_1.0



< Stylus Pen Detect S/W >



LGE Internal Use Only

LG Electronics

TITLE Schematic1

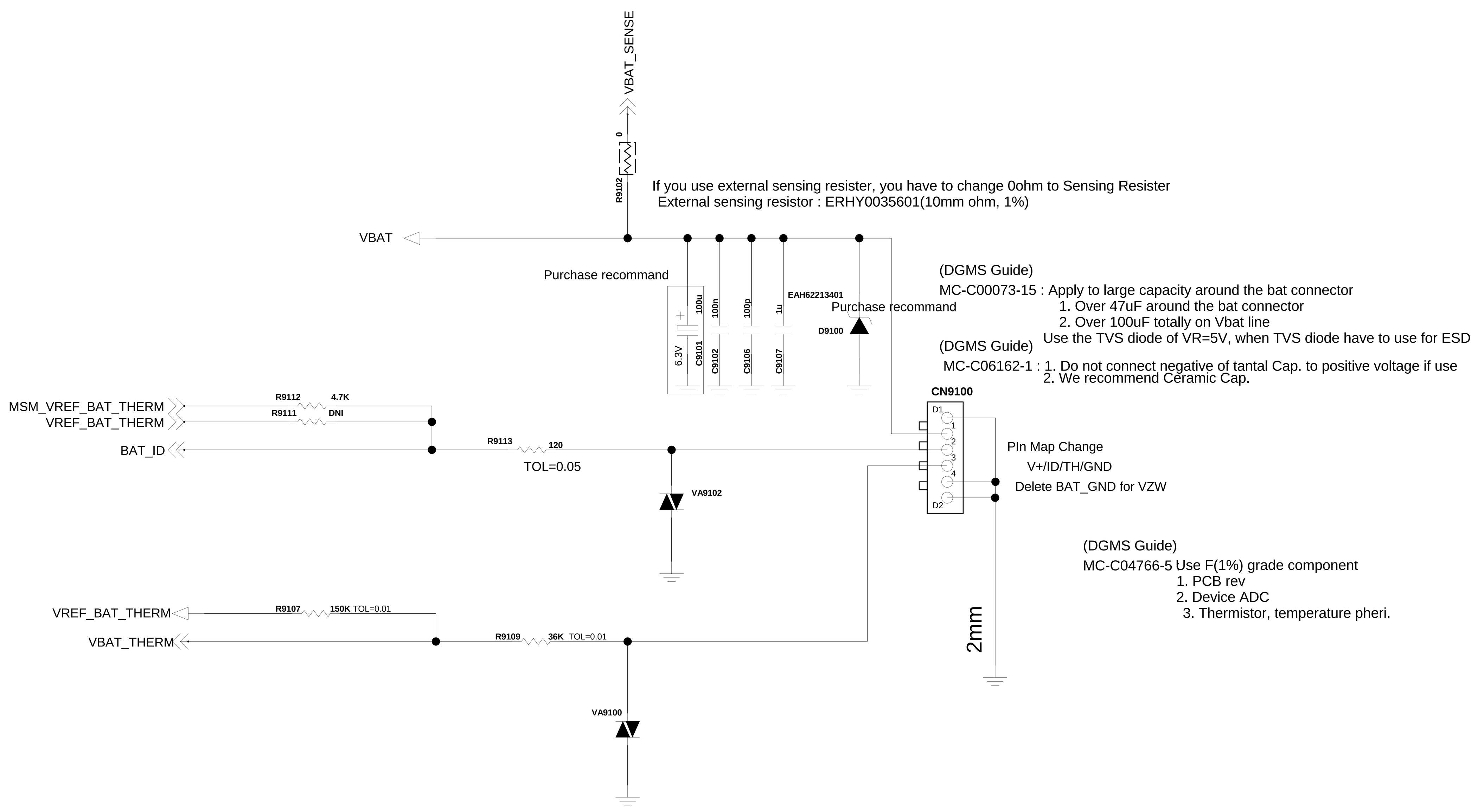
SIZE	DWG NO	REV
A2	PDM NUMBER	Rev

DRAWN BY User Name

SHEET 12 of 5 Date Drawing Date

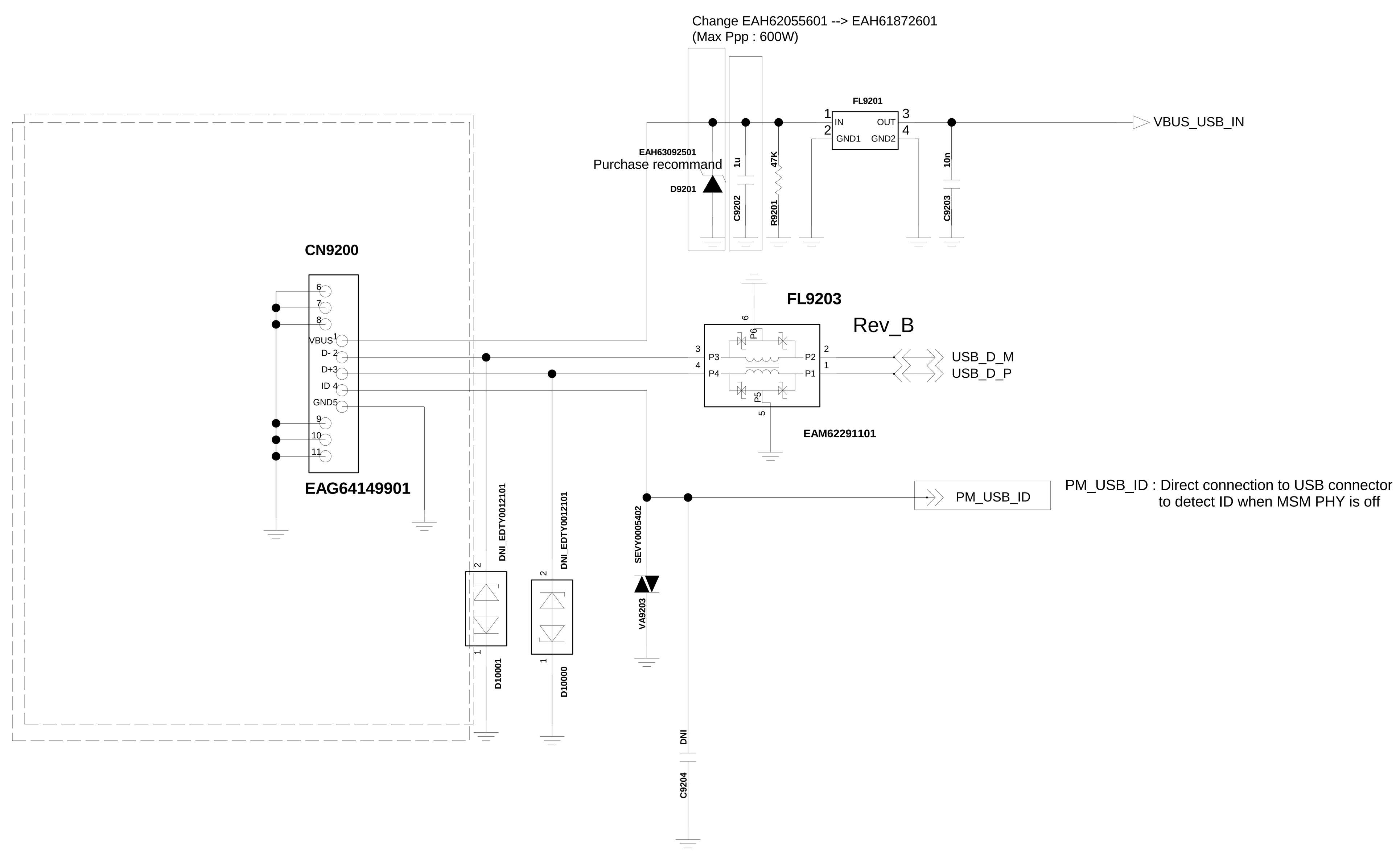
<9-1_Battery_Connector_4Pin>

Circuit 1. Batt Conn. /wo EMI Filter



<9-2-1_USB_IO> Rev.1.0

Connector 1. USB 2.0

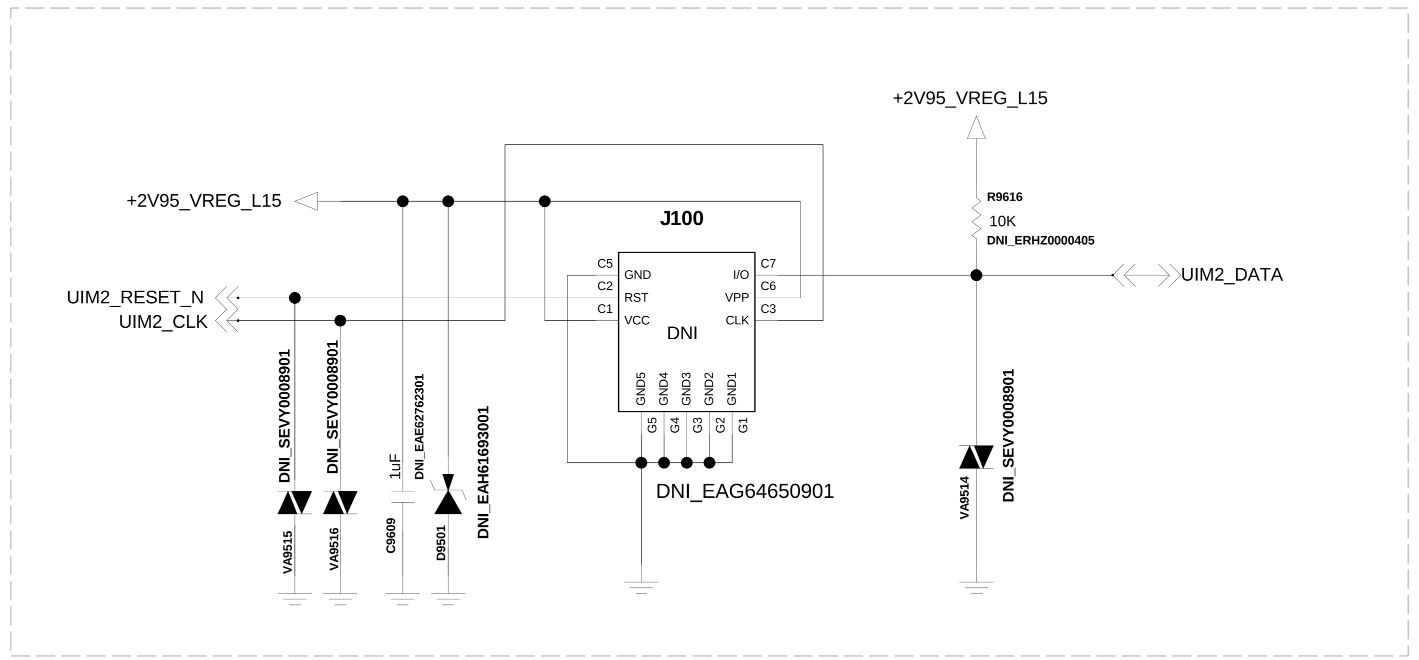


LGE Internal Use Only

LG Electronics		
TITLE	MSM8926_X5_VZW	
SIZE A2	PDM NUMBER	REV Rev
SHEET 50 of 38	Drawing Date	DRAWN BY User Name

Added for EU_Global

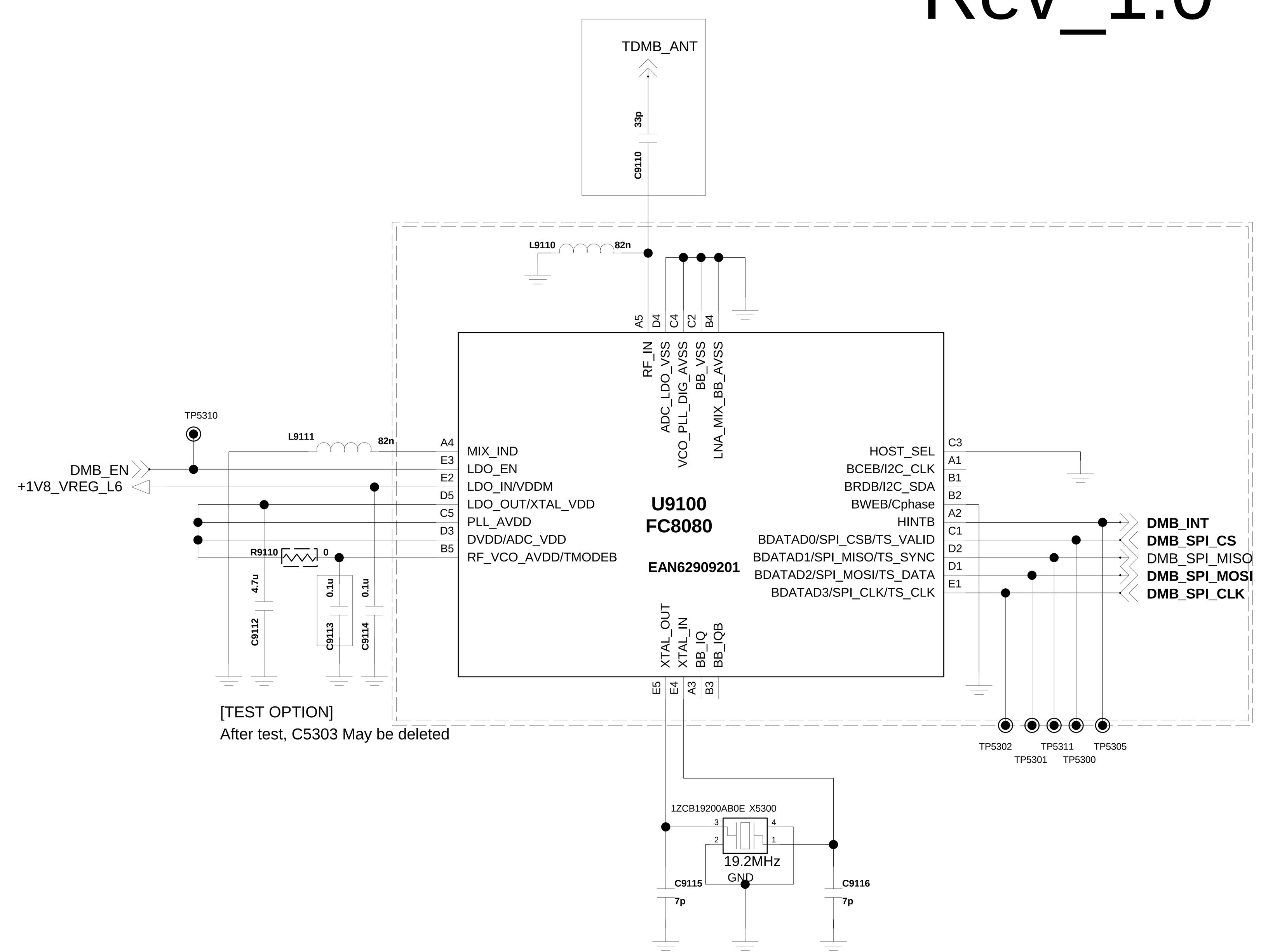
Dual Nano SIM(EAG64650901)



< 5-3-1-5_TDMB_FC8080 >

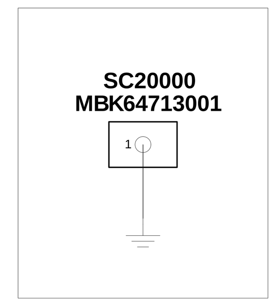
Rev_1.0

EAR_ANT



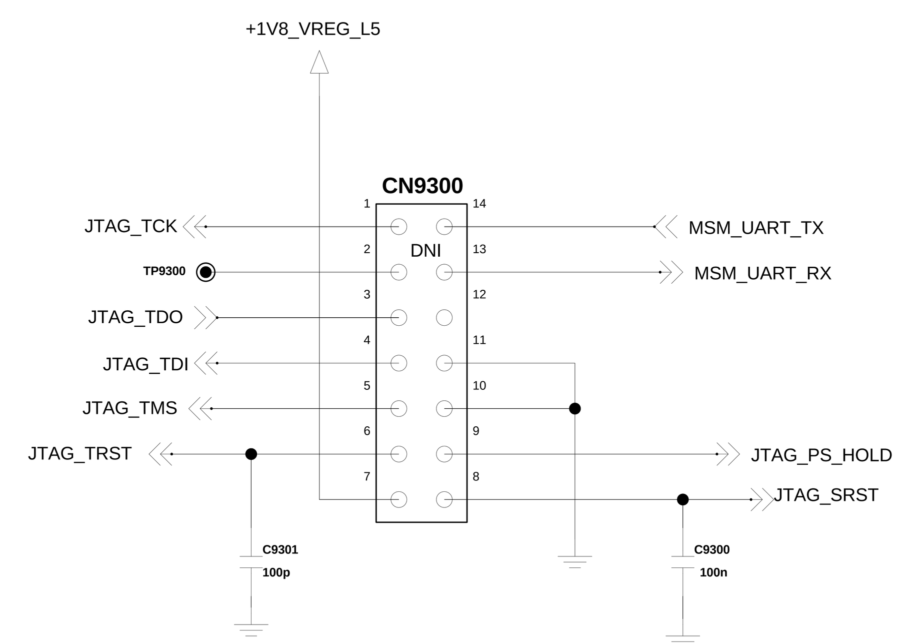
[TEST OPTION]
After test, C5303 May be deleted

TDMB ShieldCan



< 9-3-1_JTAG >

Rev_0.4



< 9-6_u_SDCARD_Socket > Rev.1.1

Circuit 1. u-SDCARD SIM Combo for QMC /w Low Detection

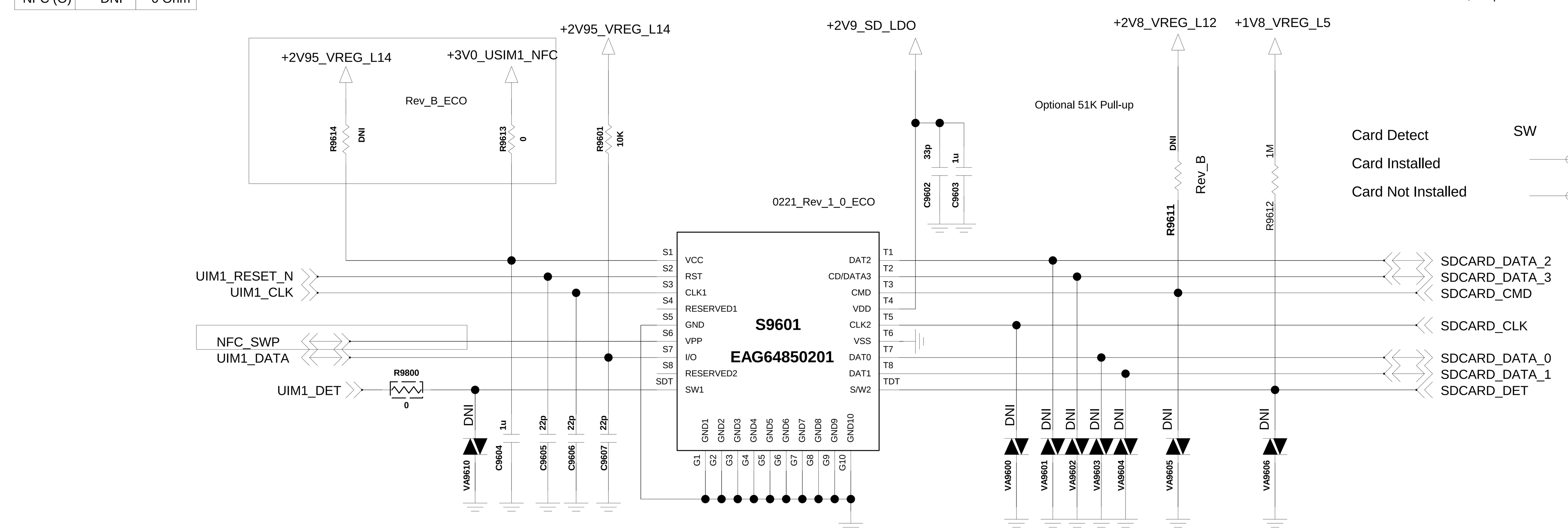
	R9614	R9613
NFC (X)	0 ohm	DNI
NFC (O)	DNI	0 Ohm

Added for NFC

Note 1. +2V95_SD_PULL_UP (1.8 or 2.95V) switches automatically per inserted SD Card version
For example, SD card Version 2.0 = 2.95V, Version 3.0 = 1.8V
Note 2. +2V95_SD_VCC is SD/MMC Power

Caution ! - According to MSM8960, APQ8064 BackPowering issue, +2V95_SD_PULL_UP was Always on during sleep mode.
However, It is possible for MSM8974, MSM8x10 .. to be turned off, so, FET circuit to turn off it is not needed any more.

Card Detect	SW	VSS(T6)	SD_CARD_DET
Card Installed			GND
Card Not Installed			1.8V



Nano SIM Combo Socket

(DGMS Guide)
MC-C04581-13 : Below items should be satisfied
1. SIM_DATA pull-up resistor
2. Use under 33pF (Cap_Varistor) on each signal line
3. When routing, SIM_DATA, SIM_CLK should be seperated
4. CAP,Varistor should Place close to connector

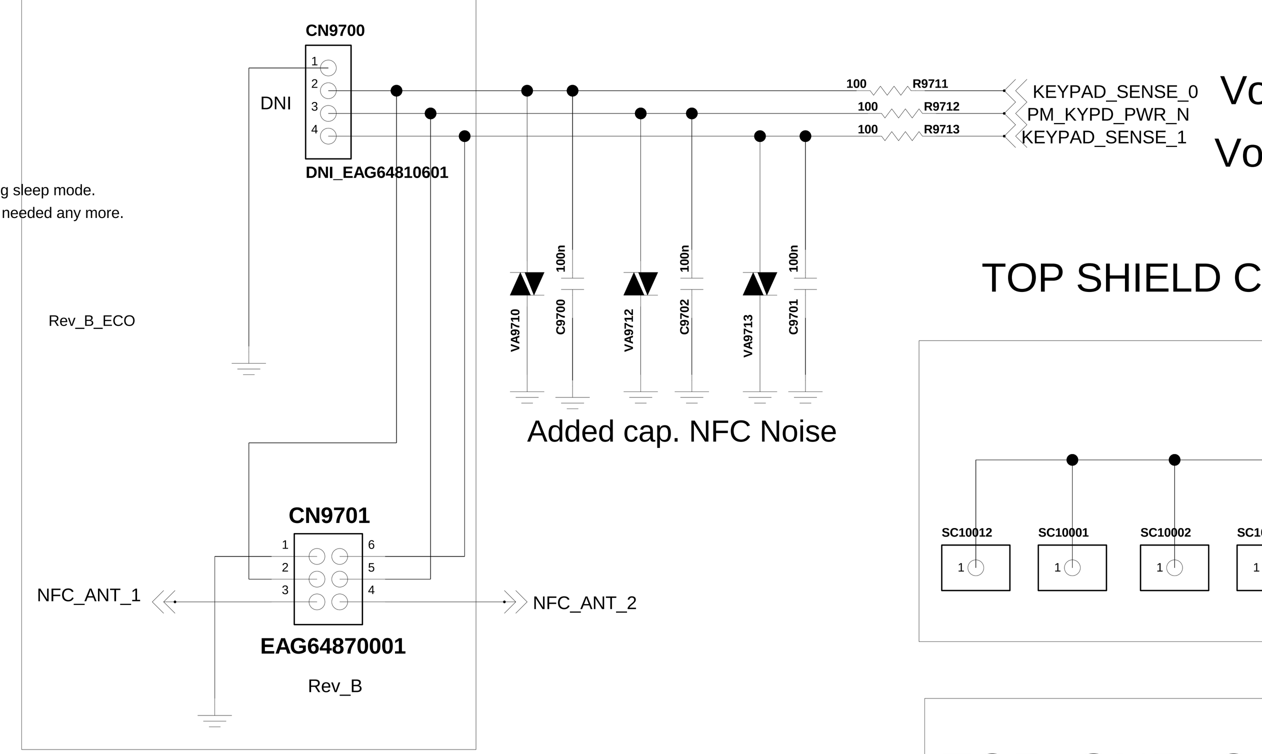
(DGMS Guide)
MC-C04582-14 : Below items should be satisfied
1. SD_DETECT Pull-up or Pull-down
2. Data/CMD line should be needed Pull-up resistor
(If chipset has internal Pull-up, comment it on circuit)
3. Use under total 10pF Capacitance on each line ; Data [0:3], CMD - Cap. TVS diode, varistor
(on other line, We recommend 10pF capacitance and use under 30pF capacitance)
4. CAP,Varistor should Place close to connector

(DGMS Guide)
MC-C05446-7 : Below items should be satisfied
1. Power of SD card should use independant Power source just like LDO/PMIC which can turn off it.
2. Per chip type which support SD card should manage current capacity.
(In case of High speed mode, SD Card Power should get margin of 200mA)
3. Do not connect serial component to SD card power, voltage drop can arise at Power line.

Circuit 3. u-SDCARD Straight /w High Detection

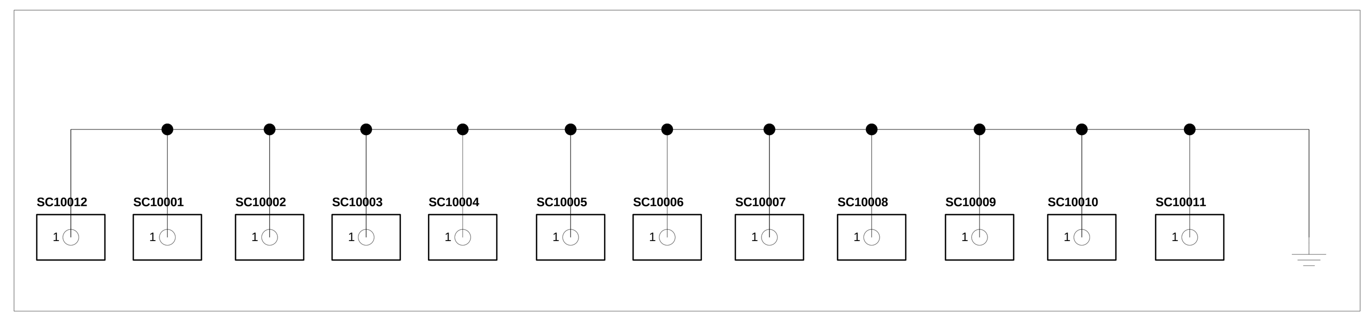
BACK_KEY & Flash LED

Added for NFC

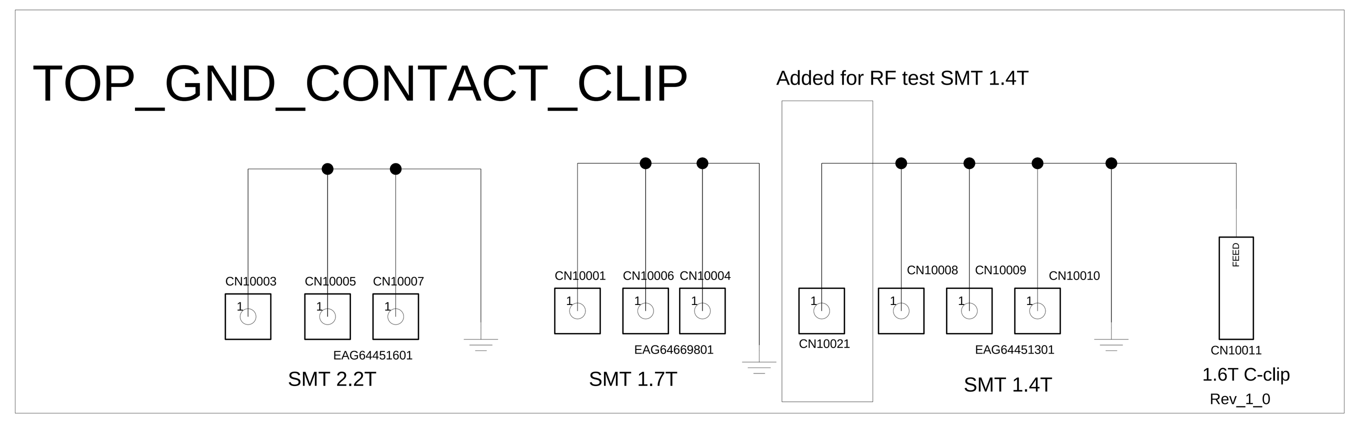


Added cap. NFC Noise

TOP SHIELD CAN CLIP

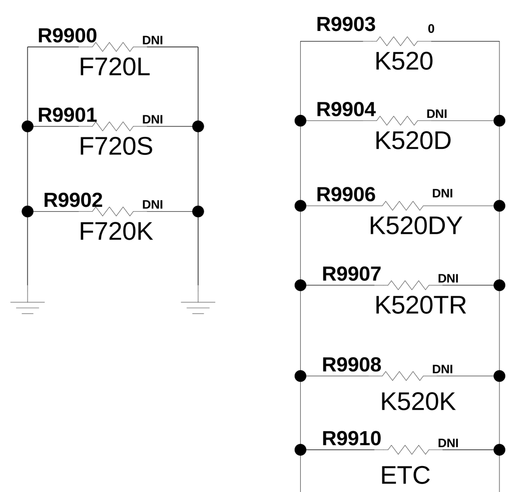


TOP_GND_CONTACT_CLIP



Domestic

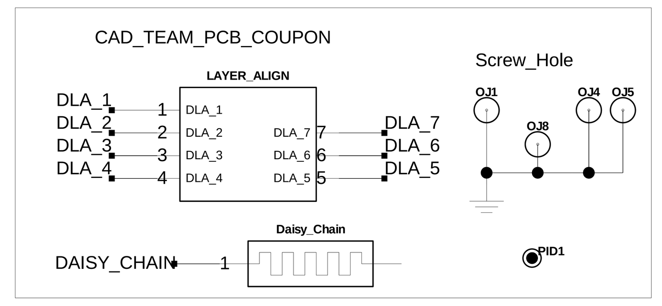
EU_Global



Model Index

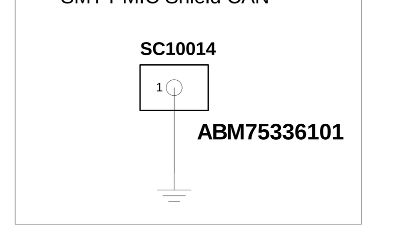
	F720L	F720S	F720K	K520	K520D	K520DY	K520TR	K520K	ETC
R9900	0 Ohm	DNI	DNI	DNI	DNI	DNI	DNI	DNI	DNI
R9901	DNI	0 Ohm	DNI	DNI	DNI	DNI	DNI	DNI	DNI
R9902	DNI	DNI	0 Ohm	DNI	DNI	DNI	DNI	DNI	DNI
R9903	DNI	DNI	DNI	0 Ohm	DNI	DNI	DNI	DNI	DNI
R9904	DNI	DNI	DNI	DNI	0 Ohm	DNI	DNI	DNI	DNI
R9906	DNI	DNI	DNI	DNI	DNI	0 Ohm	DNI	DNI	DNI
R9907	DNI	DNI	DNI	DNI	DNI	DNI	0 Ohm	DNI	DNI
R9908	DNI	DNI	DNI	DNI	DNI	DNI	DNI	0 Ohm	DNI
R9910	DNI	DNI	DNI	DNI	DNI	DNI	DNI	DNI	0 Ohm

CAD_TEAM_PCB_COUPON



ECO_0126

SMT PMIC Shield CAN



LG Electronics

MODEL NAME

PDM NUMBER

LGE Internal Use Only

1,U1500_WTR4905_RF Tranciver(TOP)

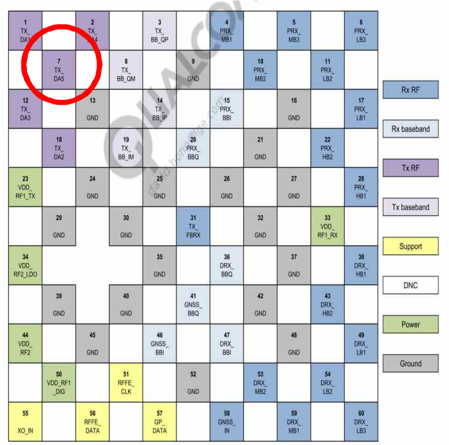


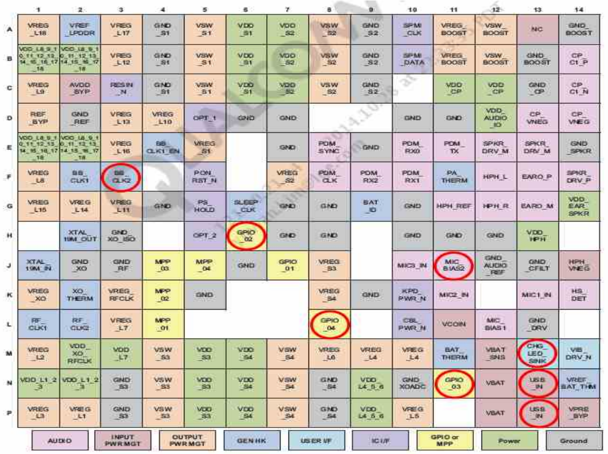
Figure 2-1 WTR4905 pin assignments – top view

O Not Used

1,U5100_WCN3620_BT/WiFi/FM IC (Top)

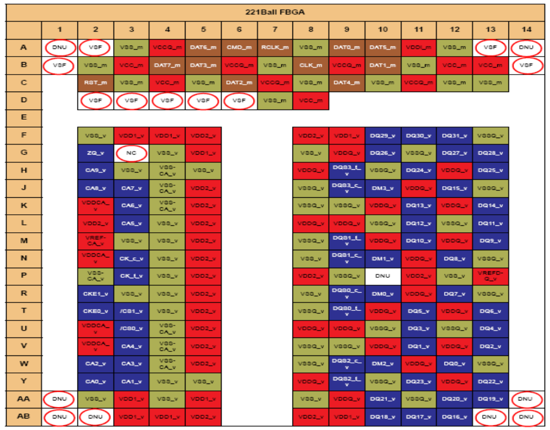
O Not Used

1. U4100_PM8916_IC,PMIC (BOT)



O Not Used

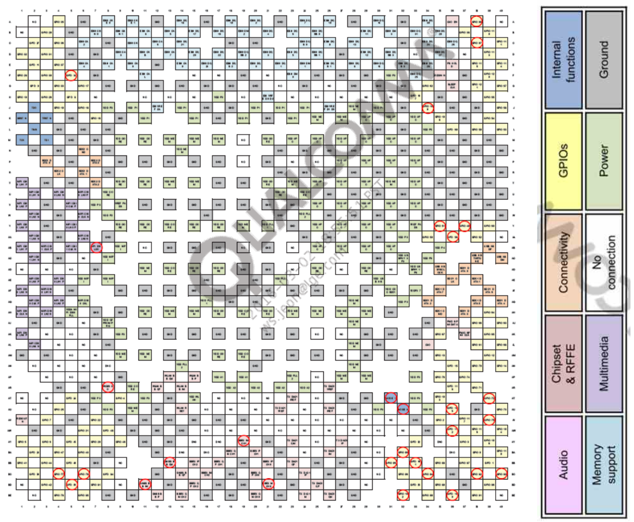
1. U3300_MCP+eMMC, 1.5GB/16GB (Top)



221 FBGA: Top View (Ball Down)

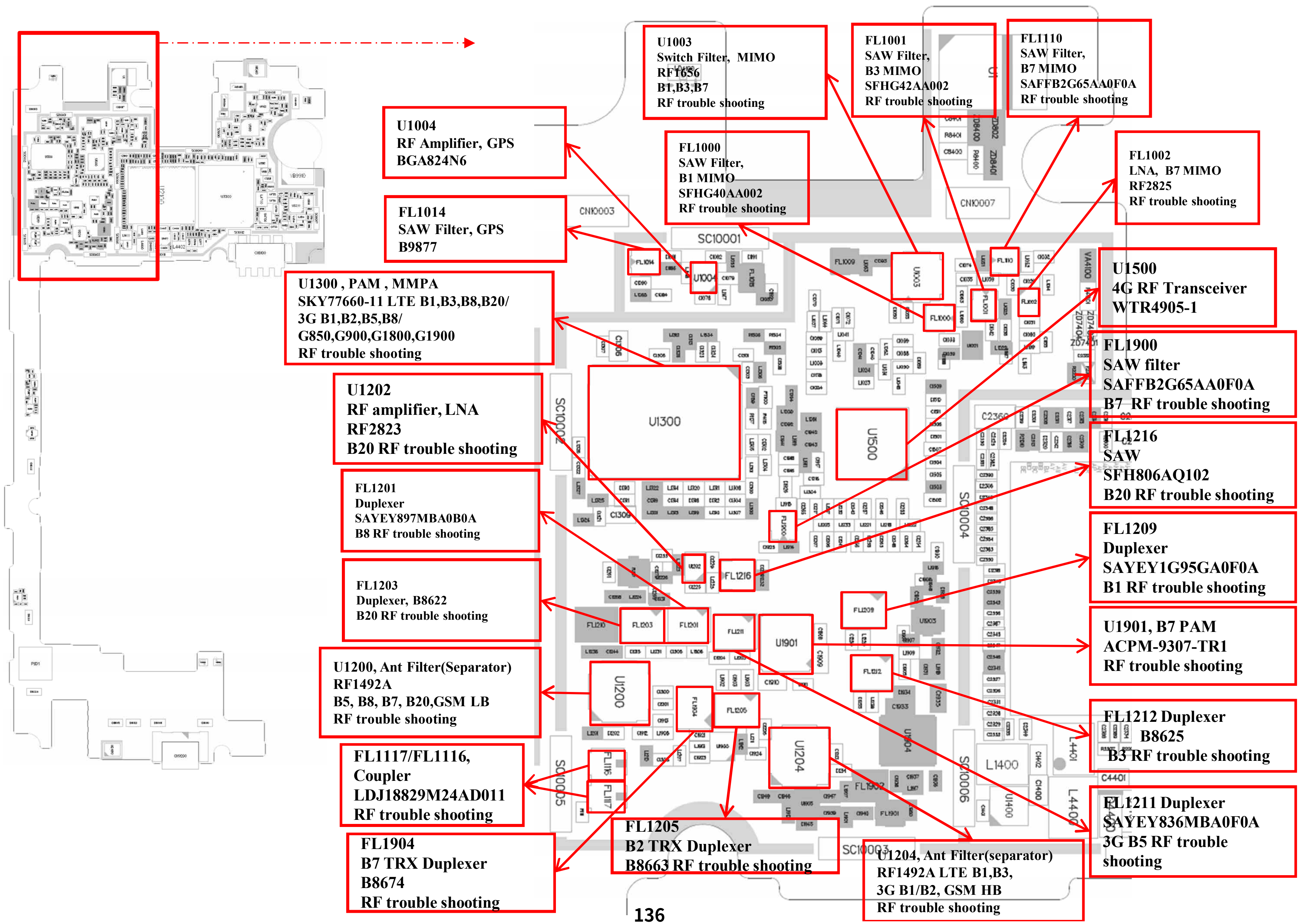
O Not Used

1. U2100_MSM8916_IC,Digital Baseband Processor (Top)



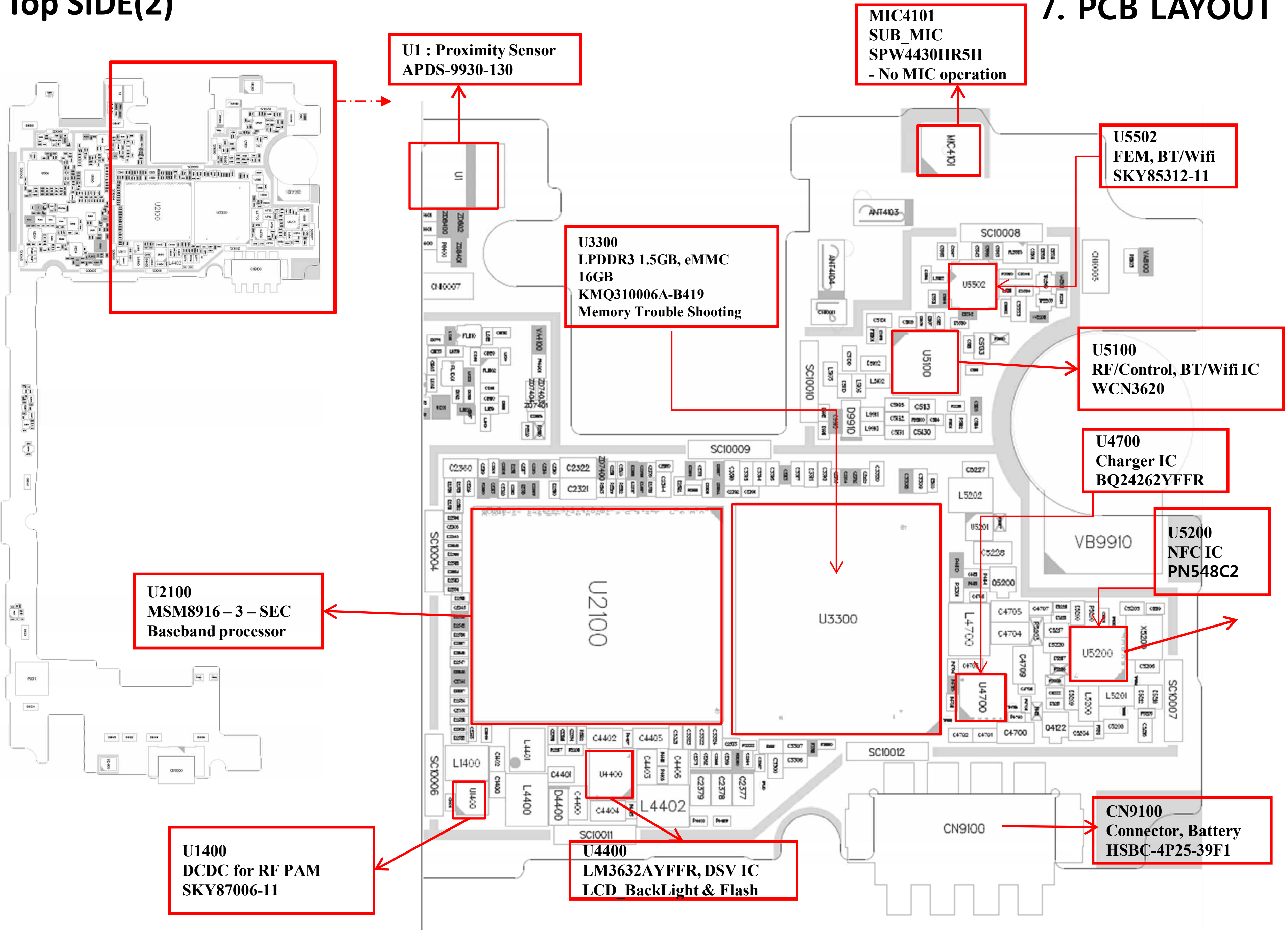
O Not Used

Top SIDE(1)

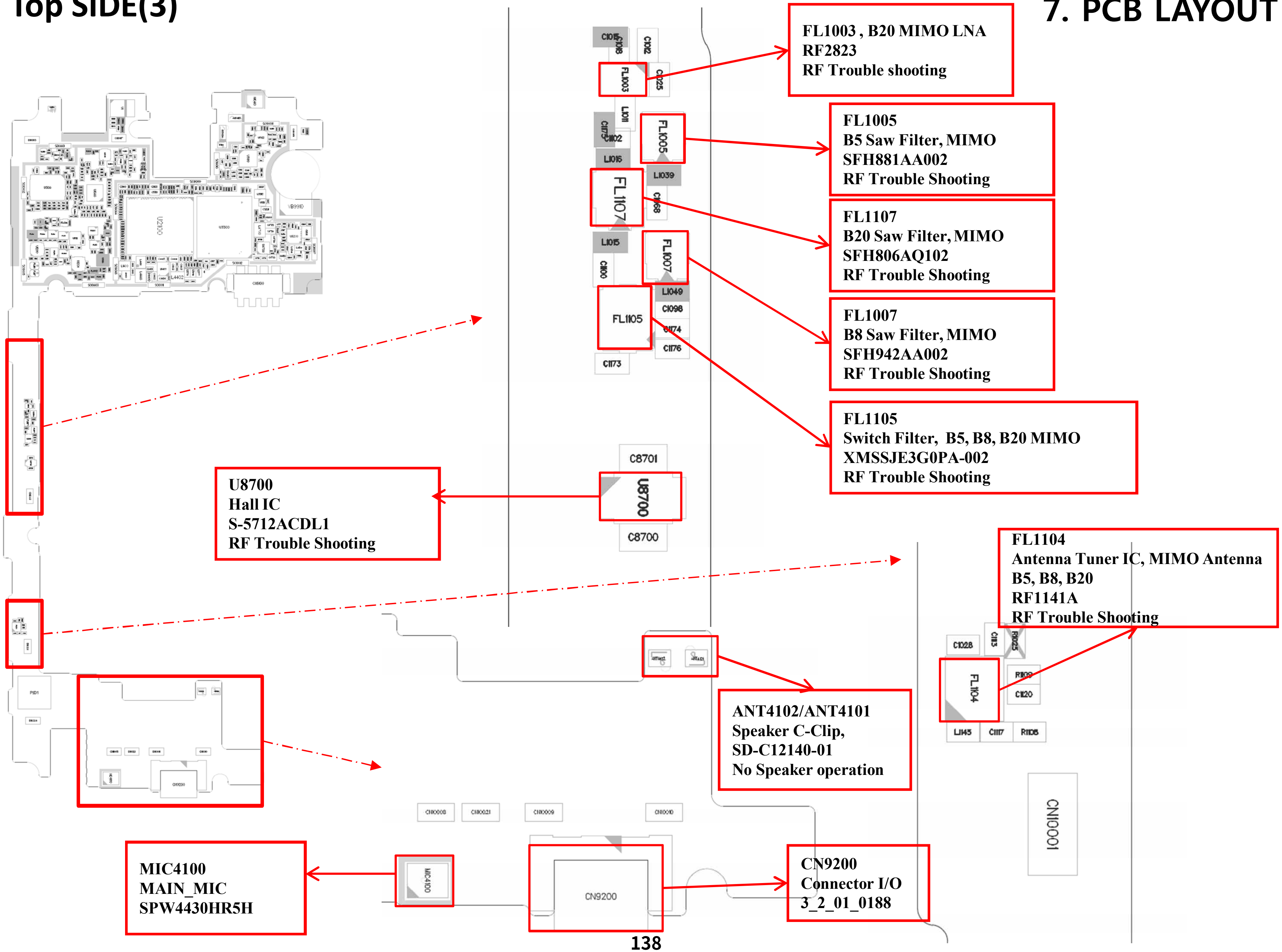

U1003
Switch Filter, MIMO
RF1656
B1,B3,B7
RF trouble shooting
FL1001
SAW Filter,
B3 MIMO
SFHG42AA002
RF trouble shooting
FL1110
SAW Filter,
B7 MIMO
SAFFB2G65AA0F0A
RF trouble shooting
U1004
RF Amplifier, GPS
BGA824N6
FL1000
SAW Filter,
B1 MIMO
SFHG40AA002
RF trouble shooting
FL1002
LNA, B7 MIMO
RF2825
RF trouble shooting
FL1014
SAW Filter, GPS
B9877
U1500
4G RF Transceiver
WTR4905-1
U1300 , PAM , MMPA
SKY77660-11 LTE B1,B3,B8,B20/
3G B1,B2,B5,B8/
G850,G900,G1800,G1900
RF trouble shooting
FL1900
SAW filter
SAFFB2G65AA0F0A
B7 RF trouble shooting
U1202
RF amplifier, LNA
RF2823
B20 RF trouble shooting
FL1216
SAW
SFH806AQ102
B20 RF trouble shooting
FL1201
Duplexer
SAYEY897MBA0B0A
B8 RF trouble shooting
FL1209
Duplexer
SAYEY1G95GA0F0A
B1 RF trouble shooting
FL1203
Duplexer, B8622
B20 RF trouble shooting
U1901, B7 PAM
ACPM-9307-TR1
RF trouble shooting
U1200, Ant Filter(Separator)
RF1492A
B5, B8, B7, B20,GSM LB
RF trouble shooting
FL1212 Duplexer
B8625
B3 RF trouble shooting
FL1117/FL1116,
Coupler
LDJ18829M24AD011
RF trouble shooting
FL1211 Duplexer
SAYEY836MBA0F0A
3G B5 RF trouble
shooting
FL1904
B7 TRX Duplexer
B8674
RF trouble shooting
FL1205
B2 TRX Duplexer
B8663 RF trouble shooting
U1204, Ant Filter(separator)
RF1492A LTE B1,B3,
3G B1/B2, GSM HB
RF trouble shooting
U1300
U1500
U1200
U1204
U1901
L1400
SC10001
SC10002
SC10003
SC10004
SC10005
SC10006
CN10003
CN10007

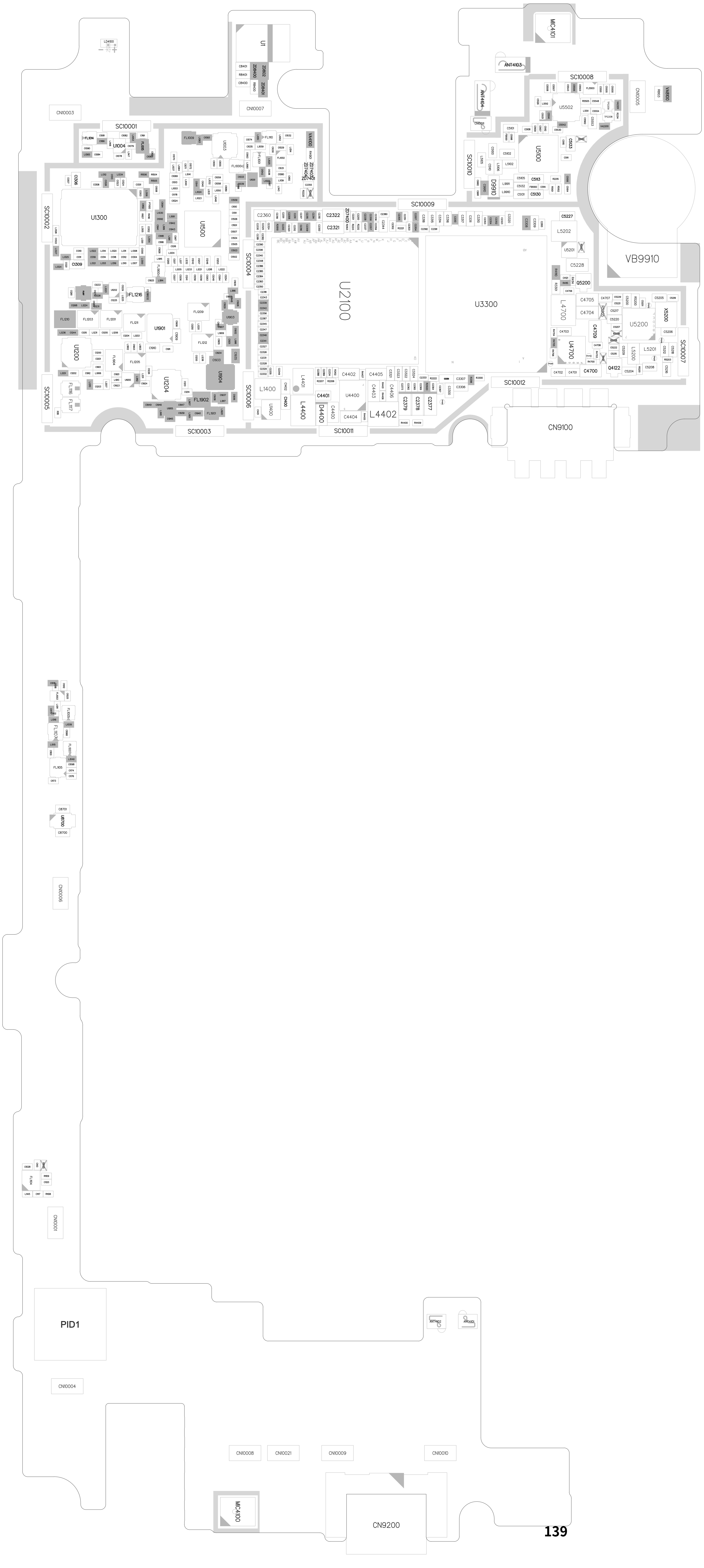
Top SIDE(2)

7. PCB LAYOUT

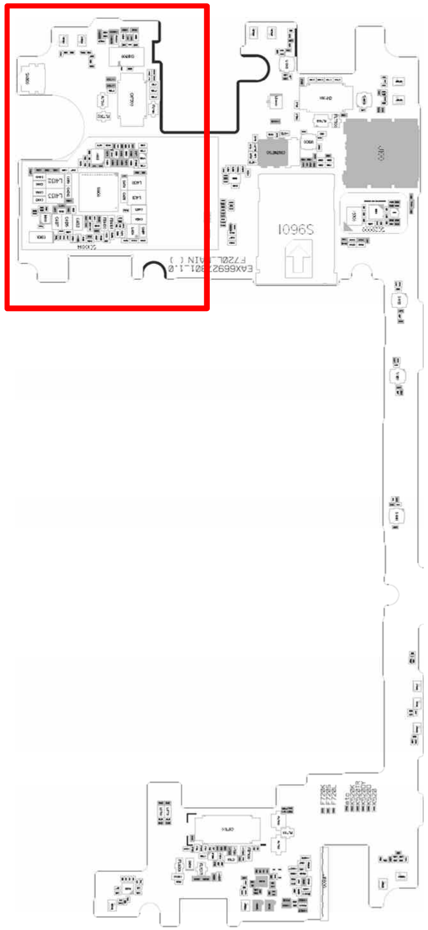


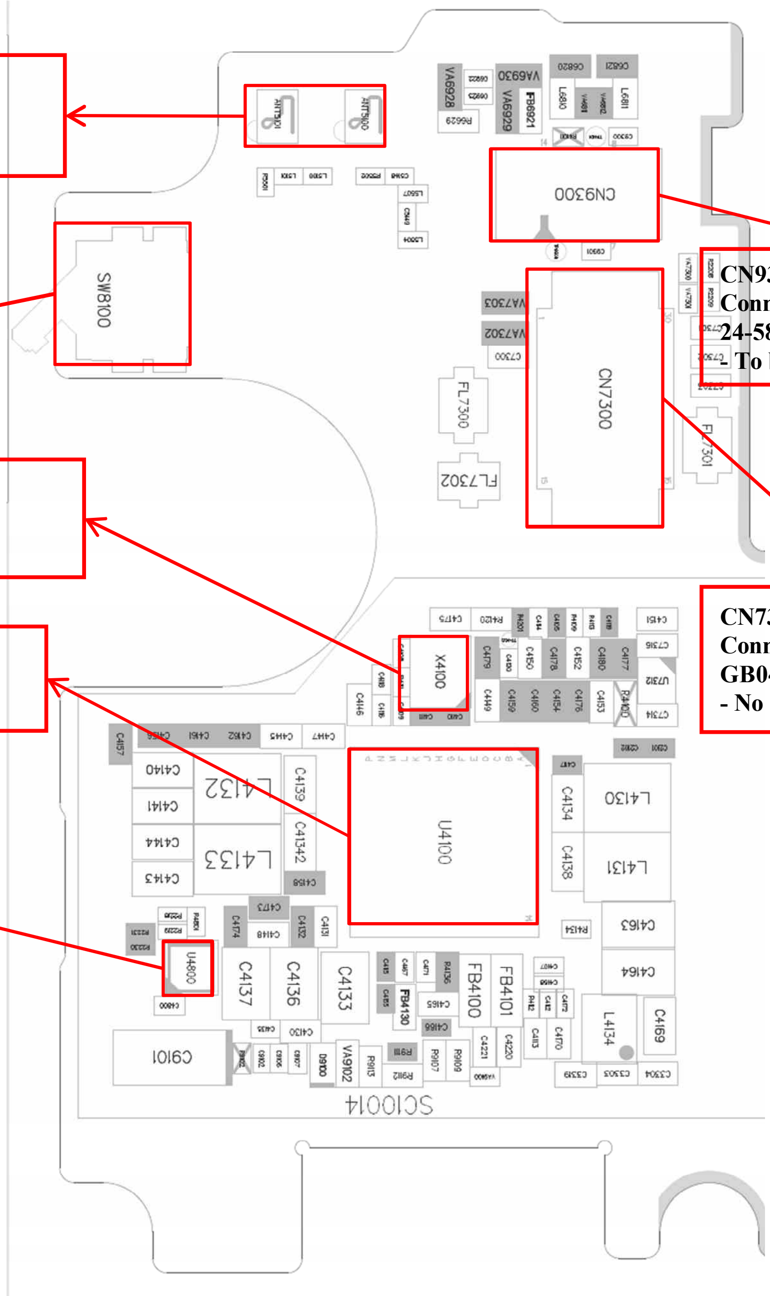
Top SIDE(3)


FL1003 , B20 MIMO LNA
RF2823
RF Trouble shooting
FL1005
B5 Saw Filter, MIMO
SFH881AA002
RF Trouble Shooting
FL1107
B20 Saw Filter, MIMO
SFH806AQ102
RF Trouble Shooting
FL1007
B8 Saw Filter, MIMO
SFH942AA002
RF Trouble Shooting
FL1105
Switch Filter, B5, B8, B20 MIMO
XMSSJE3G0PA-002
RF Trouble Shooting
U8700
Hall IC
S-5712ACDL1
RF Trouble Shooting
FL1104
Antenna Tuner IC, MIMO Antenna
B5, B8, B20
RF1141A
RF Trouble Shooting
ANT4102/ANT4101
Speaker C-Clip,
SD-C12140-01
No Speaker operation
MIC4100
MAIN_MIC
SPW4430HR5H
CN9200
Connector I/O
3_2_01_0188
FL1003
FL1005
FL1107
L1039
L1016
L1015
FL1007
L1049
C1098
FL1105
C1176
C1173
C8701
U8700
C8700
C1028
FL1104
C1120
L1145
C1117
R1108
CN10001
CN9200
MIC4100


MIC4101
ANT4103
SC10008
U1
CN10007
CN10003
SC10001
U1004
CN10005
U5502
U5100
SC10000
SC10002
U1300
U1500
SC10009
C2360
C2322
C2321
C5227
L5202
VB9910
C5228
Q5200
SC10004
U2100
U3300
L4700
C4705
C4704
U5200
SC10007
U4700
L5200
L5201
C4700
Q4122
U1901
U1200
U1204
SC10005
SC10006
L1400
L4400
D4400
U4400
C4402
C4405
C4401
C4404
L4402
C2379
C2378
C2377
SC10012
FL1902
SC10003
SC10011
CN9100
CN10006
CN10001
PID1
CN10004
CN10008
CN10021
CN10009
CN10010
MIC4100
CN9200

Bottom SIDE(1)





ANT5100/ANT5101
Antenna C-Clip, BT/Wifi
SD-C07140-01

SW8100
Detect Switch
for Stylus pen
SPVN210101_

X4100
PMIC Crystal
1RAE19200BAA

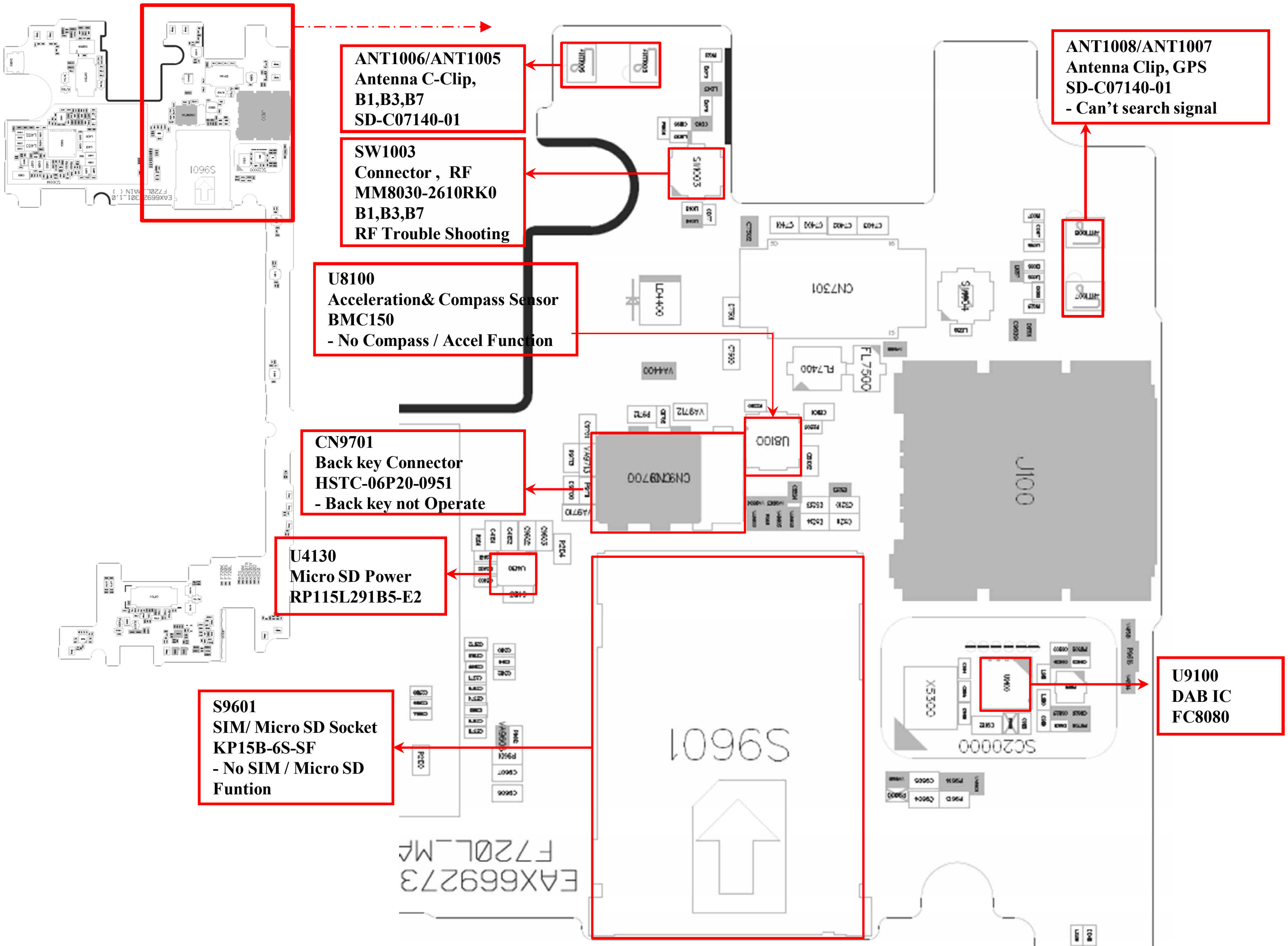
U4100
PMIC, PM8916
Power trouble shooting

U4800
Fuel Gauge
RT9428GWSC

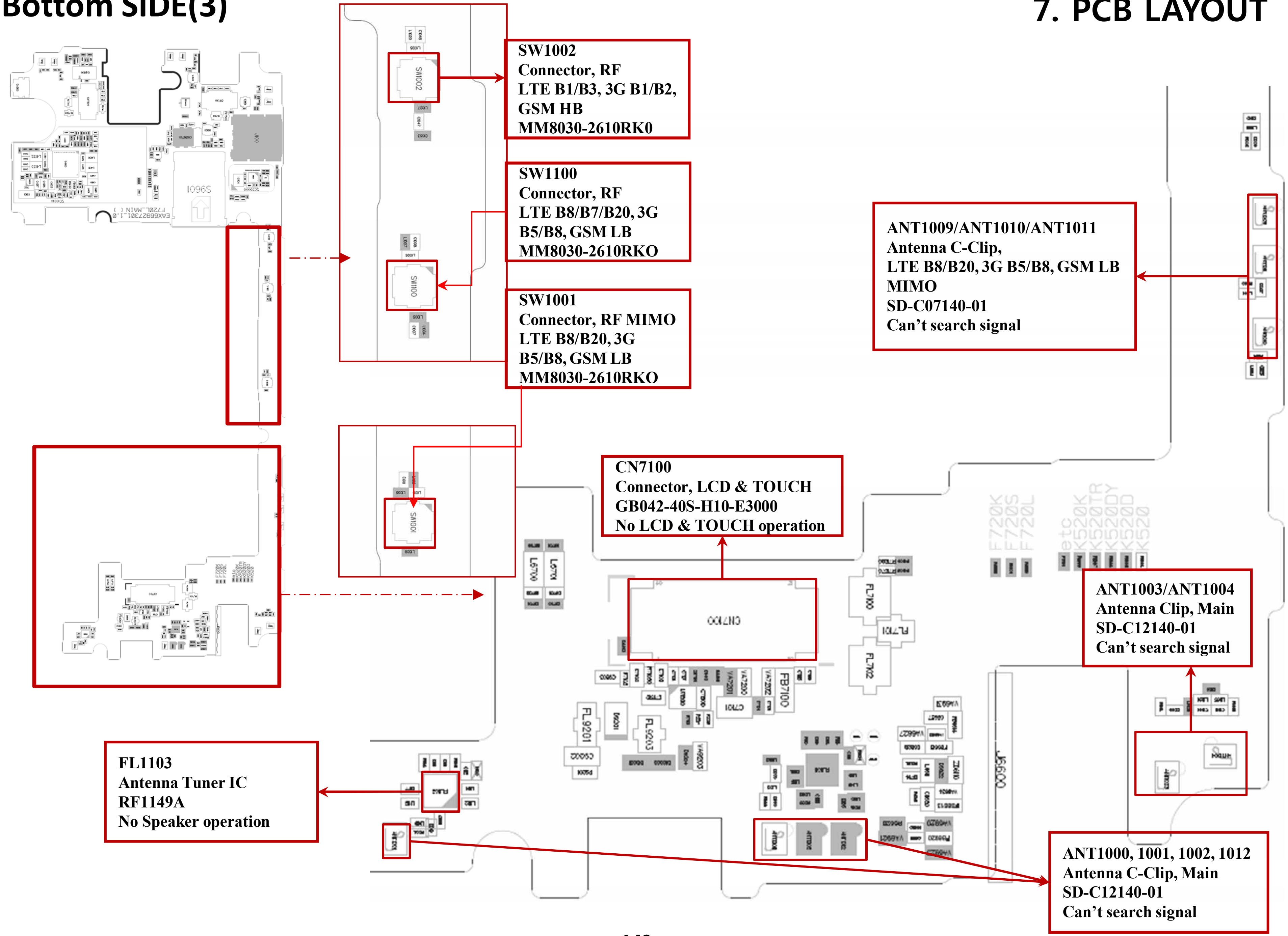
CN9300
Connector, JTAG
24-5804-014-000-829+
- To be DNI

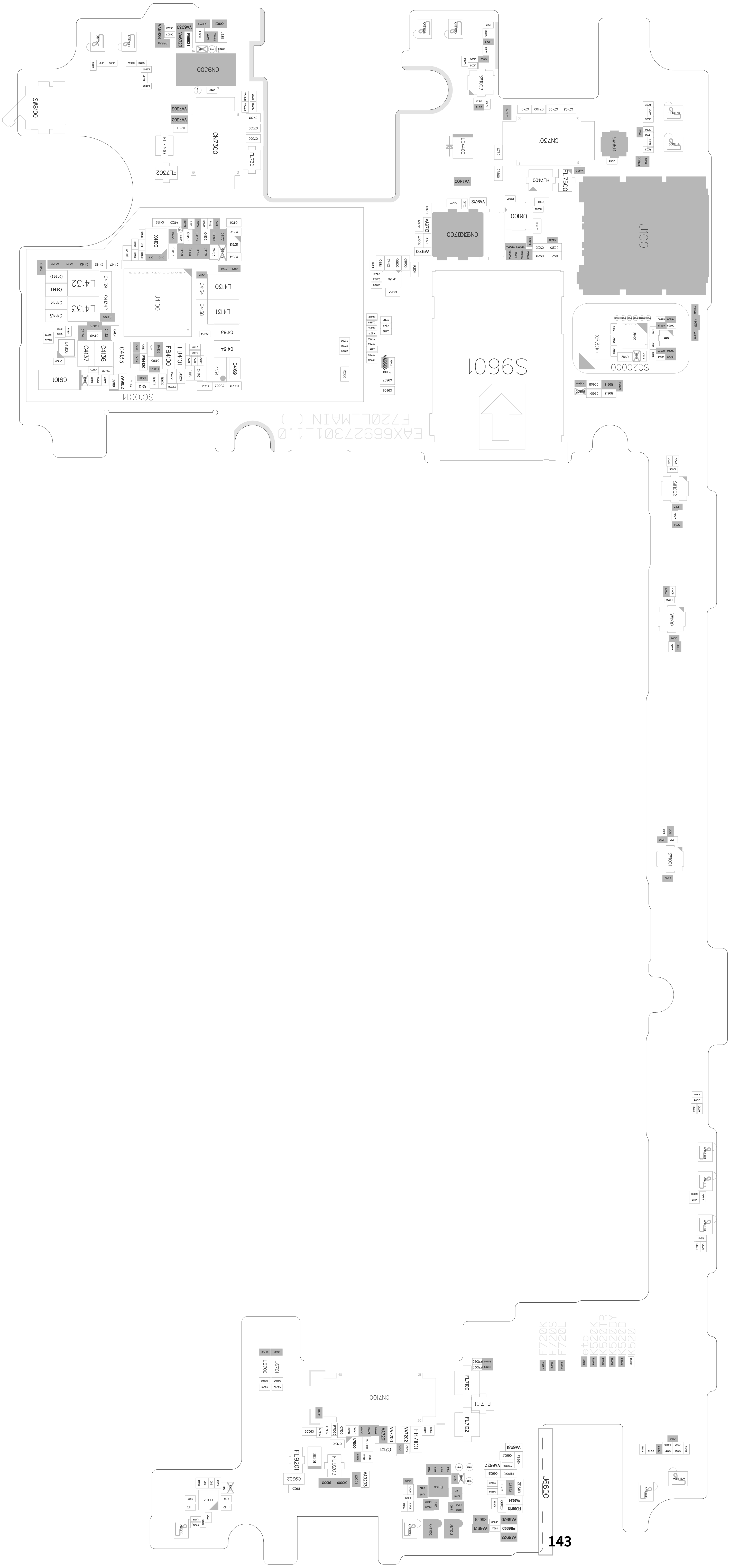
CN7300
Connector, 13M AF
GB042-30S-H10-E3000
- No Camera operation

Bottom SIDE(2)

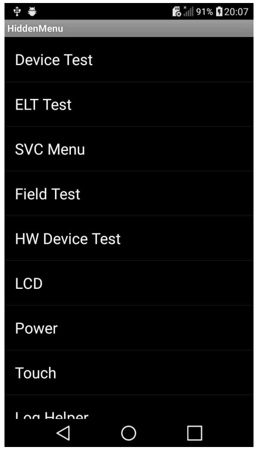


Bottom SIDE(3)



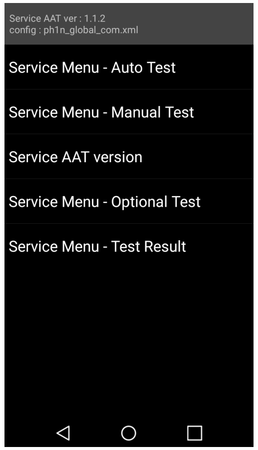

CN9300
CN7300
CN7301
J100
S9601
U4100
SC10014
SC20000
CN7100
J6600
EAX66927301_1.0
F720L_MAIN ( )

1. Hidden Menu Start



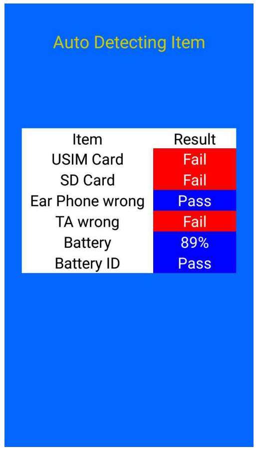
- Start shortcut key: *#546368#*520#
- Hidden Menu List
  : Start the desired menu, click

2. Device Test - SAAT



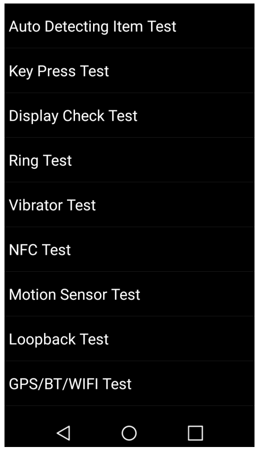
- Service Menu – Auto Test
- Service Menu – Manual Test
- Service AAT version
  → Display Hidden Menu version
- Service Menu – Optional Test
- Service Menu – Test Result

2. Device Test - SAAT



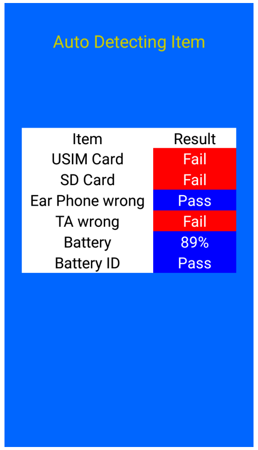
- Service Menu – Auto Test
  → All Test Items are continued one after another.

3. Device test List

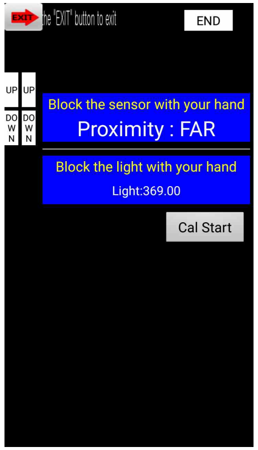


- Service Menu – Manual Test
  → Each test item can be selected and performed by user.

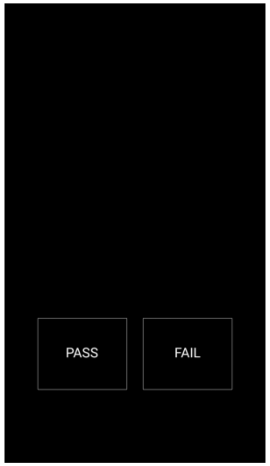
1. Auto Detecting Item Test



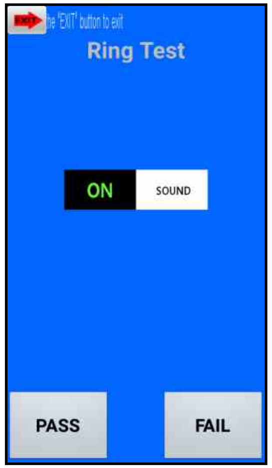
2. Key Press Test



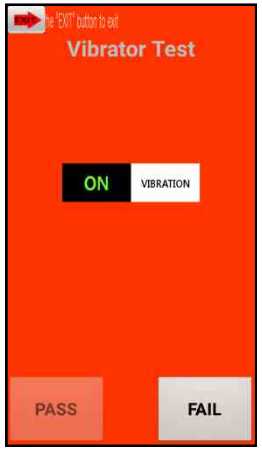
3. Display Check Test



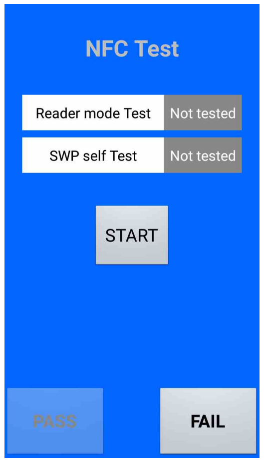
4. Ring Test



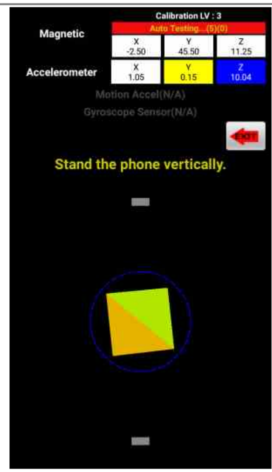
5. Vibrator Test



6. NFC Test

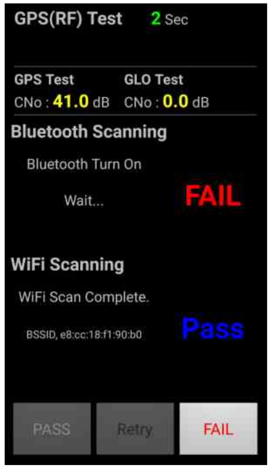


7. Motion Sensor Test

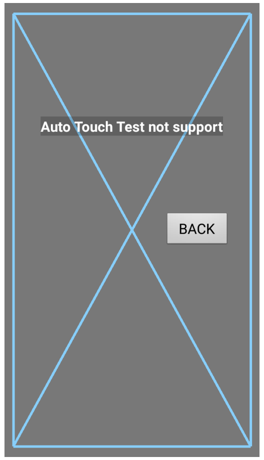


8. Loopback Test

9. GPS/BT/WIFI Test



10. Touch Draw Test - Manual

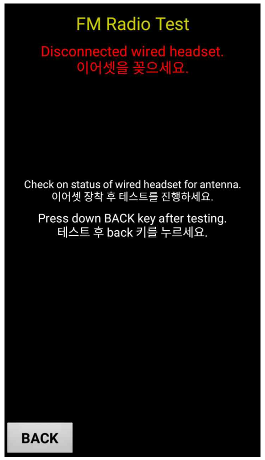


11. Camera(Main) Test

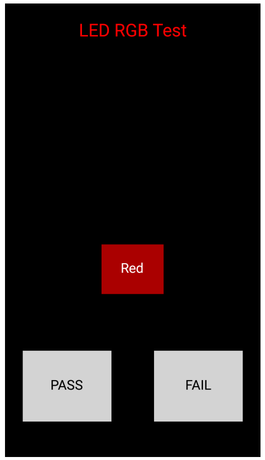
12. Camcorder Test

13. Camera(VT) Test

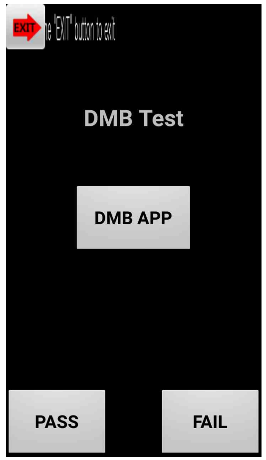
14. FMRadio Test



15. LED Test



16. DMB Test

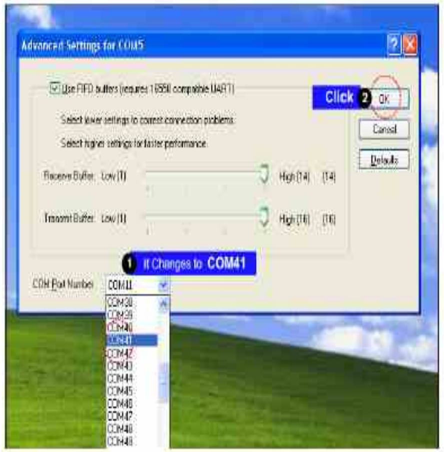


LGE Internal Use Only

1. Summary

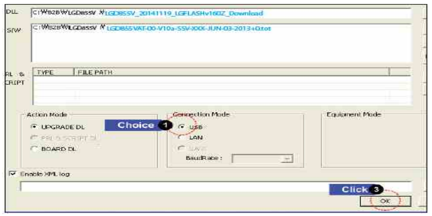
Tool Version	DLL name	USB Driver	
LGFLASH v201	LGK520_20160204_LGFLASHv215_Download	LGUnitedMobileDriver_S52MAN314AP22_ML_WHQL_Ver_3.14.1	
Please Check the Version to "LGST ServiceCenter Tool"			
H/W			
	Name	Part No.	SW
D/L Cable	Micro 5P (56-open-910K) USB DLC	RAD32167835	KDZ

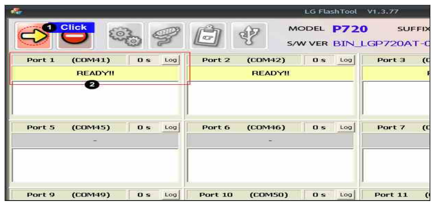
2. . USB COM port Setting

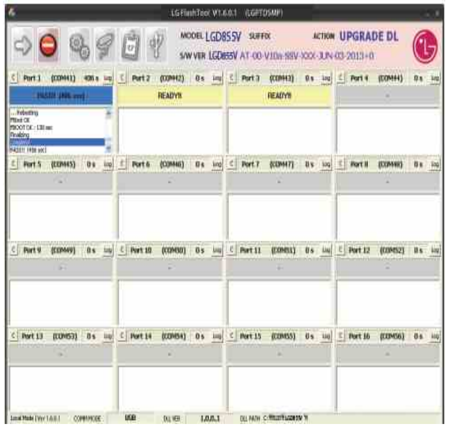


3. USB D/L Cable setting

4. Flash tool D/L setting





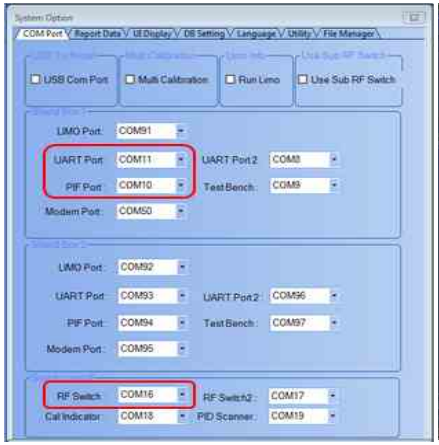


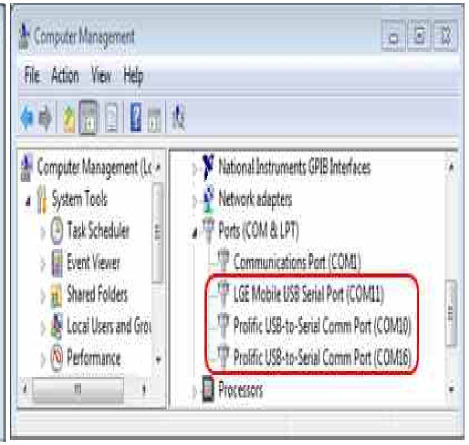
※ If you want more information, please refer LGST ServiceCenter Tool's Notification"Download User Guide".

1. Summary

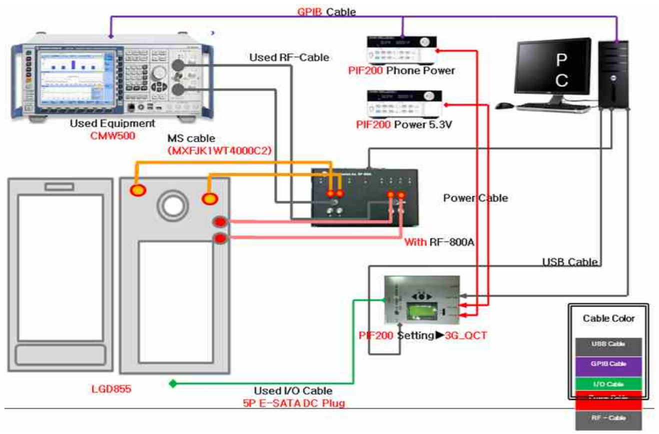
CAL INFORMATION		
S/W VERSION		
[TachyonV2013] LGK520 20160401 ALL File Calibration		
Please Check the Version to "B2B"		
H/W		
	Name	Part No.
PIF	PIF200	BJAY0024021
USB Cable	USB Cable	RAD32247898
Power Cable	DC Power Cable	RAD32247878
I/O Cable	5P E-SATA DC Plug	RAD32167861
RF Cable Main	MXFJM3WX6000	RAD32827895
Power Supply PIF		
Power Supply Phone	Power supply 5.0V	
PF Test Equipment	CMW500	
Notice	1. Use the dummy Battery (Refer to Attached image) 1) Phone states: Power off 2) If do not use the dummy battery, GSM TX fails. 2. Port Setting (Refer to Attached image) 1) Uart Port1 : Use the "LGE Mobile USB Serial Port"	
CMW 500 RF Cable connection	Reference to Attached ppt ▷How to connect between phone RF switch and RF 800A:It has to be connected as below ▷How to connect between CMW500 and RF800A ; It has to connected as below - CMW500 [RF1COM] to RF-800A [left COM], - CMW500 [RF2COM] to RF-800A [Right COM]	

2. . USB COM port Setting

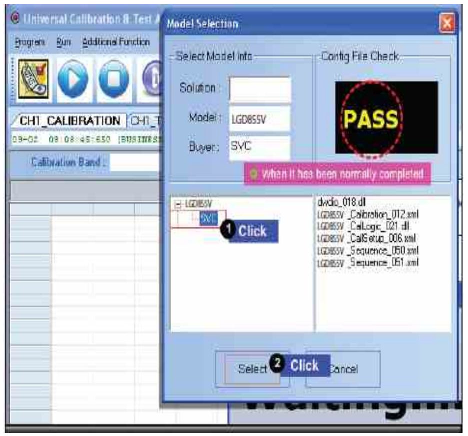




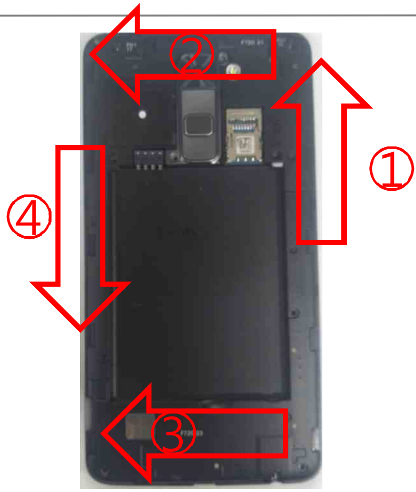
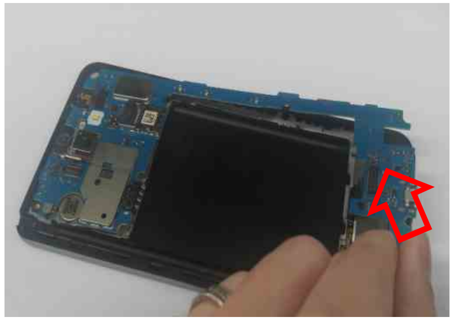
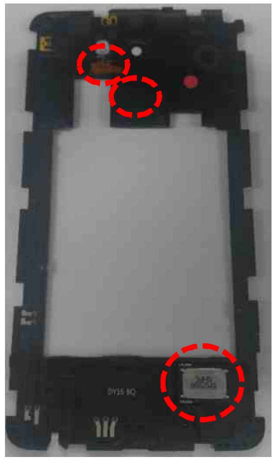
3. Calibration Cable setting

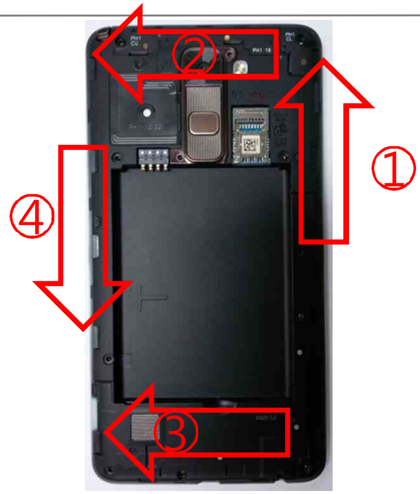


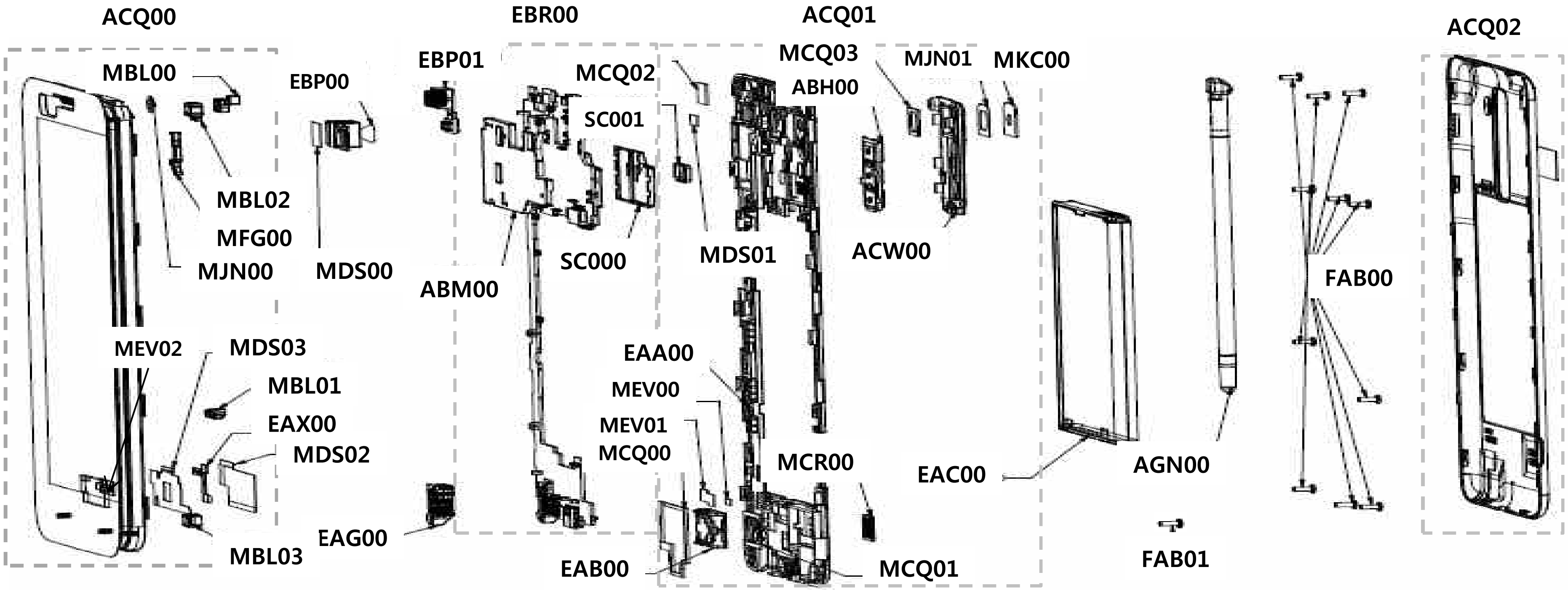
4. Tachyon setting



※ If you want more information, please refer LGST ServiceCenter Tool's Notification "RF Calibration User Guide".

1. Disassemble Screw (11ea)	2. Disassemble Rear cover	3. Disassemble Rear Cover	4. Disassemble Connector
			
5. Disassemble Main PCB	**6. Disassemble HW Parts (5ea)**	**7. Disassemble HW Parts (2ea)**	**8. Complete Disassembling**
			

1. Disassemble Screw (11ea)	2. Disassemble Rear cover	3. Disassemble Rear Cover	4. Disassemble Connector
			
5. Disassemble Main PCB	**6. Disassemble HW Parts (5ea)**	**7. Disassemble HW Parts (2ea)**	**8. Complete Disassembling**
			



Location no	Description	Location no	Description	Location no	Description	Location no	Description
ACQ00	Cover Assembly	EAG00	Jack, Phone	ACQ01	Cover Assembly, Rear	AGN00	Pen Assembly,Stylus
MJN00	Tape (VT)	EBP01	Camera Module	ABH00	Button Assembly	FAB00	Screw, Machine
MBL00	Cap (Sensor)	EBP00	Camera Module	ACW00	Decor Assembly ,SVC	EAC00	Rechargeable Battery, Lithium Ion
MBL01	Cap(Rubber pen)	MDS00	Gasket (main)	MKC00	Window, Camera	ACQ02	Cover Assembly ,Battery
MBL02	Cap (Mic BTM)	MEV00	Insulator	MCQ03	Damper, Camera	FAB01	Screw, Tapping
MBL03	Cap (top)	MEV01	Insulator	MJN01	Tape, Window	MDS01	Gasket
MDS02	Gasket	MCR00	Decor	MCQ02	Damper, Camera	EAA00	PIFA Antenna, Multiple
MFG00	Locker	EBR00	PCB main	EAB00	Speaker, Dual Mode	MEV02	Insulator
EAX00	PCB, Flexible	SC000	Can Assembly, (BTM)	MCQ00	Damper, Speaker(sealing)	ABM00	Can Assembly, Shield
MDS03	Gasket	SC001	Can, Shield(MMPA)	MCQ01	Damper, Speaker		

No	Part No	Description	Qty	Location No.
1	AGQ88998401	Phone Assembly	1	AGQ000000
2	ACQ88972401	Cover Assembly,EMS	1	ACQ100400
3	EBR82835301	PCB Assembly,Main	1	EBR00
4	EBR82091201	PCB Assembly,Main,Insert	1	EBR071500
5	ABM75256111	Can Assembly,Shield	1	ABM00
6	MBK64773611	Can,Shield	1	MBK070300
7	MEV65931401	Insulator	1	MEV000000
8	MEV65913201	Insulator	1	MEV000001
9	MEV65891601	Insulator	1	MEV000002
10	MDS65710401	Gasket	2	MDS000000
11	MDS65691301	Gasket	1	MDS000001
12	MCQ68845301	Damper	1	MCQ000000
13	MDS65693401	Gasket	1	MDS000002
14	MHK65627501	Sheet	1	MHK000000
15	RAA34549101	Resin,PC	0	RAA050100
16	EBR82871501	PCB Assembly,Main,SMT	1	EBR071800
17	EBR82871601	PCB Assembly,Main,SMT Bottom	1	EBR071600
18	EAE62282201	Capacitor,Ceramic,Chip	3	C4109,C6621,C6704
19	ABM75336101	Can Assembly,Shield	1	SC000
20	MBK64773001	Can,Shield	1	MBK070300
21	MEV65871301	Insulator	1	MEV000000
22	EAE62286801	Capacitor,Ceramic,Chip	3	C9102,C9300,C9702
23	EAE62505701	Capacitor,Ceramic,Chip	1	C9202
24	EAE62542701	Capacitor,Ceramic,Chip	4	C4136,C4137,C4163,C4164
25	EAE62685701	Capacitor,Ceramic,Chip	1	C6620

No	Part No	Description	Qty	Location No.
26	EAE62726601	Capacitor,Ceramic,Chip	1	C2361
27	EAE62762301	Capacitor,Ceramic,Chip	2	C4800,C9107
28	EAE62767801	Capacitor,Ceramic,Chip	1	C4133
29	EAE62884201	Capacitor(High Frequency),Ceramic,Chip	1	C5149
30	EAE62924001	Capacitor(High Frequency),Ceramic,Chip	2	C9115,C9116
31	EAE62969601	Capacitor,Ceramic,Chip	1	C4169
32	EAE63541901	Capacitor,Ceramic,Chip	1	C7101
33	EAE63841601	Capacitor,TA,Polymer	1	C9101
34	EAF61450601	Varistor	3	C6700,C6701,VA9100
35	EAG63772101	Connector,RF	4	SW1001,SW1002,SW1003,SW1100
36	EAG64651301	C-Clip	4	ANT1000,ANT1001,ANT1003,ANT1004
37	EAG64672301	C-Clip	9	ANT1005,ANT1006,ANT1007,ANT1008,ANT1009,ANT1010,ANT1011, ANT5100,ANT5101
38	EAG64850201	Socket,DIMM/SIMM	1	S9601
39	EAG64870001	Connector,Terminal Block	1	CN9701
40	EAH61693001	Diode,TVS	2	C6702,C6703
41	EAH62213401	Diode,TVS	1	D9100
42	EAH63092501	Diode,TVS	1	D9201
43	EAM62071001	Filter,Bead	2	FB4100,FB4101
44	EAM62150501	Filter,Bead	1	FB7100
45	EAM62291101	Filter,EMI/Power	2	FL7101,FL9203
46	EAM62530001	Filter,EMI/Power	2	FL7100,FL7102
47	EAM62630501	Filter,EMI/Power	2	FL7300,FL7301
48	EAM62730001	Filter,Separator,Switch	1	FL6610
49	EAM62770201	Filter,EMI/Power	1	FL7500
50	EAM62770301	Filter,EMI/Power	1	FL7400

No	Part No	Description	Qty	Location No.
51	EAM63030601	Filter,Separator,Switch	1	FL1103
52	EAN62637201	IC,Acceleration Sensor	1	U8100
53	EAN62909201	IC,TDMB	1	U9100
54	EAN63430901	IC,Fuel Gauge	1	U4800
55	EAN63606501	IC,LDO Voltage Regulator	1	U71500
56	EAN63707501	IC,LDO Voltage Regulator	1	U4130
57	EAN64007601	IC,LDO Voltage Regulator	1	U7312
58	EAP61866601	Inductor,Multilayer,Chip	3	L1019,L1144,L5507
59	EAN64306001	IC,PMIC	1	U4100
60	EAP61866701	Inductor,Multilayer,Chip	7	L1006,L1010,L1029,L1045,L1058,L9110,L9111
61	EAP61866801	Inductor,Multilayer,Chip	1	L1114
62	EAP62107901	Inductor,Multilayer,Chip	1	L1035
63	EAP62108101	Inductor,Multilayer,Chip	1	L1112
64	EAP62108301	Inductor,Multilayer,Chip	1	C1177
65	EAP62109801	Inductor,Wire Wound,Chip	2	L6700,L6701
66	EAP62186301	Inductor,Multilayer,Chip	1	C1077
67	EAP62225901	Inductor,Multilayer,Chip	2	L1033,L1034
68	EAP62226001	Inductor,Multilayer,Chip	1	C1044
69	EAP62226201	Inductor,Multilayer,Chip	1	L1001
70	EAP62226401	Inductor,Multilayer,Chip	1	L1008
71	EAP62266501	Inductor,Wire Wound,Chip	2	L4130,L4131
72	EAP62526401	Inductor,Multilayer,Chip	1	C1048
73	EAP62666001	Inductor,Wire Wound,Chip	2	L4132,L4133
74	EAP62886101	Inductor,Multilayer,Chip	1	L1056
75	EAP62906001	Inductor,Wire Wound,Chip	2	L6810,L6811

No	Part No	Description	Qty	Location No.
76	EAP63485901	Inductor,Multilayer,Chip	1	C1008
77	EAP63626501	Inductor,Wire Wound,Chip	1	L4134
78	EAV63012101	LED,Flash	1	LD4400
79	EAW61703601	Crystal	1	X5300
80	EAW62883701	Crystal	1	X4100
81	EBC61856201	Resistor,Chip	1	R2101
82	EBF62334501	Switch,Hook	1	SW8100
83	ECCH0000115	Capacitor,Ceramic,Chip	3	C9605,C9606,C9607
84	ECCH0000127	Capacitor,Ceramic,Chip	2	C5211,C5214
85	ECCH0000143	Capacitor,Ceramic,Chip	2	C4220,C4221
86	ECCH0000151	Capacitor,Ceramic,Chip	2	C6627,C6628
87	ECCH0000182	Capacitor,Ceramic,Chip	1	C4130
88	ECCH0000198	Capacitor,Ceramic,Chip	4	C3303,C4170,C7400,C7401
89	ECCH0000155	Capacitor,Ceramic,Chip	1	C9203
90	ECCH0002001	Capacitor,Ceramic,Chip	5	C3304,C7500,C7501,C8101,C8102
91	ECCH0004904	Capacitor,Ceramic,Chip	15	C4151,C4175,C4182,C7100,C7102,C71500,C71510,C7300,C7301,C7302,C7303,C7314,C7316,C9603,C9604
92	ECCH0005603	Capacitor,Ceramic,Chip	4	C4134,C41342,C4138,C4139
93	ECCH0007804	Capacitor,Ceramic,Chip	2	C4131,C4165
94	ECCH0009101	Capacitor,Ceramic,Chip	8	C4108,C4120,C4135,C4172,C6920,C6922,C9113,C9114
95	ECCH0009104	Capacitor,Ceramic,Chip	2	C1115,C1116
96	ECCH0009105	Capacitor,Ceramic,Chip	5	C7103,C7105,C7106,C7107,C7108
97	ECCH0009103	Capacitor,Ceramic,Chip	27	C1005,C1006,C1007,C1009,C1010,C1011,C1019,C1021,C1026,C1027,C1046,C1047,C1061,C1075,C1076,C1085,C1086,C1096,C1097,C5148,C6625,C9106,C9301,L1036,L1113,L5100,L5101
98	ECCH0009106	Capacitor,Ceramic,Chip	1	C1112
99	ECCH0009107	Capacitor,Ceramic,Chip	6	C2363,C2371,C2374,C2375,C2376,C2398
100	ECCH0009109	Capacitor,Ceramic,Chip	1	C4112

No	Part No	Description	Qty	Location No.
101	ECCH0009110	Capacitor,Ceramic,Chip	1	C2370
102	ECCH0009208	Capacitor,Ceramic,Chip	3	L1026,L1028,L5504
103	ECCH0009228	Capacitor,Ceramic,Chip	3	C2368,C2394,C2396
104	ECCH0017301	Capacitor,Ceramic,Chip	14	C2372,C2400,C2401,C2402,C2410,C2411,C2412,C4107,C4114,C4116, C4118,C4167,C4168,C4171
105	ECCH0017501	Capacitor,Ceramic,Chip	4	C4140,C4141,C4143,C4144
106	ECCH0017601	Capacitor,Ceramic,Chip	12	C3319,C4145,C4146,C4147,C4148,C4149,C4150,C4152,C4153,C7402, C7403,C9112
107	ECZH0000830	Capacitor,Ceramic,Chip	2	C4181,C9602
108	ECZH0000849	Capacitor,Ceramic,Chip	2	C5210,C5213
109	ECZH0001121	Capacitor,Ceramic,Chip	1	C4113
110	ECZH0001215	Capacitor,Ceramic,Chip	1	C4183
111	ECZH0004402	Capacitor,Ceramic,Chip	2	C9700,C9701
112	EDTY0010101	Diode,TVS	1	ZD6110
113	ECZH0025916	Capacitor,Ceramic,Chip	3	C6921,C6923,C9110
114	EDTY0012501	Diode,TVS	1	VA6624
115	ENBY0036001	Connector,BtoB	1	CN7100
116	ELCH0001444	Inductor,Multilayer,Chip	3	FB6614,FB6615,L6611
117	ENBY0040701	Connector,BtoB	2	CN7300,CN7301
118	ERHY0000140	Resistor,Chip	1	R9109
119	ERHY0003301	Resistor,Chip	3	R9711,R9712,R9713
120	ERHY0009316	Resistor,Chip	7	R1000,R1001,R1004,R1005,R1007,R1015,R5501
121	ERHY0009501	Resistor,Chip	9	C6500,R4109,R4112,R4113,R4134,R6624,R7102,R71030,R9903
122	ERHY0009505	Resistor,Chip	1	R4801
123	ERHY0009506	Resistor,Chip	3	R6501,R71070,R71080
124	ERHY0009507	Resistor,Chip	7	R1002,R1003,R1006,R1021,R1023,R5502,R9612
125	ERHY0009516	Resistor,Chip	10	R2200,R2203,R2208,R2209,R2216,R2217,R2218,R2219,R2230,R2231

No	Part No	Description	Qty	Location No.
126	ERHY0009526	Resistor,Chip	2	R1105,R1106
127	ERHY0009536	Resistor,Chip	1	R4111
128	ERHZ0000222	Resistor,Chip	1	R9107
129	ERHZ0000235	Resistor,Chip	1	R2100
130	ERHZ0000278	Resistor,Chip	1	R2104
131	ERHZ0000350	Resistor,Chip	1	R4120
132	ERHZ0000401	Resistor,Chip	1	R9613
133	ERHZ0000405	Resistor,Chip	1	R9601
134	ERHZ0000411	Resistor,Chip	1	R9113
135	ERHZ0000485	Resistor,Chip	1	R9112
136	ERHZ0000486	Resistor,Chip	1	R9201
137	MBK64713001	Can,Shield	1	SC001
138	SAFP0000401	Wire Pad,Short	7	R1024,R1100,R1116,R41011,R9102,R9110,R9800
139	SAFP0000501	Wire Pad,Short	1	R4100
140	SEVY0003601	Varistor	2	VA6627,VA6931
141	SEVY0005402	Varistor	7	VA7200,VA7202,VA9102,VA9203,VA9710,VA9712,VA9713
142	SEVY0008102	Varistor	3	VA6932,VA7300,VA7301
143	SFBH0007101	Filter,Bead	1	FB4130
144	SFBH0008105	Filter,Bead	3	FB6613,FB6920,FB6921
145	SFEY0015301	Filter,EMI/Power	1	FL9201
146	SFEY0016301	Filter,EMI/Power	1	FL7302
147	EBR82853701	PCB Assembly,Main,SMT Top	1	EBR071700
148	EAB62950801	Microphone,Condenser	2	MIC4100,MIC4101
149	EAE62282201	Capacitor,Ceramic,Chip	1	C4109,C6621,C6704
150	EAE62286801	Capacitor,Ceramic,Chip	6	C9102,C9300,C9702

No	Part No	Description	Qty	Location No.
151	EAE62502901	Capacitor,Ceramic,Chip	5	C1306,C1400,C5113,C5133,C5227
152	EAE62505701	Capacitor,Ceramic,Chip	1	C9202
153	EAE62506501	Capacitor,Ceramic,Chip	2	C5220,C5553
154	EAE62685301	Capacitor,Ceramic,Chip	1	C4707
155	EAE62762301	Capacitor,Ceramic,Chip	1	C4800,C9107
156	EAE62843001	Capacitor,Ceramic,Chip	5	C4402,C4403,C4404,C4405,C4406
157	EAE62924001	Capacitor(High Frequency),Ceramic,Chip	2	C9115,C9116
158	EAE62945801	Capacitor(High Frequency),Ceramic,Chip	2	L1301,L1307
159	EAE62946001	Capacitor(High Frequency),Ceramic,Chip	1	L1041
160	EAE62946201	Capacitor(High Frequency),Ceramic,Chip	2	L1310,L1311
161	EAE62946501	Capacitor(High Frequency),Ceramic,Chip	1	C1926
162	EAE62946701	Capacitor(High Frequency),Ceramic,Chip	1	C5501
163	EAE62946801	Capacitor(High Frequency),Ceramic,Chip	1	C1215
164	EAE62947001	Capacitor(High Frequency),Ceramic,Chip	1	C1205
165	EAE62962201	Capacitor(High Frequency),Ceramic,Chip	3	L1305,L1308,L1902
166	EAE63069301	Capacitor,Ceramic,Chip	1	C4701
167	EAE63143201	Acoustic Noise MLCC	2	C1309,C4401
168	EAE63621501	Capacitor,Ceramic,Chip	1	C5130
169	EAE63163601	Acoustic Noise MLCC	1	C1910
170	EAE63622401	Capacitor,Ceramic,Chip	1	C4700
171	EAE63702301	Acoustic Noise MLCC	1	C1402
172	EAE63724901	Capacitor,Ceramic,Chip	1	C2367
173	EAE64022001	Acoustic Noise MLCC	3	C4704,C4705,C5228
174	EAE64281601	Capacitor,TA,Polymer	3	C2377,C2378,C2379
175	EAG63652301	C-Clip	2	ANT4103,ANT4104

No	Part No	Description	Qty	Location No.
176	EAG63749801	Connector,Terminal Block	1	CN9100
177	EAG64149901	Connector,I/O	1	CN9200
178	EAG64451301	Connector,Terminal	4	CN10008,CN10009,CN10010,CN10021
179	EAG64451601	Connector,Terminal	3	CN10003,CN10005,CN10007
180	EAG64651301	C-Clip	2	ANT1000,ANT1001,ANT1003,ANT1004
181	EAG64669801	Connector,Terminal	3	CN10001,CN10004,CN10006
182	EAG64691401	C-Clip	1	CN10011
183	EAH61992701	Diode,Schottky	1	D4400
184	EAH61995401	Diode,TVS	1	ZD7400
185	EAM62471001	Filter,Bead	2	FB5100,FB5101
186	EAM62930701	Filter,Saw	2	FL1110,FL1900
187	EAM63090201	Filter,Saw	1	FL1007
188	EAM63130101	Filter,Saw	1	FL1014
189	EAM63151201	Filter,Separator,Switch	1	FL1104
190	EAM63410001	Filter,Saw	1	FL1000
191	EAM63410101	Filter,Saw	1	FL1001
192	EAM63410201	Filter,Saw	1	FL1005
193	EAM63470401	Filter,Saw	2	FL1107,FL1216
194	EAM63510201	Filter,Separator,Switch	1	U1003
195	EAM63530101	Filter,Duplexer	1	FL1203
196	EAM63670001	Filter,Saw	1	FL5503
197	EAM63790401	Filter,Separator,Switch	1	FL1105
198	EAM63830001	Filter,Duplexer	1	FL1211
199	EAM63870501	Filter,Duplexer	1	FL1212
200	EAM63871101	Filter,Duplexer	1	FL1201

No	Part No	Description	Qty	Location No.
201	EAM63910401	Filter,Duplexer	1	FL1209
202	EAM63931001	Filter,Duplexer	1	FL1205
203	EAM64072401	Filter,Duplexer	1	FL1904
204	EAN62354901	IC,Charger	1	U4700
205	EAN62357301	IC,Hall Effect Switch	1	U8700
206	EAN62698901	IC,WiFi	1	U5100
207	EAN62736101	IC,RF Amplifier	1	U1004
208	EAN62867801	IC,Power Amplifier	1	U1901
209	EAN63325801	IC,RF Amplifier	1	U1900
210	EAN63568601	IC,Digital Baseband Processor,4G	1	U2100
211	EAN63648001	IC,Proximity	1	U1
212	EAN63789701	IC,DC,DC Converter	1	U1400
213	EAN64026401	IC,NFC	1	U5200
214	EAN64030201	IC,MCP,eMMC	1	U3300
215	EAN64086901	IC,Sub PMIC	1	U4400
216	EAN64087001	IC,Power Amplifier	1	U1300
217	EAN64109201	IC,DC,DC Converter	1	U5201
218	EAN64166601	IC,RF Amplifier	1	FL1002
219	EAN64206901	IC,RF Amplifier	2	FL1003,U1202
220	EAN64246501	IC,RF Transceiver,4G	1	U1500
221	EAP61747501	Inductor,Multilayer,Chip	6	C1203,C1304,C1903,C5505,L1048,L1147
222	EAP61767701	Inductor,Multilayer,Chip	1	L1220
223	EAP61767801	Inductor,Multilayer,Chip	1	C1187
224	EAP61866401	Inductor,Multilayer,Chip	1	L1225
225	EAP61866901	Inductor,Multilayer,Chip	2	C1202,L1906

No	Part No	Description	Qty	Location No.
226	EAP62107901	Inductor,Multilayer,Chip	1	L1035
227	EAP62108001	Inductor,Multilayer,Chip	1	C1255
228	EAP62108101	Inductor,Multilayer,Chip	2	L1112
229	EAP62108201	Inductor,Multilayer,Chip	6	C1263,L1014,L1218,L1221,L1238,L1913
230	EAP62108301	Inductor,Multilayer,Chip	3	C1177
231	EAP62108401	Inductor,Multilayer,Chip	1	L1059
232	EAP62108501	Inductor,Multilayer,Chip	4	C1233,L1011,L1314,L1320
233	EAP62108601	Inductor,Multilayer,Chip	1	C1178
234	EAP62186301	Inductor,Multilayer,Chip	3	C1077
235	EAP62225901	Inductor,Multilayer,Chip	3	L1033,L1034
236	EAP62226101	Inductor,Multilayer,Chip	2	C1264,L1222
237	EAP62226201	Inductor,Multilayer,Chip	2	L1001
238	EAP62226401	Inductor,Multilayer,Chip	5	L1008
239	EAP62226301	Inductor,Multilayer,Chip	4	C1068,C1098,L1203,L1206
240	EAP62226501	Inductor,Multilayer,Chip	1	C1028
241	EAP62226601	Inductor,Multilayer,Chip	4	L1040,L1064,R2212,R2213
242	EAP62226801	Inductor,Multilayer,Chip	1	L1003
243	EAP62227101	Inductor,Multilayer,Chip	4	C1063,C1074,C1303,L1031
244	EAP62227501	Inductor,Multilayer,Chip	1	L1139
245	EAP62227601	Inductor,Multilayer,Chip	2	L1205,L1217
246	EAP62227701	Inductor,Multilayer,Chip	2	L1232,L1233
247	EAP62245901	Inductor,Multilayer,Chip	1	L1050
248	EAX66927301	PCB,Main	1	EAX010000
249	EAP62246101	Inductor,Multilayer,Chip	1	L1023
250	EAP62246201	Inductor,Multilayer,Chip	1	L1060

No	Part No	Description	Qty	Location No.
251	EAP62266401	Inductor,Multilayer,Chip	2	C1300,C1917
252	EAP62526401	Inductor,Multilayer,Chip	1	C1048
253	EAP62546201	Inductor,Multilayer,Chip	1	C1250
254	EAP62666001	Inductor,Wire Wound,Chip	1	L4132,L4133
255	EAP62886101	Inductor,Multilayer,Chip	2	L1056
256	EAP62906101	Inductor,Multilayer,Chip	3	C1257,C5503,L1032
257	EAP63146401	Inductor,Wire Wound,Chip	1	L5202
258	EAP63185901	Inductor,Multilayer,Chip	1	L4401
259	EAP63485901	Inductor,Multilayer,Chip	1	C1008
260	EAP63486401	Inductor,Multilayer,Chip	2	L5200,L5201
261	EAP63506101	Inductor,Wire Wound,Chip	1	L4400
262	EAP63526801	Inductor,Multilayer,Chip	1	L1400
263	EAP63566101	Inductor,Wire Wound,Chip	1	L4700
264	EAT61914101	Module, FEM(Front End Module)	2	U1200,U1204
265	EAT63193901	Module, FEM(Front End Module)	1	U5502
266	EAV61915501	LED,Chip	1	LD4100
267	EAW61644701	Resonator,Ceramic	1	X5200
268	EBC62236101	Resistor,Chip	2	R5201,R5203
269	EBC62581901	Resistor,Chip	2	R2102,R2103
270	EBG62446301	Thermistor,NTC	1	PT1100
271	EBK61691601	FET	2	Q4122,Q5200
272	ECA30240001	Coupler,RF Directional	2	FL1116,FL1117
273	ECCH0000112	Capacitor,Ceramic,Chip	2	C5205,C5206
274	ECCH0000122	Capacitor,Ceramic,Chip	1	C5110
275	ECCH0000143	Capacitor,Ceramic,Chip	2	C4220,C4221

No	Part No	Description	Qty	Location No.
276	ECCH0000161	Capacitor,Ceramic,Chip	1	C4703
277	ECCH0000179	Capacitor,Ceramic,Chip	1	C5105
278	ECCH0000198	Capacitor,Ceramic,Chip	1	C3303,C4170,C7400,C7401
279	ECCH0001002	Capacitor,Ceramic,Chip	2	C5208,C5216
280	ECCH0002001	Capacitor,Ceramic,Chip	12	C3304,C7500,C7501,C8101,C8102
281	ECCH0004904	Capacitor,Ceramic,Chip	5	C4151,C4175,C4182,C7100,C7102,C71500,C71510,C7300,C7301,C7302,C7303,C7314,C7316,C9603,C9604
282	ECCH0007803	Capacitor,Ceramic,Chip	1	C4709
283	ECCH0007804	Capacitor,Ceramic,Chip	1	C4131,C4165
284	ECCH0009101	Capacitor,Ceramic,Chip	35	C4108,C4120,C4135,C4172,C6920,C6922,C9113,C9114
285	ECCH0009103	Capacitor,Ceramic,Chip	50	C1005,C1006,C1007,C1009,C1010,C1011,C1019,C1021,C1026,C1027,C1046,C1047,C1061,C1075,C1076,C1085,C1086,C1096,C1097,C5148,C6625,C9106,C9301,L1036,L1113,L5100,L5101
286	ECCH0009104	Capacitor,Ceramic,Chip	8	C1115,C1116
287	ECCH0009105	Capacitor,Ceramic,Chip	1	C7103,C7105,C7106,C7107,C7108
288	ECCH0009106	Capacitor,Ceramic,Chip	8	C1112
289	ECCH0009107	Capacitor,Ceramic,Chip	4	C2363,C2371,C2374,C2375,C2376,C2398
290	ECCH0009110	Capacitor,Ceramic,Chip	2	C2370
291	ECCH0009201	Capacitor,Ceramic,Chip	3	C2317,C2320,C2343
292	ECCH0009203	Capacitor,Ceramic,Chip	1	C4121
293	ECCH0009208	Capacitor,Ceramic,Chip	3	L1026,L1028,L5504
294	ECCH0009209	Capacitor,Ceramic,Chip	1	C5198
295	ECCH0009216	Capacitor,Ceramic,Chip	6	C1024,C1040,C1058,C1246,C1253,C1930
296	ECCH0009228	Capacitor,Ceramic,Chip	4	C2368,C2394,C2396
297	ECCH0009502	Capacitor,Ceramic,Chip	1	L1037
298	ECCH0017301	Capacitor,Ceramic,Chip	18	C2372,C2400,C2401,C2402,C2410,C2411,C2412,C4107,C4114,C4116,C4118,C4167,C4168,C4171
299	ECCH0017501	Capacitor,Ceramic,Chip	2	C4140,C4141,C4143,C4144
300	ECCH0017601	Capacitor,Ceramic,Chip	6	C3319,C4145,C4146,C4147,C4148,C4149,C4150,C4152,C4153,C7402,C7403,C9112

No	Part No	Description	Qty	Location No.
301	ECCH0034801	Capacitor,Ceramic,Chip	1	C1501
302	ECZH0001122	Capacitor,Ceramic,Chip	2	C5209,C5212
303	ECZH0001216	Capacitor,Ceramic,Chip	1	C3301
304	ECZH0001217	Capacitor,Ceramic,Chip	5	C3323,C5100,C5101,C5200,C5217
305	ECZH0001421	Capacitor,Ceramic,Chip	1	C2360
306	ECZH0025908	Capacitor,Ceramic,Chip	1	C1229
307	ECZH0025911	Capacitor,Ceramic,Chip	3	C1226,C1311,C1913
308	ECZH0025920	Capacitor,Ceramic,Chip	5	C1032,C1079,C1307,C1310,C5126
309	ECZH0025916	Capacitor,Ceramic,Chip	18	C6921,C6923,C9110
310	EDSY0010501	Diode,Switching	1	D9910
311	ELCH0001425	Inductor,Multilayer,Chip	2	L5102,L5106
312	ELCH0001431	Inductor,Multilayer,Chip	1	L5105
313	ELCH0003842	Inductor,Multilayer,Chip	2	L9910,L9911
314	ERHY0000254	Resistor,Chip	1	R8400
315	ERHY0000140	Resistor,Chip	1	R9109
316	ERHY0009311	Resistor,Chip	1	R1111
317	ERHY0009501	Resistor,Chip	12	C6500,R4109,R4112,R4113,R4134,R6624,R7102,R71030,R9903
318	ERHY0009505	Resistor,Chip	4	R4801
319	ERHY0009506	Resistor,Chip	5	R6501,R71070,R71080
320	ERHY0009516	Resistor,Chip	3	R2200,R2203,R2208,R2209,R2216,R2217,R2218,R2219,R2230,R2231
321	ERHY0009526	Resistor,Chip	3	R1105,R1106
322	ERHZ0000250	Resistor,Chip	1	R3301
323	ERHZ0000441	Resistor,Chip	1	R8401
324	ERHZ0000407	Resistor,Chip	1	R5200
325	ERHZ0000412	Resistor,Chip	1	R4901

No	Part No	Description	Qty	Location No.
326	MBV62321701	Clip	12	SC10001,SC10002,SC10003,SC10004,SC10005,SC10006,SC10007,SC10008,SC10009,SC10010,SC10011,SC10012
327	MEZ65049701	Label	1	PID1
328	SAFP0000401	Wire Pad,Short	7	R1024,R1100,R1116,R41011,R9102,R9110,R9800
329	SAFP0000501	Wire Pad,Short	1	R4100
330	SEVY0008102	Varistor	3	VA6932,VA7300,VA7301
331	SAD35902701	Software,Mobile	1	SAD010000
332	RAA34548901	Resin,PC	0	RAA050100
333	ACQ88777241	Cover Assembly,Bar	1	ACQ003400
334	ACQ88758801	Cover Assembly	1	ACQ00
335	MBL66958401	Cap	1	MBL00
336	ACQ88729801	Cover Assembly,Front	1	ACQ032700
337	MHK65488401	Sheet	1	MHK000000
338	MJN69990401	Tape,Protect	1	MJN061100
339	MJN69930601	Tape	1	MJN000000
340	MJN69930701	Tape	1	MJN000001
341	ACQ88749501	Cover Assembly,Front(Sub)	1	ACQ033200
342	MCK69187901	Cover,Front	1	MCK032700
343	MICE0016905	Insert,Nut	4	MET099500
344	ADV75909401	Frame Assembly	1	ADV000000
345	MDQ64916401	Frame	1	MDQ000000
346	MCK69167801	Cover,Front	1	MCK032700
347	MJN69931001	Tape,Protect	2	MJN061102
348	MJN69930901	Tape,Protect	1	MJN061101
349	MHK65488501	Sheet	1	MHK000001
350	MBL66958501	Cap	1	MBL01

No	Part No	Description	Qty	Location No.
351	MCR66527501	Decor	1	MCR000000
352	MCQ68807501	Damper	1	MCQ000000
353	MDS65633601	Gasket	1	MDS03
354	MBL66937701	Cap	1	MBL02
355	MFG63938601	Locker	1	MFG00
356	MBL66897901	Cap	1	MBL03
357	EAX66949501	PCB,Flexible	1	EAX00
358	MJN69969401	Tape	1	MJN00
359	MDS65652701	Gasket	1	MDS02
360	EAT62793603	Module,Hybrid Touch LCD	1	EAT130000
361	MEV65917201	Insulator	1	MEV02
362	MJN69990201	Tape,Protect	1	MJN061100
363	MDS65613101	Gasket	1	MDS00
364	EAB64148501	Receiver	1	EAB010400
365	EAU62283801	Motor,DC	1	EAU010000
366	EBP62702301	Camera Module	1	EBP01
367	EBP62722101	Camera Module	1	EBP00
368	MLAZ0038301	Label	1	MEZ000000
369	EAG63849801	Jack,Phone	1	EAG00
370	MJN70075101	Tape,USP Film	1	MJN107400
371	ACQ88721171	Cover Assembly,Rear	1	ACQ01
372	ACQ88758771	Cover Assembly,Rear(SVC)	1	ACQ105800
373	ABH75799701	Button Assembly	1	ABH00
374	MBG66084301	Button	1	MBG000001
375	MBG66144901	Button	1	MBG000000

No	Part No	Description	Qty	Location No.
376	MJN69969501	Tape,Protect	1	MJN061100
377	EAA64445501	PIFA Antenna,GPS	1	EAA030100
378	EAA64445601	PIFA Antenna,WiFi	1	EAA030101
379	EAA64446001	PIFA Antenna,RF	1	EAA030102
380	EAA64486201	PIFA Antenna,RF	1	EAX010700
381	MCK69209001	Cover,Rear	1	MCK063300
382	MCQ68787101	Damper,Camera	1	MCQ009401
383	MCQ68826101	Damper,Motor	1	MCQ049800
384	MCQ68826201	Damper	1	MCQ02
385	MFB64233701	Lens,Flash	1	MFB029600
386	ACW75277611	Decor Assembly,SVC	1	ACW00
387	ACW75217701	Decor Assembly	1	ACW000000
388	MCR66508501	Decor	1	MCR000001
389	MCR66527401	Decor	1	MCR00
390	MKC65999801	Window,Camera	1	MKC00
391	MCQ68904901	Damper,Camera	1	MCQ03
392	MJN69990301	Tape,Window	1	MJN01
393	MJN69969601	Tape,Protect	1	MJN061100
394	MDS65694201	Gasket	1	MDS01
395	MEZ64319901	Label,After Service	1	MEZ000900
396	EAB64229101	Speaker,Dual Mode	1	EAB00
397	MCQ68766701	Damper,Speaker	1	MCQ01
398	MCR66510601	Decor	1	MCR000000
399	EAA64426201	PIFA Antenna,Multiple	1	EAA00
400	MEV65917301	Insulator	1	MEV01

No	Part No	Description	Qty	Location No.
401	MEV65917401	Insulator	1	MEV00
402	MCQ68807602	Damper,Speaker	1	MCQ00
403	AGN72929001	Pen Assembly,Stylus	1	AGN00
404	FAB32218901	Screw,Tapping	0	FAB01
405	GMEY0014301	Screw,Machine	11	FAB00
406	MEZ66329501	Label,Approval	1	MEZ002100
407	AAD87878101	Addition Assembly	1	AAD000000
408	EAB64269001	Earphone,Stereo	1	EAB010200
409	EAD62377921	Cable,Assembly	1	EAD010000
410	EAY62769015	Adapters	1	EAY060000
411	ACQ88779381	Cover Assembly,Battery	1	ACQ02
412	MCK69208981	Cover,Battery	1	MCK004101
413	MHK65589301	Sheet	1	MHK000001
414	MBM65425901	Card,Quick Reference	1	MBM062600
415	MFL69478101	Manual,Operation	0	MFL053800
416	EAC63158401	Rechargeable Battery,Lithium Ion	1	EAC00
417	AGF78327405	Package Assembly	1	AGF000000
418	AGJ74478301	Pallet Assembly	1	AGJ000000
419	MAY67516701	Box,Carton	0	MAY010800
420	MAY67539301	Box,Pallet Sleeve	0	MAY115200
421	MBL67000201	Cap,Box	0	MBL007000
422	MLAJ0004402	Label,Master Box	0	MEZ047200
423	MLAZ0050901	Label	0	MEZ000000
424	MGA64059201	Pallet	0	MGA000000
425	MAY67556501	Box,Master	0	MAY047100

No	Part No	Description	Qty	Location No.
426	MAY67554507	Box,Unit	1	MAY084001
427	MEZ65773003	Label,Unit Box	1	MEZ084101
428	MLAJ0004402	Label,Master Box	0	MEZ047200
429	MLAP0001138	Label,Unit	1	MEZ084000
430	MAF63267408	Bag,Vinyl	1	MAF086500

P/N : FLXXXXXXXX(X.X)